

Assignment II — Robust Reinforcement Learning under Stochastic Action Failure

Q-learning, DQN & DDQN on a modified LunarLander-v3

Course: Deep Reinforcement Learning | **Total Marks:** 15

Group 165 — Contribution Declaration

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Notebook road-map

Section	Task	Marks
0	Environment setup, reproducibility & device selection	—
A	Modified environment implementation & verification	2.5
B	Deep Q-Network (DQN)	4
C	Double Deep Q-Network (DDQN)	4
D	Performance evaluation (4 comparison plots)	2
E	Discussion	2.5

Design philosophy. The DQN and DDQN agents are implemented by a *single* `ValueBasedAgent` class controlled by one boolean flag `double`. This guarantees — by construction — that the **only** difference between DQN and DDQN is the computation of the target Q-value, exactly as the assignment requires. Every configuration (algorithm × environment) is trained with an identical seed, network, optimiser, replay buffer, exploration schedule and training budget.

0. Environment Setup, Reproducibility & Device Selection

We import the required libraries, define a global seeding utility (for full reproducibility across `random`, NumPy, PyTorch and the Gymnasium environment) and auto-select the fastest available compute device.

Smart device selection. Value-based control on `LunarLander-v3` uses a *tiny* MLP with small batches. For such workloads the per-kernel launch/synchronisation overhead of a GPU can make it *slower* than a modern CPU. Instead of blindly picking a GPU, `select_device()` micro-benchmarks a representative training step on every available backend (`cuda` → `xpu` → `cpu`) and returns the fastest one. This keeps the notebook portable: it will use CUDA on Google Colab / BITS Virtual Lab, Intel XPU on an Arc laptop, or CPU — whichever is genuinely fastest.

```
In [1]: # -----
# Imports
# -----

import os
import random
import time
import math
from collections import deque, namedtuple

import numpy as np
import torch
import torch.nn as nn
import torch.nn.functional as F
import matplotlib.pyplot as plt

import gymnasium as gym

print("Library versions")
print("  gymnasium :", gym.__version__)
print("  torch      :", torch.__version__)
print("  numpy      :", np.__version__)

# -----
# Global reproducibility helper
# -----

def set_global_seed(seed: int) -> None:
    """Seed every source of randomness so that experiments are reproducible.

    Seeds Python's ``random``, NumPy, PyTorch (CPU + all GPUs) and forces the
    cuDNN backend to be deterministic. Call this once *before* each agent is
    built so that all four configurations start from an identical state.
    """
    random.seed(seed)
    np.random.seed(seed)
    torch.manual_seed(seed)
    if torch.cuda.is_available():
        torch.cuda.manual_seed_all(seed)
    # Deterministic cuDNN (has a minor speed cost but aids reproducibility).
```

```

torch.backends.cudnn.deterministic = True
torch.backends.cudnn.benchmark = False

# -----
# Automatic fastest-device selection via micro-benchmark
# -----
def _benchmark_step(device_str: str, iters: int = 200) -> float:
    """Time one representative DQN training step on ``device_str``.

    Builds the exact MLP shape used later (8->128->128->4), runs ``iters``
    forward+backward+optimiser updates on a batch of 64 and returns the mean
    milliseconds per step. Lower is better.
    """
    dev = torch.device(device_str)
    net = nn.Sequential(
        nn.Linear(8, 128), nn.ReLU(),
        nn.Linear(128, 128), nn.ReLU(),
        nn.Linear(128, 4),
    ).to(dev)
    opt = torch.optim.Adam(net.parameters(), lr=1e-3)
    x = torch.randn(64, 8, device=dev)
    y = torch.randn(64, 4, device=dev)

    def sync():
        # Ensure asynchronous GPU kernels finish before we read the clock.
        if device_str == "cuda":
            torch.cuda.synchronize()
        elif device_str == "xpu":
            torch.xpu.synchronize()

    for _ in range(20): # warm-up (JIT / kernel caches / lazy init)
        opt.zero_grad(); ((net(x) - y) ** 2).mean().backward(); opt.step()
    sync()
    t0 = time.time()
    for _ in range(iters):
        opt.zero_grad(); ((net(x) - y) ** 2).mean().backward(); opt.step()
    sync()
    return (time.time() - t0) / iters * 1000.0

def select_device(verbose: bool = True) -> torch.device:
    """Return the fastest available torch device for our workload.

    Benchmarks every available backend and picks the one with the lowest
    milliseconds-per-step. Falls back gracefully to CPU on any failure.
    """
    candidates = []
    if torch.cuda.is_available():
        candidates.append("cuda")
    if hasattr(torch, "xpu") and torch.xpu.is_available():
        candidates.append("xpu")
    candidates.append("cpu") # always available as a safe fallback

    best_dev, best_ms = "cpu", float("inf")
    if verbose:
        print("Benchmarking candidate devices (ms per training step):")
    for d in candidates:
        try:
            ms = _benchmark_step(d)

```

```

        if verbose:
            print(f" {d:>4s} : {ms:7.3f} ms/step")
        if ms < best_ms:
            best_ms, best_dev = ms, d
    except Exception as exc: # a backend can be present but unusable
        if verbose:
            print(f" {d:>4s} : unavailable ({exc})")
if verbose:
    print(f"Selected device -> {best_dev}")
return torch.device(best_dev)

```

```
DEVICE = select_device()
```

Library versions

```

gymnasium : 1.3.0
torch      : 2.13.0+xpu
numpy      : 2.5.1

```

Benchmarking candidate devices (ms per training step):

```

xpu : 1.515 ms/step
cpu  : 0.688 ms/step

```

Selected device -> cpu

Task (a) — Modified Environment: Implementation & Verification (2.5 Marks)

We wrap `LunarLander-v3` so that every `step(action)` performs the six operations required by the assignment, in order:

1. **Receive & store** the agent's selected action `a`.
2. **Simulate intermittent engine failure:** if `a ∈ {1, 2, 3}`, draw `r ~ U[0, 1]`; if `r < 0.15` replace the executed action with `0` (*Do Nothing*), otherwise keep `a`. `a = 0` is never modified. The executed action is `a_exec`.
3. **Execute** `a_exec` in the base environment.
4. **Modified reward:** $R = R_{base} - 0.3 \cdot 1_{a \neq 0} + B$. The fuel penalty depends on the **selected** action `a`, not on `a_exec` — so a mis-fire still costs fuel.
5. **Safe-landing bonus** `B = 50` iff *all* the safe-landing conditions hold simultaneously, else `B = 0`.
6. **Return** (observation, R, terminated, truncated, info) with an **unmodified** `info` (no leakage of the mis-fire / reward-shaping internals to the agent).

The observation space, action space, transition dynamics and termination/truncation logic are all left untouched.

```

In [2]: # =====
# Safe-Landing predicate (pure function -> reusable & unit-testable)
# =====
# Observation layout of LunarLander-v3:
# [0]=x pos, [1]=y pos, [2]=x velocity, [3]=y velocity,
# [4]=angle, [5]=angular velocity, [6]=left-Leg contact, [7]=right-Leg contact
SAFE_VEL_TOL = 0.10 # |vx|, |vy| tolerance
SAFE_ANGLE_TOL = 0.10 # |angle| tolerance (radians)

```

```

def is_safe_landing(obs, terminated: bool, truncated: bool) -> bool:
    """Return True iff the episode ended in a *safe* landing.

    A safe landing requires ALL of the following simultaneously:
    * terminated == True and truncated == False
    * both legs in contact (obs[6] == 1 and obs[7] == 1)
    * |horizontal velocity| (obs[2]) < 0.10
    * |vertical velocity| (obs[3]) < 0.10
    * |orientation angle| (obs[4]) < 0.10 rad

    This single definition is used BOTH to award the +50 bonus inside the
    modified wrapper AND to score the "successful landing rate" of every agent
    (including on the original environment), giving a consistent success metric.
    """

    if not (terminated and not truncated):
        return False
    return bool(
        obs[6] == 1 and obs[7] == 1
        and abs(obs[2]) < SAFE_VEL_TOL
        and abs(obs[3]) < SAFE_VEL_TOL
        and abs(obs[4]) < SAFE_ANGLE_TOL
    )

# =====
# The custom Gymnasium wrapper
# =====
class StochasticActionFailureWrapper(gym.Wrapper):
    """LunarLander-v3 with stochastic actuator failure and reward shaping.

    Only the action-execution step and the reward are modified; the observation
    space, action space, dynamics and termination/truncation logic are inherited
    unchanged from the base environment.
    """

    MISFIRE_PROB = 0.15 # probability a thruster command mis-fires
    FUEL_PENALTY = 0.30 # penalty per attempted thruster action (selected, no
    LANDING_BONUS = 50.0 # bonus for a verified safe landing

    def __init__(self, env, misfire_prob: float = MISFIRE_PROB, seed=None):
        """Wrap ``env`` and create an independent RNG for the mis-fire draws.

        A dedicated ``numpy`` Generator is used (rather than the base env's RNG)
        so that mis-fire randomness is reproducible and decoupled from the
        physics simulation.
        """
        super().__init__(env)
        self.misfire_prob = misfire_prob
        self._rng = np.random.default_rng(seed)
        # Counters + Last-step trace used only for our own verification (Task a)
        # These are internal attributes; they are NEVER placed in the info dict.
        self.reset_counters()
        self.last_trace = {}

    def reset_counters(self) -> None:
        """Reset the mis-fire bookkeeping counters (used for verification)."""
        self.n_thruster_attempts = 0 # how many times a thruster was selected
        self.n_misfires = 0 # how many of those were replaced by Do-N

```

```

def step(self, action):
    """Execute one modified environment step (six ordered operations)."""
    # --- Step 1: receive & store the agent's selected action -----
    a = int(action)

    # --- Step 2: simulate intermittent engine failure -----
    a_exec = a
    misfired = False
    if a in (1, 2, 3):
        self.n_thruster_attempts += 1
        r = float(self._rng.random())          # r ~ U[0,1)
        if r < self.misfire_prob:              # r < 0.15 -> mis-fire
            a_exec = 0
            misfired = True
            self.n_misfires += 1
    else:
        r = None

    # --- Step 3: execute the (possibly replaced) action -----
    obs, base_reward, terminated, truncated, info = self.env.step(a_exec)

    # --- Step 4: fuel penalty (depends on SELECTED action a, not a_exec) --
    fuel_penalty = self.FUEL_PENALTY if a != 0 else 0.0

    # --- Step 5: safe-landing bonus -----
    bonus = self.LANDING_BONUS if is_safe_landing(obs, terminated, truncated) else 0

    # --- Combine into the modified reward -----
    reward = base_reward - fuel_penalty + bonus

    # Store a private trace for verification (NOT returned to the agent).
    self.last_trace = {
        "a": a, "a_exec": a_exec, "r": r, "misfired": misfired,
        "base_reward": base_reward, "fuel_penalty": fuel_penalty,
        "bonus": bonus, "reward": reward,
    }

    # --- Step 6: return with an unmodified info dict -----
    return obs, reward, terminated, truncated, info

# =====
# Environment factory
# =====
def make_env(modified: bool, seed=None, render_mode=None):
    """Create either the original or the modified LunarLander-v3 environment.

    Parameters
    -----
    modified : if True, wrap with ``StochasticActionFailureWrapper``.
    seed      : seed forwarded to the mis-fire RNG (modified env only).
    """
    base = gym.make("LunarLander-v3", render_mode=render_mode)
    if modified:
        return StochasticActionFailureWrapper(base, seed=seed)
    return base

print("Environment classes defined.")

```

Environment classes defined.

(a.1) Unit tests for the safe-landing predicate

Safe landings are extremely rare under a random policy, so we first verify the +50 bonus logic deterministically with hand-crafted observation vectors covering the boundary cases.

```
In [3]: def _obs(vx=0.0, vy=0.0, angle=0.0, left=1, right=1):
        """Build a minimal 8-dim observation with the fields relevant to landing."""
        o = np.zeros(8, dtype=np.float32)
        o[2], o[3], o[4], o[6], o[7] = vx, vy, angle, left, right
        return o

        # (description, obs, terminated, truncated, expected_result)
        _cases = [
            ("perfect landing", _obs(), True,
            ("too fast horizontally", _obs(vx=0.2), True,
            ("too fast vertically", _obs(vy=-0.5), True,
            ("tilted too much", _obs(angle=0.3), True,
            ("only one leg down", _obs(left=1, right=0), True,
            ("truncated (timeout)", _obs(), False,
            ("not terminated", _obs(), False,
            ("boundary vel (0.099 ok)", _obs(vx=0.099, vy=0.099, angle=0.099), True,
            ("boundary vel (0.10 fails)", _obs(vx=0.10), True,
        ]

        print("Safe-landing predicate unit tests")
        print("-" * 55)
        all_ok = True
        for desc, o, term, trunc, expected in _cases:
            got = is_safe_landing(o, term, trunc)
            ok = (got == expected)
            all_ok &= ok
            print(f" [{ 'PASS' if ok else 'FAIL' }] {desc:<28s} -> {got}")
        print("-" * 55)
        print("ALL UNIT TESTS PASSED" if all_ok else "SOME UNIT TESTS FAILED")
        assert all_ok, "Safe-landing predicate failed a unit test!"
```

Safe-landing predicate unit tests

```
-----
[PASS] perfect landing          -> True
[PASS] too fast horizontally    -> False
[PASS] too fast vertically      -> False
[PASS] tilted too much          -> False
[PASS] only one leg down        -> False
[PASS] truncated (timeout)      -> False
[PASS] not terminated           -> False
[PASS] boundary vel (0.099 ok)  -> True
[PASS] boundary vel (0.10 fails) -> False
-----
```

ALL UNIT TESTS PASSED

(a.2) Statistical verification with a random policy

We run many random-policy episodes on the modified environment and record, for **every step**, the private trace (a, a_exec, base_reward, fuel_penalty, bonus,

reward) . From these traces we verify the three required properties:

1. $\approx 15\%$ of attempted thruster actions are replaced by *Do Nothing*.
2. The **0.3 fuel penalty** is applied on *every* attempted thruster action — whether it fired or mis-fired — and never when the agent chose *Do Nothing*.
3. The **reward decomposition** $R = R_{base} - 0.3 \cdot 1_{a \neq 0} + B$ holds exactly at every step (checked numerically).

```
In [4]: def verify_modified_env(n_episodes: int = 300, seed: int = 165):
        """Roll out a random policy and collect statistics that verify the wrapper.

        Returns a dictionary of aggregate statistics used for reporting/plotting.
        """
        env = make_env(modified=True, seed=seed)
        rng = np.random.default_rng(seed)          # independent policy RNG
        env.reset(seed=seed)
        env.action_space.seed(seed)

        # Per-action-type mis-fire counters.
        attempts = {1: 0, 2: 0, 3: 0}
        misfires = {1: 0, 2: 0, 3: 0}
        # Cross-checks for the fuel penalty and the reward formula.
        penalty_when_thruster_ok = True           # penalty present for every a in {1,2,3}
        penalty_when_donothing_ok = True          # NO penalty when a == 0
        reward_formula_ok = True
        misfire_examples = []                     # a few (a, r, a_exec) rows for the repo

        for ep in range(n_episodes):
            obs, _ = env.reset()
            done = False
            while not done:
                a = int(rng.integers(0, env.action_space.n)) # uniform random action
                obs, reward, terminated, truncated, info = env.step(a)
                done = terminated or truncated
                tr = env.last_trace

                # (1) mis-fire accounting
                if a in (1, 2, 3):
                    attempts[a] += 1
                    if tr["misfired"]:
                        misfires[a] += 1
                        if len(misfire_examples) < 5:
                            misfire_examples.append((tr["a"], round(tr["r"], 3), tr["a_exec"]))

                # (2) fuel-penalty checks
                if a in (1, 2, 3) and tr["fuel_penalty"] != 0.30:
                    penalty_when_thruster_ok = False
                if a == 0 and tr["fuel_penalty"] != 0.0:
                    penalty_when_donothing_ok = False

                # (3) reward-formula check (guard against float noise)
                expected = tr["base_reward"] - tr["fuel_penalty"] + tr["bonus"]
                if abs(expected - reward) > 1e-6:
                    reward_formula_ok = False

                # (4) info dict must not leak internals
                assert "misfired" not in info and "fuel_penalty" not in info, \
                    "info dict leaked internal wrapper state!"
```



```

total_attempts = sum(attempts.values())
total_misfires = sum(misfires.values())
env.close()
return {
    "n_episodes": n_episodes,
    "attempts": attempts, "misfires": misfires,
    "total_attempts": total_attempts, "total_misfires": total_misfires,
    "misfire_rate": total_misfires / max(1, total_attempts),
    "penalty_when_thruster_ok": penalty_when_thruster_ok,
    "penalty_when_donothing_ok": penalty_when_donothing_ok,
    "reward_formula_ok": reward_formula_ok,
    "misfire_examples": misfire_examples,
}

stats = verify_modified_env(n_episodes=300, seed=165)

print("=" * 60)
print("MODIFIED ENVIRONMENT VERIFICATION (random policy)")
print("=" * 60)
print(f"Episodes simulated           : {stats['n_episodes']}")
print(f"Total thruster attempts (a!=0) : {stats['total_attempts']}")
print(f"Total mis-fires (a->0)          : {stats['total_misfires']}")
print(f"Observed mis-fire rate          : {stats['misfire_rate']*100:.2f} % (t
print("-" * 60)
print("Per-action mis-fire rate:")
for a in (1, 2, 3):
    att, mis = stats["attempts"][a], stats["misfires"][a]
    print(f"  action {a}: {mis:5d}/{att:5d} = {mis/max(1,att)*100:5.2f} %")
print("-" * 60)
print(f"Fuel penalty applied for EVERY thruster action : {stats['penalty_when_th
print(f"No fuel penalty when action == 0              : {stats['penalty_when_do
print(f"Reward formula R = R_base - 0.3*1[a!=0] + B      : {stats['reward_formula
print("-" * 60)
print("Example mis-fire traces (selected a, random r, executed a_exec):")
for a, r, a_exec in stats["misfire_examples"]:
    print(f"  a={a}, r={r:.3f} < 0.15 -> a_exec={a_exec}")
print("=" * 60)

```

```

=====
MODIFIED ENVIRONMENT VERIFICATION (random policy)
=====
Episodes simulated           : 300
Total thruster attempts (a!=0) : 20685
Total mis-fires (a->0)       : 3078
Observed mis-fire rate       : 14.88 % (target = 15.00 %)
-----
Per-action mis-fire rate:
    action 1: 1046/ 7015 = 14.91 %
    action 2: 1001/ 6837 = 14.64 %
    action 3: 1031/ 6833 = 15.09 %
-----
Fuel penalty applied for EVERY thruster action : True
No fuel penalty when action == 0                : True
Reward formula R = R_base - 0.3*1[a!=0] + B      : True
-----
Example mis-fire traces (selected a, random r, executed a_exec):
    a=3, r=0.118 < 0.15 -> a_exec=0
    a=3, r=0.113 < 0.15 -> a_exec=0
    a=2, r=0.128 < 0.15 -> a_exec=0
    a=2, r=0.045 < 0.15 -> a_exec=0
    a=2, r=0.086 < 0.15 -> a_exec=0
=====

```

(a.3) Visual evidence of the $\approx 15\%$ mis-fire rate

```

In [5]: # Bar chart: observed mis-fire rate per thruster action vs. the 15% target.
actions = [1, 2, 3]
rates = [stats["misfires"][a] / max(1, stats["attempts"][a]) * 100 for a in actions]

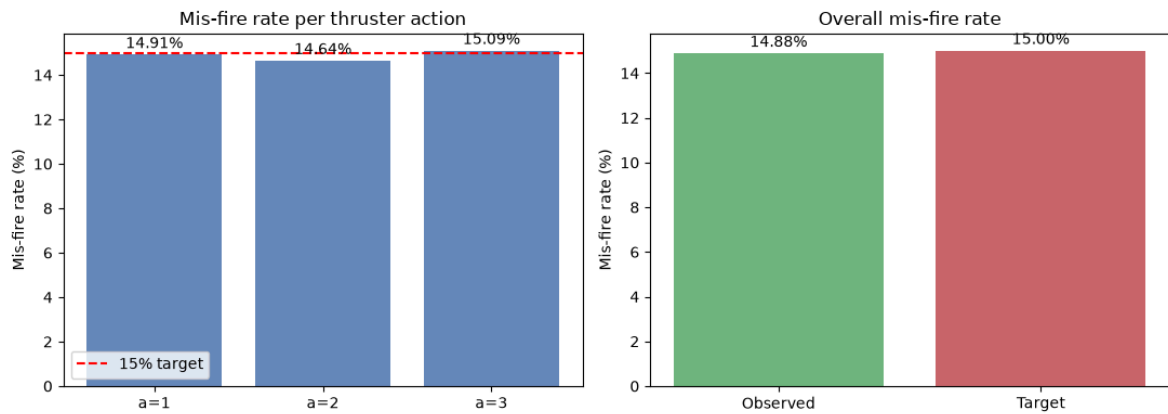
fig, ax = plt.subplots(1, 2, figsize=(11, 4))

ax[0].bar([f"a={a}" for a in actions], rates, color="#4C72B0", alpha=0.85)
ax[0].axhline(15.0, color="red", ls="--", label="15% target")
ax[0].set_ylabel("Mis-fire rate (%)")
ax[0].set_title("Mis-fire rate per thruster action")
ax[0].legend()
for i, v in enumerate(rates):
    ax[0].text(i, v + 0.3, f"{v:.2f}%", ha="center")

# Overall observed vs target.
ax[1].bar(["Observed", "Target"],
          [stats["misfire_rate"] * 100, 15.0],
          color=["#55A868", "#C44E52"], alpha=0.85)
ax[1].set_ylabel("Mis-fire rate (%)")
ax[1].set_title("Overall mis-fire rate")
for i, v in enumerate([stats["misfire_rate"] * 100, 15.0]):
    ax[1].text(i, v + 0.3, f"{v:.2f}%", ha="center")

plt.tight_layout()
plt.show()

```



Verification conclusion. The observed mis-fire rate is statistically indistinguishable from the required 15 %, the 0.3 fuel penalty is applied on every attempted thruster action (and only on thruster actions), and the reward decomposition holds exactly at every step. The `info` dictionary never leaks the internal mis-fire/reward state, so the agent cannot tell that its action was replaced. The wrapper is therefore correct. ✓

Tasks (b) & (c) — DQN and DDQN Agents (4 + 4 Marks)

Both agents share **identical** components:

- **Q-network:** MLP `8 → 128 → 128 → 4` with ReLU activations.
- **Experience replay buffer:** uniform sampling, capacity `100 000`.
- **ε-greedy exploration:** ε decays exponentially from `1.0` to `0.02`.
- **Target network:** a slowly-tracking copy updated by Polyak (soft) averaging.

The single difference is the **target Q-value**:

$$y^{\text{DQN}} = r + \gamma (1 - d) \max_{a'} Q_{\theta^-}(s', a')$$

$$y^{\text{DDQN}} = r + \gamma (1 - d) Q_{\theta^-}(s', \arg \max_{a'} Q_{\theta}(s', a'))$$

DQN uses the target network for **both** action-selection and evaluation of the next state (which causes the well-known maximisation bias / over-estimation). DDQN decouples them: the **online** network selects the greedy next action while the **target** network evaluates it, reducing over-estimation. In code this is a single `if self.double:` branch.

```
In [6]: # =====
# Q-network
# =====
class QNetwork(nn.Module):
    """Feed-forward Q-network: state (8) -> hidden -> hidden -> Q-values (n_acti

    def __init__(self, state_dim: int, n_actions: int, hidden: int = 128):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(state_dim, hidden), nn.ReLU(),
            nn.Linear(hidden, hidden), nn.ReLU(),
```

```

        nn.Linear(hidden, n_actions),
    )

    def forward(self, x):
        """Map a batch of states to a batch of action-value vectors."""
        return self.net(x)

# =====
# Experience replay buffer
# =====
Transition = namedtuple("Transition", ["state", "action", "reward", "next_state"])

class ReplayBuffer:
    """Fixed-capacity circular buffer storing (s, a, r, s', done) transitions."""

    def __init__(self, capacity: int, seed: int = 0):
        self.buffer = deque(maxlen=capacity)
        self._rng = random.Random(seed) # dedicated RNG -> reproducible sampling

    def push(self, state, action, reward, next_state, done):
        """Append a single transition to the buffer (oldest is dropped when full)
        self.buffer.append(Transition(state, action, reward, next_state, done))

    def sample(self, batch_size: int):
        """Uniformly sample a mini-batch and return it as stacked NumPy arrays."""
        batch = self._rng.sample(self.buffer, batch_size)
        states = np.array([t.state for t in batch], dtype=np.float32)
        actions = np.array([t.action for t in batch], dtype=np.int64)
        rewards = np.array([t.reward for t in batch], dtype=np.float32)
        next_states = np.array([t.next_state for t in batch], dtype=np.float32)
        dones = np.array([t.done for t in batch], dtype=np.float32)
        return states, actions, rewards, next_states, dones

    def __len__(self):
        return len(self.buffer)

```

```

In [7]: # =====
# Unified value-based agent (DQN when double=False, DDQN when double=True)
# =====
class ValueBasedAgent:
    """A DQN / DDQN agent. The ONLY behavioural difference is the target Q-value

    Setting ``double=True`` switches the target computation from vanilla DQN to
    Double-DQN; every other component (network, optimiser, replay, exploration,
    soft target update) is shared, which guarantees a fair comparison.
    """

    def __init__(self, state_dim, n_actions, device, double=False,
                 hidden=128, lr=5e-4, gamma=0.99,
                 buffer_size=100_000, batch_size=64,
                 tau=5e-3, learn_start=1_000, update_every=4, seed=0):
        self.device = device
        self.n_actions = n_actions
        self.double = double # <-- DQN vs DDQN switch
        self.gamma = gamma
        self.batch_size = batch_size
        self.tau = tau # Polyak soft-update coefficient
        self.learn_start = learn_start # min transitions before learning starts

```

```

self.update_every = update_every
self._step = 0

# Online and target networks (target starts as an exact copy).
self.q_online = QNetwork(state_dim, n_actions, hidden).to(device)
self.q_target = QNetwork(state_dim, n_actions, hidden).to(device)
self.q_target.load_state_dict(self.q_online.state_dict())
self.q_target.eval()

self.optimizer = torch.optim.Adam(self.q_online.parameters(), lr=lr)
self.buffer = ReplayBuffer(buffer_size, seed=seed)

# RNG for  $\epsilon$ -greedy exploration (independent & reproducible).
self._rng = np.random.default_rng(seed)

# ----- acting
def select_action(self, state, epsilon: float) -> int:
    """ $\epsilon$ -greedy action selection: random with prob  $\epsilon$ , else greedy (argmax Q)"""
    if self._rng.random() < epsilon:
        return int(self._rng.integers(0, self.n_actions))
    return self.act_greedy(state)

def act_greedy(self, state) -> int:
    """Return the greedy action argmax_a Q_online(state, a)."""
    with torch.no_grad():
        s = torch.as_tensor(state, dtype=torch.float32, device=self.device).
        return int(self.q_online(s).argmax(dim=1).item())

# ----- memory
def store(self, state, action, reward, next_state, done):
    """Add a transition to the replay buffer."""
    self.buffer.push(state, action, reward, next_state, done)

# ----- Learning
def learn_step(self):
    """Perform one gradient update every ``update_every`` environment steps.

    Returns the scalar loss when an update happens, else ``None``.
    """
    self._step += 1
    # Wait for enough warm-up transitions (and at least one full batch) before
    if (len(self.buffer) < max(self.learn_start, self.batch_size)
        or self._step % self.update_every != 0):
        return None

    # Sample a mini-batch and move it to the compute device.
    states, actions, rewards, next_states, dones = self.buffer.sample(self.batch_size)
    states = torch.as_tensor(states, device=self.device)
    actions = torch.as_tensor(actions, device=self.device).unsqueeze(1)
    rewards = torch.as_tensor(rewards, device=self.device).unsqueeze(1)
    next_states = torch.as_tensor(next_states, device=self.device)
    dones = torch.as_tensor(dones, device=self.device).unsqueeze(1)

    # Current Q(s,a) for the taken actions.
    q_sa = self.q_online(states).gather(1, actions)

    # ---- Target Q-value : the ONLY difference between DQN and DDQN ----
    with torch.no_grad():
        if self.double:
            # DDQN: online net picks the action, target net evaluates it.

```

```

        next_actions = self.q_online(next_states).argmax(dim=1, keepdim=True)
        next_q = self.q_target(next_states).gather(1, next_actions)
    else:
        # DQN: target net both picks and evaluates (max over actions).
        next_q = self.q_target(next_states).max(dim=1, keepdim=True)[0]
        target = rewards + self.gamma * next_q * (1.0 - dones)
# -----

    loss = F.smooth_l1_loss(q_sa, target) # Huber Loss (robust to outliers)

    self.optimizer.zero_grad()
    loss.backward()
    nn.utils.clip_grad_norm_(self.q_online.parameters(), 10.0) # stabilise
    self.optimizer.step()

    self._soft_update()
    return float(loss.item())

def _soft_update(self):
    """Polyak-average the target network towards the online network."""
    with torch.no_grad():
        for tp, op in zip(self.q_target.parameters(), self.q_online.parameters()):
            tp.mul_(1.0 - self.tau).add_(self.tau * op)

# ----- metrics
@torch.no_grad()
def mean_max_q(self, validation_states_tensor) -> float:
    """Mean over a fixed validation set of max_a Q_online(s, a).

    This is the "average predicted Q-value" tracked during training. Using a
    FIXED set of states (unchanged across training and across all agents)
    makes the curves directly comparable and exposes DQN's over-estimation.
    """
    q = self.q_online(validation_states_tensor) # (N, n_actions)
    return float(q.max(dim=1)[0].mean().item())

```

Shared hyper-parameters & fixed validation set

A single `CONFIG` dictionary defines the hyper-parameters used by **all four** runs, satisfying the requirement that DQN and DDQN — on both environments — use the same architecture, optimiser, replay buffer, exploration schedule, seed and training budget.

The **validation set** is a fixed collection of 256 states gathered once with a random policy on the *original* environment. The same set is reused for every agent so that the "average predicted Q-value" curves are directly comparable.

```

In [8]: # ----- shared config
CONFIG = dict(
    state_dim=8,
    n_actions=4,
    hidden=128,
    lr=5e-4,
    gamma=0.99,
    buffer_size=100_000,
    batch_size=64,
    tau=5e-3, # soft target-update coefficient

```

```

    learn_start=1_000, # warm-up transitions before learning
    update_every=4,    # one gradient step every 4 env steps
)
SEED = 165             # group number -> reproducible & memorable
NUM_EPISODES = 1000    # training budget per configuration
EPS_START, EPS_END, EPS_DECAY = 1.0, 0.02, 0.995 #  $\epsilon$ -greedy schedule (per epis
VALIDATION_SIZE = 256  # fixed validation set size

def collect_validation_states(n_states: int, seed: int) -> np.ndarray:
    """Collect a fixed set of states via a random policy on the ORIGINAL env.

    The set is deliberately drawn from the unmodified environment so it is a
    neutral reference shared by all four agents.
    """
    env = make_env(modified=False)
    rng = np.random.default_rng(seed)
    env.reset(seed=seed)
    env.action_space.seed(seed)
    states = []
    obs, _ = env.reset()
    while len(states) < n_states:
        states.append(obs.copy())
        a = int(rng.integers(0, env.action_space.n))
        obs, _, terminated, truncated, _ = env.step(a)
        if terminated or truncated:
            obs, _ = env.reset()
    env.close()
    return np.array(states[:n_states], dtype=np.float32)

VALIDATION_STATES = collect_validation_states(VALIDATION_SIZE, SEED)
VALIDATION_TENSOR = torch.as_tensor(VALIDATION_STATES, device=DEVICE)
print(f"Collected fixed validation set: {VALIDATION_STATES.shape}")

```

Collected fixed validation set: (256, 8)

```

In [9]: # =====
# Training loop (shared by all four configurations)
# =====
def train_agent(modified: bool, double: bool, tag: str,
                num_episodes: int = NUM_EPISODES, seed: int = SEED,
                log_every: int = 1):
    """Train one agent on one environment and return per-episode metric arrays.

    Parameters
    -----
    modified : train on the modified (True) or original (False) environment.
    double   : DDQN (True) or DQN (False).
    tag      : short label used in logs and plots.

    Returns a dict with per-episode lists:
        rewards, avg_q, landings (0/1), thrusters, steps.
    """
    # Reset ALL randomness so every configuration is identically initialised.
    set_global_seed(seed)

    env = make_env(modified=modified, seed=seed)
    env.reset(seed=seed)
    env.action_space.seed(seed)

```

```

agent = ValueBasedAgent(device=DEVICE, double=double, seed=seed, **CONFIG)

ep_rewards, ep_avg_q, ep_landings, ep_thrusters, ep_steps = [], [], [], [],
epsilon = EPS_START
t_start = time.time()

for ep in range(1, num_episodes + 1):
    obs, _ = env.reset()
    done = False
    ep_reward, thrusters, steps = 0.0, 0, 0
    terminated = truncated = False

    while not done:
        #  $\epsilon$ -greedy action; count thruster activations (selected, not execute
        a = agent.select_action(obs, epsilon)
        if a in (1, 2, 3):
            thrusters += 1

        next_obs, reward, terminated, truncated, _ = env.step(a)
        done = terminated or truncated

        # Bootstrap only on genuine termination, not on time-limit truncatio
        agent.store(obs, a, reward, next_obs, float(terminated))
        agent.learn_step()

        obs = next_obs
        ep_reward += reward
        steps += 1

    # --- per-episode metrics -----
    landed = 1 if is_safe_landing(obs, terminated, truncated) else 0
    avg_q = agent.mean_max_q(VALIDATION_TENSOR)

    ep_rewards.append(ep_reward)
    ep_avg_q.append(avg_q)
    ep_landings.append(landed)
    ep_thrusters.append(thrusters)
    ep_steps.append(steps)

    # Decay exploration once per episode.
    epsilon = max(EPS_END, epsilon * EPS_DECAY)

    # --- concise per-iteration logging -----
    if ep % log_every == 0 or ep == num_episodes:
        avg_r100 = np.mean(ep_rewards[-100:])
        land100 = np.mean(ep_landings[-100:])
        print(f"[{tag}] Ep {ep:4d} | R {ep_reward:8.2f} | avgR100 {avg_r100:8.2f} | eps {epsilon:.3f} | Qval {avg_q:7.2f} | land100 {land100:4.1f} | thr {thrusters:3d}")

env.close()
elapsed = time.time() - t_start
print(f"[{tag}] finished {num_episodes} episodes in {elapsed/60:.1f} min "
      f"(final avgR100 = {np.mean(ep_rewards[-100:]):.2f})\n")

return dict(tag=tag, modified=modified, double=double,
            rewards=ep_rewards, avg_q=ep_avg_q, landings=ep_landings,
            thrusters=ep_thrusters, steps=ep_steps, minutes=elapsed / 60,
            agent=agent)

```


Train all four configurations

We train the four required agents with **identical** settings (seed, network, optimiser, replay, ϵ -schedule and 1000-episode budget), differing only in *algorithm* (DQN/DDQN) and *environment* (original/modified):

1. **DQN — Original**
2. **DDQN — Original**
3. **DQN — Modified**
4. **DDQN — Modified**

Per-episode progress is printed for full transparency (reward, 100-episode moving averages, ϵ , predicted Q-value, landing rate, thruster count).

```
In [10]: # Train the four configurations sequentially and collect their metrics.
RESULTS = {}

configs = [
    dict(modified=False, double=False, tag="DQN-Orig"),
    dict(modified=False, double=True, tag="DDQN-Orig"),
    dict(modified=True, double=False, tag="DQN-Mod"),
    dict(modified=True, double=True, tag="DDQN-Mod"),
]

for cfg in configs:
    print("#" * 70)
    print(f"# TRAINING: {cfg['tag']} (modified={cfg['modified']}, double={cfg['double']})")
    print("#" * 70)
    RESULTS[cfg["tag"]] = train_agent(**cfg)

print("All four configurations trained.")
```

```
#####
# TRAINING: DQN-Orig (modified=False, double=False)
#####
[DQN-Orig] Ep    1 | R    -76.76 | avgR100    -76.76 | eps 0.995 | Qval    0.05 | 1
and100 0.00 | thr  43
[DQN-Orig] Ep    2 | R   -111.39 | avgR100   -94.07 | eps 0.990 | Qval    0.05 | 1
and100 0.00 | thr  65
[DQN-Orig] Ep    3 | R   -248.17 | avgR100  -145.44 | eps 0.985 | Qval    0.05 | 1
and100 0.00 | thr  64
[DQN-Orig] Ep    4 | R   -122.59 | avgR100  -139.73 | eps 0.980 | Qval    0.05 | 1
and100 0.00 | thr  56
[DQN-Orig] Ep    5 | R   -292.21 | avgR100  -170.22 | eps 0.975 | Qval    0.05 | 1
and100 0.00 | thr  73
[DQN-Orig] Ep    6 | R   -118.75 | avgR100  -161.64 | eps 0.970 | Qval    0.05 | 1
and100 0.00 | thr  79
[DQN-Orig] Ep    7 | R   -500.86 | avgR100  -210.10 | eps 0.966 | Qval    0.05 | 1
and100 0.00 | thr  74
[DQN-Orig] Ep    8 | R   -112.02 | avgR100  -197.84 | eps 0.961 | Qval    0.05 | 1
and100 0.00 | thr  51
[DQN-Orig] Ep    9 | R    -95.80 | avgR100  -186.51 | eps 0.956 | Qval    0.05 | 1
and100 0.00 | thr  59
[DQN-Orig] Ep   10 | R   -224.33 | avgR100  -190.29 | eps 0.951 | Qval    0.05 | 1
and100 0.00 | thr  79
[DQN-Orig] Ep   11 | R   -289.42 | avgR100  -199.30 | eps 0.946 | Qval    0.05 | 1
and100 0.00 | thr  81
[DQN-Orig] Ep   12 | R   -102.24 | avgR100  -191.21 | eps 0.942 | Qval    0.09 | 1
and100 0.00 | thr  55
[DQN-Orig] Ep   13 | R    -48.74 | avgR100  -180.25 | eps 0.937 | Qval    0.15 | 1
and100 0.00 | thr  63
[DQN-Orig] Ep   14 | R   -498.84 | avgR100  -203.01 | eps 0.932 | Qval    0.25 | 1
and100 0.00 | thr 104
[DQN-Orig] Ep   15 | R    -93.01 | avgR100  -195.67 | eps 0.928 | Qval    0.15 | 1
and100 0.00 | thr  56
[DQN-Orig] Ep   16 | R   -185.01 | avgR100  -195.01 | eps 0.923 | Qval    0.18 | 1
and100 0.00 | thr  51
[DQN-Orig] Ep   17 | R   -183.11 | avgR100  -194.31 | eps 0.918 | Qval    0.15 | 1
and100 0.00 | thr 106
[DQN-Orig] Ep   18 | R   -301.05 | avgR100  -200.24 | eps 0.914 | Qval    0.03 | 1
and100 0.00 | thr  94
[DQN-Orig] Ep   19 | R   -401.95 | avgR100  -210.85 | eps 0.909 | Qval   -0.19 | 1
and100 0.00 | thr  97
[DQN-Orig] Ep   20 | R   -278.16 | avgR100  -214.22 | eps 0.905 | Qval   -0.16 | 1
and100 0.00 | thr  75
[DQN-Orig] Ep   21 | R   -187.16 | avgR100  -212.93 | eps 0.900 | Qval   -0.11 | 1
and100 0.00 | thr 109
[DQN-Orig] Ep   22 | R    -84.46 | avgR100  -207.09 | eps 0.896 | Qval    0.24 | 1
and100 0.00 | thr  50
[DQN-Orig] Ep   23 | R   -242.43 | avgR100  -208.63 | eps 0.891 | Qval    0.46 | 1
and100 0.00 | thr  79
[DQN-Orig] Ep   24 | R    29.91 | avgR100  -198.69 | eps 0.887 | Qval    0.33 | 1
and100 0.00 | thr  80
[DQN-Orig] Ep   25 | R   -297.48 | avgR100  -202.64 | eps 0.882 | Qval    0.50 | 1
and100 0.00 | thr  75
```

[DQN-Orig]	Ep 26	R	-120.85	avgR100	-199.50	eps 0.878	Qval	0.59	1
and100 0.00	thr 68								
[DQN-Orig]	Ep 27	R	-104.17	avgR100	-195.96	eps 0.873	Qval	0.44	1
and100 0.00	thr 80								
[DQN-Orig]	Ep 28	R	-273.63	avgR100	-198.74	eps 0.869	Qval	0.57	1
and100 0.00	thr 57								
[DQN-Orig]	Ep 29	R	-131.88	avgR100	-196.43	eps 0.865	Qval	0.79	1
and100 0.00	thr 68								
[DQN-Orig]	Ep 30	R	-103.57	avgR100	-193.34	eps 0.860	Qval	0.62	1
and100 0.00	thr 64								
[DQN-Orig]	Ep 31	R	-112.60	avgR100	-190.73	eps 0.856	Qval	0.77	1
and100 0.00	thr 46								
[DQN-Orig]	Ep 32	R	-96.21	avgR100	-187.78	eps 0.852	Qval	0.95	1
and100 0.00	thr 88								
[DQN-Orig]	Ep 33	R	-46.33	avgR100	-183.49	eps 0.848	Qval	0.95	1
and100 0.00	thr 56								
[DQN-Orig]	Ep 34	R	-192.85	avgR100	-183.77	eps 0.843	Qval	1.15	1
and100 0.00	thr 95								
[DQN-Orig]	Ep 35	R	-174.32	avgR100	-183.50	eps 0.839	Qval	1.10	1
and100 0.00	thr 86								
[DQN-Orig]	Ep 36	R	-111.92	avgR100	-181.51	eps 0.835	Qval	1.21	1
and100 0.00	thr 83								
[DQN-Orig]	Ep 37	R	-205.51	avgR100	-182.16	eps 0.831	Qval	1.30	1
and100 0.00	thr 74								
[DQN-Orig]	Ep 38	R	-64.97	avgR100	-179.07	eps 0.827	Qval	1.46	1
and100 0.00	thr 59								
[DQN-Orig]	Ep 39	R	-102.08	avgR100	-177.10	eps 0.822	Qval	1.44	1
and100 0.00	thr 76								
[DQN-Orig]	Ep 40	R	-149.77	avgR100	-176.42	eps 0.818	Qval	1.28	1
and100 0.00	thr 92								
[DQN-Orig]	Ep 41	R	-206.38	avgR100	-177.15	eps 0.814	Qval	1.69	1
and100 0.00	thr 89								
[DQN-Orig]	Ep 42	R	-102.53	avgR100	-175.37	eps 0.810	Qval	1.68	1
and100 0.00	thr 80								
[DQN-Orig]	Ep 43	R	-93.98	avgR100	-173.48	eps 0.806	Qval	1.66	1
and100 0.00	thr 53								
[DQN-Orig]	Ep 44	R	-58.83	avgR100	-170.87	eps 0.802	Qval	1.92	1
and100 0.00	thr 65								
[DQN-Orig]	Ep 45	R	-75.51	avgR100	-168.75	eps 0.798	Qval	1.64	1
and100 0.00	thr 75								
[DQN-Orig]	Ep 46	R	-112.71	avgR100	-167.54	eps 0.794	Qval	1.95	1
and100 0.00	thr 194								
[DQN-Orig]	Ep 47	R	-363.17	avgR100	-171.70	eps 0.790	Qval	2.10	1
and100 0.00	thr 107								
[DQN-Orig]	Ep 48	R	-112.66	avgR100	-170.47	eps 0.786	Qval	1.90	1
and100 0.00	thr 81								
[DQN-Orig]	Ep 49	R	-124.73	avgR100	-169.53	eps 0.782	Qval	1.95	1
and100 0.00	thr 98								
[DQN-Orig]	Ep 50	R	-133.19	avgR100	-168.81	eps 0.778	Qval	2.16	1
and100 0.00	thr 66								
[DQN-Orig]	Ep 51	R	-233.02	avgR100	-170.07	eps 0.774	Qval	2.15	1
and100 0.00	thr 105								
[DQN-Orig]	Ep 52	R	-81.03	avgR100	-168.35	eps 0.771	Qval	2.39	1
and100 0.00	thr 70								
[DQN-Orig]	Ep 53	R	-64.28	avgR100	-166.39	eps 0.767	Qval	2.23	1
and100 0.00	thr 96								

[DQN-Orig]	Ep 54 R	-77.10	avgR100	-164.74	eps 0.763	Qval	2.30	1
and100 0.00	thr 92							
[DQN-Orig]	Ep 55 R	-109.81	avgR100	-163.74	eps 0.759	Qval	2.79	1
and100 0.00	thr 94							
[DQN-Orig]	Ep 56 R	-51.20	avgR100	-161.73	eps 0.755	Qval	3.08	1
and100 0.00	thr 91							
[DQN-Orig]	Ep 57 R	-215.82	avgR100	-162.68	eps 0.751	Qval	2.67	1
and100 0.00	thr 93							
[DQN-Orig]	Ep 58 R	-84.99	avgR100	-161.34	eps 0.748	Qval	2.86	1
and100 0.00	thr 117							
[DQN-Orig]	Ep 59 R	-82.62	avgR100	-160.00	eps 0.744	Qval	3.05	1
and100 0.00	thr 118							
[DQN-Orig]	Ep 60 R	-183.93	avgR100	-160.40	eps 0.740	Qval	3.03	1
and100 0.00	thr 93							
[DQN-Orig]	Ep 61 R	-109.44	avgR100	-159.57	eps 0.737	Qval	3.04	1
and100 0.00	thr 99							
[DQN-Orig]	Ep 62 R	-76.78	avgR100	-158.23	eps 0.733	Qval	3.24	1
and100 0.00	thr 63							
[DQN-Orig]	Ep 63 R	-99.44	avgR100	-157.30	eps 0.729	Qval	3.32	1
and100 0.00	thr 86							
[DQN-Orig]	Ep 64 R	-182.40	avgR100	-157.69	eps 0.726	Qval	3.42	1
and100 0.00	thr 107							
[DQN-Orig]	Ep 65 R	-126.75	avgR100	-157.21	eps 0.722	Qval	3.48	1
and100 0.00	thr 73							
[DQN-Orig]	Ep 66 R	-241.92	avgR100	-158.50	eps 0.718	Qval	3.54	1
and100 0.00	thr 107							
[DQN-Orig]	Ep 67 R	-112.43	avgR100	-157.81	eps 0.715	Qval	3.67	1
and100 0.00	thr 101							
[DQN-Orig]	Ep 68 R	-67.17	avgR100	-156.48	eps 0.711	Qval	3.93	1
and100 0.00	thr 96							
[DQN-Orig]	Ep 69 R	-117.35	avgR100	-155.91	eps 0.708	Qval	3.65	1
and100 0.00	thr 77							
[DQN-Orig]	Ep 70 R	-44.94	avgR100	-154.33	eps 0.704	Qval	4.04	1
and100 0.00	thr 113							
[DQN-Orig]	Ep 71 R	-140.93	avgR100	-154.14	eps 0.701	Qval	4.10	1
and100 0.00	thr 101							
[DQN-Orig]	Ep 72 R	-61.34	avgR100	-152.85	eps 0.697	Qval	4.01	1
and100 0.00	thr 79							
[DQN-Orig]	Ep 73 R	-21.64	avgR100	-151.05	eps 0.694	Qval	4.36	1
and100 0.00	thr 116							
[DQN-Orig]	Ep 74 R	-90.49	avgR100	-150.23	eps 0.690	Qval	4.37	1
and100 0.00	thr 96							
[DQN-Orig]	Ep 75 R	-106.37	avgR100	-149.65	eps 0.687	Qval	4.61	1
and100 0.00	thr 97							
[DQN-Orig]	Ep 76 R	-86.73	avgR100	-148.82	eps 0.683	Qval	4.62	1
and100 0.00	thr 91							
[DQN-Orig]	Ep 77 R	-35.63	avgR100	-147.35	eps 0.680	Qval	4.56	1
and100 0.00	thr 82							
[DQN-Orig]	Ep 78 R	-90.59	avgR100	-146.62	eps 0.676	Qval	4.57	1
and100 0.00	thr 60							
[DQN-Orig]	Ep 79 R	-38.73	avgR100	-145.26	eps 0.673	Qval	4.88	1
and100 0.00	thr 69							
[DQN-Orig]	Ep 80 R	-62.12	avgR100	-144.22	eps 0.670	Qval	4.75	1
and100 0.00	thr 95							
[DQN-Orig]	Ep 81 R	-114.61	avgR100	-143.85	eps 0.666	Qval	5.00	1
and100 0.00	thr 105							
[DQN-Orig]	Ep 82 R	-68.70	avgR100	-142.93	eps 0.663	Qval	5.24	1
and100 0.00	thr 66							
[DQN-Orig]	Ep 83 R	-63.27	avgR100	-141.97	eps 0.660	Qval	5.19	1
and100 0.00	thr 154							

[DQN-Orig]	Ep 84 R	-144.46	avgR100	-142.00	eps 0.656	Qval	5.53	1
and100 0.00	thr 166							
[DQN-Orig]	Ep 85 R	-93.25	avgR100	-141.43	eps 0.653	Qval	5.46	1
and100 0.00	thr 70							
[DQN-Orig]	Ep 86 R	-69.36	avgR100	-140.59	eps 0.650	Qval	5.57	1
and100 0.00	thr 72							
[DQN-Orig]	Ep 87 R	-152.27	avgR100	-140.73	eps 0.647	Qval	5.84	1
and100 0.00	thr 103							
[DQN-Orig]	Ep 88 R	-67.92	avgR100	-139.90	eps 0.643	Qval	5.57	1
and100 0.00	thr 110							
[DQN-Orig]	Ep 89 R	-54.45	avgR100	-138.94	eps 0.640	Qval	5.89	1
and100 0.00	thr 56							
[DQN-Orig]	Ep 90 R	-123.01	avgR100	-138.76	eps 0.637	Qval	6.08	1
and100 0.00	thr 94							
[DQN-Orig]	Ep 91 R	-52.19	avgR100	-137.81	eps 0.634	Qval	6.26	1
and100 0.00	thr 103							
[DQN-Orig]	Ep 92 R	-64.84	avgR100	-137.02	eps 0.631	Qval	6.21	1
and100 0.00	thr 71							
[DQN-Orig]	Ep 93 R	-58.67	avgR100	-136.18	eps 0.627	Qval	6.55	1
and100 0.00	thr 93							
[DQN-Orig]	Ep 94 R	-12.35	avgR100	-134.86	eps 0.624	Qval	6.49	1
and100 0.00	thr 140							
[DQN-Orig]	Ep 95 R	-79.13	avgR100	-134.27	eps 0.621	Qval	6.71	1
and100 0.00	thr 105							
[DQN-Orig]	Ep 96 R	-13.36	avgR100	-133.01	eps 0.618	Qval	6.96	1
and100 0.00	thr 139							
[DQN-Orig]	Ep 97 R	-52.51	avgR100	-132.18	eps 0.615	Qval	7.48	1
and100 0.00	thr 110							
[DQN-Orig]	Ep 98 R	-43.36	avgR100	-131.28	eps 0.612	Qval	7.41	1
and100 0.00	thr 145							
[DQN-Orig]	Ep 99 R	-58.29	avgR100	-130.54	eps 0.609	Qval	7.84	1
and100 0.00	thr 196							
[DQN-Orig]	Ep 100 R	-64.56	avgR100	-129.88	eps 0.606	Qval	7.79	1
and100 0.00	thr 73							
[DQN-Orig]	Ep 101 R	-222.96	avgR100	-131.34	eps 0.603	Qval	8.48	1
and100 0.00	thr 133							
[DQN-Orig]	Ep 102 R	-41.65	avgR100	-130.64	eps 0.600	Qval	8.37	1
and100 0.00	thr 139							
[DQN-Orig]	Ep 103 R	-92.04	avgR100	-129.08	eps 0.597	Qval	8.38	1
and100 0.00	thr 160							
[DQN-Orig]	Ep 104 R	-110.35	avgR100	-128.96	eps 0.594	Qval	9.06	1
and100 0.00	thr 296							
[DQN-Orig]	Ep 105 R	-7.70	avgR100	-126.11	eps 0.591	Qval	9.06	1
and100 0.00	thr 74							
[DQN-Orig]	Ep 106 R	-67.65	avgR100	-125.60	eps 0.588	Qval	9.09	1
and100 0.00	thr 106							
[DQN-Orig]	Ep 107 R	-41.07	avgR100	-121.01	eps 0.585	Qval	9.30	1
and100 0.00	thr 75							
[DQN-Orig]	Ep 108 R	-44.08	avgR100	-120.33	eps 0.582	Qval	9.66	1
and100 0.00	thr 99							
[DQN-Orig]	Ep 109 R	-68.10	avgR100	-120.05	eps 0.579	Qval	10.00	1
and100 0.00	thr 121							
[DQN-Orig]	Ep 110 R	-73.15	avgR100	-118.54	eps 0.576	Qval	10.01	1
and100 0.00	thr 114							

[DQN-Orig]	Ep 111 R	-37.58	avgR100	-116.02	eps 0.573	Qval 10.27	1
and100 0.00	thr 62						
[DQN-Orig]	Ep 112 R	-109.32	avgR100	-116.09	eps 0.570	Qval 9.96	1
and100 0.00	thr 104						
[DQN-Orig]	Ep 113 R	-33.58	avgR100	-115.94	eps 0.568	Qval 10.13	1
and100 0.00	thr 135						
[DQN-Orig]	Ep 114 R	-35.60	avgR100	-111.31	eps 0.565	Qval 10.42	1
and100 0.00	thr 101						
[DQN-Orig]	Ep 115 R	-83.69	avgR100	-111.21	eps 0.562	Qval 10.56	1
and100 0.00	thr 127						
[DQN-Orig]	Ep 116 R	-68.56	avgR100	-110.05	eps 0.559	Qval 10.65	1
and100 0.00	thr 55						
[DQN-Orig]	Ep 117 R	-122.48	avgR100	-109.44	eps 0.556	Qval 10.50	1
and100 0.00	thr 100						
[DQN-Orig]	Ep 118 R	18.89	avgR100	-106.24	eps 0.554	Qval 11.31	1
and100 0.00	thr 193						
[DQN-Orig]	Ep 119 R	-31.25	avgR100	-102.54	eps 0.551	Qval 11.72	1
and100 0.00	thr 125						
[DQN-Orig]	Ep 120 R	-54.63	avgR100	-100.30	eps 0.548	Qval 11.75	1
and100 0.00	thr 161						
[DQN-Orig]	Ep 121 R	-49.99	avgR100	-98.93	eps 0.545	Qval 11.94	1
and100 0.00	thr 172						
[DQN-Orig]	Ep 122 R	22.22	avgR100	-97.86	eps 0.543	Qval 12.44	1
and100 0.00	thr 202						
[DQN-Orig]	Ep 123 R	-18.29	avgR100	-95.62	eps 0.540	Qval 12.46	1
and100 0.00	thr 111						
[DQN-Orig]	Ep 124 R	19.05	avgR100	-95.73	eps 0.537	Qval 12.99	1
and100 0.00	thr 147						
[DQN-Orig]	Ep 125 R	-26.63	avgR100	-93.02	eps 0.534	Qval 13.34	1
and100 0.00	thr 164						
[DQN-Orig]	Ep 126 R	-66.35	avgR100	-92.48	eps 0.532	Qval 13.22	1
and100 0.00	thr 108						
[DQN-Orig]	Ep 127 R	-30.76	avgR100	-91.74	eps 0.529	Qval 13.66	1
and100 0.00	thr 127						
[DQN-Orig]	Ep 128 R	-193.05	avgR100	-90.94	eps 0.526	Qval 13.67	1
and100 0.00	thr 118						
[DQN-Orig]	Ep 129 R	-25.45	avgR100	-89.87	eps 0.524	Qval 14.01	1
and100 0.00	thr 70						
[DQN-Orig]	Ep 130 R	-15.44	avgR100	-88.99	eps 0.521	Qval 14.27	1
and100 0.00	thr 115						
[DQN-Orig]	Ep 131 R	20.91	avgR100	-87.65	eps 0.519	Qval 14.76	1
and100 0.00	thr 154						
[DQN-Orig]	Ep 132 R	-38.82	avgR100	-87.08	eps 0.516	Qval 14.54	1
and100 0.00	thr 152						
[DQN-Orig]	Ep 133 R	13.63	avgR100	-86.48	eps 0.513	Qval 14.85	1
and100 0.00	thr 131						
[DQN-Orig]	Ep 134 R	-35.20	avgR100	-84.90	eps 0.511	Qval 14.99	1
and100 0.00	thr 140						
[DQN-Orig]	Ep 135 R	29.26	avgR100	-82.87	eps 0.508	Qval 15.18	1
and100 0.00	thr 108						
[DQN-Orig]	Ep 136 R	-6.74	avgR100	-81.82	eps 0.506	Qval 15.37	1
and100 0.00	thr 102						
[DQN-Orig]	Ep 137 R	13.72	avgR100	-79.63	eps 0.503	Qval 17.10	1
and100 0.00	thr 854						
[DQN-Orig]	Ep 138 R	-43.12	avgR100	-79.41	eps 0.501	Qval 17.21	1
and100 0.00	thr 95						
[DQN-Orig]	Ep 139 R	62.09	avgR100	-77.76	eps 0.498	Qval 18.55	1
and100 0.00	thr 869						
[DQN-Orig]	Ep 140 R	32.37	avgR100	-75.94	eps 0.496	Qval 19.84	1
and100 0.00	thr 866						

[DQN-Orig]	Ep 141 R	43.83	avgR100	-73.44	eps 0.493	Qval	21.65	1
and100 0.00	thr 869							
[DQN-Orig]	Ep 142 R	-17.04	avgR100	-72.59	eps 0.491	Qval	21.99	1
and100 0.00	thr 129							
[DQN-Orig]	Ep 143 R	33.63	avgR100	-71.31	eps 0.488	Qval	23.17	1
and100 0.00	thr 814							
[DQN-Orig]	Ep 144 R	12.46	avgR100	-70.60	eps 0.486	Qval	23.38	1
and100 0.00	thr 130							
[DQN-Orig]	Ep 145 R	-331.03	avgR100	-73.15	eps 0.483	Qval	23.67	1
and100 0.00	thr 152							
[DQN-Orig]	Ep 146 R	-77.59	avgR100	-72.80	eps 0.481	Qval	25.03	1
and100 0.00	thr 875							
[DQN-Orig]	Ep 147 R	-31.16	avgR100	-69.48	eps 0.479	Qval	25.27	1
and100 0.00	thr 280							
[DQN-Orig]	Ep 148 R	-0.23	avgR100	-68.36	eps 0.476	Qval	25.80	1
and100 0.00	thr 84							
[DQN-Orig]	Ep 149 R	8.13	avgR100	-67.03	eps 0.474	Qval	26.74	1
and100 0.00	thr 784							
[DQN-Orig]	Ep 150 R	-69.73	avgR100	-66.39	eps 0.471	Qval	27.87	1
and100 0.00	thr 834							
[DQN-Orig]	Ep 151 R	45.05	avgR100	-63.61	eps 0.469	Qval	28.01	1
and100 0.00	thr 109							
[DQN-Orig]	Ep 152 R	28.11	avgR100	-62.52	eps 0.467	Qval	27.59	1
and100 0.00	thr 109							
[DQN-Orig]	Ep 153 R	-63.78	avgR100	-62.52	eps 0.464	Qval	27.90	1
and100 0.00	thr 190							
[DQN-Orig]	Ep 154 R	-36.82	avgR100	-62.11	eps 0.462	Qval	28.95	1
and100 0.00	thr 872							
[DQN-Orig]	Ep 155 R	-21.72	avgR100	-61.23	eps 0.460	Qval	29.84	1
and100 0.00	thr 849							
[DQN-Orig]	Ep 156 R	-127.48	avgR100	-62.00	eps 0.458	Qval	30.04	1
and100 0.00	thr 142							
[DQN-Orig]	Ep 157 R	-53.95	avgR100	-60.38	eps 0.455	Qval	30.72	1
and100 0.00	thr 860							
[DQN-Orig]	Ep 158 R	-34.92	avgR100	-59.88	eps 0.453	Qval	30.24	1
and100 0.00	thr 80							
[DQN-Orig]	Ep 159 R	-22.11	avgR100	-59.27	eps 0.451	Qval	30.48	1
and100 0.00	thr 252							
[DQN-Orig]	Ep 160 R	-37.66	avgR100	-57.81	eps 0.448	Qval	31.67	1
and100 0.00	thr 861							
[DQN-Orig]	Ep 161 R	-41.34	avgR100	-57.13	eps 0.446	Qval	31.30	1
and100 0.00	thr 95							
[DQN-Orig]	Ep 162 R	-34.46	avgR100	-56.70	eps 0.444	Qval	30.99	1
and100 0.00	thr 142							
[DQN-Orig]	Ep 163 R	7.92	avgR100	-55.63	eps 0.442	Qval	31.59	1
and100 0.00	thr 97							
[DQN-Orig]	Ep 164 R	-41.87	avgR100	-54.23	eps 0.440	Qval	31.89	1
and100 0.00	thr 874							
[DQN-Orig]	Ep 165 R	-30.35	avgR100	-53.26	eps 0.437	Qval	31.71	1
and100 0.00	thr 117							
[DQN-Orig]	Ep 166 R	-40.69	avgR100	-51.25	eps 0.435	Qval	31.56	1
and100 0.00	thr 57							
[DQN-Orig]	Ep 167 R	-298.62	avgR100	-53.11	eps 0.433	Qval	31.55	1
and100 0.00	thr 703							
[DQN-Orig]	Ep 168 R	-218.49	avgR100	-54.62	eps 0.431	Qval	31.57	1
and100 0.00	thr 161							
[DQN-Orig]	Ep 169 R	-29.94	avgR100	-53.75	eps 0.429	Qval	31.95	1
and100 0.00	thr 139							

[DQN-Orig]	Ep 170 R	-159.87	avgR100	-54.90	eps 0.427	Qval	31.92	1
and100 0.00	thr 786							
[DQN-Orig]	Ep 171 R	-1.86	avgR100	-53.51	eps 0.424	Qval	32.04	1
and100 0.00	thr 113							
[DQN-Orig]	Ep 172 R	-6.62	avgR100	-52.96	eps 0.422	Qval	31.35	1
and100 0.00	thr 856							
[DQN-Orig]	Ep 173 R	-2.90	avgR100	-52.77	eps 0.420	Qval	31.81	1
and100 0.00	thr 130							
[DQN-Orig]	Ep 174 R	-8.28	avgR100	-51.95	eps 0.418	Qval	31.69	1
and100 0.00	thr 867							
[DQN-Orig]	Ep 175 R	-25.16	avgR100	-51.14	eps 0.416	Qval	31.77	1
and100 0.00	thr 69							
[DQN-Orig]	Ep 176 R	-140.71	avgR100	-51.68	eps 0.414	Qval	31.48	1
and100 0.00	thr 134							
[DQN-Orig]	Ep 177 R	-8.81	avgR100	-51.41	eps 0.412	Qval	31.56	1
and100 0.00	thr 111							
[DQN-Orig]	Ep 178 R	-44.39	avgR100	-50.95	eps 0.410	Qval	31.78	1
and100 0.00	thr 856							
[DQN-Orig]	Ep 179 R	-33.12	avgR100	-50.89	eps 0.408	Qval	31.36	1
and100 0.00	thr 87							
[DQN-Orig]	Ep 180 R	-21.61	avgR100	-50.49	eps 0.406	Qval	31.03	1
and100 0.00	thr 862							
[DQN-Orig]	Ep 181 R	-3.94	avgR100	-49.38	eps 0.404	Qval	31.11	1
and100 0.00	thr 90							
[DQN-Orig]	Ep 182 R	-46.75	avgR100	-49.16	eps 0.402	Qval	30.75	1
and100 0.00	thr 100							
[DQN-Orig]	Ep 183 R	-26.72	avgR100	-48.80	eps 0.400	Qval	31.57	1
and100 0.00	thr 73							
[DQN-Orig]	Ep 184 R	-57.39	avgR100	-47.93	eps 0.398	Qval	31.13	1
and100 0.00	thr 241							
[DQN-Orig]	Ep 185 R	-15.70	avgR100	-47.15	eps 0.396	Qval	31.32	1
and100 0.00	thr 76							
[DQN-Orig]	Ep 186 R	-15.64	avgR100	-46.61	eps 0.394	Qval	31.13	1
and100 0.00	thr 120							
[DQN-Orig]	Ep 187 R	-59.28	avgR100	-45.68	eps 0.392	Qval	30.89	1
and100 0.00	thr 833							
[DQN-Orig]	Ep 188 R	16.48	avgR100	-44.84	eps 0.390	Qval	29.81	1
and100 0.00	thr 849							
[DQN-Orig]	Ep 189 R	-44.04	avgR100	-44.74	eps 0.388	Qval	29.68	1
and100 0.00	thr 865							
[DQN-Orig]	Ep 190 R	-118.41	avgR100	-44.69	eps 0.386	Qval	29.49	1
and100 0.00	thr 150							
[DQN-Orig]	Ep 191 R	-20.28	avgR100	-44.37	eps 0.384	Qval	28.99	1
and100 0.00	thr 875							
[DQN-Orig]	Ep 192 R	4.91	avgR100	-43.67	eps 0.382	Qval	28.26	1
and100 0.00	thr 116							
[DQN-Orig]	Ep 193 R	-34.23	avgR100	-43.43	eps 0.380	Qval	27.87	1
and100 0.00	thr 824							
[DQN-Orig]	Ep 194 R	-3.42	avgR100	-43.34	eps 0.378	Qval	27.76	1
and100 0.00	thr 841							
[DQN-Orig]	Ep 195 R	-1.21	avgR100	-42.56	eps 0.376	Qval	27.21	1
and100 0.00	thr 867							
[DQN-Orig]	Ep 196 R	-13.91	avgR100	-42.57	eps 0.374	Qval	26.26	1
and100 0.00	thr 826							
[DQN-Orig]	Ep 197 R	-69.26	avgR100	-42.73	eps 0.373	Qval	26.22	1
and100 0.00	thr 862							
[DQN-Orig]	Ep 198 R	-33.93	avgR100	-42.64	eps 0.371	Qval	25.28	1
and100 0.00	thr 865							
[DQN-Orig]	Ep 199 R	-110.69	avgR100	-43.16	eps 0.369	Qval	24.01	1
and100 0.00	thr 885							

[DQN-Orig]	Ep 200 R	-28.07	avgR100	-42.80	eps 0.367	Qval	23.91	1
and100 0.00	thr 870							
[DQN-Orig]	Ep 201 R	-7.72	avgR100	-40.65	eps 0.365	Qval	22.66	1
and100 0.00	thr 873							
[DQN-Orig]	Ep 202 R	-58.00	avgR100	-40.81	eps 0.363	Qval	21.88	1
and100 0.00	thr 861							
[DQN-Orig]	Ep 203 R	-140.43	avgR100	-41.29	eps 0.361	Qval	22.28	1
and100 0.00	thr 146							
[DQN-Orig]	Ep 204 R	-0.87	avgR100	-40.20	eps 0.360	Qval	21.66	1
and100 0.00	thr 96							
[DQN-Orig]	Ep 205 R	-33.89	avgR100	-40.46	eps 0.358	Qval	22.53	1
and100 0.00	thr 880							
[DQN-Orig]	Ep 206 R	-2.22	avgR100	-39.81	eps 0.356	Qval	21.07	1
and100 0.00	thr 829							
[DQN-Orig]	Ep 207 R	-36.59	avgR100	-39.76	eps 0.354	Qval	20.28	1
and100 0.00	thr 881							
[DQN-Orig]	Ep 208 R	-20.54	avgR100	-39.53	eps 0.353	Qval	19.65	1
and100 0.00	thr 878							
[DQN-Orig]	Ep 209 R	-25.35	avgR100	-39.10	eps 0.351	Qval	18.41	1
and100 0.00	thr 883							
[DQN-Orig]	Ep 210 R	30.41	avgR100	-38.06	eps 0.349	Qval	18.63	1
and100 0.00	thr 893							
[DQN-Orig]	Ep 211 R	-10.44	avgR100	-37.79	eps 0.347	Qval	18.68	1
and100 0.00	thr 870							
[DQN-Orig]	Ep 212 R	7.16	avgR100	-36.63	eps 0.346	Qval	17.45	1
and100 0.00	thr 885							
[DQN-Orig]	Ep 213 R	0.68	avgR100	-36.28	eps 0.344	Qval	17.38	1
and100 0.00	thr 206							
[DQN-Orig]	Ep 214 R	-25.86	avgR100	-36.19	eps 0.342	Qval	16.49	1
and100 0.00	thr 878							
[DQN-Orig]	Ep 215 R	42.52	avgR100	-34.92	eps 0.340	Qval	17.15	1
and100 0.00	thr 868							
[DQN-Orig]	Ep 216 R	-9.61	avgR100	-34.33	eps 0.339	Qval	15.25	1
and100 0.00	thr 872							
[DQN-Orig]	Ep 217 R	4.22	avgR100	-33.07	eps 0.337	Qval	15.35	1
and100 0.00	thr 869							
[DQN-Orig]	Ep 218 R	-19.07	avgR100	-33.45	eps 0.335	Qval	14.52	1
and100 0.00	thr 875							
[DQN-Orig]	Ep 219 R	34.30	avgR100	-32.79	eps 0.334	Qval	13.31	1
and100 0.00	thr 840							
[DQN-Orig]	Ep 220 R	21.46	avgR100	-32.03	eps 0.332	Qval	13.52	1
and100 0.00	thr 848							
[DQN-Orig]	Ep 221 R	-50.53	avgR100	-32.04	eps 0.330	Qval	13.22	1
and100 0.00	thr 901							
[DQN-Orig]	Ep 222 R	39.06	avgR100	-31.87	eps 0.329	Qval	12.43	1
and100 0.00	thr 860							
[DQN-Orig]	Ep 223 R	9.85	avgR100	-31.59	eps 0.327	Qval	12.51	1
and100 0.00	thr 836							
[DQN-Orig]	Ep 224 R	17.25	avgR100	-31.60	eps 0.325	Qval	10.95	1
and100 0.00	thr 860							
[DQN-Orig]	Ep 225 R	39.61	avgR100	-30.94	eps 0.324	Qval	10.88	1
and100 0.00	thr 860							
[DQN-Orig]	Ep 226 R	20.86	avgR100	-30.07	eps 0.322	Qval	10.51	1
and100 0.00	thr 851							
[DQN-Orig]	Ep 227 R	-3.83	avgR100	-29.80	eps 0.321	Qval	10.77	1
and100 0.00	thr 802							
[DQN-Orig]	Ep 228 R	-28.35	avgR100	-28.15	eps 0.319	Qval	10.11	1
and100 0.00	thr 855							
[DQN-Orig]	Ep 229 R	63.16	avgR100	-27.27	eps 0.317	Qval	9.24	1
and100 0.00	thr 872							

[DQN-Orig]	Ep 230 R	3.50 avgR100	-27.08 eps 0.316 Qval	9.02 1
and100 0.00	thr 871			
[DQN-Orig]	Ep 231 R	-48.90 avgR100	-27.78 eps 0.314 Qval	9.29 1
and100 0.00	thr 882			
[DQN-Orig]	Ep 232 R	-20.76 avgR100	-27.60 eps 0.313 Qval	8.89 1
and100 0.00	thr 88			
[DQN-Orig]	Ep 233 R	35.87 avgR100	-27.37 eps 0.311 Qval	9.86 1
and100 0.00	thr 825			
[DQN-Orig]	Ep 234 R	53.24 avgR100	-26.49 eps 0.309 Qval	9.11 1
and100 0.00	thr 801			
[DQN-Orig]	Ep 235 R	12.77 avgR100	-26.65 eps 0.308 Qval	9.60 1
and100 0.00	thr 810			
[DQN-Orig]	Ep 236 R	-15.91 avgR100	-26.75 eps 0.306 Qval	9.94 1
and100 0.00	thr 848			
[DQN-Orig]	Ep 237 R	-1.67 avgR100	-26.90 eps 0.305 Qval	10.15 1
and100 0.00	thr 132			
[DQN-Orig]	Ep 238 R	5.67 avgR100	-26.41 eps 0.303 Qval	9.62 1
and100 0.00	thr 796			
[DQN-Orig]	Ep 239 R	-0.78 avgR100	-27.04 eps 0.302 Qval	9.24 1
and100 0.00	thr 120			
[DQN-Orig]	Ep 240 R	33.50 avgR100	-27.03 eps 0.300 Qval	8.89 1
and100 0.00	thr 114			
[DQN-Orig]	Ep 241 R	26.54 avgR100	-27.20 eps 0.299 Qval	9.10 1
and100 0.00	thr 838			
[DQN-Orig]	Ep 242 R	-4.85 avgR100	-27.08 eps 0.297 Qval	9.40 1
and100 0.00	thr 768			
[DQN-Orig]	Ep 243 R	-32.95 avgR100	-27.75 eps 0.296 Qval	10.62 1
and100 0.00	thr 147			
[DQN-Orig]	Ep 244 R	-4.74 avgR100	-27.92 eps 0.294 Qval	10.04 1
and100 0.00	thr 82			
[DQN-Orig]	Ep 245 R	6.41 avgR100	-24.54 eps 0.293 Qval	9.56 1
and100 0.00	thr 135			
[DQN-Orig]	Ep 246 R	85.21 avgR100	-22.92 eps 0.291 Qval	10.29 1
and100 0.00	thr 919			
[DQN-Orig]	Ep 247 R	-1.57 avgR100	-22.62 eps 0.290 Qval	8.98 1
and100 0.00	thr 798			
[DQN-Orig]	Ep 248 R	-43.42 avgR100	-23.05 eps 0.288 Qval	11.06 1
and100 0.00	thr 96			
[DQN-Orig]	Ep 249 R	-48.49 avgR100	-23.62 eps 0.287 Qval	9.09 1
and100 0.00	thr 776			
[DQN-Orig]	Ep 250 R	67.41 avgR100	-22.25 eps 0.286 Qval	9.75 1
and100 0.00	thr 818			
[DQN-Orig]	Ep 251 R	48.39 avgR100	-22.21 eps 0.284 Qval	9.12 1
and100 0.00	thr 845			
[DQN-Orig]	Ep 252 R	17.61 avgR100	-22.32 eps 0.283 Qval	9.50 1
and100 0.00	thr 856			
[DQN-Orig]	Ep 253 R	56.20 avgR100	-21.12 eps 0.281 Qval	9.59 1
and100 0.00	thr 731			
[DQN-Orig]	Ep 254 R	34.46 avgR100	-20.40 eps 0.280 Qval	9.84 1
and100 0.00	thr 773			
[DQN-Orig]	Ep 255 R	50.39 avgR100	-19.68 eps 0.279 Qval	9.17 1
and100 0.00	thr 812			
[DQN-Orig]	Ep 256 R	37.14 avgR100	-18.04 eps 0.277 Qval	9.51 1
and100 0.00	thr 774			
[DQN-Orig]	Ep 257 R	22.59 avgR100	-17.27 eps 0.276 Qval	9.31 1
and100 0.00	thr 796			
[DQN-Orig]	Ep 258 R	-24.56 avgR100	-17.17 eps 0.274 Qval	10.35 1
and100 0.00	thr 96			
[DQN-Orig]	Ep 259 R	8.35 avgR100	-16.86 eps 0.273 Qval	9.97 1
and100 0.00	thr 772			

[DQN-Orig]	Ep 260 R	30.13	avgR100	-16.19	eps 0.272	Qval 10.45	1
and100 0.00	thr 801						
[DQN-Orig]	Ep 261 R	73.97	avgR100	-15.03	eps 0.270	Qval 11.28	1
and100 0.00	thr 814						
[DQN-Orig]	Ep 262 R	187.44	avgR100	-12.81	eps 0.269	Qval 11.40	1
and100 0.01	thr 485						
[DQN-Orig]	Ep 263 R	2.17	avgR100	-12.87	eps 0.268	Qval 10.06	1
and100 0.01	thr 855						
[DQN-Orig]	Ep 264 R	-32.29	avgR100	-12.78	eps 0.266	Qval 9.66	1
and100 0.01	thr 858						
[DQN-Orig]	Ep 265 R	12.73	avgR100	-12.34	eps 0.265	Qval 10.36	1
and100 0.01	thr 802						
[DQN-Orig]	Ep 266 R	64.69	avgR100	-11.29	eps 0.264	Qval 10.28	1
and100 0.01	thr 672						
[DQN-Orig]	Ep 267 R	11.73	avgR100	-8.19	eps 0.262	Qval 9.84	1
and100 0.01	thr 841						
[DQN-Orig]	Ep 268 R	45.86	avgR100	-5.54	eps 0.261	Qval 11.65	1
and100 0.01	thr 110						
[DQN-Orig]	Ep 269 R	-12.61	avgR100	-5.37	eps 0.260	Qval 10.67	1
and100 0.01	thr 711						
[DQN-Orig]	Ep 270 R	60.12	avgR100	-3.17	eps 0.258	Qval 11.85	1
and100 0.01	thr 669						
[DQN-Orig]	Ep 271 R	92.52	avgR100	-2.23	eps 0.257	Qval 12.38	1
and100 0.01	thr 655						
[DQN-Orig]	Ep 272 R	73.12	avgR100	-1.43	eps 0.256	Qval 12.09	1
and100 0.01	thr 724						
[DQN-Orig]	Ep 273 R	28.65	avgR100	-1.11	eps 0.255	Qval 11.80	1
and100 0.01	thr 769						
[DQN-Orig]	Ep 274 R	18.87	avgR100	-0.84	eps 0.253	Qval 11.25	1
and100 0.01	thr 101						
[DQN-Orig]	Ep 275 R	40.97	avgR100	-0.18	eps 0.252	Qval 12.72	1
and100 0.01	thr 574						
[DQN-Orig]	Ep 276 R	74.35	avgR100	1.97	eps 0.251	Qval 12.76	1
and100 0.01	thr 740						
[DQN-Orig]	Ep 277 R	171.98	avgR100	3.78	eps 0.249	Qval 13.62	1
and100 0.01	thr 557						
[DQN-Orig]	Ep 278 R	30.06	avgR100	4.52	eps 0.248	Qval 13.29	1
and100 0.01	thr 751						
[DQN-Orig]	Ep 279 R	75.97	avgR100	5.61	eps 0.247	Qval 13.52	1
and100 0.01	thr 722						
[DQN-Orig]	Ep 280 R	27.60	avgR100	6.10	eps 0.246	Qval 12.96	1
and100 0.01	thr 803						
[DQN-Orig]	Ep 281 R	33.00	avgR100	6.47	eps 0.245	Qval 13.33	1
and100 0.01	thr 134						
[DQN-Orig]	Ep 282 R	67.14	avgR100	7.61	eps 0.243	Qval 14.05	1
and100 0.01	thr 678						
[DQN-Orig]	Ep 283 R	-8.81	avgR100	7.79	eps 0.242	Qval 15.36	1
and100 0.01	thr 770						
[DQN-Orig]	Ep 284 R	79.90	avgR100	9.16	eps 0.241	Qval 15.23	1
and100 0.01	thr 820						
[DQN-Orig]	Ep 285 R	109.03	avgR100	10.41	eps 0.240	Qval 15.10	1
and100 0.01	thr 676						
[DQN-Orig]	Ep 286 R	200.92	avgR100	12.58	eps 0.238	Qval 14.97	1
and100 0.02	thr 514						
[DQN-Orig]	Ep 287 R	197.04	avgR100	15.14	eps 0.237	Qval 15.26	1
and100 0.03	thr 486						
[DQN-Orig]	Ep 288 R	52.05	avgR100	15.50	eps 0.236	Qval 13.18	1
and100 0.03	thr 815						
[DQN-Orig]	Ep 289 R	4.02	avgR100	15.98	eps 0.235	Qval 15.51	1
and100 0.03	thr 818						

[DQN-Orig]	Ep 290 R	98.28 avgR100	18.14 eps 0.234 Qval	13.80 1
and100 0.03	thr 545			
[DQN-Orig]	Ep 291 R	-30.50 avgR100	18.04 eps 0.233 Qval	15.20 1
and100 0.03	thr 93			
[DQN-Orig]	Ep 292 R	-45.82 avgR100	17.53 eps 0.231 Qval	13.93 1
and100 0.03	thr 820			
[DQN-Orig]	Ep 293 R	10.58 avgR100	17.98 eps 0.230 Qval	15.49 1
and100 0.03	thr 873			
[DQN-Orig]	Ep 294 R	-61.31 avgR100	17.40 eps 0.229 Qval	14.80 1
and100 0.03	thr 884			
[DQN-Orig]	Ep 295 R	251.30 avgR100	19.93 eps 0.228 Qval	14.01 1
and100 0.04	thr 469			
[DQN-Orig]	Ep 296 R	223.36 avgR100	22.30 eps 0.227 Qval	14.56 1
and100 0.05	thr 440			
[DQN-Orig]	Ep 297 R	121.17 avgR100	24.21 eps 0.226 Qval	15.09 1
and100 0.05	thr 604			
[DQN-Orig]	Ep 298 R	296.73 avgR100	27.51 eps 0.225 Qval	16.75 1
and100 0.06	thr 279			
[DQN-Orig]	Ep 299 R	-10.73 avgR100	28.51 eps 0.223 Qval	16.80 1
and100 0.06	thr 890			
[DQN-Orig]	Ep 300 R	66.62 avgR100	29.46 eps 0.222 Qval	17.07 1
and100 0.06	thr 867			
[DQN-Orig]	Ep 301 R	0.76 avgR100	29.54 eps 0.221 Qval	16.28 1
and100 0.06	thr 100			
[DQN-Orig]	Ep 302 R	154.26 avgR100	31.67 eps 0.220 Qval	17.26 1
and100 0.06	thr 502			
[DQN-Orig]	Ep 303 R	20.38 avgR100	33.27 eps 0.219 Qval	17.32 1
and100 0.06	thr 890			
[DQN-Orig]	Ep 304 R	230.02 avgR100	35.58 eps 0.218 Qval	18.04 1
and100 0.07	thr 299			
[DQN-Orig]	Ep 305 R	110.53 avgR100	37.03 eps 0.217 Qval	18.33 1
and100 0.07	thr 517			
[DQN-Orig]	Ep 306 R	65.95 avgR100	37.71 eps 0.216 Qval	18.70 1
and100 0.07	thr 709			
[DQN-Orig]	Ep 307 R	234.80 avgR100	40.42 eps 0.215 Qval	19.26 1
and100 0.08	thr 393			
[DQN-Orig]	Ep 308 R	207.94 avgR100	42.71 eps 0.214 Qval	20.23 1
and100 0.09	thr 406			
[DQN-Orig]	Ep 309 R	5.68 avgR100	43.02 eps 0.212 Qval	18.96 1
and100 0.09	thr 75			
[DQN-Orig]	Ep 310 R	164.33 avgR100	44.36 eps 0.211 Qval	21.95 1
and100 0.09	thr 401			
[DQN-Orig]	Ep 311 R	25.49 avgR100	44.72 eps 0.210 Qval	21.36 1
and100 0.09	thr 120			
[DQN-Orig]	Ep 312 R	78.92 avgR100	45.43 eps 0.209 Qval	22.52 1
and100 0.09	thr 703			
[DQN-Orig]	Ep 313 R	103.22 avgR100	46.46 eps 0.208 Qval	23.33 1
and100 0.09	thr 447			
[DQN-Orig]	Ep 314 R	264.82 avgR100	49.37 eps 0.207 Qval	25.06 1
and100 0.10	thr 207			
[DQN-Orig]	Ep 315 R	256.74 avgR100	51.51 eps 0.206 Qval	25.29 1
and100 0.11	thr 435			
[DQN-Orig]	Ep 316 R	274.78 avgR100	54.35 eps 0.205 Qval	25.58 1
and100 0.12	thr 320			
[DQN-Orig]	Ep 317 R	110.36 avgR100	55.41 eps 0.204 Qval	26.50 1
and100 0.12	thr 695			
[DQN-Orig]	Ep 318 R	83.41 avgR100	56.44 eps 0.203 Qval	27.01 1
and100 0.12	thr 550			
[DQN-Orig]	Ep 319 R	147.23 avgR100	57.57 eps 0.202 Qval	26.89 1
and100 0.12	thr 675			

[DQN-Orig]	Ep 320 R	68.01 avgR100	58.03 eps 0.201 Qval	28.67 1
and100 0.12	thr 905			
[DQN-Orig]	Ep 321 R	108.76 avgR100	59.63 eps 0.200 Qval	29.63 1
and100 0.12	thr 407			
[DQN-Orig]	Ep 322 R	144.39 avgR100	60.68 eps 0.199 Qval	31.74 1
and100 0.12	thr 435			
[DQN-Orig]	Ep 323 R	223.51 avgR100	62.82 eps 0.198 Qval	29.08 1
and100 0.13	thr 266			
[DQN-Orig]	Ep 324 R	259.17 avgR100	65.24 eps 0.197 Qval	30.72 1
and100 0.14	thr 408			
[DQN-Orig]	Ep 325 R	210.16 avgR100	66.94 eps 0.196 Qval	31.61 1
and100 0.14	thr 352			
[DQN-Orig]	Ep 326 R	235.93 avgR100	69.09 eps 0.195 Qval	33.48 1
and100 0.15	thr 382			
[DQN-Orig]	Ep 327 R	258.44 avgR100	71.71 eps 0.194 Qval	32.51 1
and100 0.16	thr 336			
[DQN-Orig]	Ep 328 R	246.00 avgR100	74.46 eps 0.193 Qval	32.28 1
and100 0.17	thr 451			
[DQN-Orig]	Ep 329 R	136.96 avgR100	75.20 eps 0.192 Qval	33.68 1
and100 0.17	thr 425			
[DQN-Orig]	Ep 330 R	252.47 avgR100	77.69 eps 0.191 Qval	33.70 1
and100 0.18	thr 320			
[DQN-Orig]	Ep 331 R	262.23 avgR100	80.80 eps 0.190 Qval	33.18 1
and100 0.19	thr 409			
[DQN-Orig]	Ep 332 R	217.85 avgR100	83.18 eps 0.189 Qval	32.03 1
and100 0.20	thr 258			
[DQN-Orig]	Ep 333 R	172.80 avgR100	84.55 eps 0.188 Qval	35.36 1
and100 0.20	thr 422			
[DQN-Orig]	Ep 334 R	125.39 avgR100	85.27 eps 0.187 Qval	34.18 1
and100 0.20	thr 765			
[DQN-Orig]	Ep 335 R	-94.66 avgR100	84.20 eps 0.187 Qval	33.72 1
and100 0.20	thr 137			
[DQN-Orig]	Ep 336 R	232.36 avgR100	86.68 eps 0.186 Qval	34.03 1
and100 0.21	thr 487			
[DQN-Orig]	Ep 337 R	250.66 avgR100	89.21 eps 0.185 Qval	34.44 1
and100 0.22	thr 271			
[DQN-Orig]	Ep 338 R	108.20 avgR100	90.23 eps 0.184 Qval	35.44 1
and100 0.22	thr 424			
[DQN-Orig]	Ep 339 R	253.77 avgR100	92.78 eps 0.183 Qval	35.96 1
and100 0.23	thr 395			
[DQN-Orig]	Ep 340 R	23.51 avgR100	92.68 eps 0.182 Qval	34.79 1
and100 0.23	thr 257			
[DQN-Orig]	Ep 341 R	148.21 avgR100	93.89 eps 0.181 Qval	35.85 1
and100 0.23	thr 746			
[DQN-Orig]	Ep 342 R	210.77 avgR100	96.05 eps 0.180 Qval	34.89 1
and100 0.24	thr 339			
[DQN-Orig]	Ep 343 R	83.79 avgR100	97.22 eps 0.179 Qval	35.72 1
and100 0.24	thr 570			
[DQN-Orig]	Ep 344 R	107.07 avgR100	98.33 eps 0.178 Qval	36.71 1
and100 0.24	thr 654			
[DQN-Orig]	Ep 345 R	236.90 avgR100	100.64 eps 0.177 Qval	36.57 1
and100 0.25	thr 338			
[DQN-Orig]	Ep 346 R	-20.46 avgR100	99.58 eps 0.177 Qval	35.42 1
and100 0.25	thr 166			
[DQN-Orig]	Ep 347 R	283.90 avgR100	102.44 eps 0.176 Qval	36.79 1
and100 0.26	thr 208			
[DQN-Orig]	Ep 348 R	232.23 avgR100	105.19 eps 0.175 Qval	36.27 1
and100 0.27	thr 293			
[DQN-Orig]	Ep 349 R	260.96 avgR100	108.29 eps 0.174 Qval	35.12 1
and100 0.28	thr 421			

[DQN-Orig]	Ep 350 R	287.80	avgR100	110.49	eps 0.173	Qval 35.92	1
and100 0.29	thr 264						
[DQN-Orig]	Ep 351 R	153.14	avgR100	111.54	eps 0.172	Qval 36.56	1
and100 0.29	thr 491						
[DQN-Orig]	Ep 352 R	243.01	avgR100	113.79	eps 0.171	Qval 37.23	1
and100 0.29	thr 554						
[DQN-Orig]	Ep 353 R	243.51	avgR100	115.67	eps 0.170	Qval 36.72	1
and100 0.30	thr 378						
[DQN-Orig]	Ep 354 R	94.89	avgR100	116.27	eps 0.170	Qval 37.44	1
and100 0.30	thr 748						
[DQN-Orig]	Ep 355 R	237.18	avgR100	118.14	eps 0.169	Qval 37.22	1
and100 0.30	thr 420						
[DQN-Orig]	Ep 356 R	208.74	avgR100	119.86	eps 0.168	Qval 34.78	1
and100 0.31	thr 311						
[DQN-Orig]	Ep 357 R	243.56	avgR100	122.06	eps 0.167	Qval 36.52	1
and100 0.32	thr 273						
[DQN-Orig]	Ep 358 R	271.50	avgR100	125.03	eps 0.166	Qval 37.84	1
and100 0.33	thr 239						
[DQN-Orig]	Ep 359 R	92.81	avgR100	125.87	eps 0.165	Qval 36.67	1
and100 0.33	thr 765						
[DQN-Orig]	Ep 360 R	271.31	avgR100	128.28	eps 0.165	Qval 36.49	1
and100 0.34	thr 293						
[DQN-Orig]	Ep 361 R	103.07	avgR100	128.57	eps 0.164	Qval 35.67	1
and100 0.34	thr 548						
[DQN-Orig]	Ep 362 R	144.26	avgR100	128.14	eps 0.163	Qval 35.76	1
and100 0.33	thr 583						
[DQN-Orig]	Ep 363 R	239.66	avgR100	130.52	eps 0.162	Qval 36.23	1
and100 0.34	thr 498						
[DQN-Orig]	Ep 364 R	258.93	avgR100	133.43	eps 0.161	Qval 35.70	1
and100 0.34	thr 309						
[DQN-Orig]	Ep 365 R	249.63	avgR100	135.80	eps 0.160	Qval 36.27	1
and100 0.35	thr 193						
[DQN-Orig]	Ep 366 R	203.11	avgR100	137.18	eps 0.160	Qval 36.78	1
and100 0.35	thr 535						
[DQN-Orig]	Ep 367 R	285.20	avgR100	139.92	eps 0.159	Qval 34.66	1
and100 0.36	thr 241						
[DQN-Orig]	Ep 368 R	206.61	avgR100	141.52	eps 0.158	Qval 36.17	1
and100 0.36	thr 458						
[DQN-Orig]	Ep 369 R	245.99	avgR100	144.11	eps 0.157	Qval 34.08	1
and100 0.37	thr 471						
[DQN-Orig]	Ep 370 R	149.24	avgR100	145.00	eps 0.157	Qval 37.35	1
and100 0.37	thr 591						
[DQN-Orig]	Ep 371 R	234.04	avgR100	146.42	eps 0.156	Qval 35.82	1
and100 0.38	thr 215						
[DQN-Orig]	Ep 372 R	123.13	avgR100	146.92	eps 0.155	Qval 35.87	1
and100 0.38	thr 694						
[DQN-Orig]	Ep 373 R	238.51	avgR100	149.01	eps 0.154	Qval 36.02	1
and100 0.39	thr 149						
[DQN-Orig]	Ep 374 R	283.75	avgR100	151.66	eps 0.153	Qval 35.44	1
and100 0.40	thr 181						
[DQN-Orig]	Ep 375 R	239.19	avgR100	153.65	eps 0.153	Qval 34.98	1
and100 0.41	thr 263						
[DQN-Orig]	Ep 376 R	133.85	avgR100	154.24	eps 0.152	Qval 35.37	1
and100 0.41	thr 512						
[DQN-Orig]	Ep 377 R	240.48	avgR100	154.93	eps 0.151	Qval 36.47	1
and100 0.42	thr 474						
[DQN-Orig]	Ep 378 R	244.91	avgR100	157.07	eps 0.150	Qval 36.58	1
and100 0.43	thr 306						

[DQN-Orig]	Ep 379 R	210.54	avgR100	158.42	eps 0.150	Qval 37.07	1
and100 0.44	thr 476						
[DQN-Orig]	Ep 380 R	309.70	avgR100	161.24	eps 0.149	Qval 36.37	1
and100 0.45	thr 202						
[DQN-Orig]	Ep 381 R	242.19	avgR100	163.33	eps 0.148	Qval 36.32	1
and100 0.46	thr 260						
[DQN-Orig]	Ep 382 R	258.59	avgR100	165.25	eps 0.147	Qval 37.58	1
and100 0.47	thr 235						
[DQN-Orig]	Ep 383 R	266.23	avgR100	168.00	eps 0.147	Qval 37.64	1
and100 0.48	thr 214						
[DQN-Orig]	Ep 384 R	297.07	avgR100	170.17	eps 0.146	Qval 36.89	1
and100 0.49	thr 201						
[DQN-Orig]	Ep 385 R	259.85	avgR100	171.68	eps 0.145	Qval 38.67	1
and100 0.50	thr 251						
[DQN-Orig]	Ep 386 R	248.16	avgR100	172.15	eps 0.144	Qval 37.00	1
and100 0.50	thr 512						
[DQN-Orig]	Ep 387 R	277.53	avgR100	172.96	eps 0.144	Qval 37.87	1
and100 0.50	thr 269						
[DQN-Orig]	Ep 388 R	-8.26	avgR100	172.35	eps 0.143	Qval 37.43	1
and100 0.50	thr 248						
[DQN-Orig]	Ep 389 R	269.40	avgR100	175.01	eps 0.142	Qval 36.81	1
and100 0.51	thr 165						
[DQN-Orig]	Ep 390 R	275.30	avgR100	176.78	eps 0.142	Qval 37.55	1
and100 0.52	thr 262						
[DQN-Orig]	Ep 391 R	266.74	avgR100	179.75	eps 0.141	Qval 38.18	1
and100 0.53	thr 138						
[DQN-Orig]	Ep 392 R	278.98	avgR100	183.00	eps 0.140	Qval 37.48	1
and100 0.54	thr 235						
[DQN-Orig]	Ep 393 R	232.38	avgR100	185.21	eps 0.139	Qval 36.76	1
and100 0.55	thr 327						
[DQN-Orig]	Ep 394 R	226.64	avgR100	188.09	eps 0.139	Qval 37.13	1
and100 0.56	thr 356						
[DQN-Orig]	Ep 395 R	233.17	avgR100	187.91	eps 0.138	Qval 36.00	1
and100 0.56	thr 242						
[DQN-Orig]	Ep 396 R	185.32	avgR100	187.53	eps 0.137	Qval 39.27	1
and100 0.55	thr 590						
[DQN-Orig]	Ep 397 R	215.10	avgR100	188.47	eps 0.137	Qval 37.50	1
and100 0.56	thr 269						
[DQN-Orig]	Ep 398 R	250.68	avgR100	188.01	eps 0.136	Qval 37.55	1
and100 0.56	thr 345						
[DQN-Orig]	Ep 399 R	-0.10	avgR100	188.12	eps 0.135	Qval 37.22	1
and100 0.56	thr 215						
[DQN-Orig]	Ep 400 R	290.15	avgR100	190.35	eps 0.135	Qval 37.08	1
and100 0.57	thr 208						
[DQN-Orig]	Ep 401 R	233.50	avgR100	192.68	eps 0.134	Qval 37.79	1
and100 0.57	thr 325						
[DQN-Orig]	Ep 402 R	0.06	avgR100	191.14	eps 0.133	Qval 38.60	1
and100 0.57	thr 364						
[DQN-Orig]	Ep 403 R	237.28	avgR100	193.31	eps 0.133	Qval 38.02	1
and100 0.57	thr 244						
[DQN-Orig]	Ep 404 R	240.17	avgR100	193.41	eps 0.132	Qval 39.90	1
and100 0.57	thr 398						
[DQN-Orig]	Ep 405 R	211.59	avgR100	194.42	eps 0.131	Qval 39.59	1
and100 0.58	thr 358						
[DQN-Orig]	Ep 406 R	228.10	avgR100	196.04	eps 0.131	Qval 39.91	1
and100 0.59	thr 347						
[DQN-Orig]	Ep 407 R	296.72	avgR100	196.66	eps 0.130	Qval 40.60	1
and100 0.59	thr 171						
[DQN-Orig]	Ep 408 R	238.16	avgR100	196.96	eps 0.129	Qval 39.51	1
and100 0.59	thr 307						

[DQN-Orig]	Ep 409 R	244.22	avgR100	199.35	eps 0.129	Qval 40.29	1
and100 0.60	thr 347						
[DQN-Orig]	Ep 410 R	253.03	avgR100	200.23	eps 0.128	Qval 40.74	1
and100 0.61	thr 202						
[DQN-Orig]	Ep 411 R	240.40	avgR100	202.38	eps 0.127	Qval 40.18	1
and100 0.62	thr 187						
[DQN-Orig]	Ep 412 R	244.72	avgR100	204.04	eps 0.127	Qval 41.33	1
and100 0.62	thr 256						
[DQN-Orig]	Ep 413 R	276.11	avgR100	205.77	eps 0.126	Qval 40.65	1
and100 0.63	thr 168						
[DQN-Orig]	Ep 414 R	255.78	avgR100	205.68	eps 0.126	Qval 39.18	1
and100 0.63	thr 228						
[DQN-Orig]	Ep 415 R	294.47	avgR100	206.06	eps 0.125	Qval 40.61	1
and100 0.63	thr 216						
[DQN-Orig]	Ep 416 R	278.20	avgR100	206.09	eps 0.124	Qval 40.37	1
and100 0.63	thr 137						
[DQN-Orig]	Ep 417 R	105.37	avgR100	206.04	eps 0.124	Qval 39.34	1
and100 0.63	thr 626						
[DQN-Orig]	Ep 418 R	277.66	avgR100	207.98	eps 0.123	Qval 40.74	1
and100 0.64	thr 286						
[DQN-Orig]	Ep 419 R	233.87	avgR100	208.85	eps 0.122	Qval 41.95	1
and100 0.65	thr 181						
[DQN-Orig]	Ep 420 R	248.06	avgR100	210.65	eps 0.122	Qval 42.14	1
and100 0.66	thr 463						
[DQN-Orig]	Ep 421 R	282.48	avgR100	212.39	eps 0.121	Qval 42.41	1
and100 0.66	thr 249						
[DQN-Orig]	Ep 422 R	253.46	avgR100	213.48	eps 0.121	Qval 42.30	1
and100 0.67	thr 218						
[DQN-Orig]	Ep 423 R	223.87	avgR100	213.48	eps 0.120	Qval 43.21	1
and100 0.67	thr 313						
[DQN-Orig]	Ep 424 R	97.48	avgR100	211.87	eps 0.119	Qval 43.78	1
and100 0.66	thr 610						
[DQN-Orig]	Ep 425 R	288.77	avgR100	212.65	eps 0.119	Qval 43.81	1
and100 0.67	thr 226						
[DQN-Orig]	Ep 426 R	258.62	avgR100	212.88	eps 0.118	Qval 44.10	1
and100 0.67	thr 221						
[DQN-Orig]	Ep 427 R	274.05	avgR100	213.03	eps 0.118	Qval 43.69	1
and100 0.67	thr 183						
[DQN-Orig]	Ep 428 R	250.27	avgR100	213.08	eps 0.117	Qval 43.86	1
and100 0.67	thr 216						
[DQN-Orig]	Ep 429 R	232.91	avgR100	214.04	eps 0.116	Qval 44.10	1
and100 0.68	thr 258						
[DQN-Orig]	Ep 430 R	216.45	avgR100	213.68	eps 0.116	Qval 43.64	1
and100 0.67	thr 522						
[DQN-Orig]	Ep 431 R	259.84	avgR100	213.65	eps 0.115	Qval 42.62	1
and100 0.67	thr 191						
[DQN-Orig]	Ep 432 R	284.78	avgR100	214.32	eps 0.115	Qval 43.89	1
and100 0.67	thr 196						
[DQN-Orig]	Ep 433 R	246.63	avgR100	215.06	eps 0.114	Qval 45.05	1
and100 0.68	thr 356						
[DQN-Orig]	Ep 434 R	237.40	avgR100	216.18	eps 0.114	Qval 45.46	1
and100 0.69	thr 403						
[DQN-Orig]	Ep 435 R	289.16	avgR100	220.02	eps 0.113	Qval 43.87	1
and100 0.70	thr 181						
[DQN-Orig]	Ep 436 R	110.22	avgR100	218.80	eps 0.112	Qval 44.47	1
and100 0.69	thr 606						
[DQN-Orig]	Ep 437 R	256.46	avgR100	218.86	eps 0.112	Qval 44.75	1
and100 0.69	thr 237						

[DQN-Orig]	Ep 438 R	255.55	avgR100	220.33	eps 0.111	Qval	45.75	1
and100 0.69	thr 510							
[DQN-Orig]	Ep 439 R	244.52	avgR100	220.24	eps 0.111	Qval	45.34	1
and100 0.69	thr 267							
[DQN-Orig]	Ep 440 R	33.06	avgR100	220.33	eps 0.110	Qval	46.28	1
and100 0.69	thr 570							
[DQN-Orig]	Ep 441 R	232.14	avgR100	221.17	eps 0.110	Qval	45.60	1
and100 0.70	thr 187							
[DQN-Orig]	Ep 442 R	253.46	avgR100	221.60	eps 0.109	Qval	46.32	1
and100 0.70	thr 339							
[DQN-Orig]	Ep 443 R	224.20	avgR100	223.00	eps 0.109	Qval	44.92	1
and100 0.71	thr 267							
[DQN-Orig]	Ep 444 R	258.55	avgR100	224.52	eps 0.108	Qval	45.90	1
and100 0.72	thr 349							
[DQN-Orig]	Ep 445 R	262.57	avgR100	224.77	eps 0.107	Qval	46.25	1
and100 0.72	thr 182							
[DQN-Orig]	Ep 446 R	232.18	avgR100	227.30	eps 0.107	Qval	46.99	1
and100 0.73	thr 325							
[DQN-Orig]	Ep 447 R	256.42	avgR100	227.03	eps 0.106	Qval	47.03	1
and100 0.73	thr 189							
[DQN-Orig]	Ep 448 R	250.92	avgR100	227.21	eps 0.106	Qval	46.40	1
and100 0.73	thr 263							
[DQN-Orig]	Ep 449 R	228.68	avgR100	226.89	eps 0.105	Qval	47.54	1
and100 0.73	thr 512							
[DQN-Orig]	Ep 450 R	254.03	avgR100	226.55	eps 0.105	Qval	48.12	1
and100 0.73	thr 206							
[DQN-Orig]	Ep 451 R	173.39	avgR100	226.75	eps 0.104	Qval	47.26	1
and100 0.74	thr 622							
[DQN-Orig]	Ep 452 R	259.61	avgR100	226.92	eps 0.104	Qval	46.68	1
and100 0.75	thr 279							
[DQN-Orig]	Ep 453 R	253.49	avgR100	227.02	eps 0.103	Qval	47.82	1
and100 0.75	thr 268							
[DQN-Orig]	Ep 454 R	236.50	avgR100	228.44	eps 0.103	Qval	48.22	1
and100 0.76	thr 207							
[DQN-Orig]	Ep 455 R	250.91	avgR100	228.57	eps 0.102	Qval	46.82	1
and100 0.77	thr 208							
[DQN-Orig]	Ep 456 R	212.17	avgR100	228.61	eps 0.102	Qval	47.71	1
and100 0.76	thr 475							
[DQN-Orig]	Ep 457 R	246.84	avgR100	228.64	eps 0.101	Qval	48.58	1
and100 0.76	thr 276							
[DQN-Orig]	Ep 458 R	269.15	avgR100	228.62	eps 0.101	Qval	48.80	1
and100 0.76	thr 178							
[DQN-Orig]	Ep 459 R	213.77	avgR100	229.83	eps 0.100	Qval	48.40	1
and100 0.77	thr 615							
[DQN-Orig]	Ep 460 R	240.37	avgR100	229.52	eps 0.100	Qval	48.53	1
and100 0.77	thr 250							
[DQN-Orig]	Ep 461 R	241.10	avgR100	230.90	eps 0.099	Qval	48.73	1
and100 0.78	thr 146							
[DQN-Orig]	Ep 462 R	254.71	avgR100	232.00	eps 0.099	Qval	48.03	1
and100 0.79	thr 173							
[DQN-Orig]	Ep 463 R	237.38	avgR100	231.98	eps 0.098	Qval	48.60	1
and100 0.79	thr 291							
[DQN-Orig]	Ep 464 R	244.18	avgR100	231.83	eps 0.098	Qval	47.92	1
and100 0.80	thr 187							
[DQN-Orig]	Ep 465 R	249.60	avgR100	231.83	eps 0.097	Qval	48.92	1
and100 0.80	thr 198							
[DQN-Orig]	Ep 466 R	274.26	avgR100	232.54	eps 0.097	Qval	48.30	1
and100 0.81	thr 221							

[DQN-Orig]	Ep 467 R	280.33	avgR100	232.49	eps 0.096	Qval 48.83	1
and100 0.81	thr 143						
[DQN-Orig]	Ep 468 R	255.90	avgR100	232.99	eps 0.096	Qval 48.04	1
and100 0.82	thr 212						
[DQN-Orig]	Ep 469 R	219.47	avgR100	232.72	eps 0.095	Qval 47.60	1
and100 0.82	thr 227						
[DQN-Orig]	Ep 470 R	235.75	avgR100	233.59	eps 0.095	Qval 48.99	1
and100 0.83	thr 308						
[DQN-Orig]	Ep 471 R	146.28	avgR100	232.71	eps 0.094	Qval 48.76	1
and100 0.82	thr 712						
[DQN-Orig]	Ep 472 R	267.42	avgR100	234.15	eps 0.094	Qval 48.80	1
and100 0.83	thr 301						
[DQN-Orig]	Ep 473 R	298.66	avgR100	234.75	eps 0.093	Qval 50.66	1
and100 0.83	thr 200						
[DQN-Orig]	Ep 474 R	228.35	avgR100	234.20	eps 0.093	Qval 49.86	1
and100 0.83	thr 422						
[DQN-Orig]	Ep 475 R	255.13	avgR100	234.36	eps 0.092	Qval 51.41	1
and100 0.83	thr 299						
[DQN-Orig]	Ep 476 R	275.98	avgR100	235.78	eps 0.092	Qval 51.21	1
and100 0.84	thr 364						
[DQN-Orig]	Ep 477 R	231.38	avgR100	235.69	eps 0.092	Qval 50.35	1
and100 0.84	thr 325						
[DQN-Orig]	Ep 478 R	226.02	avgR100	235.50	eps 0.091	Qval 52.05	1
and100 0.84	thr 312						
[DQN-Orig]	Ep 479 R	281.20	avgR100	236.21	eps 0.091	Qval 50.28	1
and100 0.84	thr 137						
[DQN-Orig]	Ep 480 R	286.73	avgR100	235.98	eps 0.090	Qval 50.54	1
and100 0.84	thr 241						
[DQN-Orig]	Ep 481 R	216.30	avgR100	235.72	eps 0.090	Qval 48.48	1
and100 0.83	thr 436						
[DQN-Orig]	Ep 482 R	242.56	avgR100	235.56	eps 0.089	Qval 49.95	1
and100 0.83	thr 274						
[DQN-Orig]	Ep 483 R	16.88	avgR100	233.06	eps 0.089	Qval 49.07	1
and100 0.82	thr 766						
[DQN-Orig]	Ep 484 R	197.79	avgR100	232.07	eps 0.088	Qval 49.64	1
and100 0.82	thr 429						
[DQN-Orig]	Ep 485 R	244.26	avgR100	231.92	eps 0.088	Qval 49.57	1
and100 0.82	thr 232						
[DQN-Orig]	Ep 486 R	278.35	avgR100	232.22	eps 0.088	Qval 50.80	1
and100 0.81	thr 100						
[DQN-Orig]	Ep 487 R	298.63	avgR100	232.43	eps 0.087	Qval 50.43	1
and100 0.81	thr 187						
[DQN-Orig]	Ep 488 R	285.76	avgR100	235.37	eps 0.087	Qval 50.88	1
and100 0.82	thr 152						
[DQN-Orig]	Ep 489 R	277.06	avgR100	235.45	eps 0.086	Qval 49.80	1
and100 0.82	thr 215						
[DQN-Orig]	Ep 490 R	207.56	avgR100	234.77	eps 0.086	Qval 49.65	1
and100 0.82	thr 351						
[DQN-Orig]	Ep 491 R	277.40	avgR100	234.87	eps 0.085	Qval 48.18	1
and100 0.82	thr 190						
[DQN-Orig]	Ep 492 R	243.75	avgR100	234.52	eps 0.085	Qval 50.11	1
and100 0.82	thr 346						
[DQN-Orig]	Ep 493 R	282.11	avgR100	235.02	eps 0.084	Qval 48.26	1
and100 0.82	thr 139						
[DQN-Orig]	Ep 494 R	271.68	avgR100	235.47	eps 0.084	Qval 48.29	1
and100 0.82	thr 268						
[DQN-Orig]	Ep 495 R	229.16	avgR100	235.43	eps 0.084	Qval 48.22	1
and100 0.82	thr 183						

[DQN-Orig]	Ep 496 R	248.13	avgR100	236.06	eps 0.083	Qval 49.13	1
and100 0.83	thr 242						
[DQN-Orig]	Ep 497 R	280.60	avgR100	236.71	eps 0.083	Qval 48.38	1
and100 0.83	thr 166						
[DQN-Orig]	Ep 498 R	246.16	avgR100	236.67	eps 0.082	Qval 47.53	1
and100 0.83	thr 240						
[DQN-Orig]	Ep 499 R	255.63	avgR100	239.23	eps 0.082	Qval 47.35	1
and100 0.84	thr 182						
[DQN-Orig]	Ep 500 R	237.91	avgR100	238.70	eps 0.082	Qval 48.22	1
and100 0.84	thr 160						
[DQN-Orig]	Ep 501 R	220.22	avgR100	238.57	eps 0.081	Qval 46.52	1
and100 0.85	thr 271						
[DQN-Orig]	Ep 502 R	221.96	avgR100	240.79	eps 0.081	Qval 47.85	1
and100 0.86	thr 367						
[DQN-Orig]	Ep 503 R	25.70	avgR100	238.67	eps 0.080	Qval 47.43	1
and100 0.86	thr 796						
[DQN-Orig]	Ep 504 R	299.92	avgR100	239.27	eps 0.080	Qval 46.40	1
and100 0.86	thr 215						
[DQN-Orig]	Ep 505 R	255.84	avgR100	239.71	eps 0.080	Qval 46.89	1
and100 0.86	thr 143						
[DQN-Orig]	Ep 506 R	245.48	avgR100	239.89	eps 0.079	Qval 46.31	1
and100 0.86	thr 266						
[DQN-Orig]	Ep 507 R	259.27	avgR100	239.51	eps 0.079	Qval 45.63	1
and100 0.86	thr 213						
[DQN-Orig]	Ep 508 R	261.65	avgR100	239.75	eps 0.078	Qval 45.21	1
and100 0.86	thr 209						
[DQN-Orig]	Ep 509 R	262.11	avgR100	239.93	eps 0.078	Qval 44.04	1
and100 0.86	thr 288						
[DQN-Orig]	Ep 510 R	241.80	avgR100	239.81	eps 0.078	Qval 45.19	1
and100 0.86	thr 117						
[DQN-Orig]	Ep 511 R	261.51	avgR100	240.02	eps 0.077	Qval 44.74	1
and100 0.86	thr 198						
[DQN-Orig]	Ep 512 R	290.81	avgR100	240.49	eps 0.077	Qval 44.25	1
and100 0.87	thr 207						
[DQN-Orig]	Ep 513 R	21.92	avgR100	237.94	eps 0.076	Qval 44.59	1
and100 0.86	thr 829						
[DQN-Orig]	Ep 514 R	220.60	avgR100	237.59	eps 0.076	Qval 44.36	1
and100 0.86	thr 406						
[DQN-Orig]	Ep 515 R	246.58	avgR100	237.11	eps 0.076	Qval 44.14	1
and100 0.86	thr 215						
[DQN-Orig]	Ep 516 R	233.12	avgR100	236.66	eps 0.075	Qval 43.10	1
and100 0.86	thr 212						
[DQN-Orig]	Ep 517 R	223.06	avgR100	237.84	eps 0.075	Qval 44.11	1
and100 0.87	thr 196						
[DQN-Orig]	Ep 518 R	226.93	avgR100	237.33	eps 0.075	Qval 43.85	1
and100 0.87	thr 362						
[DQN-Orig]	Ep 519 R	271.11	avgR100	237.70	eps 0.074	Qval 42.63	1
and100 0.87	thr 252						
[DQN-Orig]	Ep 520 R	245.86	avgR100	237.68	eps 0.074	Qval 41.58	1
and100 0.87	thr 233						
[DQN-Orig]	Ep 521 R	221.84	avgR100	237.08	eps 0.073	Qval 42.01	1
and100 0.88	thr 241						
[DQN-Orig]	Ep 522 R	240.06	avgR100	236.94	eps 0.073	Qval 40.45	1
and100 0.88	thr 273						
[DQN-Orig]	Ep 523 R	252.79	avgR100	237.23	eps 0.073	Qval 42.46	1
and100 0.88	thr 302						
[DQN-Orig]	Ep 524 R	233.12	avgR100	238.59	eps 0.072	Qval 42.05	1
and100 0.89	thr 325						

[DQN-Orig]	Ep 525 R	252.55	avgR100	238.23	eps 0.072	Qval 39.72	1
and100 0.89	thr 337						
[DQN-Orig]	Ep 526 R	246.26	avgR100	238.10	eps 0.072	Qval 41.67	1
and100 0.89	thr 314						
[DQN-Orig]	Ep 527 R	252.54	avgR100	237.89	eps 0.071	Qval 40.69	1
and100 0.89	thr 225						
[DQN-Orig]	Ep 528 R	230.17	avgR100	237.69	eps 0.071	Qval 41.24	1
and100 0.89	thr 236						
[DQN-Orig]	Ep 529 R	299.93	avgR100	238.36	eps 0.071	Qval 41.62	1
and100 0.89	thr 184						
[DQN-Orig]	Ep 530 R	247.50	avgR100	238.67	eps 0.070	Qval 41.86	1
and100 0.90	thr 288						
[DQN-Orig]	Ep 531 R	283.55	avgR100	238.90	eps 0.070	Qval 40.33	1
and100 0.90	thr 186						
[DQN-Orig]	Ep 532 R	273.40	avgR100	238.79	eps 0.069	Qval 40.72	1
and100 0.90	thr 227						
[DQN-Orig]	Ep 533 R	309.44	avgR100	239.42	eps 0.069	Qval 41.31	1
and100 0.90	thr 175						
[DQN-Orig]	Ep 534 R	227.27	avgR100	239.32	eps 0.069	Qval 40.73	1
and100 0.90	thr 261						
[DQN-Orig]	Ep 535 R	258.00	avgR100	239.01	eps 0.068	Qval 40.28	1
and100 0.90	thr 270						
[DQN-Orig]	Ep 536 R	277.41	avgR100	240.68	eps 0.068	Qval 39.50	1
and100 0.91	thr 395						
[DQN-Orig]	Ep 537 R	300.43	avgR100	241.12	eps 0.068	Qval 39.57	1
and100 0.91	thr 175						
[DQN-Orig]	Ep 538 R	225.69	avgR100	240.82	eps 0.067	Qval 39.44	1
and100 0.92	thr 266						
[DQN-Orig]	Ep 539 R	246.27	avgR100	240.84	eps 0.067	Qval 41.00	1
and100 0.92	thr 350						
[DQN-Orig]	Ep 540 R	227.80	avgR100	242.78	eps 0.067	Qval 39.29	1
and100 0.93	thr 351						
[DQN-Orig]	Ep 541 R	259.61	avgR100	243.06	eps 0.066	Qval 39.92	1
and100 0.93	thr 191						
[DQN-Orig]	Ep 542 R	233.75	avgR100	242.86	eps 0.066	Qval 39.11	1
and100 0.93	thr 320						
[DQN-Orig]	Ep 543 R	262.37	avgR100	243.24	eps 0.066	Qval 37.88	1
and100 0.93	thr 246						
[DQN-Orig]	Ep 544 R	230.76	avgR100	242.96	eps 0.065	Qval 39.09	1
and100 0.93	thr 250						
[DQN-Orig]	Ep 545 R	247.30	avgR100	242.81	eps 0.065	Qval 40.29	1
and100 0.93	thr 383						
[DQN-Orig]	Ep 546 R	251.53	avgR100	243.01	eps 0.065	Qval 40.26	1
and100 0.93	thr 269						
[DQN-Orig]	Ep 547 R	262.50	avgR100	243.07	eps 0.064	Qval 38.75	1
and100 0.93	thr 162						
[DQN-Orig]	Ep 548 R	240.59	avgR100	242.96	eps 0.064	Qval 38.96	1
and100 0.93	thr 185						
[DQN-Orig]	Ep 549 R	251.73	avgR100	243.19	eps 0.064	Qval 39.92	1
and100 0.93	thr 234						
[DQN-Orig]	Ep 550 R	285.40	avgR100	243.51	eps 0.063	Qval 39.92	1
and100 0.93	thr 231						
[DQN-Orig]	Ep 551 R	244.63	avgR100	244.22	eps 0.063	Qval 40.01	1
and100 0.92	thr 227						
[DQN-Orig]	Ep 552 R	261.94	avgR100	244.24	eps 0.063	Qval 38.73	1
and100 0.92	thr 226						
[DQN-Orig]	Ep 553 R	262.00	avgR100	244.33	eps 0.063	Qval 38.83	1
and100 0.92	thr 186						
[DQN-Orig]	Ep 554 R	298.00	avgR100	244.94	eps 0.062	Qval 39.03	1
and100 0.92	thr 211						

[DQN-Orig]	Ep 555 R	255.94	avgR100	244.99	eps 0.062	Qval 38.15	1
and100 0.92	thr 183						
[DQN-Orig]	Ep 556 R	244.86	avgR100	245.32	eps 0.062	Qval 37.88	1
and100 0.93	thr 276						
[DQN-Orig]	Ep 557 R	239.80	avgR100	245.25	eps 0.061	Qval 38.25	1
and100 0.93	thr 135						
[DQN-Orig]	Ep 558 R	287.64	avgR100	245.43	eps 0.061	Qval 39.18	1
and100 0.93	thr 171						
[DQN-Orig]	Ep 559 R	264.32	avgR100	245.94	eps 0.061	Qval 38.59	1
and100 0.92	thr 247						
[DQN-Orig]	Ep 560 R	243.44	avgR100	245.97	eps 0.060	Qval 38.48	1
and100 0.92	thr 217						
[DQN-Orig]	Ep 561 R	301.44	avgR100	246.57	eps 0.060	Qval 38.34	1
and100 0.92	thr 164						
[DQN-Orig]	Ep 562 R	260.38	avgR100	246.63	eps 0.060	Qval 38.83	1
and100 0.92	thr 300						
[DQN-Orig]	Ep 563 R	273.00	avgR100	246.99	eps 0.059	Qval 38.72	1
and100 0.92	thr 184						
[DQN-Orig]	Ep 564 R	247.19	avgR100	247.02	eps 0.059	Qval 38.42	1
and100 0.92	thr 265						
[DQN-Orig]	Ep 565 R	253.80	avgR100	247.06	eps 0.059	Qval 38.37	1
and100 0.92	thr 199						
[DQN-Orig]	Ep 566 R	241.85	avgR100	246.74	eps 0.059	Qval 38.28	1
and100 0.92	thr 200						
[DQN-Orig]	Ep 567 R	237.53	avgR100	246.31	eps 0.058	Qval 39.37	1
and100 0.92	thr 267						
[DQN-Orig]	Ep 568 R	307.72	avgR100	246.83	eps 0.058	Qval 38.69	1
and100 0.92	thr 156						
[DQN-Orig]	Ep 569 R	227.61	avgR100	246.91	eps 0.058	Qval 38.29	1
and100 0.92	thr 221						
[DQN-Orig]	Ep 570 R	259.32	avgR100	247.14	eps 0.057	Qval 39.41	1
and100 0.92	thr 254						
[DQN-Orig]	Ep 571 R	219.19	avgR100	247.87	eps 0.057	Qval 38.59	1
and100 0.92	thr 210						
[DQN-Orig]	Ep 572 R	228.27	avgR100	247.48	eps 0.057	Qval 39.60	1
and100 0.92	thr 216						
[DQN-Orig]	Ep 573 R	258.56	avgR100	247.08	eps 0.057	Qval 38.31	1
and100 0.92	thr 153						
[DQN-Orig]	Ep 574 R	225.44	avgR100	247.05	eps 0.056	Qval 38.87	1
and100 0.92	thr 321						
[DQN-Orig]	Ep 575 R	250.88	avgR100	247.01	eps 0.056	Qval 37.35	1
and100 0.92	thr 279						
[DQN-Orig]	Ep 576 R	249.19	avgR100	246.74	eps 0.056	Qval 37.53	1
and100 0.92	thr 166						
[DQN-Orig]	Ep 577 R	235.07	avgR100	246.78	eps 0.055	Qval 37.06	1
and100 0.91	thr 371						
[DQN-Orig]	Ep 578 R	274.86	avgR100	247.27	eps 0.055	Qval 36.23	1
and100 0.91	thr 283						
[DQN-Orig]	Ep 579 R	244.90	avgR100	246.90	eps 0.055	Qval 38.02	1
and100 0.91	thr 260						
[DQN-Orig]	Ep 580 R	215.89	avgR100	246.19	eps 0.055	Qval 37.55	1
and100 0.91	thr 309						
[DQN-Orig]	Ep 581 R	268.52	avgR100	246.72	eps 0.054	Qval 36.96	1
and100 0.92	thr 141						
[DQN-Orig]	Ep 582 R	239.03	avgR100	246.68	eps 0.054	Qval 36.95	1
and100 0.92	thr 367						
[DQN-Orig]	Ep 583 R	236.41	avgR100	248.88	eps 0.054	Qval 38.03	1
and100 0.92	thr 252						
[DQN-Orig]	Ep 584 R	246.37	avgR100	249.36	eps 0.054	Qval 36.68	1
and100 0.92	thr 177						

[DQN-Orig]	Ep 585 R	223.58	avgR100	249.15	eps 0.053	Qval 36.51	1
and100 0.92	thr 429						
[DQN-Orig]	Ep 586 R	254.50	avgR100	248.92	eps 0.053	Qval 37.77	1
and100 0.93	thr 164						
[DQN-Orig]	Ep 587 R	231.52	avgR100	248.25	eps 0.053	Qval 37.58	1
and100 0.93	thr 167						
[DQN-Orig]	Ep 588 R	253.09	avgR100	247.92	eps 0.052	Qval 37.05	1
and100 0.93	thr 159						
[DQN-Orig]	Ep 589 R	225.75	avgR100	247.41	eps 0.052	Qval 34.84	1
and100 0.93	thr 332						
[DQN-Orig]	Ep 590 R	219.89	avgR100	247.53	eps 0.052	Qval 36.03	1
and100 0.93	thr 324						
[DQN-Orig]	Ep 591 R	278.59	avgR100	247.54	eps 0.052	Qval 35.95	1
and100 0.93	thr 199						
[DQN-Orig]	Ep 592 R	279.99	avgR100	247.90	eps 0.051	Qval 35.91	1
and100 0.92	thr 222						
[DQN-Orig]	Ep 593 R	266.96	avgR100	247.75	eps 0.051	Qval 35.17	1
and100 0.92	thr 362						
[DQN-Orig]	Ep 594 R	285.33	avgR100	247.89	eps 0.051	Qval 36.04	1
and100 0.91	thr 291						
[DQN-Orig]	Ep 595 R	261.63	avgR100	248.21	eps 0.051	Qval 34.84	1
and100 0.91	thr 315						
[DQN-Orig]	Ep 596 R	237.07	avgR100	248.10	eps 0.050	Qval 36.05	1
and100 0.90	thr 333						
[DQN-Orig]	Ep 597 R	286.43	avgR100	248.16	eps 0.050	Qval 35.89	1
and100 0.90	thr 138						
[DQN-Orig]	Ep 598 R	214.14	avgR100	247.84	eps 0.050	Qval 36.44	1
and100 0.90	thr 270						
[DQN-Orig]	Ep 599 R	254.82	avgR100	247.83	eps 0.050	Qval 35.24	1
and100 0.90	thr 299						
[DQN-Orig]	Ep 600 R	245.63	avgR100	247.91	eps 0.049	Qval 34.97	1
and100 0.89	thr 313						
[DQN-Orig]	Ep 601 R	263.10	avgR100	248.34	eps 0.049	Qval 34.85	1
and100 0.89	thr 282						
[DQN-Orig]	Ep 602 R	254.97	avgR100	248.67	eps 0.049	Qval 34.24	1
and100 0.88	thr 303						
[DQN-Orig]	Ep 603 R	258.82	avgR100	251.00	eps 0.049	Qval 34.55	1
and100 0.89	thr 186						
[DQN-Orig]	Ep 604 R	252.44	avgR100	250.52	eps 0.048	Qval 35.00	1
and100 0.89	thr 297						
[DQN-Orig]	Ep 605 R	252.98	avgR100	250.50	eps 0.048	Qval 35.08	1
and100 0.89	thr 237						
[DQN-Orig]	Ep 606 R	268.58	avgR100	250.73	eps 0.048	Qval 34.96	1
and100 0.89	thr 177						
[DQN-Orig]	Ep 607 R	203.93	avgR100	250.17	eps 0.048	Qval 34.69	1
and100 0.88	thr 367						
[DQN-Orig]	Ep 608 R	207.47	avgR100	249.63	eps 0.047	Qval 34.91	1
and100 0.87	thr 241						
[DQN-Orig]	Ep 609 R	264.85	avgR100	249.66	eps 0.047	Qval 35.11	1
and100 0.86	thr 303						
[DQN-Orig]	Ep 610 R	274.39	avgR100	249.98	eps 0.047	Qval 34.77	1
and100 0.86	thr 221						
[DQN-Orig]	Ep 611 R	222.12	avgR100	249.59	eps 0.047	Qval 34.08	1
and100 0.86	thr 323						
[DQN-Orig]	Ep 612 R	250.77	avgR100	249.19	eps 0.047	Qval 35.25	1
and100 0.86	thr 137						
[DQN-Orig]	Ep 613 R	222.78	avgR100	251.20	eps 0.046	Qval 35.53	1
and100 0.87	thr 165						

[DQN-Orig] Ep 614 R and100 0.86 thr 500	230.14 avgR100	251.29 eps 0.046 Qval	36.48 1
[DQN-Orig] Ep 615 R and100 0.86 thr 157	259.89 avgR100	251.43 eps 0.046 Qval	36.42 1
[DQN-Orig] Ep 616 R and100 0.86 thr 128	240.35 avgR100	251.50 eps 0.046 Qval	36.19 1
[DQN-Orig] Ep 617 R and100 0.86 thr 170	282.15 avgR100	252.09 eps 0.045 Qval	36.37 1
[DQN-Orig] Ep 618 R and100 0.85 thr 309	209.30 avgR100	251.91 eps 0.045 Qval	35.40 1
[DQN-Orig] Ep 619 R and100 0.85 thr 130	277.74 avgR100	251.98 eps 0.045 Qval	35.59 1
[DQN-Orig] Ep 620 R and100 0.85 thr 305	296.36 avgR100	252.49 eps 0.045 Qval	34.89 1
[DQN-Orig] Ep 621 R and100 0.85 thr 155	237.56 avgR100	252.64 eps 0.044 Qval	34.32 1
[DQN-Orig] Ep 622 R and100 0.85 thr 141	246.41 avgR100	252.71 eps 0.044 Qval	33.90 1
[DQN-Orig] Ep 623 R and100 0.85 thr 240	243.76 avgR100	252.62 eps 0.044 Qval	35.21 1
[DQN-Orig] Ep 624 R and100 0.85 thr 106	246.47 avgR100	252.75 eps 0.044 Qval	35.05 1
[DQN-Orig] Ep 625 R and100 0.85 thr 153	247.70 avgR100	252.70 eps 0.044 Qval	34.37 1
[DQN-Orig] Ep 626 R and100 0.85 thr 138	240.28 avgR100	252.64 eps 0.043 Qval	34.39 1
[DQN-Orig] Ep 627 R and100 0.85 thr 201	273.02 avgR100	252.85 eps 0.043 Qval	34.46 1
[DQN-Orig] Ep 628 R and100 0.85 thr 199	247.68 avgR100	253.02 eps 0.043 Qval	34.21 1
[DQN-Orig] Ep 629 R and100 0.85 thr 227	277.48 avgR100	252.80 eps 0.043 Qval	32.83 1
[DQN-Orig] Ep 630 R and100 0.85 thr 147	289.19 avgR100	253.21 eps 0.043 Qval	34.15 1
[DQN-Orig] Ep 631 R and100 0.84 thr 271	240.97 avgR100	252.79 eps 0.042 Qval	32.53 1
[DQN-Orig] Ep 632 R and100 0.84 thr 205	301.03 avgR100	253.06 eps 0.042 Qval	34.17 1
[DQN-Orig] Ep 633 R and100 0.84 thr 517	249.55 avgR100	252.47 eps 0.042 Qval	34.38 1
[DQN-Orig] Ep 634 R and100 0.84 thr 139	282.11 avgR100	253.01 eps 0.042 Qval	34.11 1
[DQN-Orig] Ep 635 R and100 0.83 thr 620	152.72 avgR100	251.96 eps 0.041 Qval	34.55 1
[DQN-Orig] Ep 636 R and100 0.82 thr 433	167.75 avgR100	250.86 eps 0.041 Qval	33.11 1
[DQN-Orig] Ep 637 R and100 0.82 thr 181	291.05 avgR100	250.77 eps 0.041 Qval	32.43 1
[DQN-Orig] Ep 638 R and100 0.82 thr 384	302.68 avgR100	251.54 eps 0.041 Qval	32.31 1
[DQN-Orig] Ep 639 R and100 0.81 thr 163	253.14 avgR100	251.61 eps 0.041 Qval	32.61 1
[DQN-Orig] Ep 640 R and100 0.81 thr 218	305.87 avgR100	252.39 eps 0.040 Qval	32.14 1
[DQN-Orig] Ep 641 R and100 0.80 thr 347	236.32 avgR100	252.16 eps 0.040 Qval	32.44 1
[DQN-Orig] Ep 642 R and100 0.80 thr 155	254.74 avgR100	252.37 eps 0.040 Qval	32.46 1
[DQN-Orig] Ep 643 R and100 0.80 thr 251	265.11 avgR100	252.39 eps 0.040 Qval	32.67 1

[DQN-Orig]	Ep 644 R	238.95	avgR100	252.48	eps 0.040	Qval 34.26	1
and100 0.80	thr 128						
[DQN-Orig]	Ep 645 R	259.74	avgR100	252.60	eps 0.039	Qval 33.86	1
and100 0.80	thr 227						
[DQN-Orig]	Ep 646 R	277.37	avgR100	252.86	eps 0.039	Qval 33.82	1
and100 0.80	thr 164						
[DQN-Orig]	Ep 647 R	236.79	avgR100	252.60	eps 0.039	Qval 33.32	1
and100 0.80	thr 137						
[DQN-Orig]	Ep 648 R	215.40	avgR100	252.35	eps 0.039	Qval 33.01	1
and100 0.80	thr 389						
[DQN-Orig]	Ep 649 R	202.14	avgR100	251.85	eps 0.039	Qval 32.61	1
and100 0.79	thr 488						
[DQN-Orig]	Ep 650 R	275.65	avgR100	251.76	eps 0.038	Qval 32.39	1
and100 0.78	thr 193						
[DQN-Orig]	Ep 651 R	250.77	avgR100	251.82	eps 0.038	Qval 32.21	1
and100 0.79	thr 181						
[DQN-Orig]	Ep 652 R	301.07	avgR100	252.21	eps 0.038	Qval 32.54	1
and100 0.79	thr 183						
[DQN-Orig]	Ep 653 R	273.08	avgR100	252.32	eps 0.038	Qval 31.65	1
and100 0.79	thr 145						
[DQN-Orig]	Ep 654 R	279.21	avgR100	252.13	eps 0.038	Qval 32.15	1
and100 0.78	thr 241						
[DQN-Orig]	Ep 655 R	303.97	avgR100	252.61	eps 0.038	Qval 32.34	1
and100 0.78	thr 183						
[DQN-Orig]	Ep 656 R	201.16	avgR100	252.18	eps 0.037	Qval 32.57	1
and100 0.77	thr 338						
[DQN-Orig]	Ep 657 R	242.84	avgR100	252.21	eps 0.037	Qval 33.06	1
and100 0.77	thr 169						
[DQN-Orig]	Ep 658 R	229.85	avgR100	251.63	eps 0.037	Qval 33.03	1
and100 0.77	thr 148						
[DQN-Orig]	Ep 659 R	283.27	avgR100	251.82	eps 0.037	Qval 32.69	1
and100 0.78	thr 277						
[DQN-Orig]	Ep 660 R	264.59	avgR100	252.03	eps 0.037	Qval 33.36	1
and100 0.78	thr 157						
[DQN-Orig]	Ep 661 R	250.52	avgR100	251.52	eps 0.036	Qval 32.84	1
and100 0.78	thr 195						
[DQN-Orig]	Ep 662 R	218.47	avgR100	251.10	eps 0.036	Qval 33.32	1
and100 0.78	thr 435						
[DQN-Orig]	Ep 663 R	241.05	avgR100	250.78	eps 0.036	Qval 32.00	1
and100 0.77	thr 322						
[DQN-Orig]	Ep 664 R	270.21	avgR100	251.01	eps 0.036	Qval 33.30	1
and100 0.77	thr 225						
[DQN-Orig]	Ep 665 R	251.19	avgR100	250.99	eps 0.036	Qval 33.84	1
and100 0.77	thr 345						
[DQN-Orig]	Ep 666 R	254.09	avgR100	251.11	eps 0.035	Qval 32.48	1
and100 0.77	thr 285						
[DQN-Orig]	Ep 667 R	247.62	avgR100	251.21	eps 0.035	Qval 31.83	1
and100 0.77	thr 491						
[DQN-Orig]	Ep 668 R	230.77	avgR100	250.44	eps 0.035	Qval 32.31	1
and100 0.77	thr 141						
[DQN-Orig]	Ep 669 R	244.19	avgR100	250.60	eps 0.035	Qval 31.56	1
and100 0.77	thr 234						
[DQN-Orig]	Ep 670 R	277.65	avgR100	250.79	eps 0.035	Qval 33.86	1
and100 0.77	thr 159						
[DQN-Orig]	Ep 671 R	245.60	avgR100	251.05	eps 0.035	Qval 32.03	1
and100 0.78	thr 236						
[DQN-Orig]	Ep 672 R	2.31	avgR100	248.79	eps 0.034	Qval 31.68	1
and100 0.77	thr 134						

[DQN-Orig] Ep 673 R and100 0.77 thr 406	227.81 avgR100	248.48 eps 0.034 Qval	31.44 1
[DQN-Orig] Ep 674 R and100 0.77 thr 193	277.05 avgR100	249.00 eps 0.034 Qval	33.07 1
[DQN-Orig] Ep 675 R and100 0.77 thr 178	243.80 avgR100	248.93 eps 0.034 Qval	31.66 1
[DQN-Orig] Ep 676 R and100 0.77 thr 303	249.22 avgR100	248.93 eps 0.034 Qval	31.58 1
[DQN-Orig] Ep 677 R and100 0.78 thr 182	276.36 avgR100	249.34 eps 0.034 Qval	31.29 1
[DQN-Orig] Ep 678 R and100 0.78 thr 252	277.35 avgR100	249.37 eps 0.033 Qval	31.53 1
[DQN-Orig] Ep 679 R and100 0.77 thr 316	266.24 avgR100	249.58 eps 0.033 Qval	32.48 1
[DQN-Orig] Ep 680 R and100 0.77 thr 432	252.71 avgR100	249.95 eps 0.033 Qval	32.20 1
[DQN-Orig] Ep 681 R and100 0.77 thr 185	243.86 avgR100	249.70 eps 0.033 Qval	31.80 1
[DQN-Orig] Ep 682 R and100 0.77 thr 130	277.02 avgR100	250.08 eps 0.033 Qval	31.55 1
[DQN-Orig] Ep 683 R and100 0.77 thr 233	233.18 avgR100	250.05 eps 0.033 Qval	30.90 1
[DQN-Orig] Ep 684 R and100 0.77 thr 173	251.23 avgR100	250.10 eps 0.032 Qval	32.40 1
[DQN-Orig] Ep 685 R and100 0.77 thr 173	265.48 avgR100	250.52 eps 0.032 Qval	33.14 1
[DQN-Orig] Ep 686 R and100 0.77 thr 122	241.79 avgR100	250.39 eps 0.032 Qval	31.78 1
[DQN-Orig] Ep 687 R and100 0.77 thr 570	212.04 avgR100	250.20 eps 0.032 Qval	32.41 1
[DQN-Orig] Ep 688 R and100 0.77 thr 238	257.78 avgR100	250.24 eps 0.032 Qval	32.47 1
[DQN-Orig] Ep 689 R and100 0.77 thr 352	265.88 avgR100	250.64 eps 0.032 Qval	32.44 1
[DQN-Orig] Ep 690 R and100 0.77 thr 162	240.60 avgR100	250.85 eps 0.031 Qval	32.44 1
[DQN-Orig] Ep 691 R and100 0.77 thr 136	272.32 avgR100	250.79 eps 0.031 Qval	33.27 1
[DQN-Orig] Ep 692 R and100 0.78 thr 153	299.27 avgR100	250.98 eps 0.031 Qval	32.38 1
[DQN-Orig] Ep 693 R and100 0.78 thr 280	235.70 avgR100	250.67 eps 0.031 Qval	32.30 1
[DQN-Orig] Ep 694 R and100 0.79 thr 173	260.61 avgR100	250.42 eps 0.031 Qval	32.89 1
[DQN-Orig] Ep 695 R and100 0.79 thr 170	261.40 avgR100	250.42 eps 0.031 Qval	32.35 1
[DQN-Orig] Ep 696 R and100 0.80 thr 139	247.12 avgR100	250.52 eps 0.031 Qval	32.18 1
[DQN-Orig] Ep 697 R and100 0.80 thr 130	282.15 avgR100	250.48 eps 0.030 Qval	31.99 1
[DQN-Orig] Ep 698 R and100 0.80 thr 256	267.61 avgR100	251.01 eps 0.030 Qval	32.67 1
[DQN-Orig] Ep 699 R and100 0.80 thr 122	262.30 avgR100	251.09 eps 0.030 Qval	32.01 1
[DQN-Orig] Ep 700 R and100 0.81 thr 172	259.02 avgR100	251.22 eps 0.030 Qval	32.75 1
[DQN-Orig] Ep 701 R and100 0.80 thr 443	203.48 avgR100	250.62 eps 0.030 Qval	31.87 1
[DQN-Orig] Ep 702 R and100 0.81 thr 334	208.67 avgR100	250.16 eps 0.030 Qval	31.81 1

[DQN-Orig] Ep 703 R and100 0.80 thr 185	268.05 avgR100	250.25 eps 0.029 Qval	32.55 1
[DQN-Orig] Ep 704 R and100 0.80 thr 270	264.77 avgR100	250.38 eps 0.029 Qval	32.59 1
[DQN-Orig] Ep 705 R and100 0.80 thr 155	247.60 avgR100	250.32 eps 0.029 Qval	32.89 1
[DQN-Orig] Ep 706 R and100 0.79 thr 430	223.25 avgR100	249.87 eps 0.029 Qval	33.45 1
[DQN-Orig] Ep 707 R and100 0.80 thr 262	275.49 avgR100	250.59 eps 0.029 Qval	33.46 1
[DQN-Orig] Ep 708 R and100 0.81 thr 164	238.37 avgR100	250.89 eps 0.029 Qval	33.86 1
[DQN-Orig] Ep 709 R and100 0.82 thr 234	245.68 avgR100	250.70 eps 0.029 Qval	32.48 1
[DQN-Orig] Ep 710 R and100 0.82 thr 286	275.58 avgR100	250.72 eps 0.028 Qval	33.11 1
[DQN-Orig] Ep 711 R and100 0.82 thr 160	262.93 avgR100	251.12 eps 0.028 Qval	33.88 1
[DQN-Orig] Ep 712 R and100 0.82 thr 210	291.94 avgR100	251.54 eps 0.028 Qval	32.67 1
[DQN-Orig] Ep 713 R and100 0.81 thr 230	270.49 avgR100	252.01 eps 0.028 Qval	33.38 1
[DQN-Orig] Ep 714 R and100 0.82 thr 434	227.73 avgR100	251.99 eps 0.028 Qval	33.21 1
[DQN-Orig] Ep 715 R and100 0.81 thr 445	263.44 avgR100	252.02 eps 0.028 Qval	33.36 1
[DQN-Orig] Ep 716 R and100 0.81 thr 191	302.49 avgR100	252.65 eps 0.028 Qval	33.44 1
[DQN-Orig] Ep 717 R and100 0.80 thr 222	254.79 avgR100	252.37 eps 0.027 Qval	33.04 1
[DQN-Orig] Ep 718 R and100 0.81 thr 257	258.37 avgR100	252.86 eps 0.027 Qval	32.77 1
[DQN-Orig] Ep 719 R and100 0.81 thr 232	282.84 avgR100	252.91 eps 0.027 Qval	32.87 1
[DQN-Orig] Ep 720 R and100 0.81 thr 279	242.04 avgR100	252.37 eps 0.027 Qval	32.05 1
[DQN-Orig] Ep 721 R and100 0.81 thr 174	294.63 avgR100	252.94 eps 0.027 Qval	32.13 1
[DQN-Orig] Ep 722 R and100 0.81 thr 166	251.84 avgR100	252.99 eps 0.027 Qval	31.78 1
[DQN-Orig] Ep 723 R and100 0.81 thr 270	260.85 avgR100	253.17 eps 0.027 Qval	33.01 1
[DQN-Orig] Ep 724 R and100 0.81 thr 256	247.86 avgR100	253.18 eps 0.027 Qval	31.81 1
[DQN-Orig] Ep 725 R and100 0.81 thr 314	272.40 avgR100	253.43 eps 0.026 Qval	31.82 1
[DQN-Orig] Ep 726 R and100 0.80 thr 120	267.09 avgR100	253.69 eps 0.026 Qval	31.02 1
[DQN-Orig] Ep 727 R and100 0.80 thr 186	262.04 avgR100	253.59 eps 0.026 Qval	31.16 1
[DQN-Orig] Ep 728 R and100 0.80 thr 148	257.53 avgR100	253.68 eps 0.026 Qval	32.08 1
[DQN-Orig] Ep 729 R and100 0.79 thr 201	284.99 avgR100	253.76 eps 0.026 Qval	31.67 1
[DQN-Orig] Ep 730 R and100 0.79 thr 133	265.93 avgR100	253.53 eps 0.026 Qval	31.45 1
[DQN-Orig] Ep 731 R and100 0.80 thr 194	241.91 avgR100	253.54 eps 0.026 Qval	32.18 1
[DQN-Orig] Ep 732 R and100 0.80 thr 210	263.95 avgR100	253.16 eps 0.025 Qval	31.86 1

[DQN-Orig]	Ep 733 R	280.47	avgR100	253.47	eps 0.025	Qval 33.37	1
and100 0.80	thr 301						
[DQN-Orig]	Ep 734 R	244.18	avgR100	253.09	eps 0.025	Qval 32.04	1
and100 0.80	thr 199						
[DQN-Orig]	Ep 735 R	255.87	avgR100	254.13	eps 0.025	Qval 32.87	1
and100 0.81	thr 240						
[DQN-Orig]	Ep 736 R	268.65	avgR100	255.14	eps 0.025	Qval 32.50	1
and100 0.82	thr 171						
[DQN-Orig]	Ep 737 R	249.82	avgR100	254.72	eps 0.025	Qval 32.90	1
and100 0.82	thr 323						
[DQN-Orig]	Ep 738 R	234.76	avgR100	254.04	eps 0.025	Qval 34.04	1
and100 0.82	thr 438						
[DQN-Orig]	Ep 739 R	281.77	avgR100	254.33	eps 0.025	Qval 33.52	1
and100 0.83	thr 194						
[DQN-Orig]	Ep 740 R	252.31	avgR100	253.79	eps 0.024	Qval 33.27	1
and100 0.83	thr 219						
[DQN-Orig]	Ep 741 R	125.94	avgR100	252.69	eps 0.024	Qval 32.89	1
and100 0.83	thr 701						
[DQN-Orig]	Ep 742 R	258.32	avgR100	252.73	eps 0.024	Qval 32.99	1
and100 0.83	thr 154						
[DQN-Orig]	Ep 743 R	228.91	avgR100	252.36	eps 0.024	Qval 32.54	1
and100 0.83	thr 281						
[DQN-Orig]	Ep 744 R	254.77	avgR100	252.52	eps 0.024	Qval 33.66	1
and100 0.83	thr 289						
[DQN-Orig]	Ep 745 R	273.29	avgR100	252.66	eps 0.024	Qval 34.09	1
and100 0.83	thr 165						
[DQN-Orig]	Ep 746 R	262.14	avgR100	252.51	eps 0.024	Qval 33.88	1
and100 0.83	thr 196						
[DQN-Orig]	Ep 747 R	239.09	avgR100	252.53	eps 0.024	Qval 33.59	1
and100 0.83	thr 185						
[DQN-Orig]	Ep 748 R	254.99	avgR100	252.92	eps 0.024	Qval 34.61	1
and100 0.83	thr 341						
[DQN-Orig]	Ep 749 R	258.30	avgR100	253.49	eps 0.023	Qval 33.72	1
and100 0.84	thr 277						
[DQN-Orig]	Ep 750 R	249.34	avgR100	253.22	eps 0.023	Qval 33.73	1
and100 0.85	thr 391						
[DQN-Orig]	Ep 751 R	258.74	avgR100	253.30	eps 0.023	Qval 33.64	1
and100 0.85	thr 185						
[DQN-Orig]	Ep 752 R	256.70	avgR100	252.86	eps 0.023	Qval 34.21	1
and100 0.85	thr 187						
[DQN-Orig]	Ep 753 R	295.19	avgR100	253.08	eps 0.023	Qval 33.39	1
and100 0.85	thr 216						
[DQN-Orig]	Ep 754 R	282.63	avgR100	253.11	eps 0.023	Qval 33.37	1
and100 0.86	thr 175						
[DQN-Orig]	Ep 755 R	260.33	avgR100	252.68	eps 0.023	Qval 33.09	1
and100 0.86	thr 282						
[DQN-Orig]	Ep 756 R	251.56	avgR100	253.18	eps 0.023	Qval 33.80	1
and100 0.87	thr 216						
[DQN-Orig]	Ep 757 R	207.64	avgR100	252.83	eps 0.022	Qval 34.22	1
and100 0.87	thr 309						
[DQN-Orig]	Ep 758 R	254.54	avgR100	253.08	eps 0.022	Qval 33.40	1
and100 0.87	thr 135						
[DQN-Orig]	Ep 759 R	247.98	avgR100	252.72	eps 0.022	Qval 34.22	1
and100 0.87	thr 247						
[DQN-Orig]	Ep 760 R	248.27	avgR100	252.56	eps 0.022	Qval 34.85	1
and100 0.87	thr 154						
[DQN-Orig]	Ep 761 R	277.77	avgR100	252.83	eps 0.022	Qval 33.06	1
and100 0.87	thr 171						
[DQN-Orig]	Ep 762 R	266.05	avgR100	253.31	eps 0.022	Qval 33.53	1
and100 0.87	thr 194						

[DQN-Orig]	Ep 763 R	273.35	avgR100	253.63	eps 0.022	Qval 34.16	1
and100 0.88	thr 205						
[DQN-Orig]	Ep 764 R	283.02	avgR100	253.76	eps 0.022	Qval 34.76	1
and100 0.88	thr 181						
[DQN-Orig]	Ep 765 R	240.16	avgR100	253.65	eps 0.022	Qval 33.15	1
and100 0.88	thr 420						
[DQN-Orig]	Ep 766 R	240.19	avgR100	253.51	eps 0.022	Qval 34.23	1
and100 0.88	thr 362						
[DQN-Orig]	Ep 767 R	240.58	avgR100	253.44	eps 0.021	Qval 33.34	1
and100 0.88	thr 517						
[DQN-Orig]	Ep 768 R	213.45	avgR100	253.27	eps 0.021	Qval 34.46	1
and100 0.87	thr 407						
[DQN-Orig]	Ep 769 R	314.00	avgR100	253.97	eps 0.021	Qval 34.18	1
and100 0.87	thr 183						
[DQN-Orig]	Ep 770 R	253.00	avgR100	253.72	eps 0.021	Qval 34.53	1
and100 0.87	thr 176						
[DQN-Orig]	Ep 771 R	262.11	avgR100	253.88	eps 0.021	Qval 35.67	1
and100 0.87	thr 144						
[DQN-Orig]	Ep 772 R	269.23	avgR100	256.55	eps 0.021	Qval 35.05	1
and100 0.88	thr 417						
[DQN-Orig]	Ep 773 R	264.33	avgR100	256.92	eps 0.021	Qval 35.51	1
and100 0.88	thr 183						
[DQN-Orig]	Ep 774 R	277.53	avgR100	256.92	eps 0.021	Qval 34.38	1
and100 0.88	thr 309						
[DQN-Orig]	Ep 775 R	277.50	avgR100	257.26	eps 0.021	Qval 35.08	1
and100 0.88	thr 133						
[DQN-Orig]	Ep 776 R	270.90	avgR100	257.48	eps 0.020	Qval 35.63	1
and100 0.88	thr 203						
[DQN-Orig]	Ep 777 R	280.92	avgR100	257.52	eps 0.020	Qval 36.59	1
and100 0.88	thr 154						
[DQN-Orig]	Ep 778 R	272.89	avgR100	257.48	eps 0.020	Qval 34.90	1
and100 0.88	thr 152						
[DQN-Orig]	Ep 779 R	298.91	avgR100	257.81	eps 0.020	Qval 35.94	1
and100 0.89	thr 167						
[DQN-Orig]	Ep 780 R	304.06	avgR100	258.32	eps 0.020	Qval 35.85	1
and100 0.89	thr 228						
[DQN-Orig]	Ep 781 R	272.44	avgR100	258.60	eps 0.020	Qval 35.39	1
and100 0.89	thr 137						
[DQN-Orig]	Ep 782 R	270.22	avgR100	258.54	eps 0.020	Qval 36.31	1
and100 0.89	thr 167						
[DQN-Orig]	Ep 783 R	261.36	avgR100	258.82	eps 0.020	Qval 36.43	1
and100 0.90	thr 157						
[DQN-Orig]	Ep 784 R	232.87	avgR100	258.63	eps 0.020	Qval 37.56	1
and100 0.90	thr 283						
[DQN-Orig]	Ep 785 R	278.76	avgR100	258.77	eps 0.020	Qval 35.68	1
and100 0.89	thr 300						
[DQN-Orig]	Ep 786 R	296.58	avgR100	259.32	eps 0.020	Qval 36.16	1
and100 0.89	thr 213						
[DQN-Orig]	Ep 787 R	255.22	avgR100	259.75	eps 0.020	Qval 36.28	1
and100 0.89	thr 220						
[DQN-Orig]	Ep 788 R	234.12	avgR100	259.51	eps 0.020	Qval 36.14	1
and100 0.89	thr 200						
[DQN-Orig]	Ep 789 R	295.59	avgR100	259.81	eps 0.020	Qval 36.61	1
and100 0.89	thr 173						
[DQN-Orig]	Ep 790 R	293.21	avgR100	260.33	eps 0.020	Qval 36.41	1
and100 0.89	thr 161						
[DQN-Orig]	Ep 791 R	268.68	avgR100	260.30	eps 0.020	Qval 37.60	1
and100 0.88	thr 386						
[DQN-Orig]	Ep 792 R	306.26	avgR100	260.37	eps 0.020	Qval 37.86	1
and100 0.88	thr 135						

[DQN-Orig]	Ep 793 R	286.27	avgR100	260.87	eps 0.020	Qval	37.20	1
and100 0.88	thr 277							
[DQN-Orig]	Ep 794 R	277.92	avgR100	261.05	eps 0.020	Qval	38.55	1
and100 0.88	thr 152							
[DQN-Orig]	Ep 795 R	299.38	avgR100	261.43	eps 0.020	Qval	38.12	1
and100 0.88	thr 153							
[DQN-Orig]	Ep 796 R	274.46	avgR100	261.70	eps 0.020	Qval	39.10	1
and100 0.88	thr 177							
[DQN-Orig]	Ep 797 R	257.58	avgR100	261.45	eps 0.020	Qval	38.99	1
and100 0.88	thr 356							
[DQN-Orig]	Ep 798 R	254.87	avgR100	261.33	eps 0.020	Qval	38.64	1
and100 0.88	thr 226							
[DQN-Orig]	Ep 799 R	259.24	avgR100	261.30	eps 0.020	Qval	39.05	1
and100 0.88	thr 167							
[DQN-Orig]	Ep 800 R	287.56	avgR100	261.58	eps 0.020	Qval	39.65	1
and100 0.88	thr 237							
[DQN-Orig]	Ep 801 R	266.52	avgR100	262.21	eps 0.020	Qval	38.48	1
and100 0.89	thr 222							
[DQN-Orig]	Ep 802 R	287.67	avgR100	263.00	eps 0.020	Qval	38.78	1
and100 0.89	thr 291							
[DQN-Orig]	Ep 803 R	307.98	avgR100	263.40	eps 0.020	Qval	38.66	1
and100 0.90	thr 132							
[DQN-Orig]	Ep 804 R	224.96	avgR100	263.00	eps 0.020	Qval	38.45	1
and100 0.90	thr 153							
[DQN-Orig]	Ep 805 R	232.82	avgR100	262.85	eps 0.020	Qval	39.04	1
and100 0.90	thr 200							
[DQN-Orig]	Ep 806 R	283.90	avgR100	263.46	eps 0.020	Qval	38.85	1
and100 0.91	thr 272							
[DQN-Orig]	Ep 807 R	308.33	avgR100	263.79	eps 0.020	Qval	39.32	1
and100 0.90	thr 175							
[DQN-Orig]	Ep 808 R	257.40	avgR100	263.98	eps 0.020	Qval	40.08	1
and100 0.90	thr 247							
[DQN-Orig]	Ep 809 R	236.78	avgR100	263.89	eps 0.020	Qval	39.77	1
and100 0.90	thr 139							
[DQN-Orig]	Ep 810 R	272.77	avgR100	263.86	eps 0.020	Qval	39.49	1
and100 0.90	thr 220							
[DQN-Orig]	Ep 811 R	241.81	avgR100	263.65	eps 0.020	Qval	39.95	1
and100 0.90	thr 184							
[DQN-Orig]	Ep 812 R	273.15	avgR100	263.46	eps 0.020	Qval	40.07	1
and100 0.90	thr 260							
[DQN-Orig]	Ep 813 R	244.44	avgR100	263.20	eps 0.020	Qval	40.25	1
and100 0.91	thr 393							
[DQN-Orig]	Ep 814 R	236.40	avgR100	263.29	eps 0.020	Qval	39.66	1
and100 0.91	thr 237							
[DQN-Orig]	Ep 815 R	269.27	avgR100	263.35	eps 0.020	Qval	39.59	1
and100 0.92	thr 204							
[DQN-Orig]	Ep 816 R	255.40	avgR100	262.88	eps 0.020	Qval	39.30	1
and100 0.92	thr 167							
[DQN-Orig]	Ep 817 R	266.11	avgR100	262.99	eps 0.020	Qval	40.56	1
and100 0.93	thr 129							
[DQN-Orig]	Ep 818 R	265.10	avgR100	263.06	eps 0.020	Qval	40.98	1
and100 0.93	thr 231							
[DQN-Orig]	Ep 819 R	267.92	avgR100	262.91	eps 0.020	Qval	40.41	1
and100 0.93	thr 372							
[DQN-Orig]	Ep 820 R	261.56	avgR100	263.10	eps 0.020	Qval	39.78	1
and100 0.93	thr 164							
[DQN-Orig]	Ep 821 R	233.78	avgR100	262.50	eps 0.020	Qval	41.08	1
and100 0.93	thr 173							
[DQN-Orig]	Ep 822 R	290.75	avgR100	262.88	eps 0.020	Qval	42.44	1
and100 0.93	thr 190							

[DQN-Orig]	Ep 823 R	295.47	avgR100	263.23	eps 0.020	Qval	42.76	1
and100 0.92	thr 255							
[DQN-Orig]	Ep 824 R	250.46	avgR100	263.26	eps 0.020	Qval	41.91	1
and100 0.92	thr 246							
[DQN-Orig]	Ep 825 R	220.36	avgR100	262.74	eps 0.020	Qval	42.33	1
and100 0.92	thr 247							
[DQN-Orig]	Ep 826 R	254.13	avgR100	262.61	eps 0.020	Qval	41.42	1
and100 0.93	thr 187							
[DQN-Orig]	Ep 827 R	264.98	avgR100	262.64	eps 0.020	Qval	41.73	1
and100 0.93	thr 172							
[DQN-Orig]	Ep 828 R	259.11	avgR100	262.65	eps 0.020	Qval	42.48	1
and100 0.93	thr 200							
[DQN-Orig]	Ep 829 R	280.53	avgR100	262.61	eps 0.020	Qval	42.74	1
and100 0.94	thr 268							
[DQN-Orig]	Ep 830 R	234.79	avgR100	262.30	eps 0.020	Qval	43.06	1
and100 0.94	thr 478							
[DQN-Orig]	Ep 831 R	251.14	avgR100	262.39	eps 0.020	Qval	43.31	1
and100 0.94	thr 188							
[DQN-Orig]	Ep 832 R	263.64	avgR100	262.39	eps 0.020	Qval	43.77	1
and100 0.94	thr 202							
[DQN-Orig]	Ep 833 R	262.23	avgR100	262.20	eps 0.020	Qval	43.32	1
and100 0.94	thr 367							
[DQN-Orig]	Ep 834 R	261.04	avgR100	262.37	eps 0.020	Qval	44.62	1
and100 0.94	thr 273							
[DQN-Orig]	Ep 835 R	250.83	avgR100	262.32	eps 0.020	Qval	44.07	1
and100 0.93	thr 224							
[DQN-Orig]	Ep 836 R	231.96	avgR100	261.95	eps 0.020	Qval	44.81	1
and100 0.93	thr 451							
[DQN-Orig]	Ep 837 R	249.06	avgR100	261.95	eps 0.020	Qval	45.22	1
and100 0.93	thr 338							
[DQN-Orig]	Ep 838 R	230.60	avgR100	261.90	eps 0.020	Qval	44.30	1
and100 0.93	thr 156							
[DQN-Orig]	Ep 839 R	262.38	avgR100	261.71	eps 0.020	Qval	45.45	1
and100 0.93	thr 419							
[DQN-Orig]	Ep 840 R	294.76	avgR100	262.14	eps 0.020	Qval	45.65	1
and100 0.93	thr 251							
[DQN-Orig]	Ep 841 R	258.72	avgR100	263.46	eps 0.020	Qval	46.37	1
and100 0.94	thr 172							
[DQN-Orig]	Ep 842 R	241.21	avgR100	263.29	eps 0.020	Qval	45.58	1
and100 0.94	thr 182							
[DQN-Orig]	Ep 843 R	288.64	avgR100	263.89	eps 0.020	Qval	46.03	1
and100 0.94	thr 168							
[DQN-Orig]	Ep 844 R	286.04	avgR100	264.20	eps 0.020	Qval	46.60	1
and100 0.94	thr 159							
[DQN-Orig]	Ep 845 R	307.75	avgR100	264.55	eps 0.020	Qval	46.12	1
and100 0.94	thr 163							
[DQN-Orig]	Ep 846 R	236.82	avgR100	264.29	eps 0.020	Qval	46.24	1
and100 0.94	thr 262							
[DQN-Orig]	Ep 847 R	268.55	avgR100	264.59	eps 0.020	Qval	45.89	1
and100 0.94	thr 187							
[DQN-Orig]	Ep 848 R	290.60	avgR100	264.94	eps 0.020	Qval	46.00	1
and100 0.94	thr 180							
[DQN-Orig]	Ep 849 R	288.34	avgR100	265.24	eps 0.020	Qval	46.88	1
and100 0.94	thr 217							
[DQN-Orig]	Ep 850 R	310.71	avgR100	265.86	eps 0.020	Qval	46.54	1
and100 0.94	thr 159							
[DQN-Orig]	Ep 851 R	236.29	avgR100	265.63	eps 0.020	Qval	46.46	1
and100 0.93	thr 328							
[DQN-Orig]	Ep 852 R	291.97	avgR100	265.99	eps 0.020	Qval	46.92	1
and100 0.93	thr 230							

[DQN-Orig]	Ep 853 R	274.16	avgR100	265.78	eps 0.020	Qval	46.13	1
and100 0.93	thr 145							
[DQN-Orig]	Ep 854 R	273.44	avgR100	265.68	eps 0.020	Qval	47.05	1
and100 0.93	thr 323							
[DQN-Orig]	Ep 855 R	262.70	avgR100	265.71	eps 0.020	Qval	46.72	1
and100 0.93	thr 356							
[DQN-Orig]	Ep 856 R	206.36	avgR100	265.26	eps 0.020	Qval	47.77	1
and100 0.93	thr 346							
[DQN-Orig]	Ep 857 R	249.50	avgR100	265.67	eps 0.020	Qval	46.99	1
and100 0.93	thr 171							
[DQN-Orig]	Ep 858 R	266.85	avgR100	265.80	eps 0.020	Qval	47.47	1
and100 0.93	thr 325							
[DQN-Orig]	Ep 859 R	298.80	avgR100	266.31	eps 0.020	Qval	47.48	1
and100 0.93	thr 199							
[DQN-Orig]	Ep 860 R	278.46	avgR100	266.61	eps 0.020	Qval	47.73	1
and100 0.93	thr 202							
[DQN-Orig]	Ep 861 R	264.10	avgR100	266.47	eps 0.020	Qval	47.01	1
and100 0.93	thr 165							
[DQN-Orig]	Ep 862 R	263.80	avgR100	266.45	eps 0.020	Qval	47.14	1
and100 0.93	thr 330							
[DQN-Orig]	Ep 863 R	271.71	avgR100	266.43	eps 0.020	Qval	47.34	1
and100 0.93	thr 382							
[DQN-Orig]	Ep 864 R	272.94	avgR100	266.33	eps 0.020	Qval	47.10	1
and100 0.93	thr 154							
[DQN-Orig]	Ep 865 R	252.10	avgR100	266.45	eps 0.020	Qval	47.73	1
and100 0.93	thr 136							
[DQN-Orig]	Ep 866 R	258.45	avgR100	266.63	eps 0.020	Qval	47.10	1
and100 0.93	thr 184							
[DQN-Orig]	Ep 867 R	259.49	avgR100	266.82	eps 0.020	Qval	47.15	1
and100 0.93	thr 184							
[DQN-Orig]	Ep 868 R	229.54	avgR100	266.98	eps 0.020	Qval	47.07	1
and100 0.94	thr 182							
[DQN-Orig]	Ep 869 R	268.38	avgR100	266.53	eps 0.020	Qval	46.97	1
and100 0.93	thr 273							
[DQN-Orig]	Ep 870 R	275.99	avgR100	266.76	eps 0.020	Qval	46.82	1
and100 0.93	thr 156							
[DQN-Orig]	Ep 871 R	265.17	avgR100	266.79	eps 0.020	Qval	47.39	1
and100 0.93	thr 172							
[DQN-Orig]	Ep 872 R	276.94	avgR100	266.86	eps 0.020	Qval	47.23	1
and100 0.93	thr 161							
[DQN-Orig]	Ep 873 R	226.81	avgR100	266.49	eps 0.020	Qval	46.89	1
and100 0.93	thr 275							
[DQN-Orig]	Ep 874 R	217.42	avgR100	265.89	eps 0.020	Qval	48.14	1
and100 0.93	thr 296							
[DQN-Orig]	Ep 875 R	251.87	avgR100	265.63	eps 0.020	Qval	47.58	1
and100 0.93	thr 150							
[DQN-Orig]	Ep 876 R	273.41	avgR100	265.66	eps 0.020	Qval	47.72	1
and100 0.93	thr 288							
[DQN-Orig]	Ep 877 R	268.86	avgR100	265.54	eps 0.020	Qval	47.35	1
and100 0.93	thr 217							
[DQN-Orig]	Ep 878 R	264.53	avgR100	265.45	eps 0.020	Qval	48.21	1
and100 0.93	thr 137							
[DQN-Orig]	Ep 879 R	289.41	avgR100	265.36	eps 0.020	Qval	47.22	1
and100 0.93	thr 168							
[DQN-Orig]	Ep 880 R	250.90	avgR100	264.83	eps 0.020	Qval	47.42	1
and100 0.92	thr 277							
[DQN-Orig]	Ep 881 R	243.84	avgR100	264.54	eps 0.020	Qval	47.68	1
and100 0.92	thr 225							

[DQN-Orig]	Ep 882 R	268.70	avgR100	264.53	eps 0.020	Qval	47.10	1
and100 0.92	thr 205							
[DQN-Orig]	Ep 883 R	241.90	avgR100	264.33	eps 0.020	Qval	47.55	1
and100 0.92	thr 152							
[DQN-Orig]	Ep 884 R	260.93	avgR100	264.61	eps 0.020	Qval	47.99	1
and100 0.92	thr 184							
[DQN-Orig]	Ep 885 R	233.67	avgR100	264.16	eps 0.020	Qval	48.95	1
and100 0.93	thr 332							
[DQN-Orig]	Ep 886 R	254.23	avgR100	263.74	eps 0.020	Qval	47.78	1
and100 0.93	thr 197							
[DQN-Orig]	Ep 887 R	233.98	avgR100	263.52	eps 0.020	Qval	47.07	1
and100 0.93	thr 145							
[DQN-Orig]	Ep 888 R	260.71	avgR100	263.79	eps 0.020	Qval	48.23	1
and100 0.93	thr 284							
[DQN-Orig]	Ep 889 R	246.98	avgR100	263.30	eps 0.020	Qval	48.61	1
and100 0.92	thr 188							
[DQN-Orig]	Ep 890 R	223.60	avgR100	262.61	eps 0.020	Qval	48.47	1
and100 0.92	thr 257							
[DQN-Orig]	Ep 891 R	270.25	avgR100	262.62	eps 0.020	Qval	49.27	1
and100 0.93	thr 175							
[DQN-Orig]	Ep 892 R	239.70	avgR100	261.96	eps 0.020	Qval	49.08	1
and100 0.93	thr 254							
[DQN-Orig]	Ep 893 R	293.63	avgR100	262.03	eps 0.020	Qval	48.91	1
and100 0.93	thr 290							
[DQN-Orig]	Ep 894 R	221.42	avgR100	261.47	eps 0.020	Qval	48.94	1
and100 0.93	thr 244							
[DQN-Orig]	Ep 895 R	285.45	avgR100	261.33	eps 0.020	Qval	48.73	1
and100 0.93	thr 195							
[DQN-Orig]	Ep 896 R	235.05	avgR100	260.93	eps 0.020	Qval	47.85	1
and100 0.93	thr 243							
[DQN-Orig]	Ep 897 R	267.98	avgR100	261.04	eps 0.020	Qval	48.87	1
and100 0.93	thr 142							
[DQN-Orig]	Ep 898 R	260.47	avgR100	261.09	eps 0.020	Qval	48.55	1
and100 0.92	thr 278							
[DQN-Orig]	Ep 899 R	259.72	avgR100	261.10	eps 0.020	Qval	48.38	1
and100 0.91	thr 405							
[DQN-Orig]	Ep 900 R	255.88	avgR100	260.78	eps 0.020	Qval	49.30	1
and100 0.91	thr 167							
[DQN-Orig]	Ep 901 R	251.08	avgR100	260.63	eps 0.020	Qval	48.30	1
and100 0.91	thr 174							
[DQN-Orig]	Ep 902 R	257.91	avgR100	260.33	eps 0.020	Qval	47.74	1
and100 0.91	thr 187							
[DQN-Orig]	Ep 903 R	239.81	avgR100	259.65	eps 0.020	Qval	48.78	1
and100 0.91	thr 281							
[DQN-Orig]	Ep 904 R	275.06	avgR100	260.15	eps 0.020	Qval	48.36	1
and100 0.91	thr 259							
[DQN-Orig]	Ep 905 R	249.26	avgR100	260.31	eps 0.020	Qval	49.60	1
and100 0.91	thr 160							
[DQN-Orig]	Ep 906 R	275.76	avgR100	260.23	eps 0.020	Qval	47.89	1
and100 0.91	thr 182							
[DQN-Orig]	Ep 907 R	282.42	avgR100	259.97	eps 0.020	Qval	47.66	1
and100 0.92	thr 161							
[DQN-Orig]	Ep 908 R	269.75	avgR100	260.10	eps 0.020	Qval	47.81	1
and100 0.92	thr 143							
[DQN-Orig]	Ep 909 R	249.11	avgR100	260.22	eps 0.020	Qval	48.01	1
and100 0.92	thr 295							
[DQN-Orig]	Ep 910 R	257.89	avgR100	260.07	eps 0.020	Qval	48.94	1
and100 0.92	thr 156							

[DQN-Orig]	Ep 911 R	286.54 avgR100	260.52 eps 0.020	Qval 48.14 1
and100 0.92	thr 264			
[DQN-Orig]	Ep 912 R	263.96 avgR100	260.43 eps 0.020	Qval 48.59 1
and100 0.92	thr 168			
[DQN-Orig]	Ep 913 R	252.42 avgR100	260.51 eps 0.020	Qval 48.40 1
and100 0.92	thr 131			
[DQN-Orig]	Ep 914 R	264.25 avgR100	260.78 eps 0.020	Qval 48.40 1
and100 0.92	thr 169			
[DQN-Orig]	Ep 915 R	274.06 avgR100	260.83 eps 0.020	Qval 48.15 1
and100 0.92	thr 142			
[DQN-Orig]	Ep 916 R	259.31 avgR100	260.87 eps 0.020	Qval 48.43 1
and100 0.92	thr 176			
[DQN-Orig]	Ep 917 R	221.89 avgR100	260.43 eps 0.020	Qval 47.95 1
and100 0.92	thr 206			
[DQN-Orig]	Ep 918 R	263.54 avgR100	260.41 eps 0.020	Qval 49.64 1
and100 0.92	thr 226			
[DQN-Orig]	Ep 919 R	248.40 avgR100	260.22 eps 0.020	Qval 48.33 1
and100 0.92	thr 172			
[DQN-Orig]	Ep 920 R	261.50 avgR100	260.22 eps 0.020	Qval 48.28 1
and100 0.92	thr 185			
[DQN-Orig]	Ep 921 R	241.27 avgR100	260.29 eps 0.020	Qval 47.99 1
and100 0.92	thr 247			
[DQN-Orig]	Ep 922 R	284.19 avgR100	260.23 eps 0.020	Qval 49.19 1
and100 0.92	thr 164			
[DQN-Orig]	Ep 923 R	280.76 avgR100	260.08 eps 0.020	Qval 48.59 1
and100 0.93	thr 133			
[DQN-Orig]	Ep 924 R	286.94 avgR100	260.44 eps 0.020	Qval 49.95 1
and100 0.93	thr 202			
[DQN-Orig]	Ep 925 R	70.96 avgR100	258.95 eps 0.020	Qval 48.90 1
and100 0.92	thr 126			
[DQN-Orig]	Ep 926 R	286.58 avgR100	259.28 eps 0.020	Qval 48.74 1
and100 0.92	thr 269			
[DQN-Orig]	Ep 927 R	246.12 avgR100	259.09 eps 0.020	Qval 48.77 1
and100 0.92	thr 134			
[DQN-Orig]	Ep 928 R	254.86 avgR100	259.04 eps 0.020	Qval 48.81 1
and100 0.92	thr 160			
[DQN-Orig]	Ep 929 R	275.65 avgR100	259.00 eps 0.020	Qval 48.76 1
and100 0.92	thr 208			
[DQN-Orig]	Ep 930 R	241.35 avgR100	259.06 eps 0.020	Qval 48.47 1
and100 0.92	thr 152			
[DQN-Orig]	Ep 931 R	298.51 avgR100	259.53 eps 0.020	Qval 48.31 1
and100 0.92	thr 201			
[DQN-Orig]	Ep 932 R	263.73 avgR100	259.54 eps 0.020	Qval 48.53 1
and100 0.92	thr 186			
[DQN-Orig]	Ep 933 R	245.55 avgR100	259.37 eps 0.020	Qval 49.76 1
and100 0.92	thr 168			
[DQN-Orig]	Ep 934 R	262.98 avgR100	259.39 eps 0.020	Qval 49.66 1
and100 0.92	thr 132			
[DQN-Orig]	Ep 935 R	257.82 avgR100	259.46 eps 0.020	Qval 49.83 1
and100 0.93	thr 313			
[DQN-Orig]	Ep 936 R	254.52 avgR100	259.68 eps 0.020	Qval 49.66 1
and100 0.93	thr 170			
[DQN-Orig]	Ep 937 R	303.54 avgR100	260.23 eps 0.020	Qval 49.43 1
and100 0.93	thr 135			
[DQN-Orig]	Ep 938 R	246.97 avgR100	260.39 eps 0.020	Qval 49.55 1
and100 0.92	thr 165			
[DQN-Orig]	Ep 939 R	279.58 avgR100	260.56 eps 0.020	Qval 50.48 1
and100 0.92	thr 170			
[DQN-Orig]	Ep 940 R	265.67 avgR100	260.27 eps 0.020	Qval 50.05 1
and100 0.91	thr 149			

[DQN-Orig]	Ep 941 R	235.74	avgR100	260.04	eps 0.020	Qval 50.26	1
and100 0.91	thr 163						
[DQN-Orig]	Ep 942 R	252.19	avgR100	260.15	eps 0.020	Qval 49.77	1
and100 0.90	thr 153						
[DQN-Orig]	Ep 943 R	276.49	avgR100	260.03	eps 0.020	Qval 50.16	1
and100 0.89	thr 168						
[DQN-Orig]	Ep 944 R	299.77	avgR100	260.17	eps 0.020	Qval 50.24	1
and100 0.89	thr 164						
[DQN-Orig]	Ep 945 R	274.54	avgR100	259.84	eps 0.020	Qval 49.76	1
and100 0.89	thr 201						
[DQN-Orig]	Ep 946 R	265.29	avgR100	260.12	eps 0.020	Qval 50.10	1
and100 0.89	thr 371						
[DQN-Orig]	Ep 947 R	301.27	avgR100	260.45	eps 0.020	Qval 50.66	1
and100 0.89	thr 162						
[DQN-Orig]	Ep 948 R	265.88	avgR100	260.20	eps 0.020	Qval 49.78	1
and100 0.88	thr 286						
[DQN-Orig]	Ep 949 R	274.54	avgR100	260.06	eps 0.020	Qval 50.18	1
and100 0.88	thr 166						
[DQN-Orig]	Ep 950 R	287.00	avgR100	259.83	eps 0.020	Qval 50.90	1
and100 0.88	thr 204						
[DQN-Orig]	Ep 951 R	273.56	avgR100	260.20	eps 0.020	Qval 50.77	1
and100 0.89	thr 207						
[DQN-Orig]	Ep 952 R	242.75	avgR100	259.71	eps 0.020	Qval 50.22	1
and100 0.88	thr 323						
[DQN-Orig]	Ep 953 R	289.13	avgR100	259.86	eps 0.020	Qval 50.09	1
and100 0.88	thr 222						
[DQN-Orig]	Ep 954 R	271.30	avgR100	259.84	eps 0.020	Qval 50.56	1
and100 0.88	thr 243						
[DQN-Orig]	Ep 955 R	247.36	avgR100	259.68	eps 0.020	Qval 50.03	1
and100 0.88	thr 238						
[DQN-Orig]	Ep 956 R	278.67	avgR100	260.40	eps 0.020	Qval 49.74	1
and100 0.88	thr 282						
[DQN-Orig]	Ep 957 R	250.47	avgR100	260.41	eps 0.020	Qval 50.60	1
and100 0.88	thr 143						
[DQN-Orig]	Ep 958 R	282.23	avgR100	260.57	eps 0.020	Qval 49.89	1
and100 0.88	thr 181						
[DQN-Orig]	Ep 959 R	254.93	avgR100	260.13	eps 0.020	Qval 49.67	1
and100 0.88	thr 165						
[DQN-Orig]	Ep 960 R	291.18	avgR100	260.26	eps 0.020	Qval 50.68	1
and100 0.87	thr 180						
[DQN-Orig]	Ep 961 R	242.87	avgR100	260.04	eps 0.020	Qval 49.85	1
and100 0.87	thr 190						
[DQN-Orig]	Ep 962 R	272.47	avgR100	260.13	eps 0.020	Qval 50.10	1
and100 0.87	thr 188						
[DQN-Orig]	Ep 963 R	292.53	avgR100	260.34	eps 0.020	Qval 49.15	1
and100 0.87	thr 227						
[DQN-Orig]	Ep 964 R	261.35	avgR100	260.22	eps 0.020	Qval 49.78	1
and100 0.87	thr 129						
[DQN-Orig]	Ep 965 R	308.95	avgR100	260.79	eps 0.020	Qval 49.23	1
and100 0.87	thr 186						
[DQN-Orig]	Ep 966 R	235.55	avgR100	260.56	eps 0.020	Qval 49.38	1
and100 0.87	thr 167						
[DQN-Orig]	Ep 967 R	254.18	avgR100	260.51	eps 0.020	Qval 49.52	1
and100 0.87	thr 279						
[DQN-Orig]	Ep 968 R	272.24	avgR100	260.94	eps 0.020	Qval 49.61	1
and100 0.87	thr 151						
[DQN-Orig]	Ep 969 R	263.54	avgR100	260.89	eps 0.020	Qval 50.69	1
and100 0.88	thr 318						
[DQN-Orig]	Ep 970 R	305.26	avgR100	261.18	eps 0.020	Qval 50.67	1
and100 0.88	thr 190						

[DQN-Orig]	Ep 971 R	281.39	avgR100	261.34	eps 0.020	Qval 50.26	1
and100 0.88	thr 204						
[DQN-Orig]	Ep 972 R	307.07	avgR100	261.64	eps 0.020	Qval 50.24	1
and100 0.88	thr 245						
[DQN-Orig]	Ep 973 R	272.77	avgR100	262.10	eps 0.020	Qval 50.56	1
and100 0.87	thr 195						
[DQN-Orig]	Ep 974 R	263.26	avgR100	262.56	eps 0.020	Qval 49.67	1
and100 0.87	thr 172						
[DQN-Orig]	Ep 975 R	268.07	avgR100	262.72	eps 0.020	Qval 50.16	1
and100 0.86	thr 190						
[DQN-Orig]	Ep 976 R	263.57	avgR100	262.63	eps 0.020	Qval 49.24	1
and100 0.86	thr 173						
[DQN-Orig]	Ep 977 R	255.95	avgR100	262.50	eps 0.020	Qval 49.76	1
and100 0.86	thr 148						
[DQN-Orig]	Ep 978 R	257.00	avgR100	262.42	eps 0.020	Qval 49.66	1
and100 0.86	thr 126						
[DQN-Orig]	Ep 979 R	285.13	avgR100	262.38	eps 0.020	Qval 48.82	1
and100 0.86	thr 236						
[DQN-Orig]	Ep 980 R	278.93	avgR100	262.66	eps 0.020	Qval 49.40	1
and100 0.87	thr 155						
[DQN-Orig]	Ep 981 R	246.39	avgR100	262.68	eps 0.020	Qval 48.79	1
and100 0.86	thr 197						
[DQN-Orig]	Ep 982 R	244.75	avgR100	262.45	eps 0.020	Qval 48.40	1
and100 0.86	thr 235						
[DQN-Orig]	Ep 983 R	262.56	avgR100	262.65	eps 0.020	Qval 48.41	1
and100 0.86	thr 203						
[DQN-Orig]	Ep 984 R	303.11	avgR100	263.07	eps 0.020	Qval 47.86	1
and100 0.86	thr 208						
[DQN-Orig]	Ep 985 R	290.48	avgR100	263.64	eps 0.020	Qval 48.74	1
and100 0.86	thr 221						
[DQN-Orig]	Ep 986 R	295.41	avgR100	264.05	eps 0.020	Qval 47.25	1
and100 0.86	thr 253						
[DQN-Orig]	Ep 987 R	265.31	avgR100	264.37	eps 0.020	Qval 47.90	1
and100 0.86	thr 154						
[DQN-Orig]	Ep 988 R	284.00	avgR100	264.60	eps 0.020	Qval 47.81	1
and100 0.86	thr 191						
[DQN-Orig]	Ep 989 R	275.09	avgR100	264.88	eps 0.020	Qval 47.24	1
and100 0.87	thr 131						
[DQN-Orig]	Ep 990 R	263.98	avgR100	265.28	eps 0.020	Qval 46.59	1
and100 0.87	thr 209						
[DQN-Orig]	Ep 991 R	278.17	avgR100	265.36	eps 0.020	Qval 47.33	1
and100 0.87	thr 145						
[DQN-Orig]	Ep 992 R	273.67	avgR100	265.70	eps 0.020	Qval 47.64	1
and100 0.87	thr 201						
[DQN-Orig]	Ep 993 R	285.93	avgR100	265.63	eps 0.020	Qval 46.23	1
and100 0.87	thr 219						
[DQN-Orig]	Ep 994 R	265.09	avgR100	266.06	eps 0.020	Qval 47.35	1
and100 0.87	thr 169						
[DQN-Orig]	Ep 995 R	287.47	avgR100	266.08	eps 0.020	Qval 46.92	1
and100 0.87	thr 165						
[DQN-Orig]	Ep 996 R	276.49	avgR100	266.50	eps 0.020	Qval 46.95	1
and100 0.87	thr 177						
[DQN-Orig]	Ep 997 R	278.07	avgR100	266.60	eps 0.020	Qval 47.62	1
and100 0.87	thr 171						
[DQN-Orig]	Ep 998 R	294.41	avgR100	266.94	eps 0.020	Qval 46.26	1
and100 0.88	thr 296						
[DQN-Orig]	Ep 999 R	282.20	avgR100	267.16	eps 0.020	Qval 46.48	1
and100 0.89	thr 155						

[DQN-Orig] Ep 1000 | R 280.34 | avgR100 267.41 | eps 0.020 | Qval 47.62 | 1
and100 0.89 | thr 164
[DQN-Orig] finished 1000 episodes in 3.3 min (final avgR100 = 267.41)

TRAINING: DDQN-Orig (modified=False, double=True)
#####

[DDQN-Orig] Ep	1	R	-76.76	avgR100	-76.76	eps	0.995	Qval	0.05	
land100	0.00	thr	43							
[DDQN-Orig] Ep	2	R	-111.39	avgR100	-94.07	eps	0.990	Qval	0.05	
land100	0.00	thr	65							
[DDQN-Orig] Ep	3	R	-248.17	avgR100	-145.44	eps	0.985	Qval	0.05	
land100	0.00	thr	64							
[DDQN-Orig] Ep	4	R	-122.59	avgR100	-139.73	eps	0.980	Qval	0.05	
land100	0.00	thr	56							
[DDQN-Orig] Ep	5	R	-292.21	avgR100	-170.22	eps	0.975	Qval	0.05	
land100	0.00	thr	73							
[DDQN-Orig] Ep	6	R	-118.75	avgR100	-161.64	eps	0.970	Qval	0.05	
land100	0.00	thr	79							
[DDQN-Orig] Ep	7	R	-500.86	avgR100	-210.10	eps	0.966	Qval	0.05	
land100	0.00	thr	74							
[DDQN-Orig] Ep	8	R	-112.02	avgR100	-197.84	eps	0.961	Qval	0.05	
land100	0.00	thr	51							
[DDQN-Orig] Ep	9	R	-95.80	avgR100	-186.51	eps	0.956	Qval	0.05	
land100	0.00	thr	59							
[DDQN-Orig] Ep	10	R	-224.33	avgR100	-190.29	eps	0.951	Qval	0.05	
land100	0.00	thr	79							
[DDQN-Orig] Ep	11	R	-289.42	avgR100	-199.30	eps	0.946	Qval	0.05	
land100	0.00	thr	81							
[DDQN-Orig] Ep	12	R	-102.24	avgR100	-191.21	eps	0.942	Qval	0.09	
land100	0.00	thr	55							
[DDQN-Orig] Ep	13	R	-48.74	avgR100	-180.25	eps	0.937	Qval	0.15	
land100	0.00	thr	63							
[DDQN-Orig] Ep	14	R	-498.84	avgR100	-203.01	eps	0.932	Qval	0.25	
land100	0.00	thr	104							
[DDQN-Orig] Ep	15	R	-93.01	avgR100	-195.67	eps	0.928	Qval	0.15	
land100	0.00	thr	56							
[DDQN-Orig] Ep	16	R	-185.01	avgR100	-195.01	eps	0.923	Qval	0.18	
land100	0.00	thr	51							
[DDQN-Orig] Ep	17	R	-183.11	avgR100	-194.31	eps	0.918	Qval	0.10	
land100	0.00	thr	106							
[DDQN-Orig] Ep	18	R	-301.05	avgR100	-200.24	eps	0.914	Qval	-0.03	
land100	0.00	thr	94							
[DDQN-Orig] Ep	19	R	-401.95	avgR100	-210.85	eps	0.909	Qval	-0.38	
land100	0.00	thr	97							
[DDQN-Orig] Ep	20	R	-278.16	avgR100	-214.22	eps	0.905	Qval	-0.33	
land100	0.00	thr	75							
[DDQN-Orig] Ep	21	R	-187.16	avgR100	-212.93	eps	0.900	Qval	-0.23	
land100	0.00	thr	109							
[DDQN-Orig] Ep	22	R	-84.46	avgR100	-207.09	eps	0.896	Qval	0.12	
land100	0.00	thr	50							

[DDQN-Orig]	Ep	23	R	-242.43	avgR100	-208.63	eps	0.891	Qval	0.37
land100 0.00	thr	79								
[DDQN-Orig]	Ep	24	R	29.91	avgR100	-198.69	eps	0.887	Qval	0.22
land100 0.00	thr	80								
[DDQN-Orig]	Ep	25	R	-297.48	avgR100	-202.64	eps	0.882	Qval	0.45
land100 0.00	thr	75								
[DDQN-Orig]	Ep	26	R	-120.85	avgR100	-199.50	eps	0.878	Qval	0.55
land100 0.00	thr	68								
[DDQN-Orig]	Ep	27	R	-104.17	avgR100	-195.96	eps	0.873	Qval	0.38
land100 0.00	thr	80								
[DDQN-Orig]	Ep	28	R	-273.63	avgR100	-198.74	eps	0.869	Qval	0.50
land100 0.00	thr	57								
[DDQN-Orig]	Ep	29	R	-131.88	avgR100	-196.43	eps	0.865	Qval	0.73
land100 0.00	thr	68								
[DDQN-Orig]	Ep	30	R	-103.57	avgR100	-193.34	eps	0.860	Qval	0.54
land100 0.00	thr	64								
[DDQN-Orig]	Ep	31	R	-112.60	avgR100	-190.73	eps	0.856	Qval	0.73
land100 0.00	thr	46								
[DDQN-Orig]	Ep	32	R	-96.21	avgR100	-187.78	eps	0.852	Qval	0.87
land100 0.00	thr	88								
[DDQN-Orig]	Ep	33	R	-46.33	avgR100	-183.49	eps	0.848	Qval	0.89
land100 0.00	thr	56								
[DDQN-Orig]	Ep	34	R	-201.90	avgR100	-184.03	eps	0.843	Qval	1.06
land100 0.00	thr	95								
[DDQN-Orig]	Ep	35	R	-166.58	avgR100	-183.54	eps	0.839	Qval	1.09
land100 0.00	thr	89								
[DDQN-Orig]	Ep	36	R	-201.89	avgR100	-184.05	eps	0.835	Qval	1.45
land100 0.00	thr	119								
[DDQN-Orig]	Ep	37	R	-169.09	avgR100	-183.64	eps	0.831	Qval	1.25
land100 0.00	thr	71								
[DDQN-Orig]	Ep	38	R	-91.20	avgR100	-181.21	eps	0.827	Qval	1.42
land100 0.00	thr	82								
[DDQN-Orig]	Ep	39	R	-153.22	avgR100	-180.49	eps	0.822	Qval	1.25
land100 0.00	thr	94								
[DDQN-Orig]	Ep	40	R	-150.97	avgR100	-179.75	eps	0.818	Qval	1.40
land100 0.00	thr	84								
[DDQN-Orig]	Ep	41	R	-69.51	avgR100	-177.06	eps	0.814	Qval	1.43
land100 0.00	thr	61								
[DDQN-Orig]	Ep	42	R	-103.54	avgR100	-175.31	eps	0.810	Qval	1.34
land100 0.00	thr	64								
[DDQN-Orig]	Ep	43	R	-238.93	avgR100	-176.79	eps	0.806	Qval	1.84
land100 0.00	thr	92								
[DDQN-Orig]	Ep	44	R	-257.43	avgR100	-178.63	eps	0.802	Qval	1.52
land100 0.00	thr	80								
[DDQN-Orig]	Ep	45	R	-81.54	avgR100	-176.47	eps	0.798	Qval	1.88
land100 0.00	thr	107								
[DDQN-Orig]	Ep	46	R	-107.61	avgR100	-174.97	eps	0.794	Qval	1.86
land100 0.00	thr	99								
[DDQN-Orig]	Ep	47	R	-179.79	avgR100	-175.07	eps	0.790	Qval	1.96
land100 0.00	thr	81								
[DDQN-Orig]	Ep	48	R	-60.07	avgR100	-172.68	eps	0.786	Qval	2.27
land100 0.00	thr	65								

[DDQN-Orig]	Ep	52	R	-126.31	avgR100	-170.19	eps	0.771	Qval	2.24
land100	0.00	thr	95							
[DDQN-Orig]	Ep	53	R	-95.11	avgR100	-168.78	eps	0.767	Qval	2.21
land100	0.00	thr	74							
[DDQN-Orig]	Ep	54	R	-133.31	avgR100	-168.12	eps	0.763	Qval	2.68
land100	0.00	thr	114							
[DDQN-Orig]	Ep	55	R	-143.27	avgR100	-167.67	eps	0.759	Qval	2.85
land100	0.00	thr	113							
[DDQN-Orig]	Ep	56	R	-114.73	avgR100	-166.72	eps	0.755	Qval	2.89
land100	0.00	thr	120							
[DDQN-Orig]	Ep	57	R	-99.51	avgR100	-165.54	eps	0.751	Qval	2.97
land100	0.00	thr	83							
[DDQN-Orig]	Ep	58	R	-132.34	avgR100	-164.97	eps	0.748	Qval	2.72
land100	0.00	thr	89							
[DDQN-Orig]	Ep	59	R	-103.95	avgR100	-163.94	eps	0.744	Qval	3.10
land100	0.00	thr	54							
[DDQN-Orig]	Ep	60	R	-64.18	avgR100	-162.27	eps	0.740	Qval	2.95
land100	0.00	thr	79							
[DDQN-Orig]	Ep	61	R	-35.41	avgR100	-160.19	eps	0.737	Qval	3.12
land100	0.00	thr	64							
[DDQN-Orig]	Ep	62	R	-130.06	avgR100	-159.71	eps	0.733	Qval	3.59
land100	0.00	thr	83							
[DDQN-Orig]	Ep	63	R	-204.61	avgR100	-160.42	eps	0.729	Qval	3.07
land100	0.00	thr	81							
[DDQN-Orig]	Ep	64	R	-214.63	avgR100	-161.27	eps	0.726	Qval	3.42
land100	0.00	thr	98							
[DDQN-Orig]	Ep	65	R	-118.58	avgR100	-160.61	eps	0.722	Qval	3.44
land100	0.00	thr	69							
[DDQN-Orig]	Ep	66	R	-89.37	avgR100	-159.53	eps	0.718	Qval	3.53
land100	0.00	thr	65							
[DDQN-Orig]	Ep	67	R	-84.80	avgR100	-158.42	eps	0.715	Qval	3.62
land100	0.00	thr	133							
[DDQN-Orig]	Ep	68	R	-45.84	avgR100	-156.76	eps	0.711	Qval	3.85
land100	0.00	thr	74							
[DDQN-Orig]	Ep	69	R	-117.58	avgR100	-156.19	eps	0.708	Qval	3.90
land100	0.00	thr	102							
[DDQN-Orig]	Ep	70	R	-139.43	avgR100	-155.95	eps	0.704	Qval	3.75
land100	0.00	thr	68							
[DDQN-Orig]	Ep	71	R	-108.35	avgR100	-155.28	eps	0.701	Qval	4.05
land100	0.00	thr	103							
[DDQN-Orig]	Ep	72	R	-48.46	avgR100	-153.80	eps	0.697	Qval	3.90
land100	0.00	thr	108							
[DDQN-Orig]	Ep	73	R	-110.45	avgR100	-153.21	eps	0.694	Qval	4.39
land100	0.00	thr	98							
[DDQN-Orig]	Ep	74	R	-84.57	avgR100	-152.28	eps	0.690	Qval	4.14
land100	0.00	thr	123							
[DDQN-Orig]	Ep	75	R	-39.96	avgR100	-150.78	eps	0.687	Qval	4.53
land100	0.01	thr	73							
[DDQN-Orig]	Ep	76	R	-127.17	avgR100	-150.47	eps	0.683	Qval	4.64
land100	0.01	thr	103							
[DDQN-Orig]	Ep	77	R	-113.90	avgR100	-150.00	eps	0.680	Qval	4.66
land100	0.01	thr</								

[DDQN-Orig]	Ep	81 R	26.99	avgR100	-144.84	eps	0.666	Qval	4.91
land100 0.01	thr	122							
[DDQN-Orig]	Ep	82 R	-46.81	avgR100	-143.65	eps	0.663	Qval	5.58
land100 0.01	thr	125							
[DDQN-Orig]	Ep	83 R	-75.14	avgR100	-142.82	eps	0.660	Qval	5.60
land100 0.01	thr	93							
[DDQN-Orig]	Ep	84 R	-90.25	avgR100	-142.20	eps	0.656	Qval	5.70
land100 0.01	thr	134							
[DDQN-Orig]	Ep	85 R	-71.80	avgR100	-141.37	eps	0.653	Qval	6.19
land100 0.01	thr	72							
[DDQN-Orig]	Ep	86 R	-118.91	avgR100	-141.11	eps	0.650	Qval	6.15
land100 0.01	thr	86							
[DDQN-Orig]	Ep	87 R	-69.51	avgR100	-140.28	eps	0.647	Qval	6.14
land100 0.01	thr	155							
[DDQN-Orig]	Ep	88 R	-144.67	avgR100	-140.33	eps	0.643	Qval	6.32
land100 0.01	thr	93							
[DDQN-Orig]	Ep	89 R	-54.38	avgR100	-139.37	eps	0.640	Qval	6.21
land100 0.01	thr	69							
[DDQN-Orig]	Ep	90 R	-90.37	avgR100	-138.82	eps	0.637	Qval	6.68
land100 0.01	thr	83							
[DDQN-Orig]	Ep	91 R	-67.94	avgR100	-138.04	eps	0.634	Qval	6.70
land100 0.01	thr	127							
[DDQN-Orig]	Ep	92 R	15.58	avgR100	-136.37	eps	0.631	Qval	7.23
land100 0.01	thr	168							
[DDQN-Orig]	Ep	93 R	-229.43	avgR100	-137.38	eps	0.627	Qval	7.44
land100 0.01	thr	162							
[DDQN-Orig]	Ep	94 R	-59.70	avgR100	-136.55	eps	0.624	Qval	7.75
land100 0.01	thr	116							
[DDQN-Orig]	Ep	95 R	-185.71	avgR100	-137.07	eps	0.621	Qval	7.47
land100 0.01	thr	118							
[DDQN-Orig]	Ep	96 R	-84.11	avgR100	-136.51	eps	0.618	Qval	7.73
land100 0.01	thr	141							
[DDQN-Orig]	Ep	97 R	-171.69	avgR100	-136.88	eps	0.615	Qval	7.96
land100 0.01	thr	111							
[DDQN-Orig]	Ep	98 R	-125.47	avgR100	-136.76	eps	0.612	Qval	7.92
land100 0.01	thr	100							
[DDQN-Orig]	Ep	99 R	-77.58	avgR100	-136.16	eps	0.609	Qval	8.24
land100 0.01	thr	150							
[DDQN-Orig]	Ep	100 R	-47.63	avgR100	-135.28	eps	0.606	Qval	8.41
land100 0.01	thr	89							
[DDQN-Orig]	Ep	101 R	-78.44	avgR100	-135.29	eps	0.603	Qval	8.70
land100 0.01	thr	136							
[DDQN-Orig]	Ep	102 R	-137.34	avgR100	-135.55	eps	0.600	Qval	8.98
land100 0.01	thr	197							
[DDQN-Orig]	Ep	103 R	-54.27	avgR100	-133.62	eps	0.597	Qval	9.23
land100 0.01	thr	174							
[DDQN-Orig]	Ep	104 R	5.01	avgR100	-132.34	eps	0.594	Qval	9.20
land100 0.01	thr	96							
[DDQN-Orig]	Ep	105 R	-103.10	avgR100	-130.45	eps	0.591	Qval	9.53
land100 0.01	thr	188							
[DDQN-Orig]	Ep	106 R	-67.76	avgR100	-129.94	eps	0.588	Qval	9.87
land100 0.01	thr	77							
[DDQN-Orig]	Ep	107 R	-57.62	avgR100	-125.51	eps	0.585	Qval	9.47
land100 0.01	thr	122							
[DDQN-Orig]									

[DDQN-Orig]	Ep 111 R	-75.79	avgR100	-120.27	eps 0.573	Qval	10.71
land100 0.01	thr 148						
[DDQN-Orig]	Ep 112 R	-5.64	avgR100	-119.30	eps 0.570	Qval	10.82
land100 0.01	thr 132						
[DDQN-Orig]	Ep 113 R	-59.25	avgR100	-119.40	eps 0.568	Qval	10.85
land100 0.01	thr 146						
[DDQN-Orig]	Ep 114 R	-186.79	avgR100	-116.28	eps 0.565	Qval	11.19
land100 0.01	thr 98						
[DDQN-Orig]	Ep 115 R	-38.63	avgR100	-115.74	eps 0.562	Qval	11.34
land100 0.01	thr 156						
[DDQN-Orig]	Ep 116 R	-26.10	avgR100	-114.15	eps 0.559	Qval	12.01
land100 0.01	thr 135						
[DDQN-Orig]	Ep 117 R	-7.99	avgR100	-112.40	eps 0.556	Qval	11.48
land100 0.01	thr 91						
[DDQN-Orig]	Ep 118 R	-21.72	avgR100	-109.61	eps 0.554	Qval	12.26
land100 0.01	thr 143						
[DDQN-Orig]	Ep 119 R	-88.49	avgR100	-106.47	eps 0.551	Qval	12.70
land100 0.01	thr 170						
[DDQN-Orig]	Ep 120 R	-58.18	avgR100	-104.27	eps 0.548	Qval	12.81
land100 0.01	thr 179						
[DDQN-Orig]	Ep 121 R	-88.42	avgR100	-103.29	eps 0.545	Qval	12.51
land100 0.01	thr 104						
[DDQN-Orig]	Ep 122 R	-29.73	avgR100	-102.74	eps 0.543	Qval	14.56
land100 0.01	thr 862						
[DDQN-Orig]	Ep 123 R	-10.82	avgR100	-100.42	eps 0.540	Qval	14.60
land100 0.01	thr 109						
[DDQN-Orig]	Ep 124 R	-49.01	avgR100	-101.21	eps 0.537	Qval	14.73
land100 0.01	thr 88						
[DDQN-Orig]	Ep 125 R	-31.26	avgR100	-98.55	eps 0.534	Qval	14.99
land100 0.01	thr 66						
[DDQN-Orig]	Ep 126 R	18.92	avgR100	-97.15	eps 0.532	Qval	15.61
land100 0.01	thr 145						
[DDQN-Orig]	Ep 127 R	-82.20	avgR100	-96.93	eps 0.529	Qval	15.64
land100 0.01	thr 113						
[DDQN-Orig]	Ep 128 R	15.72	avgR100	-94.04	eps 0.526	Qval	16.74
land100 0.01	thr 852						
[DDQN-Orig]	Ep 129 R	20.17	avgR100	-92.52	eps 0.524	Qval	17.23
land100 0.01	thr 228						
[DDQN-Orig]	Ep 130 R	-1.81	avgR100	-91.50	eps 0.521	Qval	18.87
land100 0.01	thr 796						
[DDQN-Orig]	Ep 131 R	89.90	avgR100	-89.47	eps 0.519	Qval	20.41
land100 0.01	thr 701						
[DDQN-Orig]	Ep 132 R	49.33	avgR100	-88.02	eps 0.516	Qval	20.57
land100 0.01	thr 119						
[DDQN-Orig]	Ep 133 R	-85.24	avgR100	-88.41	eps 0.513	Qval	20.63
land100 0.01	thr 74						
[DDQN-Orig]	Ep 134 R	-64.86	avgR100	-87.04	eps 0.511	Qval	20.99
land100 0.01	thr 362						
[DDQN-Orig]	Ep 135 R	-48.90	avgR100	-85.86	eps 0.508	Qval	21.87
land100 0.01	thr 282						
[DDQN-Orig]	Ep 136 R	17.07	avgR100	-83.67	eps 0.506	Qval	22.58
land100 0.01	thr 256						
[DDQN-Orig]	Ep 137 R	-3.31	avgR100	-82.01	eps 0.503	Qval	22.61
land100 0.01	thr 175						
[DDQN-Orig]	Ep 138 R	-96.96	avgR100	-82.07	eps 0.501	Qval	22.77
land100 0.01	thr 138						
[DDQN-Orig]	Ep 139 R	-129.80	avgR100	-81.84	eps 0.498	Qval	23.31
land100 0.01	thr 181						

[DDQN-Orig]	Ep 140 R	-18.68	avgR100	-80.51	eps 0.496	Qval 23.39
land100 0.01 thr 201						
[DDQN-Orig]	Ep 141 R	-28.53	avgR100	-80.10	eps 0.493	Qval 23.79
land100 0.01 thr 123						
[DDQN-Orig]	Ep 142 R	-75.26	avgR100	-79.82	eps 0.491	Qval 23.79
land100 0.01 thr 135						
[DDQN-Orig]	Ep 143 R	30.20	avgR100	-77.13	eps 0.488	Qval 24.94
land100 0.01 thr 769						
[DDQN-Orig]	Ep 144 R	56.52	avgR100	-73.99	eps 0.486	Qval 26.24
land100 0.01 thr 791						
[DDQN-Orig]	Ep 145 R	2.83	avgR100	-73.15	eps 0.483	Qval 26.52
land100 0.01 thr 296						
[DDQN-Orig]	Ep 146 R	-67.38	avgR100	-72.74	eps 0.481	Qval 27.01
land100 0.01 thr 147						
[DDQN-Orig]	Ep 147 R	-12.35	avgR100	-71.07	eps 0.479	Qval 26.88
land100 0.01 thr 196						
[DDQN-Orig]	Ep 148 R	7.56	avgR100	-70.39	eps 0.476	Qval 26.94
land100 0.01 thr 125						
[DDQN-Orig]	Ep 149 R	-21.46	avgR100	-69.55	eps 0.474	Qval 28.74
land100 0.01 thr 808						
[DDQN-Orig]	Ep 150 R	36.57	avgR100	-68.22	eps 0.471	Qval 28.85
land100 0.01 thr 146						
[DDQN-Orig]	Ep 151 R	-36.13	avgR100	-66.25	eps 0.469	Qval 28.74
land100 0.01 thr 253						
[DDQN-Orig]	Ep 152 R	-20.47	avgR100	-65.19	eps 0.467	Qval 29.38
land100 0.01 thr 122						
[DDQN-Orig]	Ep 153 R	-22.62	avgR100	-64.47	eps 0.464	Qval 28.95
land100 0.01 thr 64						
[DDQN-Orig]	Ep 154 R	-91.96	avgR100	-64.06	eps 0.462	Qval 29.43
land100 0.01 thr 228						
[DDQN-Orig]	Ep 155 R	-44.88	avgR100	-63.07	eps 0.460	Qval 30.22
land100 0.01 thr 882						
[DDQN-Orig]	Ep 156 R	-53.62	avgR100	-62.46	eps 0.458	Qval 30.57
land100 0.01 thr 80						
[DDQN-Orig]	Ep 157 R	0.14	avgR100	-61.46	eps 0.455	Qval 30.23
land100 0.01 thr 99						
[DDQN-Orig]	Ep 158 R	-27.54	avgR100	-60.42	eps 0.453	Qval 31.71
land100 0.01 thr 802						
[DDQN-Orig]	Ep 159 R	-22.16	avgR100	-59.60	eps 0.451	Qval 31.98
land100 0.01 thr 849						
[DDQN-Orig]	Ep 160 R	29.45	avgR100	-58.66	eps 0.448	Qval 31.05
land100 0.01 thr 122						
[DDQN-Orig]	Ep 161 R	-24.68	avgR100	-58.55	eps 0.446	Qval 32.14
land100 0.01 thr 93						
[DDQN-Orig]	Ep 162 R	-10.00	avgR100	-57.35	eps 0.444	Qval 31.72
land100 0.01 thr 148						
[DDQN-Orig]	Ep 163 R	-68.04	avgR100	-55.99	eps 0.442	Qval 31.57
land100 0.01 thr 169						
[DDQN-Orig]	Ep 164 R	-22.46	avgR100	-54.07	eps 0.440	Qval 31.99
land100 0.01 thr 248						
[DDQN-Orig]	Ep 165 R	26.23	avgR100	-52.62	eps 0.437	Qval 31.57
land100 0.01 thr 137						
[DDQN-Orig]	Ep 166 R	-150.98	avgR100	-53.23	eps 0.435	Qval 32.38
land100 0.01 thr 146						
[DDQN-Orig]	Ep 167 R	-70.09	avgR100	-53.09	eps 0.433	Qval 32.27
land100 0.01 thr 73						
[DDQN-Orig]	Ep 168 R	-63.09	avgR100	-53.26	eps 0.431	Qval 32.61
land100 0.01 thr 113						
[DDQN-Orig]	Ep 169 R	-37.89	avgR100	-52.46	eps 0.429	Qval 32.17
land100 0.01 thr 206						

[DDQN-Orig]	Ep 170 R	1.27	avgR100	-51.06	eps 0.427	Qval	32.74
land100 0.01	thr 196						
[DDQN-Orig]	Ep 171 R	-26.61	avgR100	-50.24	eps 0.424	Qval	31.93
land100 0.01	thr 113						
[DDQN-Orig]	Ep 172 R	20.35	avgR100	-49.55	eps 0.422	Qval	32.72
land100 0.01	thr 848						
[DDQN-Orig]	Ep 173 R	-42.32	avgR100	-48.87	eps 0.420	Qval	32.35
land100 0.01	thr 72						
[DDQN-Orig]	Ep 174 R	-61.93	avgR100	-48.64	eps 0.418	Qval	33.18
land100 0.01	thr 871						
[DDQN-Orig]	Ep 175 R	15.38	avgR100	-48.09	eps 0.416	Qval	32.95
land100 0.00	thr 89						
[DDQN-Orig]	Ep 176 R	-24.43	avgR100	-47.06	eps 0.414	Qval	32.93
land100 0.00	thr 68						
[DDQN-Orig]	Ep 177 R	-12.91	avgR100	-46.05	eps 0.412	Qval	33.08
land100 0.00	thr 238						
[DDQN-Orig]	Ep 178 R	3.48	avgR100	-45.12	eps 0.410	Qval	33.51
land100 0.00	thr 844						
[DDQN-Orig]	Ep 179 R	7.44	avgR100	-43.89	eps 0.408	Qval	33.34
land100 0.01	thr 104						
[DDQN-Orig]	Ep 180 R	7.65	avgR100	-43.77	eps 0.406	Qval	32.23
land100 0.01	thr 103						
[DDQN-Orig]	Ep 181 R	-52.25	avgR100	-44.56	eps 0.404	Qval	33.41
land100 0.01	thr 876						
[DDQN-Orig]	Ep 182 R	-63.44	avgR100	-44.73	eps 0.402	Qval	32.66
land100 0.01	thr 216						
[DDQN-Orig]	Ep 183 R	-44.24	avgR100	-44.42	eps 0.400	Qval	32.62
land100 0.01	thr 73						
[DDQN-Orig]	Ep 184 R	6.57	avgR100	-43.45	eps 0.398	Qval	32.92
land100 0.01	thr 865						
[DDQN-Orig]	Ep 185 R	-15.23	avgR100	-42.89	eps 0.396	Qval	33.50
land100 0.01	thr 100						
[DDQN-Orig]	Ep 186 R	-102.54	avgR100	-42.72	eps 0.394	Qval	33.28
land100 0.01	thr 160						
[DDQN-Orig]	Ep 187 R	31.64	avgR100	-41.71	eps 0.392	Qval	32.85
land100 0.01	thr 846						
[DDQN-Orig]	Ep 188 R	-3.02	avgR100	-40.29	eps 0.390	Qval	32.69
land100 0.01	thr 130						
[DDQN-Orig]	Ep 189 R	-59.56	avgR100	-40.35	eps 0.388	Qval	32.70
land100 0.01	thr 63						
[DDQN-Orig]	Ep 190 R	-36.83	avgR100	-39.81	eps 0.386	Qval	32.28
land100 0.01	thr 853						
[DDQN-Orig]	Ep 191 R	10.24	avgR100	-39.03	eps 0.384	Qval	32.04
land100 0.01	thr 159						
[DDQN-Orig]	Ep 192 R	-33.21	avgR100	-39.52	eps 0.382	Qval	32.00
land100 0.01	thr 875						
[DDQN-Orig]	Ep 193 R	-116.87	avgR100	-38.39	eps 0.380	Qval	31.16
land100 0.01	thr 726						
[DDQN-Orig]	Ep 194 R	-44.50	avgR100	-38.24	eps 0.378	Qval	31.20
land100 0.01	thr 876						
[DDQN-Orig]	Ep 195 R	-93.50	avgR100	-37.32	eps 0.376	Qval	30.69
land100 0.01	thr 103						
[DDQN-Orig]	Ep 196 R	42.35	avgR100	-36.05	eps 0.374	Qval	31.02
land100 0.01	thr 118						
[DDQN-Orig]	Ep 197 R	1.70	avgR100	-34.32	eps 0.373	Qval	30.08
land100 0.01	thr 839						
[DDQN-Orig]	Ep 198 R	-25.36	avgR100	-33.32	eps 0.371	Qval	30.11
land100 0.01	thr 79						
[DDQN-Orig]	Ep 199 R	-80.46	avgR100	-33.35	eps 0.369	Qval	29.23
land100 0.01	thr 279						

[DDQN-Orig]	Ep 200	R	101.45	avgR100	-31.86	eps 0.367	Qval	29.22	
land100 0.01	thr 870								
[DDQN-Orig]	Ep 201	R	-40.93	avgR100	-31.48	eps 0.365	Qval	29.23	
land100 0.01	thr 860								
[DDQN-Orig]	Ep 202	R	-15.74	avgR100	-30.26	eps 0.363	Qval	28.11	
land100 0.01	thr 124								
[DDQN-Orig]	Ep 203	R	-15.09	avgR100	-29.87	eps 0.361	Qval	27.90	
land100 0.01	thr 828								
[DDQN-Orig]	Ep 204	R	-43.75	avgR100	-30.36	eps 0.360	Qval	27.46	
land100 0.01	thr 123								
[DDQN-Orig]	Ep 205	R	-8.23	avgR100	-29.41	eps 0.358	Qval	28.48	
land100 0.01	thr 80								
[DDQN-Orig]	Ep 206	R	-14.01	avgR100	-28.87	eps 0.356	Qval	26.72	
land100 0.01	thr 833								
[DDQN-Orig]	Ep 207	R	-20.10	avgR100	-28.50	eps 0.354	Qval	26.88	
land100 0.01	thr 89								
[DDQN-Orig]	Ep 208	R	-20.19	avgR100	-28.23	eps 0.353	Qval	25.89	
land100 0.01	thr 873								
[DDQN-Orig]	Ep 209	R	-131.05	avgR100	-28.92	eps 0.351	Qval	26.72	
land100 0.01	thr 129								
[DDQN-Orig]	Ep 210	R	-35.53	avgR100	-29.15	eps 0.349	Qval	25.38	
land100 0.01	thr 918								
[DDQN-Orig]	Ep 211	R	-31.36	avgR100	-28.70	eps 0.347	Qval	24.70	
land100 0.01	thr 854								
[DDQN-Orig]	Ep 212	R	67.27	avgR100	-27.98	eps 0.346	Qval	23.19	
land100 0.01	thr 895								
[DDQN-Orig]	Ep 213	R	0.78	avgR100	-27.37	eps 0.344	Qval	23.15	
land100 0.01	thr 142								
[DDQN-Orig]	Ep 214	R	-49.23	avgR100	-26.00	eps 0.342	Qval	23.57	
land100 0.01	thr 87								
[DDQN-Orig]	Ep 215	R	20.91	avgR100	-25.40	eps 0.340	Qval	22.98	
land100 0.01	thr 109								
[DDQN-Orig]	Ep 216	R	-9.68	avgR100	-25.24	eps 0.339	Qval	22.67	
land100 0.01	thr 814								
[DDQN-Orig]	Ep 217	R	239.66	avgR100	-22.76	eps 0.337	Qval	22.17	
land100 0.01	thr 401								
[DDQN-Orig]	Ep 218	R	-26.37	avgR100	-22.81	eps 0.335	Qval	21.28	
land100 0.01	thr 858								
[DDQN-Orig]	Ep 219	R	-19.53	avgR100	-22.12	eps 0.334	Qval	19.66	
land100 0.01	thr 845								
[DDQN-Orig]	Ep 220	R	-60.78	avgR100	-22.15	eps 0.332	Qval	20.02	
land100 0.01	thr 71								
[DDQN-Orig]	Ep 221	R	-89.86	avgR100	-22.16	eps 0.330	Qval	20.12	
land100 0.01	thr 145								
[DDQN-Orig]	Ep 222	R	7.70	avgR100	-21.79	eps 0.329	Qval	18.64	
land100 0.01	thr 802								
[DDQN-Orig]	Ep 223	R	-36.03	avgR100	-22.04	eps 0.327	Qval	18.63	
land100 0.01	thr 838								
[DDQN-Orig]	Ep 224	R	-48.13	avgR100	-22.03	eps 0.325	Qval	16.91	
land100 0.01	thr 849								
[DDQN-Orig]	Ep 225	R	-11.76	avgR100	-21.83	eps 0.324	Qval	16.80	
land100 0.01	thr 834								
[DDQN-Orig]	Ep 226	R	-20.15	avgR100	-22.23	eps 0.322	Qval	16.19	
land100 0.01	thr 850								
[DDQN-Orig]	Ep 227	R	-20.50	avgR100	-21.61	eps 0.321	Qval	15.56	
land100 0.01	thr 855								
[DDQN-Orig]	Ep 228	R	-34.13	avgR100	-22.11	eps 0.319	Qval	13.87	
land100 0.01	thr 799								
[DDQN-Orig]	Ep 229	R	3.51	avgR100	-22.27	eps 0.317	Qval	13.61	
land100 0.01	thr 133								

[DDQN-Orig] Ep 230 R	-1.65	avgR100	-22.27	eps 0.316	Qval	12.84
land100 0.01 thr 817						
[DDQN-Orig] Ep 231 R	-20.26	avgR100	-23.37	eps 0.314	Qval	12.65
land100 0.01 thr 112						
[DDQN-Orig] Ep 232 R	-14.24	avgR100	-24.01	eps 0.313	Qval	11.65
land100 0.01 thr 794						
[DDQN-Orig] Ep 233 R	-3.38	avgR100	-23.19	eps 0.311	Qval	12.08
land100 0.01 thr 863						
[DDQN-Orig] Ep 234 R	8.84	avgR100	-22.45	eps 0.309	Qval	9.48
land100 0.01 thr 878						
[DDQN-Orig] Ep 235 R	-15.00	avgR100	-22.11	eps 0.308	Qval	9.91
land100 0.01 thr 853						
[DDQN-Orig] Ep 236 R	-20.27	avgR100	-22.49	eps 0.306	Qval	9.74
land100 0.01 thr 878						
[DDQN-Orig] Ep 237 R	-9.83	avgR100	-22.55	eps 0.305	Qval	9.80
land100 0.01 thr 821						
[DDQN-Orig] Ep 238 R	25.22	avgR100	-21.33	eps 0.303	Qval	9.19
land100 0.01 thr 167						
[DDQN-Orig] Ep 239 R	57.34	avgR100	-19.46	eps 0.302	Qval	8.94
land100 0.01 thr 742						
[DDQN-Orig] Ep 240 R	50.42	avgR100	-18.77	eps 0.300	Qval	7.44
land100 0.01 thr 800						
[DDQN-Orig] Ep 241 R	-3.93	avgR100	-18.52	eps 0.299	Qval	6.88
land100 0.01 thr 840						
[DDQN-Orig] Ep 242 R	-1.04	avgR100	-17.78	eps 0.297	Qval	6.54
land100 0.01 thr 741						
[DDQN-Orig] Ep 243 R	253.74	avgR100	-15.55	eps 0.296	Qval	7.37
land100 0.02 thr 297						
[DDQN-Orig] Ep 244 R	4.05	avgR100	-16.07	eps 0.294	Qval	7.36
land100 0.02 thr 109						
[DDQN-Orig] Ep 245 R	26.08	avgR100	-15.84	eps 0.293	Qval	5.27
land100 0.02 thr 764						
[DDQN-Orig] Ep 246 R	18.26	avgR100	-14.98	eps 0.291	Qval	5.88
land100 0.02 thr 884						
[DDQN-Orig] Ep 247 R	46.51	avgR100	-14.39	eps 0.290	Qval	5.76
land100 0.02 thr 138						
[DDQN-Orig] Ep 248 R	-38.67	avgR100	-14.85	eps 0.288	Qval	5.58
land100 0.02 thr 112						
[DDQN-Orig] Ep 249 R	-7.45	avgR100	-14.71	eps 0.287	Qval	5.99
land100 0.02 thr 833						
[DDQN-Orig] Ep 250 R	6.10	avgR100	-15.02	eps 0.286	Qval	5.13
land100 0.02 thr 821						
[DDQN-Orig] Ep 251 R	52.07	avgR100	-14.14	eps 0.284	Qval	5.79
land100 0.02 thr 740						
[DDQN-Orig] Ep 252 R	30.45	avgR100	-13.63	eps 0.283	Qval	5.46
land100 0.02 thr 790						
[DDQN-Orig] Ep 253 R	25.94	avgR100	-13.14	eps 0.281	Qval	7.69
land100 0.02 thr 726						
[DDQN-Orig] Ep 254 R	81.38	avgR100	-11.41	eps 0.280	Qval	5.03
land100 0.02 thr 863						
[DDQN-Orig] Ep 255 R	-55.70	avgR100	-11.52	eps 0.279	Qval	5.08
land100 0.02 thr 131						
[DDQN-Orig] Ep 256 R	25.18	avgR100	-10.73	eps 0.277	Qval	5.64
land100 0.02 thr 717						
[DDQN-Orig] Ep 257 R	37.55	avgR100	-10.36	eps 0.276	Qval	4.35
land100 0.02 thr 744						
[DDQN-Orig] Ep 258 R	180.52	avgR100	-8.27	eps 0.274	Qval	4.06
land100 0.03 thr 624						

[DDQN-Orig]	Ep 259 R	59.87	avgR100	-7.45	eps 0.273	Qval	4.98
land100 0.03 thr 782							
[DDQN-Orig]	Ep 260 R	13.23	avgR100	-7.62	eps 0.272	Qval	4.95
land100 0.03 thr 159							
[DDQN-Orig]	Ep 261 R	71.54	avgR100	-6.65	eps 0.270	Qval	5.00
land100 0.03 thr 168							
[DDQN-Orig]	Ep 262 R	-30.56	avgR100	-6.86	eps 0.269	Qval	4.82
land100 0.03 thr 118							
[DDQN-Orig]	Ep 263 R	51.21	avgR100	-5.67	eps 0.268	Qval	5.20
land100 0.03 thr 731							
[DDQN-Orig]	Ep 264 R	-4.57	avgR100	-5.49	eps 0.266	Qval	4.02
land100 0.03 thr 91							
[DDQN-Orig]	Ep 265 R	55.63	avgR100	-5.19	eps 0.265	Qval	4.08
land100 0.03 thr 815							
[DDQN-Orig]	Ep 266 R	40.37	avgR100	-3.28	eps 0.264	Qval	3.82
land100 0.03 thr 768							
[DDQN-Orig]	Ep 267 R	111.13	avgR100	-1.47	eps 0.262	Qval	2.65
land100 0.03 thr 612							
[DDQN-Orig]	Ep 268 R	109.22	avgR100	0.25	eps 0.261	Qval	1.84
land100 0.03 thr 586							
[DDQN-Orig]	Ep 269 R	90.99	avgR100	1.54	eps 0.260	Qval	2.19
land100 0.03 thr 727							
[DDQN-Orig]	Ep 270 R	-13.43	avgR100	1.40	eps 0.258	Qval	2.42
land100 0.03 thr 94							
[DDQN-Orig]	Ep 271 R	58.67	avgR100	2.25	eps 0.257	Qval	1.70
land100 0.03 thr 736							
[DDQN-Orig]	Ep 272 R	34.50	avgR100	2.39	eps 0.256	Qval	2.66
land100 0.03 thr 690							
[DDQN-Orig]	Ep 273 R	-1.41	avgR100	2.80	eps 0.255	Qval	1.14
land100 0.03 thr 105							
[DDQN-Orig]	Ep 274 R	64.60	avgR100	4.06	eps 0.253	Qval	2.39
land100 0.03 thr 777							
[DDQN-Orig]	Ep 275 R	42.32	avgR100	4.33	eps 0.252	Qval	1.82
land100 0.03 thr 720							
[DDQN-Orig]	Ep 276 R	12.34	avgR100	4.70	eps 0.251	Qval	2.46
land100 0.03 thr 834							
[DDQN-Orig]	Ep 277 R	71.13	avgR100	5.54	eps 0.249	Qval	2.19
land100 0.03 thr 749							
[DDQN-Orig]	Ep 278 R	25.04	avgR100	5.76	eps 0.248	Qval	2.03
land100 0.03 thr 146							
[DDQN-Orig]	Ep 279 R	91.16	avgR100	6.59	eps 0.247	Qval	2.07
land100 0.02 thr 618							
[DDQN-Orig]	Ep 280 R	5.96	avgR100	6.58	eps 0.246	Qval	2.14
land100 0.02 thr 156							
[DDQN-Orig]	Ep 281 R	61.26	avgR100	7.71	eps 0.245	Qval	2.07
land100 0.02 thr 737							
[DDQN-Orig]	Ep 282 R	-34.58	avgR100	8.00	eps 0.243	Qval	1.78
land100 0.02 thr 850							
[DDQN-Orig]	Ep 283 R	55.89	avgR100	9.00	eps 0.242	Qval	1.99
land100 0.02 thr 657							
[DDQN-Orig]	Ep 284 R	124.09	avgR100	10.18	eps 0.241	Qval	2.13
land100 0.02 thr 620							
[DDQN-Orig]	Ep 285 R	78.10	avgR100	11.11	eps 0.240	Qval	4.12
land100 0.02 thr 647							
[DDQN-Orig]	Ep 286 R	254.57	avgR100	14.68	eps 0.238	Qval	5.04
land100 0.03 thr 422							
[DDQN-Orig]	Ep 287 R	23.97	avgR100	14.61	eps 0.237	Qval	5.39
land100 0.03 thr 873							
[DDQN-Orig]	Ep 288 R	21.45	avgR100	14.85	eps 0.236	Qval	4.45
land100 0.03 thr 140							

[DDQN-Orig]	Ep 289 R	125.71 avgR100	16.70 eps 0.235 Qval	6.06
land100 0.03 thr 639				
[DDQN-Orig]	Ep 290 R	93.93 avgR100	18.01 eps 0.234 Qval	6.63
land100 0.03 thr 643				
[DDQN-Orig]	Ep 291 R	198.27 avgR100	19.89 eps 0.233 Qval	6.74
land100 0.04 thr 471				
[DDQN-Orig]	Ep 292 R	202.38 avgR100	22.25 eps 0.231 Qval	6.79
land100 0.05 thr 587				
[DDQN-Orig]	Ep 293 R	99.29 avgR100	24.41 eps 0.230 Qval	9.45
land100 0.05 thr 609				
[DDQN-Orig]	Ep 294 R	278.71 avgR100	27.64 eps 0.229 Qval	8.29
land100 0.06 thr 168				
[DDQN-Orig]	Ep 295 R	143.82 avgR100	30.01 eps 0.228 Qval	9.28
land100 0.06 thr 564				
[DDQN-Orig]	Ep 296 R	273.83 avgR100	32.33 eps 0.227 Qval	9.26
land100 0.07 thr 266				
[DDQN-Orig]	Ep 297 R	-8.96 avgR100	32.22 eps 0.226 Qval	9.72
land100 0.07 thr 95				
[DDQN-Orig]	Ep 298 R	81.54 avgR100	33.29 eps 0.225 Qval	9.54
land100 0.07 thr 597				
[DDQN-Orig]	Ep 299 R	16.75 avgR100	34.26 eps 0.223 Qval	9.88
land100 0.07 thr 137				
[DDQN-Orig]	Ep 300 R	216.18 avgR100	35.41 eps 0.222 Qval	9.55
land100 0.07 thr 253				
[DDQN-Orig]	Ep 301 R	283.45 avgR100	38.65 eps 0.221 Qval	9.04
land100 0.08 thr 349				
[DDQN-Orig]	Ep 302 R	130.92 avgR100	40.12 eps 0.220 Qval	9.86
land100 0.08 thr 653				
[DDQN-Orig]	Ep 303 R	208.82 avgR100	42.36 eps 0.219 Qval	10.00
land100 0.09 thr 396				
[DDQN-Orig]	Ep 304 R	122.10 avgR100	44.02 eps 0.218 Qval	10.70
land100 0.09 thr 552				
[DDQN-Orig]	Ep 305 R	110.29 avgR100	45.20 eps 0.217 Qval	10.20
land100 0.09 thr 651				
[DDQN-Orig]	Ep 306 R	183.46 avgR100	47.18 eps 0.216 Qval	11.64
land100 0.10 thr 541				
[DDQN-Orig]	Ep 307 R	43.38 avgR100	47.81 eps 0.215 Qval	10.90
land100 0.10 thr 717				
[DDQN-Orig]	Ep 308 R	198.84 avgR100	50.00 eps 0.214 Qval	11.43
land100 0.11 thr 442				
[DDQN-Orig]	Ep 309 R	282.72 avgR100	54.14 eps 0.212 Qval	11.93
land100 0.12 thr 227				
[DDQN-Orig]	Ep 310 R	8.52 avgR100	54.58 eps 0.211 Qval	12.03
land100 0.12 thr 171				
[DDQN-Orig]	Ep 311 R	58.77 avgR100	55.48 eps 0.210 Qval	11.45
land100 0.12 thr 723				
[DDQN-Orig]	Ep 312 R	-18.24 avgR100	54.63 eps 0.209 Qval	12.52
land100 0.12 thr 65				
[DDQN-Orig]	Ep 313 R	199.22 avgR100	56.61 eps 0.208 Qval	12.04
land100 0.12 thr 704				
[DDQN-Orig]	Ep 314 R	264.06 avgR100	59.75 eps 0.207 Qval	13.01
land100 0.13 thr 286				
[DDQN-Orig]	Ep 315 R	73.45 avgR100	60.27 eps 0.206 Qval	12.53
land100 0.13 thr 602				
[DDQN-Orig]	Ep 316 R	260.36 avgR100	62.97 eps 0.205 Qval	13.95
land100 0.14 thr 308				
[DDQN-Orig]	Ep 317 R	145.12 avgR100	62.03 eps 0.204 Qval	14.46
land100 0.14 thr 477				

[DDQN-Orig]	Ep 318	R	149.86	avgR100	63.79	eps 0.203	Qval	14.75	
land100 0.15	thr 783								
[DDQN-Orig]	Ep 319	R	237.50	avgR100	66.36	eps 0.202	Qval	14.20	
land100 0.16	thr 258								
[DDQN-Orig]	Ep 320	R	253.06	avgR100	69.50	eps 0.201	Qval	14.79	
land100 0.17	thr 282								
[DDQN-Orig]	Ep 321	R	238.64	avgR100	72.78	eps 0.200	Qval	14.90	
land100 0.18	thr 213								
[DDQN-Orig]	Ep 322	R	160.45	avgR100	74.31	eps 0.199	Qval	15.25	
land100 0.18	thr 708								
[DDQN-Orig]	Ep 323	R	252.85	avgR100	77.20	eps 0.198	Qval	14.89	
land100 0.19	thr 251								
[DDQN-Orig]	Ep 324	R	106.60	avgR100	78.75	eps 0.197	Qval	13.98	
land100 0.19	thr 575								
[DDQN-Orig]	Ep 325	R	199.92	avgR100	80.86	eps 0.196	Qval	15.47	
land100 0.20	thr 669								
[DDQN-Orig]	Ep 326	R	11.89	avgR100	81.18	eps 0.195	Qval	15.03	
land100 0.20	thr 111								
[DDQN-Orig]	Ep 327	R	277.11	avgR100	84.16	eps 0.194	Qval	15.13	
land100 0.21	thr 450								
[DDQN-Orig]	Ep 328	R	248.19	avgR100	86.98	eps 0.193	Qval	15.41	
land100 0.22	thr 447								
[DDQN-Orig]	Ep 329	R	127.79	avgR100	88.22	eps 0.192	Qval	15.54	
land100 0.22	thr 418								
[DDQN-Orig]	Ep 330	R	229.73	avgR100	90.54	eps 0.191	Qval	14.70	
land100 0.23	thr 320								
[DDQN-Orig]	Ep 331	R	275.09	avgR100	93.49	eps 0.190	Qval	16.81	
land100 0.24	thr 437								
[DDQN-Orig]	Ep 332	R	264.11	avgR100	96.28	eps 0.189	Qval	16.81	
land100 0.25	thr 296								
[DDQN-Orig]	Ep 333	R	-45.52	avgR100	95.85	eps 0.188	Qval	16.38	
land100 0.25	thr 117								
[DDQN-Orig]	Ep 334	R	185.87	avgR100	97.62	eps 0.187	Qval	17.64	
land100 0.26	thr 582								
[DDQN-Orig]	Ep 335	R	233.20	avgR100	100.11	eps 0.187	Qval	18.59	
land100 0.26	thr 274								
[DDQN-Orig]	Ep 336	R	204.03	avgR100	102.35	eps 0.186	Qval	17.79	
land100 0.26	thr 664								
[DDQN-Orig]	Ep 337	R	-53.39	avgR100	101.91	eps 0.185	Qval	17.78	
land100 0.26	thr 915								
[DDQN-Orig]	Ep 338	R	107.32	avgR100	102.73	eps 0.184	Qval	17.53	
land100 0.26	thr 712								
[DDQN-Orig]	Ep 339	R	271.56	avgR100	104.88	eps 0.183	Qval	18.05	
land100 0.27	thr 299								
[DDQN-Orig]	Ep 340	R	235.86	avgR100	106.73	eps 0.182	Qval	16.59	
land100 0.28	thr 260								
[DDQN-Orig]	Ep 341	R	57.89	avgR100	107.35	eps 0.181	Qval	17.09	
land100 0.28	thr 190								
[DDQN-Orig]	Ep 342	R	271.70	avgR100	110.08	eps 0.180	Qval	16.36	
land100 0.29	thr 329								
[DDQN-Orig]	Ep 343	R	243.29	avgR100	109.97	eps 0.179	Qval	16.59	
land100 0.29	thr 377								
[DDQN-Orig]	Ep 344	R	107.35	avgR100	111.01	eps 0.178	Qval	17.28	
land100 0.29	thr 705								
[DDQN-Orig]	Ep 345	R	43.57	avgR100	111.18	eps 0.177	Qval	18.28	
land100 0.29	thr 80								
[DDQN-Orig]	Ep 346	R	264.99	avgR100	113.65	eps 0.177	Qval	18.08	
land100 0.29	thr 268								
[DDQN-Orig]	Ep 347	R	220.36	avgR100	115.39	eps 0.176	Qval	16.96	
land100 0.30	thr 366								

[DDQN-Orig]	Ep 348	R	176.38	avgR100	117.54	eps 0.175	Qval	18.55	
land100 0.31	thr 584								
[DDQN-Orig]	Ep 349	R	300.75	avgR100	120.62	eps 0.174	Qval	19.22	
land100 0.32	thr 335								
[DDQN-Orig]	Ep 350	R	27.30	avgR100	120.83	eps 0.173	Qval	18.33	
land100 0.32	thr 84								
[DDQN-Orig]	Ep 351	R	247.06	avgR100	122.78	eps 0.172	Qval	17.64	
land100 0.33	thr 465								
[DDQN-Orig]	Ep 352	R	206.13	avgR100	124.54	eps 0.171	Qval	19.77	
land100 0.34	thr 474								
[DDQN-Orig]	Ep 353	R	231.38	avgR100	126.59	eps 0.170	Qval	19.65	
land100 0.35	thr 243								
[DDQN-Orig]	Ep 354	R	241.02	avgR100	128.19	eps 0.170	Qval	19.23	
land100 0.36	thr 282								
[DDQN-Orig]	Ep 355	R	132.34	avgR100	130.07	eps 0.169	Qval	20.73	
land100 0.36	thr 858								
[DDQN-Orig]	Ep 356	R	233.80	avgR100	132.15	eps 0.168	Qval	21.14	
land100 0.37	thr 457								
[DDQN-Orig]	Ep 357	R	264.92	avgR100	134.43	eps 0.167	Qval	19.98	
land100 0.38	thr 206								
[DDQN-Orig]	Ep 358	R	114.06	avgR100	133.76	eps 0.166	Qval	20.53	
land100 0.37	thr 407								
[DDQN-Orig]	Ep 359	R	236.53	avgR100	135.53	eps 0.165	Qval	21.11	
land100 0.38	thr 572								
[DDQN-Orig]	Ep 360	R	136.58	avgR100	136.76	eps 0.165	Qval	19.84	
land100 0.38	thr 629								
[DDQN-Orig]	Ep 361	R	212.85	avgR100	138.18	eps 0.164	Qval	21.24	
land100 0.39	thr 459								
[DDQN-Orig]	Ep 362	R	248.30	avgR100	140.97	eps 0.163	Qval	22.06	
land100 0.40	thr 253								
[DDQN-Orig]	Ep 363	R	187.95	avgR100	142.33	eps 0.162	Qval	20.96	
land100 0.41	thr 446								
[DDQN-Orig]	Ep 364	R	274.87	avgR100	145.13	eps 0.161	Qval	20.66	
land100 0.42	thr 157								
[DDQN-Orig]	Ep 365	R	238.26	avgR100	146.95	eps 0.160	Qval	20.31	
land100 0.43	thr 181								
[DDQN-Orig]	Ep 366	R	76.84	avgR100	147.32	eps 0.160	Qval	20.53	
land100 0.43	thr 731								
[DDQN-Orig]	Ep 367	R	232.24	avgR100	148.53	eps 0.159	Qval	21.49	
land100 0.43	thr 386								
[DDQN-Orig]	Ep 368	R	273.87	avgR100	150.18	eps 0.158	Qval	20.39	
land100 0.44	thr 183								
[DDQN-Orig]	Ep 369	R	263.95	avgR100	151.91	eps 0.157	Qval	21.46	
land100 0.45	thr 211								
[DDQN-Orig]	Ep 370	R	235.39	avgR100	154.39	eps 0.157	Qval	21.42	
land100 0.45	thr 368								
[DDQN-Orig]	Ep 371	R	-3.16	avgR100	153.78	eps 0.156	Qval	21.44	
land100 0.45	thr 83								
[DDQN-Orig]	Ep 372	R	0.80	avgR100	153.44	eps 0.155	Qval	20.41	
land100 0.45	thr 111								
[DDQN-Orig]	Ep 373	R	226.85	avgR100	155.72	eps 0.154	Qval	22.41	
land100 0.45	thr 194								
[DDQN-Orig]	Ep 374	R	-6.06	avgR100	155.01	eps 0.153	Qval	21.01	
land100 0.45	thr 173								
[DDQN-Orig]	Ep 375	R	245.52	avgR100	157.05	eps 0.153	Qval	22.03	
land100 0.46	thr 621								
[DDQN-Orig]	Ep 376	R	224.39	avgR100	159.17	eps 0.152	Qval	22.48	
land100 0.47	thr 412								
[DDQN-Orig]	Ep 377	R	161.02	avgR100	160.07	eps 0.151	Qval	22.54	
land100 0.47	thr 591								

[DDQN-Orig]	Ep 378 R	221.41	avgR100	162.03	eps 0.150	Qval 22.68
land100 0.47	thr 724					
[DDQN-Orig]	Ep 379 R	184.10	avgR100	162.96	eps 0.150	Qval 24.06
land100 0.48	thr 539					
[DDQN-Orig]	Ep 380 R	217.67	avgR100	165.08	eps 0.149	Qval 24.78
land100 0.48	thr 512					
[DDQN-Orig]	Ep 381 R	216.56	avgR100	166.63	eps 0.148	Qval 25.20
land100 0.49	thr 601					
[DDQN-Orig]	Ep 382 R	24.83	avgR100	167.22	eps 0.147	Qval 24.30
land100 0.49	thr 235					
[DDQN-Orig]	Ep 383 R	184.52	avgR100	168.51	eps 0.147	Qval 26.40
land100 0.50	thr 503					
[DDQN-Orig]	Ep 384 R	221.29	avgR100	169.48	eps 0.146	Qval 25.01
land100 0.50	thr 255					
[DDQN-Orig]	Ep 385 R	177.40	avgR100	170.47	eps 0.145	Qval 25.27
land100 0.50	thr 560					
[DDQN-Orig]	Ep 386 R	228.40	avgR100	170.21	eps 0.144	Qval 25.91
land100 0.50	thr 363					
[DDQN-Orig]	Ep 387 R	251.13	avgR100	172.48	eps 0.144	Qval 26.04
land100 0.51	thr 200					
[DDQN-Orig]	Ep 388 R	241.57	avgR100	174.69	eps 0.143	Qval 25.27
land100 0.52	thr 175					
[DDQN-Orig]	Ep 389 R	118.79	avgR100	174.62	eps 0.142	Qval 25.75
land100 0.52	thr 613					
[DDQN-Orig]	Ep 390 R	190.17	avgR100	175.58	eps 0.142	Qval 26.11
land100 0.52	thr 489					
[DDQN-Orig]	Ep 391 R	67.69	avgR100	174.27	eps 0.141	Qval 25.78
land100 0.51	thr 899					
[DDQN-Orig]	Ep 392 R	122.77	avgR100	173.48	eps 0.140	Qval 25.64
land100 0.50	thr 470					
[DDQN-Orig]	Ep 393 R	228.35	avgR100	174.77	eps 0.139	Qval 25.95
land100 0.51	thr 234					
[DDQN-Orig]	Ep 394 R	290.29	avgR100	174.88	eps 0.139	Qval 28.16
land100 0.51	thr 179					
[DDQN-Orig]	Ep 395 R	242.22	avgR100	175.87	eps 0.138	Qval 27.29
land100 0.52	thr 384					
[DDQN-Orig]	Ep 396 R	260.70	avgR100	175.74	eps 0.137	Qval 27.40
land100 0.52	thr 209					
[DDQN-Orig]	Ep 397 R	222.80	avgR100	178.05	eps 0.137	Qval 28.67
land100 0.52	thr 264					
[DDQN-Orig]	Ep 398 R	267.73	avgR100	179.92	eps 0.136	Qval 27.38
land100 0.53	thr 153					
[DDQN-Orig]	Ep 399 R	181.65	avgR100	181.56	eps 0.135	Qval 27.28
land100 0.53	thr 403					
[DDQN-Orig]	Ep 400 R	266.28	avgR100	182.07	eps 0.135	Qval 27.18
land100 0.54	thr 151					
[DDQN-Orig]	Ep 401 R	238.48	avgR100	181.62	eps 0.134	Qval 26.13
land100 0.54	thr 328					
[DDQN-Orig]	Ep 402 R	248.75	avgR100	182.79	eps 0.133	Qval 27.52
land100 0.55	thr 201					
[DDQN-Orig]	Ep 403 R	250.32	avgR100	183.21	eps 0.133	Qval 25.68
land100 0.55	thr 205					
[DDQN-Orig]	Ep 404 R	281.88	avgR100	184.81	eps 0.132	Qval 26.05
land100 0.56	thr 179					
[DDQN-Orig]	Ep 405 R	260.48	avgR100	186.31	eps 0.131	Qval 26.78
land100 0.57	thr 268					
[DDQN-Orig]	Ep 406 R	243.43	avgR100	186.91	eps 0.131	Qval 27.40
land100 0.57	thr 159					

[DDQN-Orig]	Ep 407 R	256.02	avgR100	189.03	eps 0.130	Qval 26.94
land100 0.58	thr 139					
[DDQN-Orig]	Ep 408 R	240.76	avgR100	189.45	eps 0.129	Qval 26.85
land100 0.58	thr 161					
[DDQN-Orig]	Ep 409 R	254.33	avgR100	189.17	eps 0.129	Qval 27.78
land100 0.57	thr 240					
[DDQN-Orig]	Ep 410 R	70.98	avgR100	189.79	eps 0.128	Qval 28.44
land100 0.57	thr 781					
[DDQN-Orig]	Ep 411 R	279.94	avgR100	192.01	eps 0.127	Qval 29.15
land100 0.58	thr 331					
[DDQN-Orig]	Ep 412 R	117.55	avgR100	193.36	eps 0.127	Qval 29.28
land100 0.58	thr 645					
[DDQN-Orig]	Ep 413 R	63.68	avgR100	192.01	eps 0.126	Qval 29.73
land100 0.58	thr 898					
[DDQN-Orig]	Ep 414 R	264.45	avgR100	192.01	eps 0.126	Qval 30.47
land100 0.58	thr 232					
[DDQN-Orig]	Ep 415 R	231.03	avgR100	193.59	eps 0.125	Qval 31.48
land100 0.59	thr 472					
[DDQN-Orig]	Ep 416 R	270.37	avgR100	193.69	eps 0.124	Qval 30.66
land100 0.59	thr 307					
[DDQN-Orig]	Ep 417 R	293.09	avgR100	195.17	eps 0.124	Qval 31.77
land100 0.60	thr 242					
[DDQN-Orig]	Ep 418 R	256.17	avgR100	196.23	eps 0.123	Qval 29.30
land100 0.60	thr 307					
[DDQN-Orig]	Ep 419 R	262.26	avgR100	196.48	eps 0.122	Qval 29.61
land100 0.60	thr 265					
[DDQN-Orig]	Ep 420 R	262.28	avgR100	196.57	eps 0.122	Qval 30.62
land100 0.60	thr 155					
[DDQN-Orig]	Ep 421 R	286.90	avgR100	197.05	eps 0.121	Qval 29.96
land100 0.60	thr 187					
[DDQN-Orig]	Ep 422 R	143.49	avgR100	196.88	eps 0.121	Qval 30.54
land100 0.60	thr 401					
[DDQN-Orig]	Ep 423 R	192.33	avgR100	196.28	eps 0.120	Qval 30.35
land100 0.60	thr 710					
[DDQN-Orig]	Ep 424 R	243.07	avgR100	197.64	eps 0.119	Qval 30.91
land100 0.61	thr 149					
[DDQN-Orig]	Ep 425 R	254.95	avgR100	198.19	eps 0.119	Qval 29.50
land100 0.61	thr 356					
[DDQN-Orig]	Ep 426 R	227.05	avgR100	200.35	eps 0.118	Qval 28.57
land100 0.62	thr 485					
[DDQN-Orig]	Ep 427 R	228.24	avgR100	199.86	eps 0.118	Qval 28.68
land100 0.61	thr 267					
[DDQN-Orig]	Ep 428 R	261.05	avgR100	199.99	eps 0.117	Qval 29.82
land100 0.61	thr 241					
[DDQN-Orig]	Ep 429 R	69.99	avgR100	199.41	eps 0.116	Qval 29.37
land100 0.61	thr 813					
[DDQN-Orig]	Ep 430 R	247.01	avgR100	199.58	eps 0.116	Qval 29.31
land100 0.61	thr 327					
[DDQN-Orig]	Ep 431 R	230.01	avgR100	199.13	eps 0.115	Qval 30.11
land100 0.61	thr 192					
[DDQN-Orig]	Ep 432 R	259.54	avgR100	199.08	eps 0.115	Qval 29.40
land100 0.61	thr 195					
[DDQN-Orig]	Ep 433 R	239.34	avgR100	201.93	eps 0.114	Qval 28.27
land100 0.61	thr 300					
[DDQN-Orig]	Ep 434 R	236.58	avgR100	202.44	eps 0.114	Qval 28.59
land100 0.61	thr 288					
[DDQN-Orig]	Ep 435 R	239.91	avgR100	202.51	eps 0.113	Qval 30.21
land100 0.61	thr 302					
[DDQN-Orig]	Ep 436 R	241.57	avgR100	202.88	eps 0.112	Qval 29.16
land100 0.62	thr 438					

[DDQN-Orig]	Ep 437 R	230.09 avgR100	205.72 eps 0.112 Qval	29.56
land100 0.63 thr 341				
[DDQN-Orig]	Ep 438 R	232.99 avgR100	206.97 eps 0.111 Qval	28.74
land100 0.64 thr 278				
[DDQN-Orig]	Ep 439 R	265.64 avgR100	206.91 eps 0.111 Qval	28.50
land100 0.64 thr 276				
[DDQN-Orig]	Ep 440 R	254.41 avgR100	207.10 eps 0.110 Qval	30.14
land100 0.64 thr 258				
[DDQN-Orig]	Ep 441 R	269.19 avgR100	209.21 eps 0.110 Qval	27.92
land100 0.65 thr 296				
[DDQN-Orig]	Ep 442 R	251.92 avgR100	209.02 eps 0.109 Qval	28.81
land100 0.64 thr 193				
[DDQN-Orig]	Ep 443 R	252.73 avgR100	209.11 eps 0.109 Qval	29.85
land100 0.64 thr 347				
[DDQN-Orig]	Ep 444 R	227.81 avgR100	210.31 eps 0.108 Qval	29.97
land100 0.65 thr 362				
[DDQN-Orig]	Ep 445 R	243.97 avgR100	212.32 eps 0.107 Qval	29.54
land100 0.66 thr 301				
[DDQN-Orig]	Ep 446 R	260.47 avgR100	212.27 eps 0.107 Qval	31.15
land100 0.67 thr 299				
[DDQN-Orig]	Ep 447 R	219.71 avgR100	212.27 eps 0.106 Qval	29.47
land100 0.67 thr 351				
[DDQN-Orig]	Ep 448 R	268.41 avgR100	213.19 eps 0.106 Qval	29.65
land100 0.67 thr 243				
[DDQN-Orig]	Ep 449 R	254.25 avgR100	212.72 eps 0.105 Qval	29.15
land100 0.67 thr 225				
[DDQN-Orig]	Ep 450 R	249.25 avgR100	214.94 eps 0.105 Qval	29.71
land100 0.68 thr 365				
[DDQN-Orig]	Ep 451 R	251.23 avgR100	214.98 eps 0.104 Qval	28.94
land100 0.68 thr 186				
[DDQN-Orig]	Ep 452 R	212.13 avgR100	215.04 eps 0.104 Qval	29.58
land100 0.67 thr 359				
[DDQN-Orig]	Ep 453 R	205.07 avgR100	214.78 eps 0.103 Qval	30.21
land100 0.67 thr 517				
[DDQN-Orig]	Ep 454 R	259.94 avgR100	214.97 eps 0.103 Qval	29.33
land100 0.67 thr 192				
[DDQN-Orig]	Ep 455 R	236.90 avgR100	216.01 eps 0.102 Qval	30.06
land100 0.67 thr 381				
[DDQN-Orig]	Ep 456 R	231.41 avgR100	215.99 eps 0.102 Qval	30.70
land100 0.67 thr 155				
[DDQN-Orig]	Ep 457 R	252.46 avgR100	215.87 eps 0.101 Qval	29.98
land100 0.67 thr 352				
[DDQN-Orig]	Ep 458 R	254.86 avgR100	217.27 eps 0.101 Qval	29.73
land100 0.68 thr 304				
[DDQN-Orig]	Ep 459 R	282.99 avgR100	217.74 eps 0.100 Qval	29.90
land100 0.68 thr 136				
[DDQN-Orig]	Ep 460 R	230.78 avgR100	218.68 eps 0.100 Qval	29.89
land100 0.69 thr 265				
[DDQN-Orig]	Ep 461 R	-45.61 avgR100	216.10 eps 0.099 Qval	31.90
land100 0.68 thr 95				
[DDQN-Orig]	Ep 462 R	169.80 avgR100	215.31 eps 0.099 Qval	31.71
land100 0.67 thr 702				
[DDQN-Orig]	Ep 463 R	284.14 avgR100	216.27 eps 0.098 Qval	31.59
land100 0.67 thr 217				
[DDQN-Orig]	Ep 464 R	246.21 avgR100	215.99 eps 0.098 Qval	31.28
land100 0.66 thr 216				
[DDQN-Orig]	Ep 465 R	272.67 avgR100	216.33 eps 0.097 Qval	29.79
land100 0.66 thr 214				
[DDQN-Orig]	Ep 466 R	249.96 avgR100	218.06 eps 0.097 Qval	30.78
land100 0.67 thr 243				

[DDQN-Orig]	Ep 467 R	217.22 avgR100	217.91 eps 0.096 Qval	30.26
land100 0.68 thr 499				
[DDQN-Orig]	Ep 468 R	197.51 avgR100	217.15 eps 0.096 Qval	30.78
land100 0.68 thr 378				
[DDQN-Orig]	Ep 469 R	248.12 avgR100	216.99 eps 0.095 Qval	29.33
land100 0.67 thr 234				
[DDQN-Orig]	Ep 470 R	253.00 avgR100	217.17 eps 0.095 Qval	31.56
land100 0.68 thr 268				
[DDQN-Orig]	Ep 471 R	234.96 avgR100	219.55 eps 0.094 Qval	28.77
land100 0.69 thr 189				
[DDQN-Orig]	Ep 472 R	208.36 avgR100	221.62 eps 0.094 Qval	32.57
land100 0.70 thr 451				
[DDQN-Orig]	Ep 473 R	269.81 avgR100	222.05 eps 0.093 Qval	30.96
land100 0.71 thr 261				
[DDQN-Orig]	Ep 474 R	32.04 avgR100	222.43 eps 0.093 Qval	31.28
land100 0.71 thr 83				
[DDQN-Orig]	Ep 475 R	172.07 avgR100	221.70 eps 0.092 Qval	32.29
land100 0.70 thr 686				
[DDQN-Orig]	Ep 476 R	128.35 avgR100	220.74 eps 0.092 Qval	31.42
land100 0.69 thr 412				
[DDQN-Orig]	Ep 477 R	271.68 avgR100	221.84 eps 0.092 Qval	32.85
land100 0.70 thr 189				
[DDQN-Orig]	Ep 478 R	-59.10 avgR100	219.04 eps 0.091 Qval	32.68
land100 0.70 thr 930				
[DDQN-Orig]	Ep 479 R	210.95 avgR100	219.31 eps 0.091 Qval	33.21
land100 0.70 thr 360				
[DDQN-Orig]	Ep 480 R	272.66 avgR100	219.86 eps 0.090 Qval	32.92
land100 0.71 thr 263				
[DDQN-Orig]	Ep 481 R	282.77 avgR100	220.52 eps 0.090 Qval	33.36
land100 0.71 thr 209				
[DDQN-Orig]	Ep 482 R	276.06 avgR100	223.03 eps 0.089 Qval	32.79
land100 0.72 thr 330				
[DDQN-Orig]	Ep 483 R	17.41 avgR100	221.36 eps 0.089 Qval	34.19
land100 0.71 thr 88				
[DDQN-Orig]	Ep 484 R	205.92 avgR100	221.21 eps 0.088 Qval	34.12
land100 0.71 thr 234				
[DDQN-Orig]	Ep 485 R	110.68 avgR100	220.54 eps 0.088 Qval	33.58
land100 0.71 thr 515				
[DDQN-Orig]	Ep 486 R	254.23 avgR100	220.80 eps 0.088 Qval	34.36
land100 0.71 thr 279				
[DDQN-Orig]	Ep 487 R	265.57 avgR100	220.94 eps 0.087 Qval	34.37
land100 0.71 thr 297				
[DDQN-Orig]	Ep 488 R	42.98 avgR100	218.96 eps 0.087 Qval	33.53
land100 0.70 thr 862				
[DDQN-Orig]	Ep 489 R	201.69 avgR100	219.79 eps 0.086 Qval	33.47
land100 0.71 thr 438				
[DDQN-Orig]	Ep 490 R	150.99 avgR100	219.39 eps 0.086 Qval	33.80
land100 0.72 thr 781				
[DDQN-Orig]	Ep 491 R	263.87 avgR100	221.36 eps 0.085 Qval	33.00
land100 0.73 thr 208				
[DDQN-Orig]	Ep 492 R	265.34 avgR100	222.78 eps 0.085 Qval	32.90
land100 0.74 thr 258				
[DDQN-Orig]	Ep 493 R	250.72 avgR100	223.01 eps 0.084 Qval	32.78
land100 0.74 thr 383				
[DDQN-Orig]	Ep 494 R	268.22 avgR100	222.79 eps 0.084 Qval	33.17
land100 0.74 thr 184				
[DDQN-Orig]	Ep 495 R	287.45 avgR100	223.24 eps 0.084 Qval	32.77
land100 0.74 thr 262				

[DDQN-Orig]	Ep 496 R	248.25 avgR100	223.11 eps 0.083 Qval	32.60
land100 0.73 thr 507				
[DDQN-Orig]	Ep 497 R	226.64 avgR100	223.15 eps 0.083 Qval	32.29
land100 0.74 thr 211				
[DDQN-Orig]	Ep 498 R	224.88 avgR100	222.72 eps 0.082 Qval	34.06
land100 0.74 thr 285				
[DDQN-Orig]	Ep 499 R	242.22 avgR100	223.33 eps 0.082 Qval	34.12
land100 0.75 thr 179				
[DDQN-Orig]	Ep 500 R	270.42 avgR100	223.37 eps 0.082 Qval	33.44
land100 0.75 thr 314				
[DDQN-Orig]	Ep 501 R	257.51 avgR100	223.56 eps 0.081 Qval	34.88
land100 0.75 thr 188				
[DDQN-Orig]	Ep 502 R	231.47 avgR100	223.39 eps 0.081 Qval	34.54
land100 0.75 thr 307				
[DDQN-Orig]	Ep 503 R	277.27 avgR100	223.66 eps 0.080 Qval	33.84
land100 0.75 thr 251				
[DDQN-Orig]	Ep 504 R	212.66 avgR100	222.96 eps 0.080 Qval	33.50
land100 0.74 thr 694				
[DDQN-Orig]	Ep 505 R	24.48 avgR100	220.60 eps 0.080 Qval	33.06
land100 0.73 thr 163				
[DDQN-Orig]	Ep 506 R	157.51 avgR100	219.75 eps 0.079 Qval	33.01
land100 0.73 thr 769				
[DDQN-Orig]	Ep 507 R	206.91 avgR100	219.25 eps 0.079 Qval	32.88
land100 0.73 thr 256				
[DDQN-Orig]	Ep 508 R	90.07 avgR100	217.75 eps 0.078 Qval	33.22
land100 0.72 thr 667				
[DDQN-Orig]	Ep 509 R	217.09 avgR100	217.38 eps 0.078 Qval	32.56
land100 0.72 thr 426				
[DDQN-Orig]	Ep 510 R	239.03 avgR100	219.06 eps 0.078 Qval	33.71
land100 0.73 thr 179				
[DDQN-Orig]	Ep 511 R	229.35 avgR100	218.55 eps 0.077 Qval	32.75
land100 0.73 thr 215				
[DDQN-Orig]	Ep 512 R	202.06 avgR100	219.39 eps 0.077 Qval	33.74
land100 0.74 thr 609				
[DDQN-Orig]	Ep 513 R	229.02 avgR100	221.05 eps 0.076 Qval	33.93
land100 0.75 thr 320				
[DDQN-Orig]	Ep 514 R	233.67 avgR100	220.74 eps 0.076 Qval	32.37
land100 0.75 thr 191				
[DDQN-Orig]	Ep 515 R	250.40 avgR100	220.93 eps 0.076 Qval	33.79
land100 0.75 thr 268				
[DDQN-Orig]	Ep 516 R	213.05 avgR100	220.36 eps 0.075 Qval	31.81
land100 0.75 thr 561				
[DDQN-Orig]	Ep 517 R	288.77 avgR100	220.32 eps 0.075 Qval	31.99
land100 0.74 thr 195				
[DDQN-Orig]	Ep 518 R	250.02 avgR100	220.26 eps 0.075 Qval	33.46
land100 0.74 thr 310				
[DDQN-Orig]	Ep 519 R	69.32 avgR100	218.33 eps 0.074 Qval	33.11
land100 0.73 thr 788				
[DDQN-Orig]	Ep 520 R	190.05 avgR100	217.60 eps 0.074 Qval	31.58
land100 0.73 thr 509				
[DDQN-Orig]	Ep 521 R	264.69 avgR100	217.38 eps 0.073 Qval	32.88
land100 0.73 thr 408				
[DDQN-Orig]	Ep 522 R	261.25 avgR100	218.56 eps 0.073 Qval	33.20
land100 0.74 thr 173				
[DDQN-Orig]	Ep 523 R	240.42 avgR100	219.04 eps 0.073 Qval	33.13
land100 0.74 thr 228				
[DDQN-Orig]	Ep 524 R	228.80 avgR100	218.90 eps 0.072 Qval	32.97
land100 0.74 thr 373				

[DDQN-Orig]	Ep 525 R	191.38	avgR100	218.26	eps 0.072	Qval	33.96
land100 0.74	thr 595						
[DDQN-Orig]	Ep 526 R	244.78	avgR100	218.44	eps 0.072	Qval	34.38
land100 0.74	thr 170						
[DDQN-Orig]	Ep 527 R	227.50	avgR100	218.43	eps 0.071	Qval	34.43
land100 0.75	thr 266						
[DDQN-Orig]	Ep 528 R	214.60	avgR100	217.97	eps 0.071	Qval	35.96
land100 0.75	thr 288						
[DDQN-Orig]	Ep 529 R	267.60	avgR100	219.94	eps 0.071	Qval	36.75
land100 0.76	thr 214						
[DDQN-Orig]	Ep 530 R	230.66	avgR100	219.78	eps 0.070	Qval	36.06
land100 0.76	thr 263						
[DDQN-Orig]	Ep 531 R	195.25	avgR100	219.43	eps 0.070	Qval	35.55
land100 0.75	thr 587						
[DDQN-Orig]	Ep 532 R	194.27	avgR100	218.78	eps 0.069	Qval	35.50
land100 0.74	thr 329						
[DDQN-Orig]	Ep 533 R	236.79	avgR100	218.75	eps 0.069	Qval	35.67
land100 0.74	thr 322						
[DDQN-Orig]	Ep 534 R	218.70	avgR100	218.58	eps 0.069	Qval	36.59
land100 0.74	thr 226						
[DDQN-Orig]	Ep 535 R	169.46	avgR100	217.87	eps 0.068	Qval	36.98
land100 0.75	thr 656						
[DDQN-Orig]	Ep 536 R	272.21	avgR100	218.18	eps 0.068	Qval	35.71
land100 0.75	thr 348						
[DDQN-Orig]	Ep 537 R	249.72	avgR100	218.37	eps 0.068	Qval	36.71
land100 0.74	thr 350						
[DDQN-Orig]	Ep 538 R	246.66	avgR100	218.51	eps 0.067	Qval	35.92
land100 0.74	thr 382						
[DDQN-Orig]	Ep 539 R	249.68	avgR100	218.35	eps 0.067	Qval	37.09
land100 0.74	thr 354						
[DDQN-Orig]	Ep 540 R	277.05	avgR100	218.58	eps 0.067	Qval	36.27
land100 0.74	thr 155						
[DDQN-Orig]	Ep 541 R	250.51	avgR100	218.39	eps 0.066	Qval	36.42
land100 0.73	thr 213						
[DDQN-Orig]	Ep 542 R	239.25	avgR100	218.26	eps 0.066	Qval	36.17
land100 0.74	thr 230						
[DDQN-Orig]	Ep 543 R	256.92	avgR100	218.31	eps 0.066	Qval	36.60
land100 0.74	thr 196						
[DDQN-Orig]	Ep 544 R	220.68	avgR100	218.23	eps 0.065	Qval	37.55
land100 0.74	thr 430						
[DDQN-Orig]	Ep 545 R	271.46	avgR100	218.51	eps 0.065	Qval	36.40
land100 0.74	thr 267						
[DDQN-Orig]	Ep 546 R	246.16	avgR100	218.37	eps 0.065	Qval	36.36
land100 0.74	thr 258						
[DDQN-Orig]	Ep 547 R	197.67	avgR100	218.15	eps 0.064	Qval	36.85
land100 0.73	thr 538						
[DDQN-Orig]	Ep 548 R	286.48	avgR100	218.33	eps 0.064	Qval	37.06
land100 0.73	thr 205						
[DDQN-Orig]	Ep 549 R	234.15	avgR100	218.13	eps 0.064	Qval	36.55
land100 0.73	thr 301						
[DDQN-Orig]	Ep 550 R	293.95	avgR100	218.57	eps 0.063	Qval	36.67
land100 0.73	thr 213						
[DDQN-Orig]	Ep 551 R	228.71	avgR100	218.35	eps 0.063	Qval	37.58
land100 0.72	thr 239						
[DDQN-Orig]	Ep 552 R	49.77	avgR100	216.72	eps 0.063	Qval	34.90
land100 0.72	thr 760						
[DDQN-Orig]	Ep 553 R	219.47	avgR100	216.87	eps 0.063	Qval	36.43
land100 0.71	thr 528						
[DDQN-Orig]	Ep 554 R	245.32	avgR100	216.72	eps 0.062	Qval	36.31
land100 0.71	thr 301						

[DDQN-Orig]	Ep 555 R	193.66 avgR100	216.29 eps 0.062 Qval	36.03
land100 0.71 thr 342				
[DDQN-Orig]	Ep 556 R	228.83 avgR100	216.26 eps 0.062 Qval	35.26
land100 0.70 thr 335				
[DDQN-Orig]	Ep 557 R	236.87 avgR100	216.11 eps 0.061 Qval	36.40
land100 0.70 thr 263				
[DDQN-Orig]	Ep 558 R	239.73 avgR100	215.96 eps 0.061 Qval	36.08
land100 0.70 thr 244				
[DDQN-Orig]	Ep 559 R	235.94 avgR100	215.49 eps 0.061 Qval	36.23
land100 0.70 thr 224				
[DDQN-Orig]	Ep 560 R	191.99 avgR100	215.10 eps 0.060 Qval	34.45
land100 0.70 thr 567				
[DDQN-Orig]	Ep 561 R	108.09 avgR100	216.64 eps 0.060 Qval	35.14
land100 0.71 thr 810				
[DDQN-Orig]	Ep 562 R	226.67 avgR100	217.20 eps 0.060 Qval	34.91
land100 0.71 thr 327				
[DDQN-Orig]	Ep 563 R	272.90 avgR100	217.09 eps 0.059 Qval	34.65
land100 0.71 thr 414				
[DDQN-Orig]	Ep 564 R	230.80 avgR100	216.94 eps 0.059 Qval	35.27
land100 0.72 thr 255				
[DDQN-Orig]	Ep 565 R	272.51 avgR100	216.94 eps 0.059 Qval	35.48
land100 0.72 thr 240				
[DDQN-Orig]	Ep 566 R	210.03 avgR100	216.54 eps 0.059 Qval	35.06
land100 0.72 thr 375				
[DDQN-Orig]	Ep 567 R	254.75 avgR100	216.91 eps 0.058 Qval	35.49
land100 0.72 thr 351				
[DDQN-Orig]	Ep 568 R	251.45 avgR100	217.45 eps 0.058 Qval	36.25
land100 0.72 thr 316				
[DDQN-Orig]	Ep 569 R	276.34 avgR100	217.73 eps 0.058 Qval	36.83
land100 0.73 thr 242				
[DDQN-Orig]	Ep 570 R	240.88 avgR100	217.61 eps 0.057 Qval	35.76
land100 0.72 thr 383				
[DDQN-Orig]	Ep 571 R	242.80 avgR100	217.69 eps 0.057 Qval	36.05
land100 0.72 thr 357				
[DDQN-Orig]	Ep 572 R	227.02 avgR100	217.88 eps 0.057 Qval	36.61
land100 0.71 thr 307				
[DDQN-Orig]	Ep 573 R	241.26 avgR100	217.59 eps 0.057 Qval	37.23
land100 0.71 thr 321				
[DDQN-Orig]	Ep 574 R	160.84 avgR100	218.88 eps 0.056 Qval	37.24
land100 0.72 thr 664				
[DDQN-Orig]	Ep 575 R	39.11 avgR100	217.55 eps 0.056 Qval	38.62
land100 0.72 thr 762				
[DDQN-Orig]	Ep 576 R	171.07 avgR100	217.98 eps 0.056 Qval	38.29
land100 0.73 thr 498				
[DDQN-Orig]	Ep 577 R	242.95 avgR100	217.69 eps 0.055 Qval	38.89
land100 0.73 thr 184				
[DDQN-Orig]	Ep 578 R	256.52 avgR100	220.85 eps 0.055 Qval	38.08
land100 0.74 thr 173				
[DDQN-Orig]	Ep 579 R	241.46 avgR100	221.15 eps 0.055 Qval	39.11
land100 0.73 thr 394				
[DDQN-Orig]	Ep 580 R	224.46 avgR100	220.67 eps 0.055 Qval	39.02
land100 0.73 thr 464				
[DDQN-Orig]	Ep 581 R	219.19 avgR100	220.03 eps 0.054 Qval	37.97
land100 0.73 thr 389				
[DDQN-Orig]	Ep 582 R	285.01 avgR100	220.12 eps 0.054 Qval	38.59
land100 0.73 thr 167				
[DDQN-Orig]	Ep 583 R	244.94 avgR100	222.40 eps 0.054 Qval	38.72
land100 0.74 thr 163				

[DDQN-Orig]	Ep 584 R	236.18 avgR100	222.70 eps 0.054 Qval	38.90
land100 0.74 thr 312				
[DDQN-Orig]	Ep 585 R	190.49 avgR100	223.50 eps 0.053 Qval	38.37
land100 0.74 thr 241				
[DDQN-Orig]	Ep 586 R	258.21 avgR100	223.54 eps 0.053 Qval	38.63
land100 0.74 thr 224				
[DDQN-Orig]	Ep 587 R	192.26 avgR100	222.81 eps 0.053 Qval	39.34
land100 0.73 thr 419				
[DDQN-Orig]	Ep 588 R	210.20 avgR100	224.48 eps 0.052 Qval	40.15
land100 0.74 thr 432				
[DDQN-Orig]	Ep 589 R	280.49 avgR100	225.27 eps 0.052 Qval	39.18
land100 0.74 thr 557				
[DDQN-Orig]	Ep 590 R	242.33 avgR100	226.18 eps 0.052 Qval	38.65
land100 0.74 thr 187				
[DDQN-Orig]	Ep 591 R	117.65 avgR100	224.72 eps 0.052 Qval	40.57
land100 0.73 thr 538				
[DDQN-Orig]	Ep 592 R	227.57 avgR100	224.34 eps 0.051 Qval	39.17
land100 0.72 thr 380				
[DDQN-Orig]	Ep 593 R	232.06 avgR100	224.15 eps 0.051 Qval	40.00
land100 0.71 thr 257				
[DDQN-Orig]	Ep 594 R	264.90 avgR100	224.12 eps 0.051 Qval	40.01
land100 0.71 thr 214				
[DDQN-Orig]	Ep 595 R	234.30 avgR100	223.59 eps 0.051 Qval	39.54
land100 0.70 thr 232				
[DDQN-Orig]	Ep 596 R	-21.09 avgR100	220.89 eps 0.050 Qval	38.63
land100 0.70 thr 135				
[DDQN-Orig]	Ep 597 R	268.49 avgR100	221.31 eps 0.050 Qval	39.14
land100 0.70 thr 195				
[DDQN-Orig]	Ep 598 R	233.37 avgR100	221.40 eps 0.050 Qval	40.46
land100 0.70 thr 526				
[DDQN-Orig]	Ep 599 R	227.91 avgR100	221.26 eps 0.050 Qval	39.66
land100 0.69 thr 533				
[DDQN-Orig]	Ep 600 R	287.24 avgR100	221.42 eps 0.049 Qval	40.73
land100 0.69 thr 166				
[DDQN-Orig]	Ep 601 R	-59.82 avgR100	218.25 eps 0.049 Qval	39.63
land100 0.68 thr 874				
[DDQN-Orig]	Ep 602 R	144.32 avgR100	217.38 eps 0.049 Qval	40.40
land100 0.67 thr 689				
[DDQN-Orig]	Ep 603 R	242.59 avgR100	217.03 eps 0.049 Qval	41.55
land100 0.67 thr 166				
[DDQN-Orig]	Ep 604 R	206.97 avgR100	216.97 eps 0.048 Qval	42.00
land100 0.68 thr 453				
[DDQN-Orig]	Ep 605 R	260.45 avgR100	219.33 eps 0.048 Qval	41.43
land100 0.69 thr 191				
[DDQN-Orig]	Ep 606 R	225.63 avgR100	220.02 eps 0.048 Qval	40.98
land100 0.69 thr 228				
[DDQN-Orig]	Ep 607 R	20.49 avgR100	218.15 eps 0.048 Qval	41.13
land100 0.68 thr 721				
[DDQN-Orig]	Ep 608 R	211.80 avgR100	219.37 eps 0.047 Qval	40.67
land100 0.68 thr 423				
[DDQN-Orig]	Ep 609 R	257.95 avgR100	219.78 eps 0.047 Qval	40.55
land100 0.69 thr 209				
[DDQN-Orig]	Ep 610 R	197.86 avgR100	219.37 eps 0.047 Qval	42.15
land100 0.69 thr 613				
[DDQN-Orig]	Ep 611 R	216.43 avgR100	219.24 eps 0.047 Qval	42.51
land100 0.69 thr 290				
[DDQN-Orig]	Ep 612 R	241.59 avgR100	219.63 eps 0.047 Qval	41.93
land100 0.68 thr 451				
[DDQN-Orig]	Ep 613 R	271.31 avgR100	220.05 eps 0.046 Qval	43.66
land100 0.68 thr 350				

[DDQN-Orig]	Ep 614 R	231.86	avgR100	220.04	eps 0.046	Qval	42.76
land100 0.68	thr 303						
[DDQN-Orig]	Ep 615 R	71.51	avgR100	218.25	eps 0.046	Qval	42.48
land100 0.67	thr 845						
[DDQN-Orig]	Ep 616 R	220.53	avgR100	218.32	eps 0.046	Qval	43.11
land100 0.67	thr 404						
[DDQN-Orig]	Ep 617 R	178.50	avgR100	217.22	eps 0.045	Qval	41.83
land100 0.68	thr 415						
[DDQN-Orig]	Ep 618 R	205.15	avgR100	216.77	eps 0.045	Qval	43.22
land100 0.68	thr 382						
[DDQN-Orig]	Ep 619 R	238.24	avgR100	218.46	eps 0.045	Qval	43.84
land100 0.69	thr 265						
[DDQN-Orig]	Ep 620 R	294.31	avgR100	219.50	eps 0.045	Qval	42.64
land100 0.69	thr 198						
[DDQN-Orig]	Ep 621 R	261.87	avgR100	219.47	eps 0.044	Qval	43.49
land100 0.69	thr 174						
[DDQN-Orig]	Ep 622 R	250.47	avgR100	219.37	eps 0.044	Qval	43.61
land100 0.69	thr 302						
[DDQN-Orig]	Ep 623 R	214.07	avgR100	219.10	eps 0.044	Qval	42.88
land100 0.69	thr 320						
[DDQN-Orig]	Ep 624 R	8.79	avgR100	216.90	eps 0.044	Qval	42.64
land100 0.68	thr 92						
[DDQN-Orig]	Ep 625 R	232.84	avgR100	217.32	eps 0.044	Qval	43.64
land100 0.67	thr 210						
[DDQN-Orig]	Ep 626 R	262.12	avgR100	217.49	eps 0.043	Qval	43.46
land100 0.67	thr 168						
[DDQN-Orig]	Ep 627 R	247.81	avgR100	217.69	eps 0.043	Qval	43.11
land100 0.67	thr 258						
[DDQN-Orig]	Ep 628 R	290.03	avgR100	218.45	eps 0.043	Qval	42.80
land100 0.67	thr 246						
[DDQN-Orig]	Ep 629 R	141.55	avgR100	217.19	eps 0.043	Qval	42.55
land100 0.67	thr 693						
[DDQN-Orig]	Ep 630 R	174.75	avgR100	216.63	eps 0.043	Qval	42.03
land100 0.67	thr 490						
[DDQN-Orig]	Ep 631 R	263.97	avgR100	217.32	eps 0.042	Qval	42.32
land100 0.68	thr 171						
[DDQN-Orig]	Ep 632 R	236.15	avgR100	217.74	eps 0.042	Qval	42.18
land100 0.69	thr 275						
[DDQN-Orig]	Ep 633 R	200.55	avgR100	217.37	eps 0.042	Qval	42.65
land100 0.70	thr 500						
[DDQN-Orig]	Ep 634 R	250.12	avgR100	217.69	eps 0.042	Qval	43.05
land100 0.70	thr 114						
[DDQN-Orig]	Ep 635 R	258.84	avgR100	218.58	eps 0.041	Qval	43.33
land100 0.70	thr 194						
[DDQN-Orig]	Ep 636 R	198.92	avgR100	217.85	eps 0.041	Qval	43.12
land100 0.70	thr 343						
[DDQN-Orig]	Ep 637 R	221.57	avgR100	217.57	eps 0.041	Qval	43.23
land100 0.70	thr 311						
[DDQN-Orig]	Ep 638 R	251.79	avgR100	217.62	eps 0.041	Qval	43.71
land100 0.70	thr 228						
[DDQN-Orig]	Ep 639 R	311.99	avgR100	218.24	eps 0.041	Qval	43.57
land100 0.70	thr 214						
[DDQN-Orig]	Ep 640 R	272.58	avgR100	218.20	eps 0.040	Qval	44.56
land100 0.70	thr 284						
[DDQN-Orig]	Ep 641 R	291.13	avgR100	218.60	eps 0.040	Qval	45.98
land100 0.71	thr 187						
[DDQN-Orig]	Ep 642 R	271.18	avgR100	218.92	eps 0.040	Qval	45.01
land100 0.71	thr 214						
[DDQN-Orig]	Ep 643 R	230.74	avgR100	218.66	eps 0.040	Qval	44.81
land100 0.71	thr 458						

[DDQN-Orig]	Ep 644 R	184.54	avgR100	218.30	eps 0.040	Qval 44.08
land100 0.71	thr 465					
[DDQN-Orig]	Ep 645 R	171.57	avgR100	217.30	eps 0.039	Qval 45.23
land100 0.71	thr 696					
[DDQN-Orig]	Ep 646 R	248.56	avgR100	217.32	eps 0.039	Qval 44.40
land100 0.71	thr 337					
[DDQN-Orig]	Ep 647 R	200.34	avgR100	217.35	eps 0.039	Qval 45.38
land100 0.72	thr 520					
[DDQN-Orig]	Ep 648 R	238.17	avgR100	216.87	eps 0.039	Qval 44.95
land100 0.72	thr 310					
[DDQN-Orig]	Ep 649 R	302.35	avgR100	217.55	eps 0.039	Qval 45.12
land100 0.72	thr 255					
[DDQN-Orig]	Ep 650 R	190.04	avgR100	216.51	eps 0.038	Qval 44.75
land100 0.72	thr 477					
[DDQN-Orig]	Ep 651 R	222.93	avgR100	216.45	eps 0.038	Qval 45.50
land100 0.72	thr 378					
[DDQN-Orig]	Ep 652 R	256.81	avgR100	218.52	eps 0.038	Qval 46.12
land100 0.73	thr 420					
[DDQN-Orig]	Ep 653 R	188.95	avgR100	218.22	eps 0.038	Qval 47.40
land100 0.74	thr 365					
[DDQN-Orig]	Ep 654 R	-30.22	avgR100	215.46	eps 0.038	Qval 47.95
land100 0.73	thr 787					
[DDQN-Orig]	Ep 655 R	291.46	avgR100	216.44	eps 0.038	Qval 45.79
land100 0.73	thr 183					
[DDQN-Orig]	Ep 656 R	223.97	avgR100	216.39	eps 0.037	Qval 47.13
land100 0.74	thr 385					
[DDQN-Orig]	Ep 657 R	266.38	avgR100	216.69	eps 0.037	Qval 46.24
land100 0.74	thr 267					
[DDQN-Orig]	Ep 658 R	261.46	avgR100	216.90	eps 0.037	Qval 46.00
land100 0.74	thr 188					
[DDQN-Orig]	Ep 659 R	256.16	avgR100	217.11	eps 0.037	Qval 47.91
land100 0.74	thr 181					
[DDQN-Orig]	Ep 660 R	237.28	avgR100	217.56	eps 0.037	Qval 47.17
land100 0.74	thr 314					
[DDQN-Orig]	Ep 661 R	261.95	avgR100	219.10	eps 0.036	Qval 46.87
land100 0.73	thr 262					
[DDQN-Orig]	Ep 662 R	266.94	avgR100	219.50	eps 0.036	Qval 46.43
land100 0.74	thr 285					
[DDQN-Orig]	Ep 663 R	256.02	avgR100	219.33	eps 0.036	Qval 47.13
land100 0.74	thr 351					
[DDQN-Orig]	Ep 664 R	252.91	avgR100	219.55	eps 0.036	Qval 47.78
land100 0.74	thr 197					
[DDQN-Orig]	Ep 665 R	299.79	avgR100	219.82	eps 0.036	Qval 46.75
land100 0.73	thr 201					
[DDQN-Orig]	Ep 666 R	236.83	avgR100	220.09	eps 0.035	Qval 47.45
land100 0.73	thr 368					
[DDQN-Orig]	Ep 667 R	230.95	avgR100	219.85	eps 0.035	Qval 46.57
land100 0.73	thr 254					
[DDQN-Orig]	Ep 668 R	253.83	avgR100	219.88	eps 0.035	Qval 46.36
land100 0.73	thr 200					
[DDQN-Orig]	Ep 669 R	295.12	avgR100	220.07	eps 0.035	Qval 47.97
land100 0.73	thr 337					
[DDQN-Orig]	Ep 670 R	53.18	avgR100	218.19	eps 0.035	Qval 48.15
land100 0.73	thr 732					
[DDQN-Orig]	Ep 671 R	272.77	avgR100	218.49	eps 0.035	Qval 47.87
land100 0.73	thr 301					
[DDQN-Orig]	Ep 672 R	252.60	avgR100	218.75	eps 0.034	Qval 47.76
land100 0.74	thr 164					

[DDQN-Orig] Ep 673 R	168.00	avgR100	218.01	eps 0.034	Qval	47.98
land100 0.74 thr 684						
[DDQN-Orig] Ep 674 R	220.10	avgR100	218.61	eps 0.034	Qval	48.63
land100 0.74 thr 344						
[DDQN-Orig] Ep 675 R	267.41	avgR100	220.89	eps 0.034	Qval	48.31
land100 0.75 thr 143						
[DDQN-Orig] Ep 676 R	274.56	avgR100	221.92	eps 0.034	Qval	47.50
land100 0.74 thr 343						
[DDQN-Orig] Ep 677 R	269.10	avgR100	222.18	eps 0.034	Qval	47.54
land100 0.74 thr 327						
[DDQN-Orig] Ep 678 R	228.59	avgR100	221.91	eps 0.033	Qval	49.08
land100 0.74 thr 424						
[DDQN-Orig] Ep 679 R	240.93	avgR100	221.90	eps 0.033	Qval	48.04
land100 0.74 thr 199						
[DDQN-Orig] Ep 680 R	238.44	avgR100	222.04	eps 0.033	Qval	47.51
land100 0.74 thr 311						
[DDQN-Orig] Ep 681 R	245.61	avgR100	222.30	eps 0.033	Qval	48.32
land100 0.74 thr 240						
[DDQN-Orig] Ep 682 R	239.53	avgR100	221.85	eps 0.033	Qval	49.06
land100 0.74 thr 424						
[DDQN-Orig] Ep 683 R	230.48	avgR100	221.70	eps 0.033	Qval	50.13
land100 0.74 thr 328						
[DDQN-Orig] Ep 684 R	206.95	avgR100	221.41	eps 0.032	Qval	49.47
land100 0.75 thr 464						
[DDQN-Orig] Ep 685 R	214.32	avgR100	221.65	eps 0.032	Qval	48.04
land100 0.76 thr 482						
[DDQN-Orig] Ep 686 R	238.18	avgR100	221.45	eps 0.032	Qval	48.43
land100 0.76 thr 173						
[DDQN-Orig] Ep 687 R	246.45	avgR100	221.99	eps 0.032	Qval	49.74
land100 0.77 thr 326						
[DDQN-Orig] Ep 688 R	266.96	avgR100	222.56	eps 0.032	Qval	49.80
land100 0.77 thr 212						
[DDQN-Orig] Ep 689 R	246.29	avgR100	222.22	eps 0.032	Qval	50.04
land100 0.77 thr 225						
[DDQN-Orig] Ep 690 R	239.94	avgR100	222.19	eps 0.031	Qval	49.46
land100 0.77 thr 310						
[DDQN-Orig] Ep 691 R	233.95	avgR100	223.36	eps 0.031	Qval	49.53
land100 0.78 thr 399						
[DDQN-Orig] Ep 692 R	270.34	avgR100	223.78	eps 0.031	Qval	50.38
land100 0.79 thr 206						
[DDQN-Orig] Ep 693 R	256.33	avgR100	224.03	eps 0.031	Qval	50.14
land100 0.80 thr 213						
[DDQN-Orig] Ep 694 R	205.24	avgR100	223.43	eps 0.031	Qval	49.71
land100 0.80 thr 388						
[DDQN-Orig] Ep 695 R	213.63	avgR100	223.22	eps 0.031	Qval	49.96
land100 0.81 thr 414						
[DDQN-Orig] Ep 696 R	226.20	avgR100	225.70	eps 0.031	Qval	52.09
land100 0.82 thr 262						
[DDQN-Orig] Ep 697 R	197.95	avgR100	224.99	eps 0.030	Qval	52.38
land100 0.82 thr 426						
[DDQN-Orig] Ep 698 R	254.01	avgR100	225.20	eps 0.030	Qval	52.69
land100 0.82 thr 206						
[DDQN-Orig] Ep 699 R	247.41	avgR100	225.39	eps 0.030	Qval	52.02
land100 0.83 thr 214						
[DDQN-Orig] Ep 700 R	254.24	avgR100	225.06	eps 0.030	Qval	52.26
land100 0.83 thr 226						
[DDQN-Orig] Ep 701 R	234.21	avgR100	228.00	eps 0.030	Qval	52.40
land100 0.84 thr 348						

[DDQN-Orig]	Ep 702 R	272.30 avgR100	229.28 eps 0.030 Qval	52.90
land100 0.85 thr 318				
[DDQN-Orig]	Ep 703 R	225.90 avgR100	229.12 eps 0.029 Qval	53.02
land100 0.85 thr 327				
[DDQN-Orig]	Ep 704 R	230.23 avgR100	229.35 eps 0.029 Qval	54.20
land100 0.85 thr 456				
[DDQN-Orig]	Ep 705 R	242.04 avgR100	229.16 eps 0.029 Qval	55.30
land100 0.85 thr 262				
[DDQN-Orig]	Ep 706 R	142.40 avgR100	228.33 eps 0.029 Qval	55.43
land100 0.85 thr 686				
[DDQN-Orig]	Ep 707 R	258.96 avgR100	230.72 eps 0.029 Qval	55.18
land100 0.86 thr 337				
[DDQN-Orig]	Ep 708 R	237.10 avgR100	230.97 eps 0.029 Qval	55.27
land100 0.87 thr 191				
[DDQN-Orig]	Ep 709 R	224.75 avgR100	230.64 eps 0.029 Qval	55.66
land100 0.87 thr 199				
[DDQN-Orig]	Ep 710 R	233.02 avgR100	230.99 eps 0.028 Qval	56.32
land100 0.87 thr 265				
[DDQN-Orig]	Ep 711 R	276.31 avgR100	231.59 eps 0.028 Qval	57.01
land100 0.87 thr 188				
[DDQN-Orig]	Ep 712 R	248.43 avgR100	231.66 eps 0.028 Qval	57.82
land100 0.88 thr 443				
[DDQN-Orig]	Ep 713 R	228.19 avgR100	231.23 eps 0.028 Qval	57.44
land100 0.88 thr 212				
[DDQN-Orig]	Ep 714 R	198.22 avgR100	230.89 eps 0.028 Qval	58.57
land100 0.87 thr 250				
[DDQN-Orig]	Ep 715 R	252.08 avgR100	232.69 eps 0.028 Qval	58.17
land100 0.88 thr 289				
[DDQN-Orig]	Ep 716 R	253.32 avgR100	233.02 eps 0.028 Qval	58.31
land100 0.87 thr 421				
[DDQN-Orig]	Ep 717 R	256.15 avgR100	233.80 eps 0.027 Qval	58.89
land100 0.87 thr 276				
[DDQN-Orig]	Ep 718 R	236.88 avgR100	234.12 eps 0.027 Qval	60.64
land100 0.87 thr 321				
[DDQN-Orig]	Ep 719 R	213.06 avgR100	233.86 eps 0.027 Qval	60.21
land100 0.87 thr 235				
[DDQN-Orig]	Ep 720 R	257.72 avgR100	233.50 eps 0.027 Qval	59.63
land100 0.87 thr 158				
[DDQN-Orig]	Ep 721 R	228.13 avgR100	233.16 eps 0.027 Qval	59.76
land100 0.87 thr 326				
[DDQN-Orig]	Ep 722 R	259.69 avgR100	233.25 eps 0.027 Qval	59.89
land100 0.87 thr 155				
[DDQN-Orig]	Ep 723 R	132.96 avgR100	232.44 eps 0.027 Qval	60.45
land100 0.87 thr 633				
[DDQN-Orig]	Ep 724 R	219.63 avgR100	234.55 eps 0.027 Qval	60.44
land100 0.87 thr 473				
[DDQN-Orig]	Ep 725 R	242.06 avgR100	234.64 eps 0.026 Qval	59.72
land100 0.88 thr 196				
[DDQN-Orig]	Ep 726 R	270.43 avgR100	234.73 eps 0.026 Qval	59.85
land100 0.88 thr 279				
[DDQN-Orig]	Ep 727 R	168.93 avgR100	233.94 eps 0.026 Qval	61.67
land100 0.88 thr 571				
[DDQN-Orig]	Ep 728 R	257.26 avgR100	233.61 eps 0.026 Qval	60.40
land100 0.88 thr 196				
[DDQN-Orig]	Ep 729 R	260.10 avgR100	234.80 eps 0.026 Qval	61.96
land100 0.88 thr 437				
[DDQN-Orig]	Ep 730 R	257.52 avgR100	235.62 eps 0.026 Qval	61.37
land100 0.88 thr 179				
[DDQN-Orig]	Ep 731 R	260.81 avgR100	235.59 eps 0.026 Qval	61.49
land100 0.88 thr 176				

[DDQN-Orig]	Ep 732 R	229.04 avgR100	235.52 eps 0.025 Qval	62.43
land100 0.88 thr 365				
[DDQN-Orig]	Ep 733 R	243.72 avgR100	235.95 eps 0.025 Qval	62.51
land100 0.88 thr 289				
[DDQN-Orig]	Ep 734 R	266.93 avgR100	236.12 eps 0.025 Qval	62.96
land100 0.88 thr 183				
[DDQN-Orig]	Ep 735 R	242.73 avgR100	235.96 eps 0.025 Qval	63.40
land100 0.87 thr 261				
[DDQN-Orig]	Ep 736 R	174.89 avgR100	235.72 eps 0.025 Qval	64.30
land100 0.87 thr 653				
[DDQN-Orig]	Ep 737 R	251.95 avgR100	236.02 eps 0.025 Qval	65.01
land100 0.88 thr 200				
[DDQN-Orig]	Ep 738 R	223.93 avgR100	235.74 eps 0.025 Qval	64.63
land100 0.88 thr 295				
[DDQN-Orig]	Ep 739 R	199.16 avgR100	234.62 eps 0.025 Qval	65.42
land100 0.88 thr 347				
[DDQN-Orig]	Ep 740 R	258.11 avgR100	234.47 eps 0.024 Qval	65.94
land100 0.87 thr 222				
[DDQN-Orig]	Ep 741 R	255.39 avgR100	234.11 eps 0.024 Qval	66.44
land100 0.87 thr 515				
[DDQN-Orig]	Ep 742 R	230.34 avgR100	233.71 eps 0.024 Qval	65.91
land100 0.87 thr 460				
[DDQN-Orig]	Ep 743 R	251.68 avgR100	233.91 eps 0.024 Qval	66.31
land100 0.87 thr 267				
[DDQN-Orig]	Ep 744 R	270.15 avgR100	234.77 eps 0.024 Qval	66.10
land100 0.87 thr 199				
[DDQN-Orig]	Ep 745 R	267.85 avgR100	235.73 eps 0.024 Qval	66.41
land100 0.87 thr 243				
[DDQN-Orig]	Ep 746 R	253.06 avgR100	235.78 eps 0.024 Qval	66.57
land100 0.86 thr 349				
[DDQN-Orig]	Ep 747 R	262.52 avgR100	236.40 eps 0.024 Qval	66.59
land100 0.86 thr 292				
[DDQN-Orig]	Ep 748 R	262.41 avgR100	236.64 eps 0.024 Qval	66.46
land100 0.86 thr 205				
[DDQN-Orig]	Ep 749 R	296.97 avgR100	236.59 eps 0.023 Qval	66.06
land100 0.86 thr 186				
[DDQN-Orig]	Ep 750 R	204.09 avgR100	236.73 eps 0.023 Qval	65.83
land100 0.86 thr 477				
[DDQN-Orig]	Ep 751 R	42.33 avgR100	234.92 eps 0.023 Qval	65.58
land100 0.86 thr 97				
[DDQN-Orig]	Ep 752 R	251.53 avgR100	234.87 eps 0.023 Qval	65.29
land100 0.86 thr 307				
[DDQN-Orig]	Ep 753 R	40.09 avgR100	233.38 eps 0.023 Qval	65.64
land100 0.85 thr 52				
[DDQN-Orig]	Ep 754 R	244.82 avgR100	236.13 eps 0.023 Qval	66.82
land100 0.85 thr 454				
[DDQN-Orig]	Ep 755 R	192.33 avgR100	235.14 eps 0.023 Qval	66.88
land100 0.86 thr 463				
[DDQN-Orig]	Ep 756 R	243.67 avgR100	235.34 eps 0.023 Qval	67.41
land100 0.85 thr 115				
[DDQN-Orig]	Ep 757 R	207.83 avgR100	234.75 eps 0.022 Qval	67.39
land100 0.85 thr 429				
[DDQN-Orig]	Ep 758 R	180.48 avgR100	233.94 eps 0.022 Qval	66.92
land100 0.85 thr 486				
[DDQN-Orig]	Ep 759 R	263.04 avgR100	234.01 eps 0.022 Qval	67.47
land100 0.84 thr 263				
[DDQN-Orig]	Ep 760 R	233.21 avgR100	233.97 eps 0.022 Qval	67.13
land100 0.84 thr 291				
[DDQN-Orig]	Ep 761 R	233.80 avgR100	233.69 eps 0.022 Qval	66.48
land100 0.85 thr 215				

[DDQN-Orig] Ep 762 R land100 0.85 thr 374	220.18 avgR100	233.22 eps 0.022 Qval	67.40
[DDQN-Orig] Ep 763 R land100 0.85 thr 292	215.39 avgR100	232.82 eps 0.022 Qval	67.29
[DDQN-Orig] Ep 764 R land100 0.84 thr 188	259.16 avgR100	232.88 eps 0.022 Qval	67.78
[DDQN-Orig] Ep 765 R land100 0.85 thr 174	298.75 avgR100	232.87 eps 0.022 Qval	67.21
[DDQN-Orig] Ep 766 R land100 0.85 thr 181	237.13 avgR100	232.87 eps 0.022 Qval	66.51
[DDQN-Orig] Ep 767 R land100 0.85 thr 100	261.18 avgR100	233.17 eps 0.021 Qval	67.17
[DDQN-Orig] Ep 768 R land100 0.85 thr 526	185.70 avgR100	232.49 eps 0.021 Qval	67.88
[DDQN-Orig] Ep 769 R land100 0.85 thr 282	261.27 avgR100	232.15 eps 0.021 Qval	66.60
[DDQN-Orig] Ep 770 R land100 0.86 thr 246	274.39 avgR100	234.37 eps 0.021 Qval	66.74
[DDQN-Orig] Ep 771 R land100 0.86 thr 193	264.05 avgR100	234.28 eps 0.021 Qval	67.42
[DDQN-Orig] Ep 772 R land100 0.85 thr 66	-4.92 avgR100	231.70 eps 0.021 Qval	67.77
[DDQN-Orig] Ep 773 R land100 0.85 thr 337	273.67 avgR100	232.76 eps 0.021 Qval	67.09
[DDQN-Orig] Ep 774 R land100 0.84 thr 278	243.44 avgR100	232.99 eps 0.021 Qval	68.05
[DDQN-Orig] Ep 775 R land100 0.84 thr 228	233.24 avgR100	232.65 eps 0.021 Qval	68.62
[DDQN-Orig] Ep 776 R land100 0.85 thr 299	217.27 avgR100	232.08 eps 0.020 Qval	68.13
[DDQN-Orig] Ep 777 R land100 0.85 thr 479	211.44 avgR100	231.50 eps 0.020 Qval	68.48
[DDQN-Orig] Ep 778 R land100 0.85 thr 538	177.23 avgR100	230.99 eps 0.020 Qval	69.99
[DDQN-Orig] Ep 779 R land100 0.86 thr 164	285.72 avgR100	231.44 eps 0.020 Qval	69.34
[DDQN-Orig] Ep 780 R land100 0.86 thr 207	253.48 avgR100	231.59 eps 0.020 Qval	69.53
[DDQN-Orig] Ep 781 R land100 0.86 thr 590	209.56 avgR100	231.23 eps 0.020 Qval	69.32
[DDQN-Orig] Ep 782 R land100 0.86 thr 460	201.34 avgR100	230.84 eps 0.020 Qval	70.05
[DDQN-Orig] Ep 783 R land100 0.86 thr 368	192.96 avgR100	230.47 eps 0.020 Qval	68.80
[DDQN-Orig] Ep 784 R land100 0.86 thr 520	182.01 avgR100	230.22 eps 0.020 Qval	68.82
[DDQN-Orig] Ep 785 R land100 0.86 thr 207	255.85 avgR100	230.63 eps 0.020 Qval	69.47
[DDQN-Orig] Ep 786 R land100 0.86 thr 436	221.27 avgR100	230.47 eps 0.020 Qval	69.76
[DDQN-Orig] Ep 787 R land100 0.86 thr 201	243.13 avgR100	230.43 eps 0.020 Qval	69.94
[DDQN-Orig] Ep 788 R land100 0.85 thr 200	265.42 avgR100	230.42 eps 0.020 Qval	70.31
[DDQN-Orig] Ep 789 R land100 0.84 thr 176	252.03 avgR100	230.47 eps 0.020 Qval	70.03
[DDQN-Orig] Ep 790 R land100 0.84 thr 233	266.33 avgR100	230.74 eps 0.020 Qval	69.32
[DDQN-Orig] Ep 791 R land100 0.84 thr 176	254.28 avgR100	230.94 eps 0.020 Qval	70.69

[DDQN-Orig]	Ep 792 R	241.65 avgR100	230.65 eps 0.020	Qval 68.94
land100 0.84	thr 224			
[DDQN-Orig]	Ep 793 R	217.26 avgR100	230.26 eps 0.020	Qval 69.54
land100 0.84	thr 315			
[DDQN-Orig]	Ep 794 R	214.47 avgR100	230.36 eps 0.020	Qval 69.49
land100 0.84	thr 252			
[DDQN-Orig]	Ep 795 R	274.60 avgR100	230.97 eps 0.020	Qval 68.85
land100 0.84	thr 290			
[DDQN-Orig]	Ep 796 R	218.79 avgR100	230.89 eps 0.020	Qval 69.80
land100 0.84	thr 388			
[DDQN-Orig]	Ep 797 R	232.00 avgR100	231.23 eps 0.020	Qval 70.13
land100 0.84	thr 366			
[DDQN-Orig]	Ep 798 R	225.57 avgR100	230.95 eps 0.020	Qval 70.35
land100 0.84	thr 364			
[DDQN-Orig]	Ep 799 R	209.73 avgR100	230.57 eps 0.020	Qval 69.91
land100 0.84	thr 354			
[DDQN-Orig]	Ep 800 R	221.47 avgR100	230.24 eps 0.020	Qval 70.55
land100 0.83	thr 535			
[DDQN-Orig]	Ep 801 R	226.89 avgR100	230.17 eps 0.020	Qval 70.55
land100 0.83	thr 312			
[DDQN-Orig]	Ep 802 R	262.20 avgR100	230.07 eps 0.020	Qval 70.46
land100 0.82	thr 154			
[DDQN-Orig]	Ep 803 R	237.88 avgR100	230.19 eps 0.020	Qval 69.51
land100 0.82	thr 363			
[DDQN-Orig]	Ep 804 R	237.57 avgR100	230.26 eps 0.020	Qval 70.10
land100 0.81	thr 309			
[DDQN-Orig]	Ep 805 R	235.75 avgR100	230.20 eps 0.020	Qval 70.84
land100 0.81	thr 252			
[DDQN-Orig]	Ep 806 R	235.72 avgR100	231.13 eps 0.020	Qval 70.63
land100 0.80	thr 340			
[DDQN-Orig]	Ep 807 R	242.27 avgR100	230.97 eps 0.020	Qval 70.85
land100 0.80	thr 175			
[DDQN-Orig]	Ep 808 R	222.97 avgR100	230.82 eps 0.020	Qval 71.04
land100 0.80	thr 276			
[DDQN-Orig]	Ep 809 R	263.34 avgR100	231.21 eps 0.020	Qval 70.88
land100 0.80	thr 223			
[DDQN-Orig]	Ep 810 R	260.39 avgR100	231.48 eps 0.020	Qval 71.71
land100 0.80	thr 290			
[DDQN-Orig]	Ep 811 R	256.43 avgR100	231.29 eps 0.020	Qval 70.27
land100 0.80	thr 224			
[DDQN-Orig]	Ep 812 R	248.52 avgR100	231.29 eps 0.020	Qval 71.12
land100 0.80	thr 274			
[DDQN-Orig]	Ep 813 R	281.05 avgR100	231.81 eps 0.020	Qval 71.19
land100 0.80	thr 285			
[DDQN-Orig]	Ep 814 R	260.96 avgR100	232.44 eps 0.020	Qval 70.98
land100 0.81	thr 311			
[DDQN-Orig]	Ep 815 R	234.21 avgR100	232.26 eps 0.020	Qval 71.10
land100 0.81	thr 362			
[DDQN-Orig]	Ep 816 R	245.30 avgR100	232.18 eps 0.020	Qval 71.44
land100 0.81	thr 399			
[DDQN-Orig]	Ep 817 R	276.57 avgR100	232.39 eps 0.020	Qval 72.23
land100 0.81	thr 272			
[DDQN-Orig]	Ep 818 R	198.23 avgR100	232.00 eps 0.020	Qval 71.31
land100 0.81	thr 374			
[DDQN-Orig]	Ep 819 R	231.25 avgR100	232.18 eps 0.020	Qval 71.92
land100 0.81	thr 269			
[DDQN-Orig]	Ep 820 R	229.86 avgR100	231.90 eps 0.020	Qval 72.63
land100 0.81	thr 306			
[DDQN-Orig]	Ep 821 R	260.94 avgR100	232.23 eps 0.020	Qval 72.11
land100 0.81	thr 162			

[DDQN-Orig]	Ep 822 R	166.56 avgR100	231.30 eps 0.020	Qval 72.02
land100 0.81 thr 562				
[DDQN-Orig]	Ep 823 R	253.73 avgR100	232.51 eps 0.020	Qval 73.10
land100 0.81 thr 299				
[DDQN-Orig]	Ep 824 R	268.56 avgR100	233.00 eps 0.020	Qval 73.21
land100 0.82 thr 143				
[DDQN-Orig]	Ep 825 R	253.05 avgR100	233.11 eps 0.020	Qval 73.67
land100 0.82 thr 382				
[DDQN-Orig]	Ep 826 R	269.52 avgR100	233.10 eps 0.020	Qval 73.32
land100 0.82 thr 223				
[DDQN-Orig]	Ep 827 R	216.53 avgR100	233.57 eps 0.020	Qval 72.03
land100 0.82 thr 356				
[DDQN-Orig]	Ep 828 R	286.51 avgR100	233.87 eps 0.020	Qval 73.67
land100 0.82 thr 257				
[DDQN-Orig]	Ep 829 R	274.81 avgR100	234.01 eps 0.020	Qval 72.86
land100 0.82 thr 295				
[DDQN-Orig]	Ep 830 R	227.17 avgR100	233.71 eps 0.020	Qval 72.72
land100 0.82 thr 503				
[DDQN-Orig]	Ep 831 R	268.89 avgR100	233.79 eps 0.020	Qval 71.37
land100 0.82 thr 231				
[DDQN-Orig]	Ep 832 R	213.47 avgR100	233.64 eps 0.020	Qval 72.74
land100 0.82 thr 374				
[DDQN-Orig]	Ep 833 R	231.85 avgR100	233.52 eps 0.020	Qval 73.61
land100 0.82 thr 425				
[DDQN-Orig]	Ep 834 R	226.64 avgR100	233.11 eps 0.020	Qval 74.04
land100 0.82 thr 180				
[DDQN-Orig]	Ep 835 R	230.82 avgR100	233.00 eps 0.020	Qval 72.49
land100 0.83 thr 248				
[DDQN-Orig]	Ep 836 R	226.24 avgR100	233.51 eps 0.020	Qval 71.98
land100 0.83 thr 193				
[DDQN-Orig]	Ep 837 R	261.31 avgR100	233.60 eps 0.020	Qval 72.76
land100 0.83 thr 192				
[DDQN-Orig]	Ep 838 R	245.78 avgR100	233.82 eps 0.020	Qval 72.49
land100 0.83 thr 328				
[DDQN-Orig]	Ep 839 R	251.95 avgR100	234.35 eps 0.020	Qval 73.02
land100 0.83 thr 210				
[DDQN-Orig]	Ep 840 R	261.52 avgR100	234.38 eps 0.020	Qval 72.65
land100 0.84 thr 177				
[DDQN-Orig]	Ep 841 R	275.22 avgR100	234.58 eps 0.020	Qval 73.16
land100 0.84 thr 201				
[DDQN-Orig]	Ep 842 R	265.63 avgR100	234.93 eps 0.020	Qval 73.03
land100 0.84 thr 194				
[DDQN-Orig]	Ep 843 R	272.51 avgR100	235.14 eps 0.020	Qval 73.28
land100 0.84 thr 273				
[DDQN-Orig]	Ep 844 R	259.94 avgR100	235.04 eps 0.020	Qval 73.31
land100 0.84 thr 214				
[DDQN-Orig]	Ep 845 R	261.32 avgR100	234.97 eps 0.020	Qval 73.88
land100 0.84 thr 229				
[DDQN-Orig]	Ep 846 R	249.50 avgR100	234.94 eps 0.020	Qval 74.59
land100 0.85 thr 202				
[DDQN-Orig]	Ep 847 R	254.83 avgR100	234.86 eps 0.020	Qval 74.68
land100 0.85 thr 290				
[DDQN-Orig]	Ep 848 R	255.87 avgR100	234.80 eps 0.020	Qval 74.18
land100 0.85 thr 338				
[DDQN-Orig]	Ep 849 R	247.91 avgR100	234.31 eps 0.020	Qval 74.01
land100 0.85 thr 228				
[DDQN-Orig]	Ep 850 R	224.52 avgR100	234.51 eps 0.020	Qval 74.56
land100 0.85 thr 256				
[DDQN-Orig]	Ep 851 R	181.27 avgR100	235.90 eps 0.020	Qval 74.33
land100 0.86 thr 467				

[DDQN-Orig]	Ep 852 R	256.45	avgR100	235.95	eps 0.020	Qval 73.59
land100 0.86	thr 280					
[DDQN-Orig]	Ep 853 R	273.16	avgR100	238.28	eps 0.020	Qval 74.39
land100 0.87	thr 195					
[DDQN-Orig]	Ep 854 R	224.73	avgR100	238.08	eps 0.020	Qval 74.08
land100 0.88	thr 361					
[DDQN-Orig]	Ep 855 R	279.19	avgR100	238.95	eps 0.020	Qval 74.30
land100 0.88	thr 283					
[DDQN-Orig]	Ep 856 R	262.19	avgR100	239.13	eps 0.020	Qval 74.34
land100 0.89	thr 318					
[DDQN-Orig]	Ep 857 R	236.64	avgR100	239.42	eps 0.020	Qval 73.37
land100 0.89	thr 196					
[DDQN-Orig]	Ep 858 R	269.54	avgR100	240.31	eps 0.020	Qval 72.98
land100 0.89	thr 221					
[DDQN-Orig]	Ep 859 R	245.85	avgR100	240.14	eps 0.020	Qval 73.91
land100 0.90	thr 148					
[DDQN-Orig]	Ep 860 R	233.01	avgR100	240.14	eps 0.020	Qval 74.55
land100 0.90	thr 179					
[DDQN-Orig]	Ep 861 R	280.37	avgR100	240.60	eps 0.020	Qval 74.92
land100 0.90	thr 241					
[DDQN-Orig]	Ep 862 R	280.92	avgR100	241.21	eps 0.020	Qval 74.47
land100 0.90	thr 283					
[DDQN-Orig]	Ep 863 R	232.12	avgR100	241.38	eps 0.020	Qval 74.31
land100 0.90	thr 486					
[DDQN-Orig]	Ep 864 R	293.48	avgR100	241.72	eps 0.020	Qval 74.68
land100 0.91	thr 217					
[DDQN-Orig]	Ep 865 R	268.46	avgR100	241.42	eps 0.020	Qval 74.47
land100 0.91	thr 237					
[DDQN-Orig]	Ep 866 R	260.87	avgR100	241.66	eps 0.020	Qval 74.86
land100 0.91	thr 194					
[DDQN-Orig]	Ep 867 R	279.54	avgR100	241.84	eps 0.020	Qval 75.16
land100 0.91	thr 208					
[DDQN-Orig]	Ep 868 R	218.88	avgR100	242.17	eps 0.020	Qval 75.32
land100 0.91	thr 302					
[DDQN-Orig]	Ep 869 R	225.71	avgR100	241.82	eps 0.020	Qval 75.86
land100 0.91	thr 494					
[DDQN-Orig]	Ep 870 R	250.52	avgR100	241.58	eps 0.020	Qval 76.11
land100 0.91	thr 189					
[DDQN-Orig]	Ep 871 R	265.69	avgR100	241.59	eps 0.020	Qval 75.55
land100 0.91	thr 171					
[DDQN-Orig]	Ep 872 R	208.65	avgR100	243.73	eps 0.020	Qval 75.63
land100 0.92	thr 188					
[DDQN-Orig]	Ep 873 R	260.46	avgR100	243.60	eps 0.020	Qval 74.92
land100 0.92	thr 253					
[DDQN-Orig]	Ep 874 R	280.03	avgR100	243.96	eps 0.020	Qval 75.41
land100 0.93	thr 344					
[DDQN-Orig]	Ep 875 R	209.18	avgR100	243.72	eps 0.020	Qval 75.00
land100 0.93	thr 430					
[DDQN-Orig]	Ep 876 R	257.20	avgR100	244.12	eps 0.020	Qval 75.79
land100 0.93	thr 236					
[DDQN-Orig]	Ep 877 R	252.80	avgR100	244.53	eps 0.020	Qval 76.02
land100 0.93	thr 349					
[DDQN-Orig]	Ep 878 R	233.63	avgR100	245.10	eps 0.020	Qval 75.64
land100 0.93	thr 218					
[DDQN-Orig]	Ep 879 R	253.04	avgR100	244.77	eps 0.020	Qval 76.14
land100 0.93	thr 186					
[DDQN-Orig]	Ep 880 R	275.08	avgR100	244.99	eps 0.020	Qval 76.11
land100 0.93	thr 151					
[DDQN-Orig]	Ep 881 R	249.93	avgR100	245.39	eps 0.020	Qval 76.29
land100 0.93	thr 354					

[DDQN-Orig]	Ep 882 R	265.95	avgR100	246.04	eps 0.020	Qval 75.88
land100 0.93	thr 290					
[DDQN-Orig]	Ep 883 R	37.64	avgR100	244.48	eps 0.020	Qval 75.71
land100 0.92	thr 53					
[DDQN-Orig]	Ep 884 R	261.16	avgR100	245.28	eps 0.020	Qval 76.57
land100 0.92	thr 175					
[DDQN-Orig]	Ep 885 R	250.97	avgR100	245.23	eps 0.020	Qval 76.36
land100 0.92	thr 408					
[DDQN-Orig]	Ep 886 R	275.55	avgR100	245.77	eps 0.020	Qval 76.96
land100 0.92	thr 302					
[DDQN-Orig]	Ep 887 R	272.67	avgR100	246.07	eps 0.020	Qval 77.41
land100 0.91	thr 140					
[DDQN-Orig]	Ep 888 R	262.23	avgR100	246.03	eps 0.020	Qval 76.68
land100 0.92	thr 215					
[DDQN-Orig]	Ep 889 R	249.57	avgR100	246.01	eps 0.020	Qval 77.86
land100 0.93	thr 348					
[DDQN-Orig]	Ep 890 R	241.01	avgR100	245.76	eps 0.020	Qval 77.87
land100 0.93	thr 131					
[DDQN-Orig]	Ep 891 R	219.10	avgR100	245.40	eps 0.020	Qval 78.24
land100 0.93	thr 249					
[DDQN-Orig]	Ep 892 R	262.06	avgR100	245.61	eps 0.020	Qval 77.33
land100 0.93	thr 160					
[DDQN-Orig]	Ep 893 R	210.45	avgR100	245.54	eps 0.020	Qval 77.41
land100 0.93	thr 264					
[DDQN-Orig]	Ep 894 R	249.05	avgR100	245.89	eps 0.020	Qval 78.11
land100 0.93	thr 187					
[DDQN-Orig]	Ep 895 R	254.37	avgR100	245.68	eps 0.020	Qval 77.92
land100 0.93	thr 229					
[DDQN-Orig]	Ep 896 R	197.45	avgR100	245.47	eps 0.020	Qval 77.75
land100 0.93	thr 280					
[DDQN-Orig]	Ep 897 R	12.04	avgR100	243.27	eps 0.020	Qval 77.76
land100 0.92	thr 43					
[DDQN-Orig]	Ep 898 R	-30.89	avgR100	240.71	eps 0.020	Qval 78.00
land100 0.91	thr 39					
[DDQN-Orig]	Ep 899 R	-18.91	avgR100	238.42	eps 0.020	Qval 78.50
land100 0.90	thr 29					
[DDQN-Orig]	Ep 900 R	276.71	avgR100	238.97	eps 0.020	Qval 77.13
land100 0.91	thr 213					
[DDQN-Orig]	Ep 901 R	301.48	avgR100	239.72	eps 0.020	Qval 78.24
land100 0.91	thr 209					
[DDQN-Orig]	Ep 902 R	252.11	avgR100	239.62	eps 0.020	Qval 77.46
land100 0.92	thr 229					
[DDQN-Orig]	Ep 903 R	259.21	avgR100	239.83	eps 0.020	Qval 76.63
land100 0.92	thr 255					
[DDQN-Orig]	Ep 904 R	279.74	avgR100	240.25	eps 0.020	Qval 76.28
land100 0.93	thr 239					
[DDQN-Orig]	Ep 905 R	268.41	avgR100	240.58	eps 0.020	Qval 76.49
land100 0.93	thr 344					
[DDQN-Orig]	Ep 906 R	268.38	avgR100	240.91	eps 0.020	Qval 76.59
land100 0.94	thr 249					
[DDQN-Orig]	Ep 907 R	212.36	avgR100	240.61	eps 0.020	Qval 76.04
land100 0.94	thr 376					
[DDQN-Orig]	Ep 908 R	62.42	avgR100	239.00	eps 0.020	Qval 76.33
land100 0.93	thr 124					
[DDQN-Orig]	Ep 909 R	255.44	avgR100	238.92	eps 0.020	Qval 77.07
land100 0.93	thr 192					
[DDQN-Orig]	Ep 910 R	242.88	avgR100	238.75	eps 0.020	Qval 76.75
land100 0.93	thr 171					

[DDQN-Orig]	Ep 911 R	231.06	avgR100	238.49	eps 0.020	Qval	76.39	
land100 0.93	thr 320							
[DDQN-Orig]	Ep 912 R	44.85	avgR100	236.46	eps 0.020	Qval	76.05	
land100 0.92	thr 97							
[DDQN-Orig]	Ep 913 R	-11.28	avgR100	233.53	eps 0.020	Qval	76.23	
land100 0.91	thr 51							
[DDQN-Orig]	Ep 914 R	247.84	avgR100	233.40	eps 0.020	Qval	77.33	
land100 0.91	thr 281							
[DDQN-Orig]	Ep 915 R	264.45	avgR100	233.70	eps 0.020	Qval	75.99	
land100 0.91	thr 191							
[DDQN-Orig]	Ep 916 R	32.44	avgR100	231.58	eps 0.020	Qval	75.96	
land100 0.91	thr 92							
[DDQN-Orig]	Ep 917 R	244.44	avgR100	231.25	eps 0.020	Qval	75.67	
land100 0.91	thr 248							
[DDQN-Orig]	Ep 918 R	244.64	avgR100	231.72	eps 0.020	Qval	75.38	
land100 0.91	thr 203							
[DDQN-Orig]	Ep 919 R	250.58	avgR100	231.91	eps 0.020	Qval	76.84	
land100 0.91	thr 182							
[DDQN-Orig]	Ep 920 R	240.66	avgR100	232.02	eps 0.020	Qval	76.44	
land100 0.90	thr 283							
[DDQN-Orig]	Ep 921 R	273.63	avgR100	232.15	eps 0.020	Qval	75.02	
land100 0.90	thr 198							
[DDQN-Orig]	Ep 922 R	245.86	avgR100	232.94	eps 0.020	Qval	75.55	
land100 0.90	thr 228							
[DDQN-Orig]	Ep 923 R	197.89	avgR100	232.38	eps 0.020	Qval	75.25	
land100 0.90	thr 355							
[DDQN-Orig]	Ep 924 R	258.81	avgR100	232.28	eps 0.020	Qval	74.98	
land100 0.90	thr 182							
[DDQN-Orig]	Ep 925 R	277.48	avgR100	232.53	eps 0.020	Qval	74.58	
land100 0.90	thr 225							
[DDQN-Orig]	Ep 926 R	250.34	avgR100	232.34	eps 0.020	Qval	75.78	
land100 0.89	thr 162							
[DDQN-Orig]	Ep 927 R	225.09	avgR100	232.42	eps 0.020	Qval	76.05	
land100 0.89	thr 265							
[DDQN-Orig]	Ep 928 R	213.14	avgR100	231.69	eps 0.020	Qval	74.28	
land100 0.89	thr 241							
[DDQN-Orig]	Ep 929 R	264.96	avgR100	231.59	eps 0.020	Qval	73.42	
land100 0.89	thr 165							
[DDQN-Orig]	Ep 930 R	214.02	avgR100	231.46	eps 0.020	Qval	73.57	
land100 0.89	thr 284							
[DDQN-Orig]	Ep 931 R	289.41	avgR100	231.66	eps 0.020	Qval	73.44	
land100 0.89	thr 151							
[DDQN-Orig]	Ep 932 R	284.29	avgR100	232.37	eps 0.020	Qval	74.00	
land100 0.89	thr 256							
[DDQN-Orig]	Ep 933 R	226.42	avgR100	232.32	eps 0.020	Qval	74.72	
land100 0.89	thr 317							
[DDQN-Orig]	Ep 934 R	253.55	avgR100	232.59	eps 0.020	Qval	74.39	
land100 0.88	thr 222							
[DDQN-Orig]	Ep 935 R	268.72	avgR100	232.97	eps 0.020	Qval	74.51	
land100 0.88	thr 278							
[DDQN-Orig]	Ep 936 R	265.52	avgR100	233.36	eps 0.020	Qval	73.89	
land100 0.88	thr 234							
[DDQN-Orig]	Ep 937 R	256.99	avgR100	233.32	eps 0.020	Qval	72.02	
land100 0.88	thr 416							
[DDQN-Orig]	Ep 938 R	279.80	avgR100	233.66	eps 0.020	Qval	72.62	
land100 0.88	thr 205							
[DDQN-Orig]	Ep 939 R	277.70	avgR100	233.91	eps 0.020	Qval	73.64	
land100 0.88	thr 217							
[DDQN-Orig]	Ep 940 R	275.17	avgR100	234.05	eps 0.020	Qval	73.98	
land100 0.88	thr 191							

[DDQN-Orig]	Ep 941 R	262.57 avgR100	233.92 eps 0.020	Qval 72.60
land100 0.88	thr 249			
[DDQN-Orig]	Ep 942 R	242.49 avgR100	233.69 eps 0.020	Qval 73.43
land100 0.88	thr 305			
[DDQN-Orig]	Ep 943 R	240.15 avgR100	233.37 eps 0.020	Qval 73.27
land100 0.88	thr 167			
[DDQN-Orig]	Ep 944 R	273.84 avgR100	233.51 eps 0.020	Qval 73.29
land100 0.88	thr 281			
[DDQN-Orig]	Ep 945 R	243.40 avgR100	233.33 eps 0.020	Qval 72.26
land100 0.88	thr 249			
[DDQN-Orig]	Ep 946 R	245.85 avgR100	233.29 eps 0.020	Qval 73.10
land100 0.88	thr 332			
[DDQN-Orig]	Ep 947 R	285.25 avgR100	233.60 eps 0.020	Qval 72.47
land100 0.88	thr 219			
[DDQN-Orig]	Ep 948 R	233.55 avgR100	233.37 eps 0.020	Qval 72.72
land100 0.88	thr 447			
[DDQN-Orig]	Ep 949 R	245.88 avgR100	233.35 eps 0.020	Qval 73.21
land100 0.88	thr 259			
[DDQN-Orig]	Ep 950 R	257.55 avgR100	233.68 eps 0.020	Qval 74.05
land100 0.87	thr 198			
[DDQN-Orig]	Ep 951 R	246.62 avgR100	234.34 eps 0.020	Qval 73.02
land100 0.86	thr 245			
[DDQN-Orig]	Ep 952 R	247.28 avgR100	234.24 eps 0.020	Qval 73.32
land100 0.86	thr 324			
[DDQN-Orig]	Ep 953 R	294.39 avgR100	234.46 eps 0.020	Qval 72.85
land100 0.86	thr 196			
[DDQN-Orig]	Ep 954 R	277.46 avgR100	234.98 eps 0.020	Qval 72.50
land100 0.86	thr 245			
[DDQN-Orig]	Ep 955 R	238.98 avgR100	234.58 eps 0.020	Qval 73.01
land100 0.86	thr 357			
[DDQN-Orig]	Ep 956 R	274.92 avgR100	234.71 eps 0.020	Qval 72.94
land100 0.86	thr 251			
[DDQN-Orig]	Ep 957 R	268.04 avgR100	235.02 eps 0.020	Qval 73.01
land100 0.86	thr 223			
[DDQN-Orig]	Ep 958 R	263.07 avgR100	234.96 eps 0.020	Qval 72.15
land100 0.86	thr 230			
[DDQN-Orig]	Ep 959 R	267.88 avgR100	235.18 eps 0.020	Qval 71.85
land100 0.86	thr 146			
[DDQN-Orig]	Ep 960 R	252.91 avgR100	235.38 eps 0.020	Qval 71.42
land100 0.86	thr 205			
[DDQN-Orig]	Ep 961 R	283.77 avgR100	235.41 eps 0.020	Qval 71.06
land100 0.86	thr 239			
[DDQN-Orig]	Ep 962 R	248.69 avgR100	235.09 eps 0.020	Qval 70.40
land100 0.86	thr 234			
[DDQN-Orig]	Ep 963 R	232.54 avgR100	235.09 eps 0.020	Qval 71.69
land100 0.85	thr 217			
[DDQN-Orig]	Ep 964 R	252.44 avgR100	234.68 eps 0.020	Qval 71.22
land100 0.85	thr 248			
[DDQN-Orig]	Ep 965 R	281.97 avgR100	234.82 eps 0.020	Qval 71.02
land100 0.85	thr 168			
[DDQN-Orig]	Ep 966 R	289.88 avgR100	235.11 eps 0.020	Qval 70.62
land100 0.85	thr 268			
[DDQN-Orig]	Ep 967 R	-0.65 avgR100	232.31 eps 0.020	Qval 70.50
land100 0.84	thr 80			
[DDQN-Orig]	Ep 968 R	287.69 avgR100	232.99 eps 0.020	Qval 70.12
land100 0.84	thr 117			

[DDQN-Orig]	Ep 969 R	284.81 avgR100	233.59 eps 0.020 Qval	70.38
land100 0.84 thr 202				
[DDQN-Orig]	Ep 970 R	256.80 avgR100	233.65 eps 0.020 Qval	70.18
land100 0.84 thr 166				
[DDQN-Orig]	Ep 971 R	-98.74 avgR100	230.00 eps 0.020 Qval	70.87
land100 0.83 thr 137				
[DDQN-Orig]	Ep 972 R	274.52 avgR100	230.66 eps 0.020 Qval	69.90
land100 0.83 thr 181				
[DDQN-Orig]	Ep 973 R	243.52 avgR100	230.49 eps 0.020 Qval	70.59
land100 0.83 thr 196				
[DDQN-Orig]	Ep 974 R	217.06 avgR100	229.86 eps 0.020 Qval	70.32
land100 0.83 thr 289				
[DDQN-Orig]	Ep 975 R	252.56 avgR100	230.30 eps 0.020 Qval	69.81
land100 0.83 thr 237				
[DDQN-Orig]	Ep 976 R	310.35 avgR100	230.83 eps 0.020 Qval	70.03
land100 0.83 thr 147				
[DDQN-Orig]	Ep 977 R	302.42 avgR100	231.33 eps 0.020 Qval	69.34
land100 0.83 thr 193				
[DDQN-Orig]	Ep 978 R	276.66 avgR100	231.76 eps 0.020 Qval	70.89
land100 0.83 thr 209				
[DDQN-Orig]	Ep 979 R	279.16 avgR100	232.02 eps 0.020 Qval	71.04
land100 0.83 thr 232				
[DDQN-Orig]	Ep 980 R	261.14 avgR100	231.88 eps 0.020 Qval	69.83
land100 0.83 thr 311				
[DDQN-Orig]	Ep 981 R	213.24 avgR100	231.51 eps 0.020 Qval	69.89
land100 0.83 thr 408				
[DDQN-Orig]	Ep 982 R	230.61 avgR100	231.16 eps 0.020 Qval	69.13
land100 0.83 thr 248				
[DDQN-Orig]	Ep 983 R	227.91 avgR100	233.06 eps 0.020 Qval	69.36
land100 0.84 thr 221				
[DDQN-Orig]	Ep 984 R	267.72 avgR100	233.13 eps 0.020 Qval	68.96
land100 0.84 thr 336				
[DDQN-Orig]	Ep 985 R	224.75 avgR100	232.86 eps 0.020 Qval	69.40
land100 0.84 thr 558				
[DDQN-Orig]	Ep 986 R	256.36 avgR100	232.67 eps 0.020 Qval	70.04
land100 0.84 thr 173				
[DDQN-Orig]	Ep 987 R	272.08 avgR100	232.67 eps 0.020 Qval	69.79
land100 0.85 thr 140				
[DDQN-Orig]	Ep 988 R	224.07 avgR100	232.28 eps 0.020 Qval	69.59
land100 0.85 thr 259				
[DDQN-Orig]	Ep 989 R	264.69 avgR100	232.44 eps 0.020 Qval	68.59
land100 0.85 thr 282				
[DDQN-Orig]	Ep 990 R	237.81 avgR100	232.40 eps 0.020 Qval	68.69
land100 0.85 thr 267				
[DDQN-Orig]	Ep 991 R	264.00 avgR100	232.85 eps 0.020 Qval	68.86
land100 0.84 thr 295				
[DDQN-Orig]	Ep 992 R	257.90 avgR100	232.81 eps 0.020 Qval	68.50
land100 0.84 thr 156				
[DDQN-Orig]	Ep 993 R	270.37 avgR100	233.41 eps 0.020 Qval	68.34
land100 0.84 thr 145				
[DDQN-Orig]	Ep 994 R	272.63 avgR100	233.65 eps 0.020 Qval	68.54
land100 0.84 thr 169				
[DDQN-Orig]	Ep 995 R	270.21 avgR100	233.80 eps 0.020 Qval	68.72
land100 0.84 thr 220				
[DDQN-Orig]	Ep 996 R	280.67 avgR100	234.64 eps 0.020 Qval	68.53
land100 0.84 thr 236				
[DDQN-Orig]	Ep 997 R	271.18 avgR100	237.23 eps 0.020 Qval	68.97
land100 0.84 thr 544				
[DDQN-Orig]	Ep 998 R	250.33 avgR100	240.04 eps 0.020 Qval	67.99
land100 0.85 thr 275				

[DDQN-Orig] Ep 999 | R 270.16 | avgR100 242.93 | eps 0.020 | Qval 68.24 |
land100 0.86 | thr 182
[DDQN-Orig] Ep 1000 | R 11.65 | avgR100 240.28 | eps 0.020 | Qval 67.92 |
land100 0.85 | thr 77
[DDQN-Orig] finished 1000 episodes in 3.6 min (final avgR100 = 240.28)

TRAINING: DQN-Mod (modified=True, double=False)
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[DQN-Mod] Ep	1	R	-102.54	avgR100	-102.54	eps	0.995	Qval	0.05	la
nd100	0.00	thr	41							
[DQN-Mod] Ep	2	R	-98.27	avgR100	-100.40	eps	0.990	Qval	0.05	la
nd100	0.00	thr	50							
[DQN-Mod] Ep	3	R	-307.70	avgR100	-169.50	eps	0.985	Qval	0.05	la
nd100	0.00	thr	87							
[DQN-Mod] Ep	4	R	-166.45	avgR100	-168.74	eps	0.980	Qval	0.05	la
nd100	0.00	thr	75							
[DQN-Mod] Ep	5	R	-122.17	avgR100	-159.43	eps	0.975	Qval	0.05	la
nd100	0.00	thr	54							
[DQN-Mod] Ep	6	R	-281.18	avgR100	-179.72	eps	0.970	Qval	0.05	la
nd100	0.00	thr	69							
[DQN-Mod] Ep	7	R	-391.00	avgR100	-209.90	eps	0.966	Qval	0.05	la
nd100	0.00	thr	58							
[DQN-Mod] Ep	8	R	-64.07	avgR100	-191.67	eps	0.961	Qval	0.05	la
nd100	0.12	thr	64							
[DQN-Mod] Ep	9	R	-240.66	avgR100	-197.12	eps	0.956	Qval	0.05	la
nd100	0.11	thr	67							
[DQN-Mod] Ep	10	R	-122.26	avgR100	-189.63	eps	0.951	Qval	0.05	la
nd100	0.10	thr	80							
[DQN-Mod] Ep	11	R	-137.70	avgR100	-184.91	eps	0.946	Qval	0.05	la
nd100	0.09	thr	78							
[DQN-Mod] Ep	12	R	-336.48	avgR100	-197.54	eps	0.942	Qval	0.14	la
nd100	0.08	thr	71							
[DQN-Mod] Ep	13	R	-243.07	avgR100	-201.04	eps	0.937	Qval	-0.18	la
nd100	0.08	thr	93							
[DQN-Mod] Ep	14	R	-195.35	avgR100	-200.64	eps	0.932	Qval	-0.48	la
nd100	0.07	thr	121							
[DQN-Mod] Ep	15	R	-172.56	avgR100	-198.76	eps	0.928	Qval	-0.57	la
nd100	0.07	thr	56							
[DQN-Mod] Ep	16	R	-491.52	avgR100	-217.06	eps	0.923	Qval	-0.43	la
nd100	0.06	thr	80							
[DQN-Mod] Ep	17	R	-324.77	avgR100	-223.40	eps	0.918	Qval	-0.52	la
nd100	0.06	thr	73							
[DQN-Mod] Ep	18	R	-275.27	avgR100	-226.28	eps	0.914	Qval	-0.62	la
nd100	0.06	thr	106							
[DQN-Mod] Ep	19	R	-348.87	avgR100	-232.73	eps	0.909	Qval	-0.57	la
nd100	0.05	thr	87							
[DQN-Mod] Ep	20	R	-42.96	avgR100	-223.24	eps	0.905	Qval	-0.19	la
nd100	0.10	thr	85							
[DQN-Mod] Ep	21	R	-136.45	avgR100	-219.11	eps	0.900	Qval	-0.40	la
nd100	0.10	thr	68							
[DQN-Mod] Ep	22	R	-115.12	avgR100	-214.38	eps	0.896	Qval	-0.34	la
nd100	0.09	thr	89							
[DQN-Mod] Ep	23	R	-305.09	avgR100	-218.33	eps	0.891	Qval	-0.28	la
nd100	0.09	thr	76							
[DQN-Mod] Ep	24	R	-184.64	avgR100	-216.92	eps	0.887	Qval	-0.34	la
nd100	0.08	thr	100							
[DQN-Mod] Ep	25	R	-94.47	avgR100	-212.02	eps	0.882	Qval	-0.25	la
nd100	0.08	thr	62							

[DQN-Mod]	Ep 26	R	-199.04	avgR100	-211.53	eps 0.878	Qval	-0.12	la
nd100 0.08	thr 78								
[DQN-Mod]	Ep 27	R	-284.66	avgR100	-214.23	eps 0.873	Qval	0.10	la
nd100 0.07	thr 52								
[DQN-Mod]	Ep 28	R	-181.22	avgR100	-213.05	eps 0.869	Qval	-0.14	la
nd100 0.07	thr 66								
[DQN-Mod]	Ep 29	R	-14.17	avgR100	-206.20	eps 0.865	Qval	-0.11	la
nd100 0.07	thr 70								
[DQN-Mod]	Ep 30	R	-20.58	avgR100	-200.01	eps 0.860	Qval	0.07	la
nd100 0.10	thr 47								
[DQN-Mod]	Ep 31	R	-93.22	avgR100	-196.56	eps 0.856	Qval	-0.01	la
nd100 0.10	thr 69								
[DQN-Mod]	Ep 32	R	-224.29	avgR100	-197.43	eps 0.852	Qval	0.24	la
nd100 0.09	thr 89								
[DQN-Mod]	Ep 33	R	-250.08	avgR100	-199.03	eps 0.848	Qval	-0.05	la
nd100 0.09	thr 60								
[DQN-Mod]	Ep 34	R	-62.95	avgR100	-195.02	eps 0.843	Qval	0.05	la
nd100 0.09	thr 80								
[DQN-Mod]	Ep 35	R	-146.18	avgR100	-193.63	eps 0.839	Qval	0.04	la
nd100 0.09	thr 64								
[DQN-Mod]	Ep 36	R	-101.66	avgR100	-191.07	eps 0.835	Qval	-0.21	la
nd100 0.08	thr 94								
[DQN-Mod]	Ep 37	R	-153.46	avgR100	-190.06	eps 0.831	Qval	0.07	la
nd100 0.08	thr 77								
[DQN-Mod]	Ep 38	R	-102.53	avgR100	-187.75	eps 0.827	Qval	-0.15	la
nd100 0.08	thr 67								
[DQN-Mod]	Ep 39	R	-49.74	avgR100	-184.21	eps 0.822	Qval	0.20	la
nd100 0.08	thr 75								
[DQN-Mod]	Ep 40	R	-185.58	avgR100	-184.25	eps 0.818	Qval	0.14	la
nd100 0.07	thr 71								
[DQN-Mod]	Ep 41	R	-173.25	avgR100	-183.98	eps 0.814	Qval	0.07	la
nd100 0.07	thr 50								
[DQN-Mod]	Ep 42	R	-136.69	avgR100	-182.85	eps 0.810	Qval	-0.02	la
nd100 0.07	thr 78								
[DQN-Mod]	Ep 43	R	-282.54	avgR100	-185.17	eps 0.806	Qval	0.34	la
nd100 0.07	thr 75								
[DQN-Mod]	Ep 44	R	-300.69	avgR100	-187.80	eps 0.802	Qval	0.08	la
nd100 0.07	thr 76								
[DQN-Mod]	Ep 45	R	-107.06	avgR100	-186.00	eps 0.798	Qval	0.25	la
nd100 0.07	thr 69								
[DQN-Mod]	Ep 46	R	-143.83	avgR100	-185.09	eps 0.794	Qval	0.13	la
nd100 0.07	thr 124								
[DQN-Mod]	Ep 47	R	-157.12	avgR100	-184.49	eps 0.790	Qval	0.13	la
nd100 0.06	thr 47								
[DQN-Mod]	Ep 48	R	-90.15	avgR100	-182.53	eps 0.786	Qval	0.03	la
nd100 0.06	thr 58								
[DQN-Mod]	Ep 49	R	-122.04	avgR100	-181.29	eps 0.782	Qval	0.23	la
nd100 0.06	thr 82								
[DQN-Mod]	Ep 50	R	-195.94	avgR100	-181.59	eps 0.778	Qval	0.15	la
nd100 0.06	thr 96								
[DQN-Mod]	Ep 51	R	-118.71	avgR100	-180.35	eps 0.774	Qval	0.07	la
nd100 0.06	thr 60								
[DQN-Mod]	Ep 52	R	-208.88	avgR100	-180.90	eps 0.771	Qval	0.04	la
nd100 0.06	thr 73	</							

[DQN-Mod] Ep	54	R	-98.50	avgR100	-182.46	eps	0.763	Qval	0.47	la
nd100 0.06	thr	57								
[DQN-Mod] Ep	55	R	-151.01	avgR100	-181.89	eps	0.759	Qval	0.09	la
nd100 0.05	thr	91								
[DQN-Mod] Ep	56	R	-126.28	avgR100	-180.90	eps	0.755	Qval	-0.01	la
nd100 0.05	thr	117								
[DQN-Mod] Ep	57	R	-132.40	avgR100	-180.05	eps	0.751	Qval	0.21	la
nd100 0.05	thr	86								
[DQN-Mod] Ep	58	R	-126.18	avgR100	-179.12	eps	0.748	Qval	0.16	la
nd100 0.05	thr	94								
[DQN-Mod] Ep	59	R	-100.44	avgR100	-177.78	eps	0.744	Qval	0.02	la
nd100 0.05	thr	72								
[DQN-Mod] Ep	60	R	-173.67	avgR100	-177.72	eps	0.740	Qval	-0.22	la
nd100 0.05	thr	89								
[DQN-Mod] Ep	61	R	-166.16	avgR100	-177.53	eps	0.737	Qval	-0.02	la
nd100 0.05	thr	93								
[DQN-Mod] Ep	62	R	-192.67	avgR100	-177.77	eps	0.733	Qval	-0.03	la
nd100 0.05	thr	85								
[DQN-Mod] Ep	63	R	-104.57	avgR100	-176.61	eps	0.729	Qval	-0.02	la
nd100 0.05	thr	49								
[DQN-Mod] Ep	64	R	-13.11	avgR100	-174.05	eps	0.726	Qval	0.02	la
nd100 0.06	thr	118								
[DQN-Mod] Ep	65	R	-81.36	avgR100	-172.63	eps	0.722	Qval	0.42	la
nd100 0.06	thr	102								
[DQN-Mod] Ep	66	R	-112.86	avgR100	-171.72	eps	0.718	Qval	0.23	la
nd100 0.06	thr	74								
[DQN-Mod] Ep	67	R	-167.11	avgR100	-171.65	eps	0.715	Qval	0.37	la
nd100 0.06	thr	54								
[DQN-Mod] Ep	68	R	-166.29	avgR100	-171.57	eps	0.711	Qval	0.23	la
nd100 0.06	thr	114								
[DQN-Mod] Ep	69	R	-97.99	avgR100	-170.51	eps	0.708	Qval	-0.07	la
nd100 0.06	thr	47								
[DQN-Mod] Ep	70	R	-162.97	avgR100	-170.40	eps	0.704	Qval	0.14	la
nd100 0.06	thr	73								
[DQN-Mod] Ep	71	R	-61.30	avgR100	-168.86	eps	0.701	Qval	0.15	la
nd100 0.06	thr	98								
[DQN-Mod] Ep	72	R	-95.25	avgR100	-167.84	eps	0.697	Qval	0.08	la
nd100 0.06	thr	80								
[DQN-Mod] Ep	73	R	-125.22	avgR100	-167.26	eps	0.694	Qval	-0.18	la
nd100 0.05	thr	91								
[DQN-Mod] Ep	74	R	-114.86	avgR100	-166.55	eps	0.690	Qval	0.24	la
nd100 0.05	thr	69								
[DQN-Mod] Ep	75	R	-291.95	avgR100	-168.22	eps	0.687	Qval	-0.05	la
nd100 0.05	thr	88								
[DQN-Mod] Ep	76	R	-125.71	avgR100	-167.66	eps	0.683	Qval	0.17	la
nd100 0.05	thr	78								
[DQN-Mod] Ep	77	R	-35.22	avgR100	-165.94	eps	0.680	Qval	0.10	la
nd100 0.06	thr	72								
[DQN-Mod] Ep	78	R	-84.15	avgR100	-164.89	eps	0.676	Qval	-0.01	la
nd100 0.06	thr	95								
[DQN-Mod] Ep	79	R	-126.53	avgR100	-164.41	eps	0.673	Qval	0.29	la
nd100 0.06	thr	73								
[DQN-Mod] Ep	80	R	-117.76	avgR100	-163.82	eps	0.670	Qval	0.21	la
nd100 0.06	thr	75								
[DQN-Mod] Ep	81	R	-119.26	avgR100	-163.27	eps	0.666	Qval	0.28	la
nd100 0.06	thr	94								
[DQN-Mod] Ep	82	R	-195.52	avgR100	-163.67	eps	0.663	Qval	0.57	la
nd100 0.06	thr	86								

[DQN-Mod] Ep	83	R	-120.55	avgR100	-163.15	eps	0.660	Qval	0.20	la
nd100 0.06		thr	129							
[DQN-Mod] Ep	84	R	-188.20	avgR100	-163.45	eps	0.656	Qval	0.18	la
nd100 0.06		thr	108							
[DQN-Mod] Ep	85	R	-171.66	avgR100	-163.54	eps	0.653	Qval	0.41	la
nd100 0.06		thr	78							
[DQN-Mod] Ep	86	R	-57.02	avgR100	-162.30	eps	0.650	Qval	0.26	la
nd100 0.06		thr	79							
[DQN-Mod] Ep	87	R	-56.17	avgR100	-161.08	eps	0.647	Qval	0.80	la
nd100 0.06		thr	110							
[DQN-Mod] Ep	88	R	-161.26	avgR100	-161.09	eps	0.643	Qval	0.79	la
nd100 0.06		thr	84							
[DQN-Mod] Ep	89	R	-92.95	avgR100	-160.32	eps	0.640	Qval	0.23	la
nd100 0.06		thr	69							
[DQN-Mod] Ep	90	R	-122.08	avgR100	-159.90	eps	0.637	Qval	0.51	la
nd100 0.06		thr	65							
[DQN-Mod] Ep	91	R	-14.85	avgR100	-158.30	eps	0.634	Qval	0.62	la
nd100 0.05		thr	69							
[DQN-Mod] Ep	92	R	-118.43	avgR100	-157.87	eps	0.631	Qval	0.48	la
nd100 0.05		thr	126							
[DQN-Mod] Ep	93	R	-59.22	avgR100	-156.81	eps	0.627	Qval	1.03	la
nd100 0.06		thr	98							
[DQN-Mod] Ep	94	R	-110.86	avgR100	-156.32	eps	0.624	Qval	1.06	la
nd100 0.06		thr	116							
[DQN-Mod] Ep	95	R	-141.16	avgR100	-156.16	eps	0.621	Qval	0.86	la
nd100 0.06		thr	64							
[DQN-Mod] Ep	96	R	-136.19	avgR100	-155.95	eps	0.618	Qval	0.82	la
nd100 0.06		thr	81							
[DQN-Mod] Ep	97	R	-103.65	avgR100	-155.41	eps	0.615	Qval	0.95	la
nd100 0.06		thr	133							
[DQN-Mod] Ep	98	R	-16.73	avgR100	-154.00	eps	0.612	Qval	1.08	la
nd100 0.06		thr	153							
[DQN-Mod] Ep	99	R	-68.83	avgR100	-153.14	eps	0.609	Qval	0.73	la
nd100 0.06		thr	105							
[DQN-Mod] Ep	100	R	18.55	avgR100	-151.42	eps	0.606	Qval	0.90	la
nd100 0.07		thr	77							
[DQN-Mod] Ep	101	R	-79.74	avgR100	-151.19	eps	0.603	Qval	1.21	la
nd100 0.07		thr	65							
[DQN-Mod] Ep	102	R	-125.60	avgR100	-151.47	eps	0.600	Qval	0.99	la
nd100 0.07		thr	121							
[DQN-Mod] Ep	103	R	-138.30	avgR100	-149.77	eps	0.597	Qval	1.12	la
nd100 0.07		thr	107							
[DQN-Mod] Ep	104	R	-81.40	avgR100	-148.92	eps	0.594	Qval	1.18	la
nd100 0.07		thr	81							
[DQN-Mod] Ep	105	R	-93.66	avgR100	-148.64	eps	0.591	Qval	1.47	la
nd100 0.07		thr	105							
[DQN-Mod] Ep	106	R	-63.63	avgR100	-146.46	eps	0.588	Qval	1.14	la
nd100 0.07		thr	56							
[DQN-Mod] Ep	107	R	-98.86	avgR100	-143.54	eps	0.585	Qval	1.63	la
nd100 0.07		thr	78							
[DQN-Mod] Ep	108	R	-90.16	avgR100	-143.80	eps	0.582	Qval	1.45	la
nd100 0.06		thr	61							
[DQN-Mod] Ep	109	R	-131.79	avgR100	-142.71	eps	0.579	Qval	1.74	la
nd100 0.06		thr	143							
[DQN-Mod] Ep	110	R	-1.37	avgR100	-141.50	eps	0.576	Qval	1.68	la
nd100 0.07		thr	68							

[DQN-Mod] Ep nd100 0.07	111 R thr 100	-105.30	avgR100	-141.18	eps 0.573	Qval	1.93	1a
[DQN-Mod] Ep nd100 0.07	112 R thr 140	-37.10	avgR100	-138.18	eps 0.570	Qval	1.72	1a
[DQN-Mod] Ep nd100 0.07	113 R thr 177	-116.45	avgR100	-136.92	eps 0.568	Qval	1.52	1a
[DQN-Mod] Ep nd100 0.07	114 R thr 114	-115.50	avgR100	-136.12	eps 0.565	Qval	1.88	1a
[DQN-Mod] Ep nd100 0.07	115 R thr 162	-73.08	avgR100	-135.12	eps 0.562	Qval	2.38	1a
[DQN-Mod] Ep nd100 0.07	116 R thr 128	-137.16	avgR100	-131.58	eps 0.559	Qval	2.56	1a
[DQN-Mod] Ep nd100 0.07	117 R thr 74	-84.21	avgR100	-129.18	eps 0.556	Qval	2.19	1a
[DQN-Mod] Ep nd100 0.07	118 R thr 73	-41.73	avgR100	-126.84	eps 0.554	Qval	2.36	1a
[DQN-Mod] Ep nd100 0.07	119 R thr 70	-37.76	avgR100	-123.73	eps 0.551	Qval	2.55	1a
[DQN-Mod] Ep nd100 0.06	120 R thr 117	-133.50	avgR100	-124.63	eps 0.548	Qval	2.54	1a
[DQN-Mod] Ep nd100 0.06	121 R thr 114	-78.83	avgR100	-124.06	eps 0.545	Qval	2.54	1a
[DQN-Mod] Ep nd100 0.06	122 R thr 74	-94.27	avgR100	-123.85	eps 0.543	Qval	2.89	1a
[DQN-Mod] Ep nd100 0.06	123 R thr 74	-91.29	avgR100	-121.71	eps 0.540	Qval	2.59	1a
[DQN-Mod] Ep nd100 0.06	124 R thr 90	-160.88	avgR100	-121.47	eps 0.537	Qval	2.99	1a
[DQN-Mod] Ep nd100 0.06	125 R thr 112	-49.45	avgR100	-121.02	eps 0.534	Qval	2.90	1a
[DQN-Mod] Ep nd100 0.06	126 R thr 110	-133.82	avgR100	-120.37	eps 0.532	Qval	2.61	1a
[DQN-Mod] Ep nd100 0.06	127 R thr 109	-132.14	avgR100	-118.85	eps 0.529	Qval	2.68	1a
[DQN-Mod] Ep nd100 0.06	128 R thr 117	-52.03	avgR100	-117.55	eps 0.526	Qval	2.77	1a
[DQN-Mod] Ep nd100 0.06	129 R thr 135	-67.15	avgR100	-118.08	eps 0.524	Qval	3.26	1a
[DQN-Mod] Ep nd100 0.05	130 R thr 63	-91.05	avgR100	-118.79	eps 0.521	Qval	3.05	1a
[DQN-Mod] Ep nd100 0.05	131 R thr 152	-68.67	avgR100	-118.54	eps 0.519	Qval	3.05	1a
[DQN-Mod] Ep nd100 0.05	132 R thr 120	-61.64	avgR100	-116.92	eps 0.516	Qval	3.53	1a
[DQN-Mod] Ep nd100 0.05	133 R thr 152	-84.70	avgR100	-115.26	eps 0.513	Qval	3.64	1a
[DQN-Mod] Ep nd100 0.05	134 R thr 168	-72.29	avgR100	-115.36	eps 0.511	Qval	3.33	1a
[DQN-Mod] Ep nd100 0.05	135 R thr 145	-0.70	avgR100	-113.90	eps 0.508	Qval	3.55	1a
[DQN-Mod] Ep nd100 0.05	136 R thr 143	-17.32	avgR100	-113.06	eps 0.506	Qval	3.77	1a
[DQN-Mod] Ep nd100 0.05	137 R thr 122	-92.47	avgR100	-112.45	eps 0.503	Qval	3.82	1a
[DQN-Mod] Ep nd100 0.05	138 R thr 137	-146.33	avgR100	-112.89	eps 0.501	Qval	3.90	1a
[DQN-Mod] Ep nd100 0.05	139 R thr 108	-56.04	avgR100	-112.95	eps 0.498	Qval	3.74	1a

[DQN-Mod] Ep nd100 0.05	140 R thr 119	-113.32	avgR100	-112.23	eps 0.496	Qval	4.19	1a
[DQN-Mod] Ep nd100 0.05	141 R thr 121	-33.95	avgR100	-110.83	eps 0.493	Qval	4.43	1a
[DQN-Mod] Ep nd100 0.05	142 R thr 182	-86.18	avgR100	-110.33	eps 0.491	Qval	3.99	1a
[DQN-Mod] Ep nd100 0.05	143 R thr 130	-90.52	avgR100	-108.41	eps 0.488	Qval	4.48	1a
[DQN-Mod] Ep nd100 0.05	144 R thr 170	-113.47	avgR100	-106.54	eps 0.486	Qval	4.95	1a
[DQN-Mod] Ep nd100 0.05	145 R thr 108	-67.31	avgR100	-106.14	eps 0.483	Qval	5.48	1a
[DQN-Mod] Ep nd100 0.05	146 R thr 102	-63.89	avgR100	-105.34	eps 0.481	Qval	5.31	1a
[DQN-Mod] Ep nd100 0.05	147 R thr 119	-92.75	avgR100	-104.70	eps 0.479	Qval	5.24	1a
[DQN-Mod] Ep nd100 0.05	148 R thr 94	-121.33	avgR100	-105.01	eps 0.476	Qval	5.08	1a
[DQN-Mod] Ep nd100 0.05	149 R thr 168	-61.75	avgR100	-104.40	eps 0.474	Qval	5.58	1a
[DQN-Mod] Ep nd100 0.05	150 R thr 128	-93.86	avgR100	-103.38	eps 0.471	Qval	5.63	1a
[DQN-Mod] Ep nd100 0.05	151 R thr 143	-58.85	avgR100	-102.79	eps 0.469	Qval	5.62	1a
[DQN-Mod] Ep nd100 0.05	152 R thr 74	-67.50	avgR100	-101.37	eps 0.467	Qval	5.80	1a
[DQN-Mod] Ep nd100 0.05	153 R thr 129	-41.09	avgR100	-98.31	eps 0.464	Qval	6.15	1a
[DQN-Mod] Ep nd100 0.05	154 R thr 191	-87.76	avgR100	-98.20	eps 0.462	Qval	6.17	1a
[DQN-Mod] Ep nd100 0.05	155 R thr 116	-37.63	avgR100	-97.07	eps 0.460	Qval	6.46	1a
[DQN-Mod] Ep nd100 0.05	156 R thr 172	-82.53	avgR100	-96.63	eps 0.458	Qval	7.08	1a
[DQN-Mod] Ep nd100 0.05	157 R thr 139	-101.68	avgR100	-96.32	eps 0.455	Qval	7.36	1a
[DQN-Mod] Ep nd100 0.05	158 R thr 82	-60.81	avgR100	-95.67	eps 0.453	Qval	6.76	1a
[DQN-Mod] Ep nd100 0.05	159 R thr 138	-94.27	avgR100	-95.61	eps 0.451	Qval	7.25	1a
[DQN-Mod] Ep nd100 0.05	160 R thr 77	-48.56	avgR100	-94.35	eps 0.448	Qval	7.29	1a
[DQN-Mod] Ep nd100 0.05	161 R thr 883	-322.07	avgR100	-95.91	eps 0.446	Qval	7.82	1a
[DQN-Mod] Ep nd100 0.05	162 R thr 102	-43.42	avgR100	-94.42	eps 0.444	Qval	8.00	1a
[DQN-Mod] Ep nd100 0.05	163 R thr 94	-55.68	avgR100	-93.93	eps 0.442	Qval	8.20	1a
[DQN-Mod] Ep nd100 0.04	164 R thr 167	-40.83	avgR100	-94.21	eps 0.440	Qval	8.82	1a
[DQN-Mod] Ep nd100 0.04	165 R thr 124	-63.18	avgR100	-94.03	eps 0.437	Qval	8.45	1a
[DQN-Mod] Ep nd100 0.04	166 R thr 82	-18.76	avgR100	-93.09	eps 0.435	Qval	8.72	1a

[DQN-Mod] Ep 167 R -234.44 avgR100 -93.76 eps 0.433 Qval 8.63 la
nd100 0.04 thr 719
[DQN-Mod] Ep 168 R -54.84 avgR100 -92.65 eps 0.431 Qval 8.72 la
nd100 0.04 thr 104
[DQN-Mod] Ep 169 R -40.69 avgR100 -92.07 eps 0.429 Qval 9.50 la
nd100 0.04 thr 94
[DQN-Mod] Ep 170 R -56.81 avgR100 -91.01 eps 0.427 Qval 9.41 la
nd100 0.04 thr 77
[DQN-Mod] Ep 171 R -12.82 avgR100 -90.53 eps 0.424 Qval 9.76 la
nd100 0.04 thr 190
[DQN-Mod] Ep 172 R -115.60 avgR100 -90.73 eps 0.422 Qval 10.12 la
nd100 0.04 thr 112
[DQN-Mod] Ep 173 R -24.78 avgR100 -89.72 eps 0.420 Qval 9.65 la
nd100 0.04 thr 105
[DQN-Mod] Ep 174 R -35.11 avgR100 -88.93 eps 0.418 Qval 9.96 la
nd100 0.04 thr 133
[DQN-Mod] Ep 175 R -45.93 avgR100 -86.47 eps 0.416 Qval 10.01 la
nd100 0.04 thr 112
[DQN-Mod] Ep 176 R 26.62 avgR100 -84.94 eps 0.414 Qval 10.02 la
nd100 0.05 thr 85
[DQN-Mod] Ep 177 R -51.67 avgR100 -85.11 eps 0.412 Qval 9.99 la
nd100 0.04 thr 116
[DQN-Mod] Ep 178 R -57.95 avgR100 -84.85 eps 0.410 Qval 10.52 la
nd100 0.04 thr 86
[DQN-Mod] Ep 179 R -51.77 avgR100 -84.10 eps 0.408 Qval 10.05 la
nd100 0.04 thr 155
[DQN-Mod] Ep 180 R -24.76 avgR100 -83.17 eps 0.406 Qval 10.80 la
nd100 0.04 thr 121
[DQN-Mod] Ep 181 R -62.28 avgR100 -82.60 eps 0.404 Qval 10.97 la
nd100 0.04 thr 103
[DQN-Mod] Ep 182 R -40.00 avgR100 -81.04 eps 0.402 Qval 11.39 la
nd100 0.04 thr 129
[DQN-Mod] Ep 183 R -54.58 avgR100 -80.38 eps 0.400 Qval 10.57 la
nd100 0.04 thr 77
[DQN-Mod] Ep 184 R -79.92 avgR100 -79.30 eps 0.398 Qval 12.03 la
nd100 0.04 thr 720
[DQN-Mod] Ep 185 R -101.88 avgR100 -78.60 eps 0.396 Qval 11.25 la
nd100 0.04 thr 191
[DQN-Mod] Ep 186 R -57.16 avgR100 -78.61 eps 0.394 Qval 12.56 la
nd100 0.04 thr 122
[DQN-Mod] Ep 187 R -24.05 avgR100 -78.28 eps 0.392 Qval 12.41 la
nd100 0.04 thr 148
[DQN-Mod] Ep 188 R -69.93 avgR100 -77.37 eps 0.390 Qval 12.45 la
nd100 0.04 thr 136
[DQN-Mod] Ep 189 R -74.11 avgR100 -77.18 eps 0.388 Qval 12.65 la
nd100 0.04 thr 99
[DQN-Mod] Ep 190 R -33.67 avgR100 -76.30 eps 0.386 Qval 12.52 la
nd100 0.04 thr 91
[DQN-Mod] Ep 191 R -67.82 avgR100 -76.83 eps 0.384 Qval 12.83 la
nd100 0.04 thr 99
[DQN-Mod] Ep 192 R -116.49 avgR100 -76.81 eps 0.382 Qval 12.97 la
nd100 0.04 thr 399
[DQN-Mod] Ep 193 R -52.76 avgR100 -76.74 eps 0.380 Qval 13.39 la
nd100 0.03 thr 191
[DQN-Mod] Ep 194 R -70.67 avgR100 -76.34 eps 0.378 Qval 13.11 la
nd100 0.03 thr 102
[DQN-Mod] Ep 195 R -98.43 avgR100 -75.91 eps 0.376 Qval 13.20 la
nd100 0.03 thr 108
[DQN-Mod] Ep 196 R -79.11 avgR100 -75.34 eps 0.374 Qval 13.38 la
nd100 0.03 thr 131

[DQN-Mod] Ep 197 R	-36.00 avgR100	-74.67 eps 0.373 Qval	13.99 la
nd100 0.03 thr 142			
[DQN-Mod] Ep 198 R	5.29 avgR100	-74.45 eps 0.371 Qval	13.28 la
nd100 0.03 thr 101			
[DQN-Mod] Ep 199 R	-105.45 avgR100	-74.81 eps 0.369 Qval	13.51 la
nd100 0.03 thr 161			
[DQN-Mod] Ep 200 R	-79.78 avgR100	-75.80 eps 0.367 Qval	13.63 la
nd100 0.02 thr 92			
[DQN-Mod] Ep 201 R	-62.55 avgR100	-75.62 eps 0.365 Qval	13.95 la
nd100 0.02 thr 119			
[DQN-Mod] Ep 202 R	-112.76 avgR100	-75.50 eps 0.363 Qval	13.47 la
nd100 0.02 thr 146			
[DQN-Mod] Ep 203 R	-66.73 avgR100	-74.78 eps 0.361 Qval	13.93 la
nd100 0.02 thr 125			
[DQN-Mod] Ep 204 R	-42.93 avgR100	-74.40 eps 0.360 Qval	14.07 la
nd100 0.02 thr 89			
[DQN-Mod] Ep 205 R	-91.84 avgR100	-74.38 eps 0.358 Qval	13.57 la
nd100 0.02 thr 118			
[DQN-Mod] Ep 206 R	-66.57 avgR100	-74.41 eps 0.356 Qval	13.84 la
nd100 0.02 thr 132			
[DQN-Mod] Ep 207 R	-19.13 avgR100	-73.61 eps 0.354 Qval	14.72 la
nd100 0.02 thr 181			
[DQN-Mod] Ep 208 R	-78.72 avgR100	-73.50 eps 0.353 Qval	14.06 la
nd100 0.02 thr 175			
[DQN-Mod] Ep 209 R	-20.88 avgR100	-72.39 eps 0.351 Qval	14.06 la
nd100 0.02 thr 137			
[DQN-Mod] Ep 210 R	-64.22 avgR100	-73.01 eps 0.349 Qval	14.69 la
nd100 0.01 thr 75			
[DQN-Mod] Ep 211 R	-83.68 avgR100	-72.80 eps 0.347 Qval	14.72 la
nd100 0.01 thr 154			
[DQN-Mod] Ep 212 R	-60.13 avgR100	-73.03 eps 0.346 Qval	14.64 la
nd100 0.01 thr 114			
[DQN-Mod] Ep 213 R	-30.52 avgR100	-72.17 eps 0.344 Qval	14.58 la
nd100 0.01 thr 83			
[DQN-Mod] Ep 214 R	-440.32 avgR100	-75.42 eps 0.342 Qval	15.46 la
nd100 0.01 thr 631			
[DQN-Mod] Ep 215 R	-72.61 avgR100	-75.41 eps 0.340 Qval	14.90 la
nd100 0.01 thr 94			
[DQN-Mod] Ep 216 R	-67.42 avgR100	-74.72 eps 0.339 Qval	15.29 la
nd100 0.01 thr 160			
[DQN-Mod] Ep 217 R	-59.74 avgR100	-74.47 eps 0.337 Qval	15.13 la
nd100 0.01 thr 71			
[DQN-Mod] Ep 218 R	-111.44 avgR100	-75.17 eps 0.335 Qval	15.37 la
nd100 0.01 thr 591			
[DQN-Mod] Ep 219 R	-23.24 avgR100	-75.02 eps 0.334 Qval	15.58 la
nd100 0.02 thr 218			
[DQN-Mod] Ep 220 R	-167.09 avgR100	-75.36 eps 0.332 Qval	15.86 la
nd100 0.02 thr 692			
[DQN-Mod] Ep 221 R	-129.32 avgR100	-75.86 eps 0.330 Qval	15.96 la
nd100 0.02 thr 95			
[DQN-Mod] Ep 222 R	-50.60 avgR100	-75.43 eps 0.329 Qval	15.50 la
nd100 0.02 thr 56			
[DQN-Mod] Ep 223 R	-42.85 avgR100	-74.94 eps 0.327 Qval	15.45 la
nd100 0.02 thr 134			

[DQN-Mod] Ep nd100 0.02	224 R thr 123	-46.57 avgR100	-73.80 eps 0.325 Qval	15.98 la
[DQN-Mod] Ep nd100 0.03	225 R thr 64	14.83 avgR100	-73.16 eps 0.324 Qval	15.12 la
[DQN-Mod] Ep nd100 0.03	226 R thr 86	-52.80 avgR100	-72.35 eps 0.322 Qval	15.67 la
[DQN-Mod] Ep nd100 0.03	227 R thr 86	-102.99 avgR100	-72.05 eps 0.321 Qval	15.98 la
[DQN-Mod] Ep nd100 0.03	228 R thr 100	-64.09 avgR100	-72.18 eps 0.319 Qval	16.05 la
[DQN-Mod] Ep nd100 0.03	229 R thr 68	-54.37 avgR100	-72.05 eps 0.317 Qval	15.95 la
[DQN-Mod] Ep nd100 0.03	230 R thr 91	-25.36 avgR100	-71.39 eps 0.316 Qval	16.38 la
[DQN-Mod] Ep nd100 0.03	231 R thr 891	-270.40 avgR100	-73.41 eps 0.314 Qval	15.89 la
[DQN-Mod] Ep nd100 0.03	232 R thr 121	-45.53 avgR100	-73.25 eps 0.313 Qval	15.69 la
[DQN-Mod] Ep nd100 0.03	233 R thr 113	-202.05 avgR100	-74.42 eps 0.311 Qval	16.22 la
[DQN-Mod] Ep nd100 0.03	234 R thr 153	-64.70 avgR100	-74.34 eps 0.309 Qval	15.02 la
[DQN-Mod] Ep nd100 0.03	235 R thr 95	-60.65 avgR100	-74.94 eps 0.308 Qval	15.24 la
[DQN-Mod] Ep nd100 0.03	236 R thr 353	-151.51 avgR100	-76.29 eps 0.306 Qval	15.72 la
[DQN-Mod] Ep nd100 0.03	237 R thr 126	-24.17 avgR100	-75.60 eps 0.305 Qval	15.09 la
[DQN-Mod] Ep nd100 0.03	238 R thr 133	-25.39 avgR100	-74.39 eps 0.303 Qval	15.27 la
[DQN-Mod] Ep nd100 0.03	239 R thr 443	-7.06 avgR100	-73.90 eps 0.302 Qval	15.03 la
[DQN-Mod] Ep nd100 0.03	240 R thr 73	-97.61 avgR100	-73.75 eps 0.300 Qval	14.96 la
[DQN-Mod] Ep nd100 0.03	241 R thr 98	-75.98 avgR100	-74.17 eps 0.299 Qval	15.61 la
[DQN-Mod] Ep nd100 0.03	242 R thr 107	-62.51 avgR100	-73.93 eps 0.297 Qval	15.05 la
[DQN-Mod] Ep nd100 0.03	243 R thr 755	-210.26 avgR100	-75.13 eps 0.296 Qval	14.53 la
[DQN-Mod] Ep nd100 0.03	244 R thr 67	-77.62 avgR100	-74.77 eps 0.294 Qval	14.33 la
[DQN-Mod] Ep nd100 0.03	245 R thr 87	-42.55 avgR100	-74.52 eps 0.293 Qval	14.57 la
[DQN-Mod] Ep nd100 0.03	246 R thr 123	-63.42 avgR100	-74.52 eps 0.291 Qval	14.26 la
[DQN-Mod] Ep nd100 0.03	247 R thr 55	-42.46 avgR100	-74.01 eps 0.290 Qval	14.16 la
[DQN-Mod] Ep nd100 0.03	248 R thr 135	-47.74 avgR100	-73.28 eps 0.288 Qval	14.21 la
[DQN-Mod] Ep nd100 0.03	249 R thr 96	-74.54 avgR100	-73.41 eps 0.287 Qval	14.18 la
[DQN-Mod] Ep nd100 0.03	250 R thr 64	-50.82 avgR100	-72.98 eps 0.286 Qval	14.00 la
[DQN-Mod] Ep nd100 0.03	251 R thr 81	-47.37 avgR100	-72.86 eps 0.284 Qval	13.90 la
[DQN-Mod] Ep nd100 0.03	252 R thr 74	-59.62 avgR100	-72.78 eps 0.283 Qval	14.38 la

[DQN-Mod] Ep nd100 0.03	253 R thr 612	-526.79	avgR100	-77.64	eps 0.281	Qval	13.90	la
[DQN-Mod] Ep nd100 0.03	254 R thr 64	-53.04	avgR100	-77.29	eps 0.280	Qval	13.54	la
[DQN-Mod] Ep nd100 0.03	255 R thr 115	-49.06	avgR100	-77.41	eps 0.279	Qval	13.69	la
[DQN-Mod] Ep nd100 0.03	256 R thr 86	7.21	avgR100	-76.51	eps 0.277	Qval	13.55	la
[DQN-Mod] Ep nd100 0.03	257 R thr 61	-83.66	avgR100	-76.33	eps 0.276	Qval	13.79	la
[DQN-Mod] Ep nd100 0.03	258 R thr 149	-22.61	avgR100	-75.95	eps 0.274	Qval	13.41	la
[DQN-Mod] Ep nd100 0.03	259 R thr 95	-18.77	avgR100	-75.19	eps 0.273	Qval	13.58	la
[DQN-Mod] Ep nd100 0.03	260 R thr 78	-54.99	avgR100	-75.26	eps 0.272	Qval	12.90	la
[DQN-Mod] Ep nd100 0.03	261 R thr 91	-25.59	avgR100	-72.29	eps 0.270	Qval	13.20	la
[DQN-Mod] Ep nd100 0.04	262 R thr 210	219.37	avgR100	-69.66	eps 0.269	Qval	12.09	la
[DQN-Mod] Ep nd100 0.04	263 R thr 843	-223.87	avgR100	-71.35	eps 0.268	Qval	12.52	la
[DQN-Mod] Ep nd100 0.04	264 R thr 110	-37.15	avgR100	-71.31	eps 0.266	Qval	12.41	la
[DQN-Mod] Ep nd100 0.04	265 R thr 408	-153.34	avgR100	-72.21	eps 0.265	Qval	12.12	la
[DQN-Mod] Ep nd100 0.04	266 R thr 99	-3.90	avgR100	-72.06	eps 0.264	Qval	11.54	la
[DQN-Mod] Ep nd100 0.04	267 R thr 97	-27.07	avgR100	-69.99	eps 0.262	Qval	11.61	la
[DQN-Mod] Ep nd100 0.04	268 R thr 76	-34.79	avgR100	-69.79	eps 0.261	Qval	11.46	la
[DQN-Mod] Ep nd100 0.04	269 R thr 863	-319.80	avgR100	-72.58	eps 0.260	Qval	10.94	la
[DQN-Mod] Ep nd100 0.04	270 R thr 88	-118.83	avgR100	-73.20	eps 0.258	Qval	10.76	la
[DQN-Mod] Ep nd100 0.04	271 R thr 219	-121.21	avgR100	-74.28	eps 0.257	Qval	9.97	la
[DQN-Mod] Ep nd100 0.04	272 R thr 79	-13.55	avgR100	-73.26	eps 0.256	Qval	10.71	la
[DQN-Mod] Ep nd100 0.04	273 R thr 156	-80.49	avgR100	-73.82	eps 0.255	Qval	9.80	la
[DQN-Mod] Ep nd100 0.04	274 R thr 890	-355.08	avgR100	-77.02	eps 0.253	Qval	8.67	la
[DQN-Mod] Ep nd100 0.04	275 R thr 88	-12.69	avgR100	-76.69	eps 0.252	Qval	9.03	la
[DQN-Mod] Ep nd100 0.03	276 R thr 285	-94.88	avgR100	-77.90	eps 0.251	Qval	8.34	la
[DQN-Mod] Ep nd100 0.03	277 R thr 115	-24.10	avgR100	-77.63	eps 0.249	Qval	8.08	la
[DQN-Mod] Ep nd100 0.03	278 R thr 132	-52.60	avgR100	-77.57	eps 0.248	Qval	8.24	la
[DQN-Mod] Ep nd100 0.03	279 R thr 245	-70.89	avgR100	-77.76	eps 0.247	Qval	7.71	la

[DQN-Mod] Ep nd100 0.03	280 R thr 835	-232.14	avgR100	-79.84	eps 0.246	Qval	7.18	1a
[DQN-Mod] Ep nd100 0.03	281 R thr 92	-71.47	avgR100	-79.93	eps 0.245	Qval	6.42	1a
[DQN-Mod] Ep nd100 0.03	282 R thr 91	-29.58	avgR100	-79.82	eps 0.243	Qval	7.16	1a
[DQN-Mod] Ep nd100 0.03	283 R thr 97	-46.21	avgR100	-79.74	eps 0.242	Qval	6.71	1a
[DQN-Mod] Ep nd100 0.03	284 R thr 74	-63.33	avgR100	-79.58	eps 0.241	Qval	6.66	1a
[DQN-Mod] Ep nd100 0.03	285 R thr 870	-294.09	avgR100	-81.50	eps 0.240	Qval	5.22	1a
[DQN-Mod] Ep nd100 0.03	286 R thr 357	-130.94	avgR100	-82.24	eps 0.238	Qval	4.34	1a
[DQN-Mod] Ep nd100 0.03	287 R thr 415	-152.18	avgR100	-83.52	eps 0.237	Qval	4.42	1a
[DQN-Mod] Ep nd100 0.03	288 R thr 78	-15.43	avgR100	-82.97	eps 0.236	Qval	3.81	1a
[DQN-Mod] Ep nd100 0.03	289 R thr 818	-259.40	avgR100	-84.82	eps 0.235	Qval	2.35	1a
[DQN-Mod] Ep nd100 0.03	290 R thr 81	-80.62	avgR100	-85.29	eps 0.234	Qval	2.38	1a
[DQN-Mod] Ep nd100 0.03	291 R thr 91	-26.86	avgR100	-84.88	eps 0.233	Qval	2.17	1a
[DQN-Mod] Ep nd100 0.03	292 R thr 127	-58.82	avgR100	-84.31	eps 0.231	Qval	1.98	1a
[DQN-Mod] Ep nd100 0.03	293 R thr 103	-64.10	avgR100	-84.42	eps 0.230	Qval	1.53	1a
[DQN-Mod] Ep nd100 0.03	294 R thr 121	-22.51	avgR100	-83.94	eps 0.229	Qval	1.41	1a
[DQN-Mod] Ep nd100 0.03	295 R thr 176	-94.92	avgR100	-83.90	eps 0.228	Qval	1.59	1a
[DQN-Mod] Ep nd100 0.03	296 R thr 57	-69.29	avgR100	-83.81	eps 0.227	Qval	1.64	1a
[DQN-Mod] Ep nd100 0.03	297 R thr 87	-26.97	avgR100	-83.72	eps 0.226	Qval	1.34	1a
[DQN-Mod] Ep nd100 0.04	298 R thr 526	105.44	avgR100	-82.71	eps 0.225	Qval	0.11	1a
[DQN-Mod] Ep nd100 0.04	299 R thr 54	-66.85	avgR100	-82.33	eps 0.223	Qval	0.23	1a
[DQN-Mod] Ep nd100 0.04	300 R thr 835	-233.67	avgR100	-83.87	eps 0.222	Qval	-1.11	1a
[DQN-Mod] Ep nd100 0.04	301 R thr 128	-194.98	avgR100	-85.19	eps 0.221	Qval	-1.13	1a
[DQN-Mod] Ep nd100 0.05	302 R thr 404	132.68	avgR100	-82.74	eps 0.220	Qval	-2.25	1a
[DQN-Mod] Ep nd100 0.05	303 R thr 86	-24.46	avgR100	-82.31	eps 0.219	Qval	-2.63	1a
[DQN-Mod] Ep nd100 0.05	304 R thr 809	-223.37	avgR100	-84.12	eps 0.218	Qval	-4.19	1a
[DQN-Mod] Ep nd100 0.05	305 R thr 887	-281.72	avgR100	-86.02	eps 0.217	Qval	-4.97	1a
[DQN-Mod] Ep nd100 0.05	306 R thr 74	-60.83	avgR100	-85.96	eps 0.216	Qval	-6.16	1a
[DQN-Mod] Ep nd100 0.05	307 R thr 141	-43.77	avgR100	-86.21	eps 0.215	Qval	-5.69	1a
[DQN-Mod] Ep nd100 0.05	308 R thr 710	-163.06	avgR100	-87.05	eps 0.214	Qval	-6.74	1a

[DQN-Mod] Ep nd100 0.05	309 R thr 806	-239.87	avgR100	-89.24	eps 0.212	Qval	-8.63	la
[DQN-Mod] Ep nd100 0.05	310 R thr 120	-21.22	avgR100	-88.81	eps 0.211	Qval	-9.04	la
[DQN-Mod] Ep nd100 0.05	311 R thr 744	-242.35	avgR100	-90.40	eps 0.210	Qval	-9.63	la
[DQN-Mod] Ep nd100 0.05	312 R thr 116	-6.79	avgR100	-89.86	eps 0.209	Qval	-10.13	la
[DQN-Mod] Ep nd100 0.05	313 R thr 776	-232.85	avgR100	-91.89	eps 0.208	Qval	-11.46	la
[DQN-Mod] Ep nd100 0.05	314 R thr 106	-19.28	avgR100	-87.68	eps 0.207	Qval	-11.49	la
[DQN-Mod] Ep nd100 0.05	315 R thr 820	-265.99	avgR100	-89.61	eps 0.206	Qval	-13.33	la
[DQN-Mod] Ep nd100 0.05	316 R thr 798	-240.26	avgR100	-91.34	eps 0.205	Qval	-13.98	la
[DQN-Mod] Ep nd100 0.05	317 R thr 401	-155.35	avgR100	-92.29	eps 0.204	Qval	-15.19	la
[DQN-Mod] Ep nd100 0.05	318 R thr 685	-169.27	avgR100	-92.87	eps 0.203	Qval	-16.36	la
[DQN-Mod] Ep nd100 0.04	319 R thr 847	-294.87	avgR100	-95.59	eps 0.202	Qval	-17.71	la
[DQN-Mod] Ep nd100 0.04	320 R thr 216	171.93	avgR100	-92.20	eps 0.201	Qval	-17.89	la
[DQN-Mod] Ep nd100 0.05	321 R thr 701	46.79	avgR100	-90.44	eps 0.200	Qval	-18.88	la
[DQN-Mod] Ep nd100 0.05	322 R thr 828	-238.44	avgR100	-92.32	eps 0.199	Qval	-19.62	la
[DQN-Mod] Ep nd100 0.05	323 R thr 111	-73.58	avgR100	-92.62	eps 0.198	Qval	-20.54	la
[DQN-Mod] Ep nd100 0.06	324 R thr 499	129.04	avgR100	-90.87	eps 0.197	Qval	-20.68	la
[DQN-Mod] Ep nd100 0.05	325 R thr 141	-53.33	avgR100	-91.55	eps 0.196	Qval	-20.43	la
[DQN-Mod] Ep nd100 0.05	326 R thr 671	-131.68	avgR100	-92.34	eps 0.195	Qval	-23.07	la
[DQN-Mod] Ep nd100 0.05	327 R thr 853	-262.37	avgR100	-93.93	eps 0.194	Qval	-23.66	la
[DQN-Mod] Ep nd100 0.05	328 R thr 841	-231.42	avgR100	-95.60	eps 0.193	Qval	-24.07	la
[DQN-Mod] Ep nd100 0.05	329 R thr 821	-234.60	avgR100	-97.41	eps 0.192	Qval	-26.83	la
[DQN-Mod] Ep nd100 0.05	330 R thr 730	-184.59	avgR100	-99.00	eps 0.191	Qval	-27.23	la
[DQN-Mod] Ep nd100 0.05	331 R thr 814	-297.96	avgR100	-99.27	eps 0.190	Qval	-29.24	la
[DQN-Mod] Ep nd100 0.06	332 R thr 297	202.38	avgR100	-96.80	eps 0.189	Qval	-28.82	la
[DQN-Mod] Ep nd100 0.06	333 R thr 639	-275.98	avgR100	-97.54	eps 0.188	Qval	-29.72	la
[DQN-Mod] Ep nd100 0.06	334 R thr 852	-280.66	avgR100	-99.69	eps 0.187	Qval	-30.97	la
[DQN-Mod] Ep nd100 0.06	335 R thr 746	-222.57	avgR100	-101.31	eps 0.187	Qval	-31.44	la
[DQN-Mod] Ep nd100 0.07	336 R thr 609	49.86	avgR100	-99.30	eps 0.186	Qval	-34.77	la
[DQN-Mod] Ep nd100 0.07	337 R thr 593	-81.49	avgR100	-99.87	eps 0.185	Qval	-35.00	la
[DQN-Mod] Ep nd100 0.08	338 R thr 584	96.39	avgR100	-98.66	eps 0.184	Qval	-35.79	la

[DQN-Mod] Ep nd100 0.08	339 R thr 745	-195.33	avgR100	-100.54	eps 0.183	Qval -36.77	la
[DQN-Mod] Ep nd100 0.08	340 R thr 821	-257.50	avgR100	-102.14	eps 0.182	Qval -37.63	la
[DQN-Mod] Ep nd100 0.08	341 R thr 859	-253.42	avgR100	-103.91	eps 0.181	Qval -38.26	la
[DQN-Mod] Ep nd100 0.08	342 R thr 121	-70.50	avgR100	-103.99	eps 0.180	Qval -39.06	la
[DQN-Mod] Ep nd100 0.08	343 R thr 853	-267.91	avgR100	-104.57	eps 0.179	Qval -38.52	la
[DQN-Mod] Ep nd100 0.09	344 R thr 713	-16.25	avgR100	-103.95	eps 0.178	Qval -40.75	la
[DQN-Mod] Ep nd100 0.10	345 R thr 217	253.38	avgR100	-101.00	eps 0.177	Qval -40.36	la
[DQN-Mod] Ep nd100 0.10	346 R thr 770	-214.35	avgR100	-102.50	eps 0.177	Qval -42.65	la
[DQN-Mod] Ep nd100 0.10	347 R thr 813	-240.29	avgR100	-104.48	eps 0.176	Qval -43.61	la
[DQN-Mod] Ep nd100 0.10	348 R thr 787	-256.41	avgR100	-106.57	eps 0.175	Qval -44.16	la
[DQN-Mod] Ep nd100 0.10	349 R thr 93	-87.63	avgR100	-106.70	eps 0.174	Qval -44.70	la
[DQN-Mod] Ep nd100 0.10	350 R thr 119	-57.07	avgR100	-106.76	eps 0.173	Qval -45.13	la
[DQN-Mod] Ep nd100 0.11	351 R thr 276	274.36	avgR100	-103.55	eps 0.172	Qval -44.22	la
[DQN-Mod] Ep nd100 0.11	352 R thr 821	-299.60	avgR100	-105.95	eps 0.171	Qval -43.72	la
[DQN-Mod] Ep nd100 0.11	353 R thr 778	-230.53	avgR100	-102.98	eps 0.170	Qval -45.78	la
[DQN-Mod] Ep nd100 0.11	354 R thr 124	-24.33	avgR100	-102.70	eps 0.170	Qval -45.14	la
[DQN-Mod] Ep nd100 0.11	355 R thr 180	-41.76	avgR100	-102.62	eps 0.169	Qval -45.01	la
[DQN-Mod] Ep nd100 0.11	356 R thr 800	-281.59	avgR100	-105.51	eps 0.168	Qval -44.92	la
[DQN-Mod] Ep nd100 0.11	357 R thr 771	-197.66	avgR100	-106.65	eps 0.167	Qval -46.57	la
[DQN-Mod] Ep nd100 0.11	358 R thr 100	-80.35	avgR100	-107.23	eps 0.166	Qval -46.29	la
[DQN-Mod] Ep nd100 0.11	359 R thr 824	-245.65	avgR100	-109.50	eps 0.165	Qval -45.60	la
[DQN-Mod] Ep nd100 0.11	360 R thr 107	-74.83	avgR100	-109.70	eps 0.165	Qval -46.62	la
[DQN-Mod] Ep nd100 0.11	361 R thr 782	-147.59	avgR100	-110.92	eps 0.164	Qval -46.99	la
[DQN-Mod] Ep nd100 0.11	362 R thr 419	147.39	avgR100	-111.63	eps 0.163	Qval -47.44	la
[DQN-Mod] Ep nd100 0.12	363 R thr 114	72.93	avgR100	-108.67	eps 0.162	Qval -47.14	la
[DQN-Mod] Ep nd100 0.12	364 R thr 748	-205.96	avgR100	-110.36	eps 0.161	Qval -48.78	la
[DQN-Mod] Ep nd100 0.12	365 R thr 809	-167.89	avgR100	-110.50	eps 0.160	Qval -49.00	la
[DQN-Mod] Ep nd100 0.12	366 R thr 797	-240.46	avgR100	-112.87	eps 0.160	Qval -50.23	la
[DQN-Mod] Ep nd100 0.13	367 R thr 665	-28.97	avgR100	-112.89	eps 0.159	Qval -48.27	la
[DQN-Mod] Ep nd100 0.13	368 R thr 165	-32.70	avgR100	-112.86	eps 0.158	Qval -48.89	la

[DQN-Mod] Ep nd100 0.14	R thr 663	30.31	avgR100	-109.36	eps 0.157	Qval -49.88	la
[DQN-Mod] Ep nd100 0.15	R thr 558	69.03	avgR100	-107.48	eps 0.157	Qval -49.67	la
[DQN-Mod] Ep nd100 0.15	R thr 764	-197.33	avgR100	-108.25	eps 0.156	Qval -49.94	la
[DQN-Mod] Ep nd100 0.16	R thr 479	88.44	avgR100	-107.23	eps 0.155	Qval -50.81	la
[DQN-Mod] Ep nd100 0.17	R thr 471	123.92	avgR100	-105.18	eps 0.154	Qval -51.15	la
[DQN-Mod] Ep nd100 0.17	R thr 691	-173.29	avgR100	-103.36	eps 0.153	Qval -52.11	la
[DQN-Mod] Ep nd100 0.18	R thr 601	35.68	avgR100	-102.88	eps 0.153	Qval -49.50	la
[DQN-Mod] Ep nd100 0.19	R thr 384	177.68	avgR100	-100.15	eps 0.152	Qval -50.18	la
[DQN-Mod] Ep nd100 0.19	R thr 773	-225.37	avgR100	-102.17	eps 0.151	Qval -49.47	la
[DQN-Mod] Ep nd100 0.19	R thr 786	-232.98	avgR100	-103.97	eps 0.150	Qval -50.63	la
[DQN-Mod] Ep nd100 0.19	R thr 774	-229.06	avgR100	-105.55	eps 0.150	Qval -50.35	la
[DQN-Mod] Ep nd100 0.20	R thr 533	107.35	avgR100	-102.16	eps 0.149	Qval -50.79	la
[DQN-Mod] Ep nd100 0.20	R thr 129	-36.26	avgR100	-101.81	eps 0.148	Qval -51.47	la
[DQN-Mod] Ep nd100 0.21	R thr 397	156.81	avgR100	-99.94	eps 0.147	Qval -50.53	la
[DQN-Mod] Ep nd100 0.21	R thr 734	-215.56	avgR100	-101.64	eps 0.147	Qval -49.66	la
[DQN-Mod] Ep nd100 0.22	R thr 460	99.46	avgR100	-100.01	eps 0.146	Qval -49.83	la
[DQN-Mod] Ep nd100 0.23	R thr 203	215.79	avgR100	-94.91	eps 0.145	Qval -50.13	la
[DQN-Mod] Ep nd100 0.23	R thr 817	-262.21	avgR100	-96.22	eps 0.144	Qval -51.66	la
[DQN-Mod] Ep nd100 0.24	R thr 269	225.65	avgR100	-92.44	eps 0.144	Qval -51.09	la
[DQN-Mod] Ep nd100 0.24	R thr 115	-8.33	avgR100	-92.37	eps 0.143	Qval -50.91	la
[DQN-Mod] Ep nd100 0.25	R thr 479	128.51	avgR100	-88.49	eps 0.142	Qval -51.04	la
[DQN-Mod] Ep nd100 0.25	R thr 189	-81.34	avgR100	-88.50	eps 0.142	Qval -50.53	la
[DQN-Mod] Ep nd100 0.25	R thr 99	-94.20	avgR100	-89.17	eps 0.141	Qval -51.66	la
[DQN-Mod] Ep nd100 0.25	R thr 832	-239.08	avgR100	-90.98	eps 0.140	Qval -50.86	la
[DQN-Mod] Ep nd100 0.25	R thr 667	-154.16	avgR100	-91.88	eps 0.139	Qval -51.93	la
[DQN-Mod] Ep nd100 0.25	R thr 106	0.54	avgR100	-91.65	eps 0.139	Qval -52.32	la
[DQN-Mod] Ep nd100 0.25	R thr 217	-33.89	avgR100	-91.04	eps 0.138	Qval -51.40	la
[DQN-Mod] Ep nd100 0.25	R thr 860	-235.01	avgR100	-92.69	eps 0.137	Qval -52.77	la
[DQN-Mod] Ep nd100 0.25	R thr 739	-240.48	avgR100	-94.83	eps 0.137	Qval -52.76	la
[DQN-Mod] Ep nd100 0.24	R thr 794	-198.16	avgR100	-97.86	eps 0.136	Qval -50.86	la

[DQN-Mod]	Ep	399	R	-200.76	avgR100	-99.20	eps	0.135	Qval	-52.39	la
nd100	0.24	thr	808								
[DQN-Mod]	Ep	400	R	-94.09	avgR100	-97.81	eps	0.135	Qval	-52.95	la
nd100	0.24	thr	83								
[DQN-Mod]	Ep	401	R	-247.67	avgR100	-98.33	eps	0.134	Qval	-51.84	la
nd100	0.24	thr	811								
[DQN-Mod]	Ep	402	R	-254.54	avgR100	-102.21	eps	0.133	Qval	-52.94	la
nd100	0.23	thr	791								
[DQN-Mod]	Ep	403	R	-260.43	avgR100	-104.57	eps	0.133	Qval	-54.61	la
nd100	0.23	thr	837								
[DQN-Mod]	Ep	404	R	-253.52	avgR100	-104.87	eps	0.132	Qval	-57.59	la
nd100	0.23	thr	804								
[DQN-Mod]	Ep	405	R	-113.52	avgR100	-103.19	eps	0.131	Qval	-56.42	la
nd100	0.23	thr	394								
[DQN-Mod]	Ep	406	R	19.68	avgR100	-102.38	eps	0.131	Qval	-57.36	la
nd100	0.23	thr	531								
[DQN-Mod]	Ep	407	R	-38.60	avgR100	-102.33	eps	0.130	Qval	-57.27	la
nd100	0.23	thr	168								
[DQN-Mod]	Ep	408	R	-64.93	avgR100	-101.35	eps	0.129	Qval	-56.57	la
nd100	0.23	thr	122								
[DQN-Mod]	Ep	409	R	-249.98	avgR100	-101.45	eps	0.129	Qval	-55.79	la
nd100	0.23	thr	801								
[DQN-Mod]	Ep	410	R	-288.44	avgR100	-104.12	eps	0.128	Qval	-58.73	la
nd100	0.23	thr	815								
[DQN-Mod]	Ep	411	R	-385.54	avgR100	-105.55	eps	0.127	Qval	-58.95	la
nd100	0.23	thr	309								
[DQN-Mod]	Ep	412	R	-264.42	avgR100	-108.13	eps	0.127	Qval	-58.25	la
nd100	0.23	thr	568								
[DQN-Mod]	Ep	413	R	-300.19	avgR100	-108.80	eps	0.126	Qval	-58.15	la
nd100	0.23	thr	815								
[DQN-Mod]	Ep	414	R	-277.33	avgR100	-111.38	eps	0.126	Qval	-58.66	la
nd100	0.23	thr	819								
[DQN-Mod]	Ep	415	R	-290.74	avgR100	-111.63	eps	0.125	Qval	-60.45	la
nd100	0.23	thr	784								
[DQN-Mod]	Ep	416	R	114.91	avgR100	-108.08	eps	0.124	Qval	-59.69	la
nd100	0.24	thr	514								
[DQN-Mod]	Ep	417	R	-178.54	avgR100	-108.31	eps	0.124	Qval	-60.63	la
nd100	0.24	thr	737								
[DQN-Mod]	Ep	418	R	-44.45	avgR100	-107.06	eps	0.123	Qval	-61.01	la
nd100	0.24	thr	128								
[DQN-Mod]	Ep	419	R	-71.05	avgR100	-104.82	eps	0.122	Qval	-61.92	la
nd100	0.24	thr	113								
[DQN-Mod]	Ep	420	R	-249.76	avgR100	-109.04	eps	0.122	Qval	-62.81	la
nd100	0.24	thr	763								
[DQN-Mod]	Ep	421	R	7.62	avgR100	-109.43	eps	0.121	Qval	-59.92	la
nd100	0.24	thr	669								
[DQN-Mod]	Ep	422	R	-243.72	avgR100	-109.49	eps	0.121	Qval	-61.10	la
nd100	0.24	thr	790								

[DQN-Mod] Ep nd100 0.24	429 R thr 795	-227.61	avgR100	-113.23	eps 0.116	Qval -65.77	la
[DQN-Mod] Ep nd100 0.24	430 R thr 797	-260.88	avgR100	-113.99	eps 0.116	Qval -68.07	la
[DQN-Mod] Ep nd100 0.24	431 R thr 781	-287.04	avgR100	-113.88	eps 0.115	Qval -68.88	la
[DQN-Mod] Ep nd100 0.24	432 R thr 617	81.38	avgR100	-115.09	eps 0.115	Qval -70.50	la
[DQN-Mod] Ep nd100 0.25	433 R thr 152	276.15	avgR100	-109.57	eps 0.114	Qval -69.65	la
[DQN-Mod] Ep nd100 0.25	434 R thr 822	-300.39	avgR100	-109.77	eps 0.114	Qval -72.49	la
[DQN-Mod] Ep nd100 0.25	435 R thr 134	-22.31	avgR100	-107.76	eps 0.113	Qval -72.34	la
[DQN-Mod] Ep nd100 0.24	436 R thr 857	-300.29	avgR100	-111.26	eps 0.112	Qval -73.62	la
[DQN-Mod] Ep nd100 0.24	437 R thr 795	-274.03	avgR100	-113.19	eps 0.112	Qval -72.33	la
[DQN-Mod] Ep nd100 0.24	438 R thr 607	9.67	avgR100	-114.06	eps 0.111	Qval -73.11	la
[DQN-Mod] Ep nd100 0.24	439 R thr 75	-3.08	avgR100	-112.14	eps 0.111	Qval -73.49	la
[DQN-Mod] Ep nd100 0.24	440 R thr 812	-233.72	avgR100	-111.90	eps 0.110	Qval -74.15	la
[DQN-Mod] Ep nd100 0.25	441 R thr 569	72.59	avgR100	-108.64	eps 0.110	Qval -74.02	la
[DQN-Mod] Ep nd100 0.25	442 R thr 115	-19.27	avgR100	-108.12	eps 0.109	Qval -74.04	la
[DQN-Mod] Ep nd100 0.25	443 R thr 643	-128.17	avgR100	-106.73	eps 0.109	Qval -74.30	la
[DQN-Mod] Ep nd100 0.24	444 R thr 803	-256.92	avgR100	-109.13	eps 0.108	Qval -75.16	la
[DQN-Mod] Ep nd100 0.23	445 R thr 175	11.35	avgR100	-111.55	eps 0.107	Qval -73.33	la
[DQN-Mod] Ep nd100 0.23	446 R thr 826	-240.29	avgR100	-111.81	eps 0.107	Qval -74.41	la
[DQN-Mod] Ep nd100 0.23	447 R thr 826	-271.97	avgR100	-112.13	eps 0.106	Qval -77.04	la
[DQN-Mod] Ep nd100 0.23	448 R thr 103	-11.80	avgR100	-109.68	eps 0.106	Qval -75.19	la
[DQN-Mod] Ep nd100 0.23	449 R thr 785	-251.17	avgR100	-111.32	eps 0.105	Qval -76.23	la
[DQN-Mod] Ep nd100 0.23	450 R thr 761	-163.69	avgR100	-112.39	eps 0.105	Qval -76.01	la
[DQN-Mod] Ep nd100 0.22	451 R thr 830	-241.85	avgR100	-117.55	eps 0.104	Qval -75.79	la
[DQN-Mod] Ep nd100 0.22	452 R thr 804	-223.24	avgR100	-116.78	eps 0.104	Qval -75.36	la
[DQN-Mod] Ep nd100 0.22	453 R thr 787	-190.08	avgR100	-116.38	eps 0.103	Qval -78.23	la
[DQN-Mod] Ep nd100 0.22	454 R thr 796	-273.02	avgR100	-118.87	eps 0.103	Qval -78.32	la
[DQN-Mod] Ep nd100 0.22	455 R thr 799	-259.78	avgR100	-121.05	eps 0.102	Qval -77.84	la
[DQN-Mod] Ep nd100 0.22	456 R thr 190	-37.15	avgR100	-118.60	eps 0.102	Qval -77.74	la
[DQN-Mod] Ep nd100 0.22	457 R thr 789	-211.25	avgR100	-118.74	eps 0.101	Qval -78.71	la
[DQN-Mod] Ep nd100 0.22	458 R thr 768	-246.47	avgR100	-120.40	eps 0.101	Qval -77.67	la

[DQN-Mod] Ep nd100 0.23	459 R thr 346	202.24	avgR100	-115.92	eps 0.100	Qval -76.78	la
[DQN-Mod] Ep nd100 0.23	460 R thr 811	-240.97	avgR100	-117.58	eps 0.100	Qval -77.17	la
[DQN-Mod] Ep nd100 0.24	461 R thr 273	228.28	avgR100	-113.82	eps 0.099	Qval -78.87	la
[DQN-Mod] Ep nd100 0.23	462 R thr 803	-235.86	avgR100	-117.66	eps 0.099	Qval -78.75	la
[DQN-Mod] Ep nd100 0.22	463 R thr 131	-29.67	avgR100	-118.68	eps 0.098	Qval -77.51	la
[DQN-Mod] Ep nd100 0.22	464 R thr 786	-219.47	avgR100	-118.82	eps 0.098	Qval -76.00	la
[DQN-Mod] Ep nd100 0.22	465 R thr 754	-212.94	avgR100	-119.27	eps 0.097	Qval -78.22	la
[DQN-Mod] Ep nd100 0.23	466 R thr 611	74.51	avgR100	-116.12	eps 0.097	Qval -75.76	la
[DQN-Mod] Ep nd100 0.23	467 R thr 552	64.06	avgR100	-115.19	eps 0.096	Qval -75.65	la
[DQN-Mod] Ep nd100 0.23	468 R thr 792	-216.22	avgR100	-117.02	eps 0.096	Qval -76.15	la
[DQN-Mod] Ep nd100 0.22	469 R thr 138	-84.95	avgR100	-118.18	eps 0.095	Qval -76.19	la
[DQN-Mod] Ep nd100 0.21	470 R thr 798	-239.05	avgR100	-121.26	eps 0.095	Qval -75.68	la
[DQN-Mod] Ep nd100 0.21	471 R thr 754	-209.88	avgR100	-121.38	eps 0.094	Qval -76.82	la
[DQN-Mod] Ep nd100 0.20	472 R thr 803	-264.33	avgR100	-124.91	eps 0.094	Qval -77.24	la
[DQN-Mod] Ep nd100 0.19	473 R thr 770	-251.34	avgR100	-128.66	eps 0.093	Qval -76.46	la
[DQN-Mod] Ep nd100 0.20	474 R thr 552	27.17	avgR100	-126.66	eps 0.093	Qval -77.54	la
[DQN-Mod] Ep nd100 0.20	475 R thr 643	68.35	avgR100	-126.33	eps 0.092	Qval -76.19	la
[DQN-Mod] Ep nd100 0.19	476 R thr 758	-170.99	avgR100	-129.82	eps 0.092	Qval -76.55	la
[DQN-Mod] Ep nd100 0.20	477 R thr 293	217.01	avgR100	-125.39	eps 0.092	Qval -75.04	la
[DQN-Mod] Ep nd100 0.21	478 R thr 449	161.55	avgR100	-121.45	eps 0.091	Qval -77.18	la
[DQN-Mod] Ep nd100 0.21	479 R thr 114	-15.88	avgR100	-119.32	eps 0.091	Qval -77.89	la
[DQN-Mod] Ep nd100 0.21	480 R thr 598	62.43	avgR100	-119.77	eps 0.090	Qval -77.66	la
[DQN-Mod] Ep nd100 0.22	481 R thr 642	12.52	avgR100	-119.28	eps 0.090	Qval -77.08	la
[DQN-Mod] Ep nd100 0.22	482 R thr 261	223.22	avgR100	-118.61	eps 0.089	Qval -77.34	la
[DQN-Mod] Ep nd100 0.23	483 R thr 406	167.70	avgR100	-114.78	eps 0.089	Qval -76.12	la
[DQN-Mod] Ep nd100 0.23	484 R thr 273	228.86	avgR100	-113.49	eps 0.088	Qval -76.74	la
[DQN-Mod] Ep nd100 0.22	485 R thr 773	-223.09	avgR100	-117.88	eps 0.088	Qval -74.37	la
[DQN-Mod] Ep nd100 0.22	486 R thr 763	-224.10	avgR100	-117.50	eps 0.088	Qval -73.98	la
[DQN-Mod] Ep nd100 0.22	487 R thr 447	121.94	avgR100	-118.53	eps 0.087	Qval -76.08	la
[DQN-Mod] Ep nd100 0.23	488 R thr 287	187.93	avgR100	-116.57	eps 0.087	Qval -75.35	la

[DQN-Mod] Ep 489 R	41.34	avgR100	-117.44	eps 0.086	Qval	-75.26	la
nd100 0.23 thr 656							
[DQN-Mod] Ep 490 R	95.33	avgR100	-115.67	eps 0.086	Qval	-73.46	la
nd100 0.24 thr 448							
[DQN-Mod] Ep 491 R	162.84	avgR100	-113.10	eps 0.085	Qval	-76.02	la
nd100 0.25 thr 379							
[DQN-Mod] Ep 492 R	101.02	avgR100	-109.70	eps 0.085	Qval	-74.33	la
nd100 0.26 thr 489							
[DQN-Mod] Ep 493 R	-53.42	avgR100	-108.70	eps 0.084	Qval	-75.21	la
nd100 0.26 thr 95							
[DQN-Mod] Ep 494 R	76.81	avgR100	-107.93	eps 0.084	Qval	-74.27	la
nd100 0.27 thr 516							
[DQN-Mod] Ep 495 R	55.58	avgR100	-107.04	eps 0.084	Qval	-72.89	la
nd100 0.28 thr 644							
[DQN-Mod] Ep 496 R	-272.48	avgR100	-107.41	eps 0.083	Qval	-72.58	la
nd100 0.28 thr 791							
[DQN-Mod] Ep 497 R	71.36	avgR100	-104.29	eps 0.083	Qval	-72.13	la
nd100 0.29 thr 505							
[DQN-Mod] Ep 498 R	68.22	avgR100	-101.63	eps 0.082	Qval	-71.13	la
nd100 0.30 thr 583							
[DQN-Mod] Ep 499 R	-227.23	avgR100	-101.90	eps 0.082	Qval	-73.22	la
nd100 0.30 thr 789							
[DQN-Mod] Ep 500 R	168.61	avgR100	-99.27	eps 0.082	Qval	-72.73	la
nd100 0.31 thr 337							
[DQN-Mod] Ep 501 R	213.19	avgR100	-94.66	eps 0.081	Qval	-71.93	la
nd100 0.32 thr 202							
[DQN-Mod] Ep 502 R	-218.04	avgR100	-94.29	eps 0.081	Qval	-71.46	la
nd100 0.32 thr 768							
[DQN-Mod] Ep 503 R	179.87	avgR100	-89.89	eps 0.080	Qval	-70.19	la
nd100 0.33 thr 407							
[DQN-Mod] Ep 504 R	123.36	avgR100	-86.12	eps 0.080	Qval	-68.62	la
nd100 0.34 thr 447							
[DQN-Mod] Ep 505 R	38.06	avgR100	-84.61	eps 0.080	Qval	-67.03	la
nd100 0.35 thr 647							
[DQN-Mod] Ep 506 R	201.99	avgR100	-82.78	eps 0.079	Qval	-65.95	la
nd100 0.36 thr 299							
[DQN-Mod] Ep 507 R	185.89	avgR100	-80.54	eps 0.079	Qval	-66.07	la
nd100 0.37 thr 309							
[DQN-Mod] Ep 508 R	132.06	avgR100	-78.57	eps 0.078	Qval	-65.00	la
nd100 0.38 thr 346							
[DQN-Mod] Ep 509 R	-286.86	avgR100	-78.94	eps 0.078	Qval	-64.83	la
nd100 0.38 thr 355							
[DQN-Mod] Ep 510 R	142.33	avgR100	-74.63	eps 0.078	Qval	-64.23	la
nd100 0.39 thr 396							
[DQN-Mod] Ep 511 R	77.16	avgR100	-70.00	eps 0.077	Qval	-62.77	la
nd100 0.40 thr 545							
[DQN-Mod] Ep 512 R	99.37	avgR100	-66.37	eps 0.077	Qval	-60.93	la
nd100 0.41 thr 548							
[DQN-Mod] Ep 513 R	222.71	avgR100	-61.14	eps 0.076	Qval	-59.80	la
nd100 0.42 thr 306							
[DQN-Mod] Ep 514 R	209.67	avgR100	-56.27	eps 0.076	Qval	-59.19	la
nd100 0.43 thr 336							
[DQN-Mod] Ep 515 R	162.45	avgR100	-51.73	eps 0.076	Qval	-58.17	la
nd100 0.44 thr 386							
[DQN-Mod] Ep 516 R	101.64	avgR100	-51.87	eps 0.075	Qval	-57.07	la
nd100 0.44 thr 446							
[DQN-Mod] Ep 517 R	208.34	avgR100	-48.00	eps 0.075	Qval	-57.29	la
nd100 0.45 thr 340							

[DQN-Mod] Ep nd100 0.46	518 R thr 341	163.60	avgR100	-45.92	eps 0.075	Qval -57.02	1a
[DQN-Mod] Ep nd100 0.47	519 R thr 234	239.92	avgR100	-42.81	eps 0.074	Qval -57.37	1a
[DQN-Mod] Ep nd100 0.48	520 R thr 456	86.33	avgR100	-39.45	eps 0.074	Qval -56.61	1a
[DQN-Mod] Ep nd100 0.48	521 R thr 286	234.60	avgR100	-37.18	eps 0.073	Qval -56.60	1a
[DQN-Mod] Ep nd100 0.49	522 R thr 201	247.15	avgR100	-32.27	eps 0.073	Qval -55.71	1a
[DQN-Mod] Ep nd100 0.50	523 R thr 278	208.80	avgR100	-27.60	eps 0.073	Qval -54.91	1a
[DQN-Mod] Ep nd100 0.51	524 R thr 599	30.21	avgR100	-26.93	eps 0.072	Qval -53.97	1a
[DQN-Mod] Ep nd100 0.52	525 R thr 421	158.48	avgR100	-22.50	eps 0.072	Qval -52.90	1a
[DQN-Mod] Ep nd100 0.52	526 R thr 262	223.29	avgR100	-21.50	eps 0.072	Qval -51.88	1a
[DQN-Mod] Ep nd100 0.53	527 R thr 241	236.91	avgR100	-16.52	eps 0.071	Qval -52.54	1a
[DQN-Mod] Ep nd100 0.54	528 R thr 342	151.67	avgR100	-12.13	eps 0.071	Qval -51.62	1a
[DQN-Mod] Ep nd100 0.55	529 R thr 309	185.15	avgR100	-8.00	eps 0.071	Qval -48.88	1a
[DQN-Mod] Ep nd100 0.56	530 R thr 440	150.21	avgR100	-3.89	eps 0.070	Qval -50.61	1a
[DQN-Mod] Ep nd100 0.57	531 R thr 390	154.61	avgR100	0.52	eps 0.070	Qval -48.97	1a
[DQN-Mod] Ep nd100 0.57	532 R thr 262	197.55	avgR100	1.69	eps 0.069	Qval -47.43	1a
[DQN-Mod] Ep nd100 0.57	533 R thr 331	207.36	avgR100	1.00	eps 0.069	Qval -48.84	1a
[DQN-Mod] Ep nd100 0.58	534 R thr 177	247.89	avgR100	6.48	eps 0.069	Qval -47.96	1a
[DQN-Mod] Ep nd100 0.59	535 R thr 371	171.58	avgR100	8.42	eps 0.068	Qval -46.57	1a
[DQN-Mod] Ep nd100 0.60	536 R thr 422	137.23	avgR100	12.79	eps 0.068	Qval -47.08	1a
[DQN-Mod] Ep nd100 0.61	537 R thr 328	187.11	avgR100	17.41	eps 0.068	Qval -46.36	1a
[DQN-Mod] Ep nd100 0.61	538 R thr 465	116.58	avgR100	18.47	eps 0.067	Qval -46.42	1a
[DQN-Mod] Ep nd100 0.62	539 R thr 359	140.56	avgR100	19.91	eps 0.067	Qval -45.91	1a
[DQN-Mod] Ep nd100 0.63	540 R thr 214	208.67	avgR100	24.34	eps 0.067	Qval -44.40	1a
[DQN-Mod] Ep nd100 0.63	541 R thr 235	186.67	avgR100	25.48	eps 0.066	Qval -44.41	1a
[DQN-Mod] Ep nd100 0.64	542 R thr 386	134.22	avgR100	27.01	eps 0.066	Qval -43.33	1a
[DQN-Mod] Ep nd100 0.64	543 R thr 563	54.15	avgR100	28.83	eps 0.066	Qval -44.43	1a
[DQN-Mod] Ep nd100 0.65	544 R thr 200	276.71	avgR100	34.17	eps 0.065	Qval -43.15	1a
[DQN-Mod] Ep nd100 0.66	545 R thr 340	139.86	avgR100	35.46	eps 0.065	Qval -42.58	1a
[DQN-Mod] Ep nd100 0.67	546 R thr 249	239.85	avgR100	40.26	eps 0.065	Qval -41.70	1a

[DQN-Mod] Ep 547 R 171.64 avgR100 44.69 eps 0.064 Qval -41.11 la
nd100 0.68 thr 403
[DQN-Mod] Ep 548 R 225.21 avgR100 47.06 eps 0.064 Qval -41.89 la
nd100 0.69 thr 294
[DQN-Mod] Ep 549 R 233.90 avgR100 51.91 eps 0.064 Qval -39.91 la
nd100 0.70 thr 235
[DQN-Mod] Ep 550 R 228.41 avgR100 55.83 eps 0.063 Qval -39.06 la
nd100 0.71 thr 293
[DQN-Mod] Ep 551 R 233.90 avgR100 60.59 eps 0.063 Qval -37.50 la
nd100 0.72 thr 190
[DQN-Mod] Ep 552 R -47.69 avgR100 62.35 eps 0.063 Qval -39.62 la
nd100 0.72 thr 288
[DQN-Mod] Ep 553 R 240.95 avgR100 66.66 eps 0.063 Qval -38.98 la
nd100 0.73 thr 251
[DQN-Mod] Ep 554 R 237.51 avgR100 71.76 eps 0.062 Qval -38.82 la
nd100 0.74 thr 177
[DQN-Mod] Ep 555 R 150.25 avgR100 75.86 eps 0.062 Qval -38.00 la
nd100 0.75 thr 365
[DQN-Mod] Ep 556 R 148.90 avgR100 77.72 eps 0.062 Qval -36.74 la
nd100 0.76 thr 369
[DQN-Mod] Ep 557 R 198.28 avgR100 81.82 eps 0.061 Qval -37.10 la
nd100 0.77 thr 247
[DQN-Mod] Ep 558 R 186.64 avgR100 86.15 eps 0.061 Qval -35.74 la
nd100 0.78 thr 248
[DQN-Mod] Ep 559 R 203.20 avgR100 86.16 eps 0.061 Qval -36.48 la
nd100 0.78 thr 301
[DQN-Mod] Ep 560 R 105.14 avgR100 89.62 eps 0.060 Qval -35.57 la
nd100 0.79 thr 532
[DQN-Mod] Ep 561 R 198.60 avgR100 89.32 eps 0.060 Qval -34.09 la
nd100 0.79 thr 324
[DQN-Mod] Ep 562 R 13.66 avgR100 91.82 eps 0.060 Qval -35.46 la
nd100 0.79 thr 104
[DQN-Mod] Ep 563 R 258.53 avgR100 94.70 eps 0.059 Qval -32.18 la
nd100 0.80 thr 181
[DQN-Mod] Ep 564 R 213.34 avgR100 99.03 eps 0.059 Qval -31.96 la
nd100 0.81 thr 333
[DQN-Mod] Ep 565 R 103.63 avgR100 102.20 eps 0.059 Qval -30.79 la
nd100 0.82 thr 514
[DQN-Mod] Ep 566 R 220.32 avgR100 103.65 eps 0.059 Qval -30.88 la
nd100 0.82 thr 320
[DQN-Mod] Ep 567 R 207.15 avgR100 105.08 eps 0.058 Qval -29.84 la
nd100 0.82 thr 327
[DQN-Mod] Ep 568 R -49.17 avgR100 106.75 eps 0.058 Qval -30.09 la
nd100 0.82 thr 108
[DQN-Mod] Ep 569 R 287.90 avgR100 110.48 eps 0.058 Qval -30.31 la
nd100 0.83 thr 148
[DQN-Mod] Ep 570 R 168.82 avgR100 114.56 eps 0.057 Qval -29.10 la
nd100 0.84 thr 425
[DQN-Mod] Ep 571 R 183.05 avgR100 118.49 eps 0.057 Qval -28.18 la
nd100 0.85 thr 347
[DQN-Mod] Ep 572 R 204.81 avgR100 123.18 eps 0.057 Qval -26.65 la
nd100 0.85 thr 256
[DQN-Mod] Ep 573 R 219.13 avgR100 127.89 eps 0.057 Qval -27.46 la
nd100 0.86 thr 323
[DQN-Mod] Ep 574 R 266.94 avgR100 130.28 eps 0.056 Qval -26.45 la
nd100 0.86 thr 210
[DQN-Mod] Ep 575 R 135.26 avgR100 130.95 eps 0.056 Qval -25.26 la
nd100 0.86 thr 382
[DQN-Mod] Ep 576 R 225.47 avgR100 134.92 eps 0.056 Qval -23.79 la
nd100 0.87 thr 177

[DQN-Mod] Ep 577 R	70.37	avgR100	133.45	eps 0.055	Qval	-23.65	la
nd100 0.87 thr 585							
[DQN-Mod] Ep 578 R	52.42	avgR100	132.36	eps 0.055	Qval	-22.79	la
nd100 0.87 thr 625							
[DQN-Mod] Ep 579 R	232.88	avgR100	134.85	eps 0.055	Qval	-22.59	la
nd100 0.88 thr 272							
[DQN-Mod] Ep 580 R	207.75	avgR100	136.30	eps 0.055	Qval	-22.32	la
nd100 0.88 thr 361							
[DQN-Mod] Ep 581 R	289.91	avgR100	139.08	eps 0.054	Qval	-22.43	la
nd100 0.88 thr 126							
[DQN-Mod] Ep 582 R	-41.78	avgR100	136.43	eps 0.054	Qval	-20.30	la
nd100 0.87 thr 151							
[DQN-Mod] Ep 583 R	217.84	avgR100	136.93	eps 0.054	Qval	-22.38	la
nd100 0.87 thr 228							
[DQN-Mod] Ep 584 R	-37.44	avgR100	134.26	eps 0.054	Qval	-21.90	la
nd100 0.86 thr 119							
[DQN-Mod] Ep 585 R	174.32	avgR100	138.24	eps 0.053	Qval	-21.52	la
nd100 0.86 thr 211							
[DQN-Mod] Ep 586 R	-12.02	avgR100	140.36	eps 0.053	Qval	-21.93	la
nd100 0.86 thr 132							
[DQN-Mod] Ep 587 R	207.33	avgR100	141.21	eps 0.053	Qval	-21.98	la
nd100 0.86 thr 200							
[DQN-Mod] Ep 588 R	221.91	avgR100	141.55	eps 0.052	Qval	-20.86	la
nd100 0.85 thr 201							
[DQN-Mod] Ep 589 R	214.51	avgR100	143.28	eps 0.052	Qval	-21.53	la
nd100 0.85 thr 233							
[DQN-Mod] Ep 590 R	132.67	avgR100	143.66	eps 0.052	Qval	-20.92	la
nd100 0.85 thr 396							
[DQN-Mod] Ep 591 R	262.23	avgR100	144.65	eps 0.052	Qval	-20.09	la
nd100 0.85 thr 233							
[DQN-Mod] Ep 592 R	222.00	avgR100	145.86	eps 0.051	Qval	-21.75	la
nd100 0.85 thr 185							
[DQN-Mod] Ep 593 R	245.39	avgR100	148.85	eps 0.051	Qval	-19.98	la
nd100 0.86 thr 249							
[DQN-Mod] Ep 594 R	73.92	avgR100	148.82	eps 0.051	Qval	-20.32	la
nd100 0.85 thr 370							
[DQN-Mod] Ep 595 R	209.99	avgR100	150.36	eps 0.051	Qval	-19.38	la
nd100 0.85 thr 270							
[DQN-Mod] Ep 596 R	282.06	avgR100	155.91	eps 0.050	Qval	-19.23	la
nd100 0.86 thr 150							
[DQN-Mod] Ep 597 R	-17.52	avgR100	155.02	eps 0.050	Qval	-18.35	la
nd100 0.85 thr 157							
[DQN-Mod] Ep 598 R	263.46	avgR100	156.97	eps 0.050	Qval	-18.11	la
nd100 0.85 thr 187							
[DQN-Mod] Ep 599 R	-92.72	avgR100	158.32	eps 0.050	Qval	-18.11	la
nd100 0.85 thr 646							
[DQN-Mod] Ep 600 R	179.13	avgR100	158.42	eps 0.049	Qval	-16.55	la
nd100 0.85 thr 315							
[DQN-Mod] Ep 601 R	175.77	avgR100	158.05	eps 0.049	Qval	-16.61	la
nd100 0.85 thr 289							
[DQN-Mod] Ep 602 R	126.16	avgR100	161.49	eps 0.049	Qval	-15.05	la
nd100 0.86 thr 451							
[DQN-Mod] Ep 603 R	204.57	avgR100	161.74	eps 0.049	Qval	-15.91	la
nd100 0.85 thr 215							
[DQN-Mod] Ep 604 R	244.94	avgR100	162.95	eps 0.048	Qval	-15.07	la
nd100 0.85 thr 240							
[DQN-Mod] Ep 605 R	232.40	avgR100	164.90	eps 0.048	Qval	-16.04	la
nd100 0.85 thr 193							

[DQN-Mod] Ep 606 R	6.97	avgR100	162.95	eps 0.048	Qval	-14.94	la
nd100 0.84 thr 148							
[DQN-Mod] Ep 607 R	224.99	avgR100	163.34	eps 0.048	Qval	-16.12	la
nd100 0.84 thr 260							
[DQN-Mod] Ep 608 R	286.40	avgR100	164.88	eps 0.047	Qval	-15.64	la
nd100 0.84 thr 161							
[DQN-Mod] Ep 609 R	-6.61	avgR100	167.68	eps 0.047	Qval	-15.08	la
nd100 0.84 thr 126							
[DQN-Mod] Ep 610 R	240.09	avgR100	168.66	eps 0.047	Qval	-14.72	la
nd100 0.84 thr 210							
[DQN-Mod] Ep 611 R	267.55	avgR100	170.57	eps 0.047	Qval	-13.52	la
nd100 0.84 thr 195							
[DQN-Mod] Ep 612 R	149.46	avgR100	171.07	eps 0.047	Qval	-14.43	la
nd100 0.84 thr 331							
[DQN-Mod] Ep 613 R	211.79	avgR100	170.96	eps 0.046	Qval	-12.41	la
nd100 0.84 thr 283							
[DQN-Mod] Ep 614 R	-177.11	avgR100	167.09	eps 0.046	Qval	-13.35	la
nd100 0.83 thr 830							
[DQN-Mod] Ep 615 R	199.91	avgR100	167.46	eps 0.046	Qval	-12.39	la
nd100 0.83 thr 329							
[DQN-Mod] Ep 616 R	298.61	avgR100	169.43	eps 0.046	Qval	-14.14	la
nd100 0.83 thr 195							
[DQN-Mod] Ep 617 R	199.34	avgR100	169.34	eps 0.045	Qval	-12.33	la
nd100 0.83 thr 261							
[DQN-Mod] Ep 618 R	240.14	avgR100	170.11	eps 0.045	Qval	-12.98	la
nd100 0.83 thr 184							
[DQN-Mod] Ep 619 R	215.89	avgR100	169.87	eps 0.045	Qval	-11.64	la
nd100 0.83 thr 275							
[DQN-Mod] Ep 620 R	212.98	avgR100	171.14	eps 0.045	Qval	-12.40	la
nd100 0.83 thr 255							
[DQN-Mod] Ep 621 R	202.44	avgR100	170.81	eps 0.044	Qval	-12.94	la
nd100 0.83 thr 317							
[DQN-Mod] Ep 622 R	251.62	avgR100	170.86	eps 0.044	Qval	-12.49	la
nd100 0.83 thr 216							
[DQN-Mod] Ep 623 R	243.91	avgR100	171.21	eps 0.044	Qval	-11.57	la
nd100 0.83 thr 240							
[DQN-Mod] Ep 624 R	217.12	avgR100	173.08	eps 0.044	Qval	-12.23	la
nd100 0.83 thr 191							
[DQN-Mod] Ep 625 R	167.30	avgR100	173.17	eps 0.044	Qval	-11.18	la
nd100 0.83 thr 339							
[DQN-Mod] Ep 626 R	251.36	avgR100	173.45	eps 0.043	Qval	-10.99	la
nd100 0.83 thr 233							
[DQN-Mod] Ep 627 R	216.26	avgR100	173.24	eps 0.043	Qval	-12.13	la
nd100 0.83 thr 293							
[DQN-Mod] Ep 628 R	73.27	avgR100	172.46	eps 0.043	Qval	-10.78	la
nd100 0.83 thr 569							
[DQN-Mod] Ep 629 R	243.12	avgR100	173.04	eps 0.043	Qval	-11.09	la
nd100 0.83 thr 185							
[DQN-Mod] Ep 630 R	50.46	avgR100	172.04	eps 0.043	Qval	-10.10	la
nd100 0.83 thr 591							
[DQN-Mod] Ep 631 R	212.73	avgR100	172.62	eps 0.042	Qval	-9.11	la
nd100 0.83 thr 195							
[DQN-Mod] Ep 632 R	213.77	avgR100	172.78	eps 0.042	Qval	-9.17	la
nd100 0.83 thr 257							
[DQN-Mod] Ep 633 R	282.12	avgR100	173.53	eps 0.042	Qval	-9.24	la
nd100 0.83 thr 176							
[DQN-Mod] Ep 634 R	223.35	avgR100	173.29	eps 0.042	Qval	-8.51	la
nd100 0.83 thr 241							
[DQN-Mod] Ep 635 R	241.90	avgR100	173.99	eps 0.041	Qval	-9.38	la
nd100 0.83 thr 281							

[DQN-Mod] Ep 636 R	230.24 avgR100	174.92 eps 0.041 Qval	-8.89 la
nd100 0.83 thr 262			
[DQN-Mod] Ep 637 R	131.66 avgR100	174.36 eps 0.041 Qval	-7.93 la
nd100 0.83 thr 353			
[DQN-Mod] Ep 638 R	117.28 avgR100	174.37 eps 0.041 Qval	-8.84 la
nd100 0.83 thr 503			
[DQN-Mod] Ep 639 R	238.74 avgR100	175.35 eps 0.041 Qval	-9.44 la
nd100 0.83 thr 227			
[DQN-Mod] Ep 640 R	254.06 avgR100	175.81 eps 0.040 Qval	-8.83 la
nd100 0.83 thr 218			
[DQN-Mod] Ep 641 R	181.57 avgR100	175.76 eps 0.040 Qval	-8.24 la
nd100 0.83 thr 252			
[DQN-Mod] Ep 642 R	240.16 avgR100	176.81 eps 0.040 Qval	-8.70 la
nd100 0.83 thr 290			
[DQN-Mod] Ep 643 R	235.10 avgR100	178.62 eps 0.040 Qval	-8.65 la
nd100 0.84 thr 286			
[DQN-Mod] Ep 644 R	244.78 avgR100	178.31 eps 0.040 Qval	-7.17 la
nd100 0.84 thr 195			
[DQN-Mod] Ep 645 R	252.40 avgR100	179.43 eps 0.039 Qval	-9.03 la
nd100 0.84 thr 161			
[DQN-Mod] Ep 646 R	227.33 avgR100	179.31 eps 0.039 Qval	-8.52 la
nd100 0.84 thr 175			
[DQN-Mod] Ep 647 R	0.72 avgR100	177.60 eps 0.039 Qval	-8.26 la
nd100 0.83 thr 135			
[DQN-Mod] Ep 648 R	183.44 avgR100	177.18 eps 0.039 Qval	-8.05 la
nd100 0.83 thr 294			
[DQN-Mod] Ep 649 R	-173.59 avgR100	173.10 eps 0.039 Qval	-8.69 la
nd100 0.82 thr 915			
[DQN-Mod] Ep 650 R	172.98 avgR100	172.55 eps 0.038 Qval	-8.90 la
nd100 0.81 thr 243			
[DQN-Mod] Ep 651 R	-1.50 avgR100	170.20 eps 0.038 Qval	-9.98 la
nd100 0.81 thr 723			
[DQN-Mod] Ep 652 R	144.39 avgR100	172.12 eps 0.038 Qval	-8.28 la
nd100 0.82 thr 432			
[DQN-Mod] Ep 653 R	225.46 avgR100	171.96 eps 0.038 Qval	-8.86 la
nd100 0.82 thr 229			
[DQN-Mod] Ep 654 R	245.70 avgR100	172.04 eps 0.038 Qval	-8.77 la
nd100 0.82 thr 241			
[DQN-Mod] Ep 655 R	267.89 avgR100	173.22 eps 0.038 Qval	-9.96 la
nd100 0.82 thr 152			
[DQN-Mod] Ep 656 R	288.53 avgR100	174.62 eps 0.037 Qval	-8.99 la
nd100 0.82 thr 213			
[DQN-Mod] Ep 657 R	247.37 avgR100	175.11 eps 0.037 Qval	-9.18 la
nd100 0.82 thr 241			
[DQN-Mod] Ep 658 R	188.94 avgR100	175.13 eps 0.037 Qval	-9.01 la
nd100 0.82 thr 318			
[DQN-Mod] Ep 659 R	-162.82 avgR100	171.47 eps 0.037 Qval	-6.93 la
nd100 0.81 thr 746			
[DQN-Mod] Ep 660 R	308.61 avgR100	173.50 eps 0.037 Qval	-7.27 la
nd100 0.81 thr 134			
[DQN-Mod] Ep 661 R	255.61 avgR100	174.07 eps 0.036 Qval	-7.15 la
nd100 0.81 thr 191			
[DQN-Mod] Ep 662 R	200.75 avgR100	175.95 eps 0.036 Qval	-6.05 la
nd100 0.82 thr 230			
[DQN-Mod] Ep 663 R	214.07 avgR100	175.50 eps 0.036 Qval	-6.90 la
nd100 0.82 thr 214			
[DQN-Mod] Ep 664 R	116.74 avgR100	174.53 eps 0.036 Qval	-7.91 la
nd100 0.82 thr 400			

[DQN-Mod] Ep 665 R -240.20 avgR100 171.10 eps 0.036 Qval -5.97 la
nd100 0.81 thr 752
[DQN-Mod] Ep 666 R 213.92 avgR100 171.03 eps 0.035 Qval -5.78 la
nd100 0.81 thr 280
[DQN-Mod] Ep 667 R 178.33 avgR100 170.74 eps 0.035 Qval -5.46 la
nd100 0.81 thr 389
[DQN-Mod] Ep 668 R 126.32 avgR100 172.50 eps 0.035 Qval -4.84 la
nd100 0.82 thr 495
[DQN-Mod] Ep 669 R 200.54 avgR100 171.63 eps 0.035 Qval -5.48 la
nd100 0.82 thr 309
[DQN-Mod] Ep 670 R 68.91 avgR100 170.63 eps 0.035 Qval -4.82 la
nd100 0.82 thr 563
[DQN-Mod] Ep 671 R 178.36 avgR100 170.58 eps 0.035 Qval -4.61 la
nd100 0.82 thr 311
[DQN-Mod] Ep 672 R 159.38 avgR100 170.13 eps 0.034 Qval -4.19 la
nd100 0.83 thr 367
[DQN-Mod] Ep 673 R 168.90 avgR100 169.62 eps 0.034 Qval -5.03 la
nd100 0.83 thr 369
[DQN-Mod] Ep 674 R 276.03 avgR100 169.71 eps 0.034 Qval -4.31 la
nd100 0.83 thr 170
[DQN-Mod] Ep 675 R 251.21 avgR100 170.87 eps 0.034 Qval -4.68 la
nd100 0.83 thr 205
[DQN-Mod] Ep 676 R 245.56 avgR100 171.07 eps 0.034 Qval -4.63 la
nd100 0.83 thr 196
[DQN-Mod] Ep 677 R 139.35 avgR100 171.76 eps 0.034 Qval -4.03 la
nd100 0.83 thr 473
[DQN-Mod] Ep 678 R 232.77 avgR100 173.57 eps 0.033 Qval -4.05 la
nd100 0.82 thr 199
[DQN-Mod] Ep 679 R 163.04 avgR100 172.87 eps 0.033 Qval -4.57 la
nd100 0.82 thr 322
[DQN-Mod] Ep 680 R 223.49 avgR100 173.03 eps 0.033 Qval -3.24 la
nd100 0.82 thr 239
[DQN-Mod] Ep 681 R 151.55 avgR100 171.64 eps 0.033 Qval -3.76 la
nd100 0.82 thr 381
[DQN-Mod] Ep 682 R 204.84 avgR100 174.11 eps 0.033 Qval -4.17 la
nd100 0.83 thr 244
[DQN-Mod] Ep 683 R 135.58 avgR100 173.29 eps 0.033 Qval -4.13 la
nd100 0.83 thr 455
[DQN-Mod] Ep 684 R 229.85 avgR100 175.96 eps 0.032 Qval -4.19 la
nd100 0.84 thr 246
[DQN-Mod] Ep 685 R 1.21 avgR100 174.23 eps 0.032 Qval -3.55 la
nd100 0.84 thr 686
[DQN-Mod] Ep 686 R -41.23 avgR100 173.94 eps 0.032 Qval -4.66 la
nd100 0.84 thr 154
[DQN-Mod] Ep 687 R -108.15 avgR100 170.78 eps 0.032 Qval -4.53 la
nd100 0.83 thr 580
[DQN-Mod] Ep 688 R 196.28 avgR100 170.53 eps 0.032 Qval -3.80 la
nd100 0.84 thr 338
[DQN-Mod] Ep 689 R 250.52 avgR100 170.89 eps 0.032 Qval -3.28 la
nd100 0.84 thr 237
[DQN-Mod] Ep 690 R 239.02 avgR100 171.95 eps 0.031 Qval -3.88 la
nd100 0.84 thr 192
[DQN-Mod] Ep 691 R 253.35 avgR100 171.86 eps 0.031 Qval -5.28 la
nd100 0.84 thr 133
[DQN-Mod] Ep 692 R 227.88 avgR100 171.92 eps 0.031 Qval -4.13 la
nd100 0.84 thr 181
[DQN-Mod] Ep 693 R 260.25 avgR100 172.07 eps 0.031 Qval -4.54 la
nd100 0.84 thr 177

[DQN-Mod] Ep 694 R	219.85 avgR100	173.53 eps 0.031 Qval	-3.53 la
nd100 0.85 thr 199			
[DQN-Mod] Ep 695 R	188.00 avgR100	173.31 eps 0.031 Qval	-5.75 la
nd100 0.84 thr 130			
[DQN-Mod] Ep 696 R	101.54 avgR100	171.50 eps 0.031 Qval	-5.49 la
nd100 0.84 thr 514			
[DQN-Mod] Ep 697 R	-42.84 avgR100	171.25 eps 0.030 Qval	-4.86 la
nd100 0.84 thr 145			
[DQN-Mod] Ep 698 R	266.43 avgR100	171.28 eps 0.030 Qval	-3.21 la
nd100 0.84 thr 254			
[DQN-Mod] Ep 699 R	9.18 avgR100	172.30 eps 0.030 Qval	-3.64 la
nd100 0.84 thr 490			
[DQN-Mod] Ep 700 R	263.75 avgR100	173.14 eps 0.030 Qval	-2.91 la
nd100 0.84 thr 171			
[DQN-Mod] Ep 701 R	170.33 avgR100	173.09 eps 0.030 Qval	-3.67 la
nd100 0.84 thr 293			
[DQN-Mod] Ep 702 R	23.74 avgR100	172.06 eps 0.030 Qval	-5.03 la
nd100 0.84 thr 672			
[DQN-Mod] Ep 703 R	226.54 avgR100	172.28 eps 0.029 Qval	-4.09 la
nd100 0.85 thr 319			
[DQN-Mod] Ep 704 R	227.98 avgR100	172.12 eps 0.029 Qval	-3.98 la
nd100 0.85 thr 255			
[DQN-Mod] Ep 705 R	220.35 avgR100	171.99 eps 0.029 Qval	-3.55 la
nd100 0.84 thr 171			
[DQN-Mod] Ep 706 R	70.93 avgR100	172.63 eps 0.029 Qval	-3.55 la
nd100 0.84 thr 543			
[DQN-Mod] Ep 707 R	7.00 avgR100	170.45 eps 0.029 Qval	-3.44 la
nd100 0.83 thr 150			
[DQN-Mod] Ep 708 R	267.46 avgR100	170.26 eps 0.029 Qval	-2.82 la
nd100 0.83 thr 164			
[DQN-Mod] Ep 709 R	253.69 avgR100	172.87 eps 0.029 Qval	-3.31 la
nd100 0.84 thr 159			
[DQN-Mod] Ep 710 R	244.60 avgR100	172.91 eps 0.028 Qval	-4.38 la
nd100 0.84 thr 267			
[DQN-Mod] Ep 711 R	161.35 avgR100	171.85 eps 0.028 Qval	-4.08 la
nd100 0.84 thr 388			
[DQN-Mod] Ep 712 R	101.27 avgR100	171.37 eps 0.028 Qval	-4.44 la
nd100 0.84 thr 136			
[DQN-Mod] Ep 713 R	202.97 avgR100	171.28 eps 0.028 Qval	-4.56 la
nd100 0.84 thr 248			
[DQN-Mod] Ep 714 R	235.30 avgR100	175.41 eps 0.028 Qval	-4.15 la
nd100 0.85 thr 258			
[DQN-Mod] Ep 715 R	176.05 avgR100	175.17 eps 0.028 Qval	-4.21 la
nd100 0.85 thr 355			
[DQN-Mod] Ep 716 R	276.77 avgR100	174.95 eps 0.028 Qval	-4.04 la
nd100 0.85 thr 191			
[DQN-Mod] Ep 717 R	231.79 avgR100	175.27 eps 0.027 Qval	-4.01 la
nd100 0.85 thr 237			
[DQN-Mod] Ep 718 R	254.94 avgR100	175.42 eps 0.027 Qval	-4.53 la
nd100 0.85 thr 216			
[DQN-Mod] Ep 719 R	231.57 avgR100	175.58 eps 0.027 Qval	-4.53 la
nd100 0.85 thr 198			
[DQN-Mod] Ep 720 R	218.32 avgR100	175.63 eps 0.027 Qval	-5.34 la
nd100 0.85 thr 252			
[DQN-Mod] Ep 721 R	172.93 avgR100	175.34 eps 0.027 Qval	-4.15 la
nd100 0.85 thr 355			
[DQN-Mod] Ep 722 R	232.47 avgR100	175.14 eps 0.027 Qval	-3.35 la
nd100 0.85 thr 242			

[DQN-Mod] Ep nd100 0.85	723 R thr 523	92.30	avgR100	173.63	eps 0.027	Qval	-4.78	1a
[DQN-Mod] Ep nd100 0.85	724 R thr 273	209.22	avgR100	173.55	eps 0.027	Qval	-3.97	1a
[DQN-Mod] Ep nd100 0.85	725 R thr 172	249.28	avgR100	174.37	eps 0.026	Qval	-2.90	1a
[DQN-Mod] Ep nd100 0.84	726 R thr 279	185.30	avgR100	173.71	eps 0.026	Qval	-3.54	1a
[DQN-Mod] Ep nd100 0.84	727 R thr 104	286.75	avgR100	174.41	eps 0.026	Qval	-3.92	1a
[DQN-Mod] Ep nd100 0.84	728 R thr 117	252.53	avgR100	176.21	eps 0.026	Qval	-3.66	1a
[DQN-Mod] Ep nd100 0.84	729 R thr 287	193.81	avgR100	175.71	eps 0.026	Qval	-2.55	1a
[DQN-Mod] Ep nd100 0.84	730 R thr 189	246.38	avgR100	177.67	eps 0.026	Qval	-3.96	1a
[DQN-Mod] Ep nd100 0.84	731 R thr 195	245.81	avgR100	178.00	eps 0.026	Qval	-4.05	1a
[DQN-Mod] Ep nd100 0.83	732 R thr 568	-70.70	avgR100	175.16	eps 0.025	Qval	-3.53	1a
[DQN-Mod] Ep nd100 0.83	733 R thr 193	258.30	avgR100	174.92	eps 0.025	Qval	-4.17	1a
[DQN-Mod] Ep nd100 0.83	734 R thr 294	170.72	avgR100	174.39	eps 0.025	Qval	-2.53	1a
[DQN-Mod] Ep nd100 0.83	735 R thr 453	84.60	avgR100	172.82	eps 0.025	Qval	-4.14	1a
[DQN-Mod] Ep nd100 0.83	736 R thr 227	213.16	avgR100	172.65	eps 0.025	Qval	-4.23	1a
[DQN-Mod] Ep nd100 0.82	737 R thr 146	-43.95	avgR100	170.89	eps 0.025	Qval	-3.87	1a
[DQN-Mod] Ep nd100 0.81	738 R thr 139	-8.04	avgR100	169.64	eps 0.025	Qval	-3.76	1a
[DQN-Mod] Ep nd100 0.81	739 R thr 249	263.64	avgR100	169.89	eps 0.025	Qval	-4.78	1a
[DQN-Mod] Ep nd100 0.81	740 R thr 177	256.14	avgR100	169.91	eps 0.024	Qval	-3.56	1a
[DQN-Mod] Ep nd100 0.80	741 R thr 149	-23.63	avgR100	167.86	eps 0.024	Qval	-4.35	1a
[DQN-Mod] Ep nd100 0.80	742 R thr 227	207.15	avgR100	167.53	eps 0.024	Qval	-4.96	1a
[DQN-Mod] Ep nd100 0.80	743 R thr 271	223.98	avgR100	167.42	eps 0.024	Qval	-4.52	1a
[DQN-Mod] Ep nd100 0.79	744 R thr 150	-28.68	avgR100	164.68	eps 0.024	Qval	-4.96	1a
[DQN-Mod] Ep nd100 0.78	745 R thr 166	-27.19	avgR100	161.89	eps 0.024	Qval	-5.74	1a
[DQN-Mod] Ep nd100 0.78	746 R thr 218	218.45	avgR100	161.80	eps 0.024	Qval	-4.52	1a
[DQN-Mod] Ep nd100 0.78	747 R thr 119	-36.34	avgR100	161.43	eps 0.024	Qval	-5.18	1a
[DQN-Mod] Ep nd100 0.78	748 R thr 231	212.16	avgR100	161.71	eps 0.024	Qval	-4.86	1a
[DQN-Mod] Ep nd100 0.79	749 R thr 181	242.05	avgR100	165.87	eps 0.023	Qval	-5.14	1a
[DQN-Mod] Ep nd100 0.80	750 R thr 221	272.31	avgR100	166.86	eps 0.023	Qval	-5.75	1a
[DQN-Mod] Ep nd100 0.80	751 R thr 112	253.90	avgR100	169.42	eps 0.023	Qval	-5.15	1a

[DQN-Mod] Ep nd100 0.80	752 R thr 265	226.95	avgR100	170.24	eps 0.023	Qval	-4.12	1a
[DQN-Mod] Ep nd100 0.80	753 R thr 206	243.33	avgR100	170.42	eps 0.023	Qval	-5.33	1a
[DQN-Mod] Ep nd100 0.80	754 R thr 168	273.63	avgR100	170.70	eps 0.023	Qval	-6.15	1a
[DQN-Mod] Ep nd100 0.79	755 R thr 103	238.17	avgR100	170.40	eps 0.023	Qval	-5.37	1a
[DQN-Mod] Ep nd100 0.79	756 R thr 164	256.23	avgR100	170.08	eps 0.023	Qval	-5.09	1a
[DQN-Mod] Ep nd100 0.79	757 R thr 173	285.87	avgR100	170.47	eps 0.022	Qval	-4.01	1a
[DQN-Mod] Ep nd100 0.79	758 R thr 224	244.05	avgR100	171.02	eps 0.022	Qval	-4.79	1a
[DQN-Mod] Ep nd100 0.79	759 R thr 125	-41.10	avgR100	172.23	eps 0.022	Qval	-4.76	1a
[DQN-Mod] Ep nd100 0.79	760 R thr 276	204.71	avgR100	171.20	eps 0.022	Qval	-4.80	1a
[DQN-Mod] Ep nd100 0.79	761 R thr 242	214.87	avgR100	170.79	eps 0.022	Qval	-4.47	1a
[DQN-Mod] Ep nd100 0.79	762 R thr 345	219.37	avgR100	170.97	eps 0.022	Qval	-4.92	1a
[DQN-Mod] Ep nd100 0.79	763 R thr 168	262.57	avgR100	171.46	eps 0.022	Qval	-4.97	1a
[DQN-Mod] Ep nd100 0.79	764 R thr 340	210.49	avgR100	172.40	eps 0.022	Qval	-3.04	1a
[DQN-Mod] Ep nd100 0.80	765 R thr 127	278.75	avgR100	177.59	eps 0.022	Qval	-4.59	1a
[DQN-Mod] Ep nd100 0.80	766 R thr 211	228.26	avgR100	177.73	eps 0.022	Qval	-3.72	1a
[DQN-Mod] Ep nd100 0.80	767 R thr 188	259.04	avgR100	178.54	eps 0.021	Qval	-4.07	1a
[DQN-Mod] Ep nd100 0.80	768 R thr 407	129.57	avgR100	178.57	eps 0.021	Qval	-4.61	1a
[DQN-Mod] Ep nd100 0.80	769 R thr 169	268.25	avgR100	179.25	eps 0.021	Qval	-5.03	1a
[DQN-Mod] Ep nd100 0.80	770 R thr 116	295.29	avgR100	181.51	eps 0.021	Qval	-3.46	1a
[DQN-Mod] Ep nd100 0.79	771 R thr 151	-22.27	avgR100	179.50	eps 0.021	Qval	-4.27	1a
[DQN-Mod] Ep nd100 0.78	772 R thr 803	-133.77	avgR100	176.57	eps 0.021	Qval	-2.80	1a
[DQN-Mod] Ep nd100 0.78	773 R thr 167	247.85	avgR100	177.36	eps 0.021	Qval	-2.41	1a
[DQN-Mod] Ep nd100 0.77	774 R thr 171	-65.46	avgR100	173.95	eps 0.021	Qval	-3.27	1a
[DQN-Mod] Ep nd100 0.77	775 R thr 137	254.27	avgR100	173.98	eps 0.021	Qval	-3.26	1a
[DQN-Mod] Ep nd100 0.77	776 R thr 208	264.25	avgR100	174.16	eps 0.020	Qval	-4.47	1a
[DQN-Mod] Ep nd100 0.77	777 R thr 391	169.91	avgR100	174.47	eps 0.020	Qval	-4.11	1a
[DQN-Mod] Ep nd100 0.78	778 R thr 156	271.79	avgR100	174.86	eps 0.020	Qval	-4.44	1a
[DQN-Mod] Ep nd100 0.78	779 R thr 223	242.04	avgR100	175.65	eps 0.020	Qval	-3.91	1a
[DQN-Mod] Ep nd100 0.78	780 R thr 157	254.23	avgR100	175.96	eps 0.020	Qval	-3.79	1a

[DQN-Mod] Ep 781 R	-4.44	avgR100	174.40	eps 0.020	Qval	-3.73	1a
nd100 0.77 thr 122							
[DQN-Mod] Ep 782 R	7.99	avgR100	172.43	eps 0.020	Qval	-2.02	1a
nd100 0.76 thr 123							
[DQN-Mod] Ep 783 R	-56.44	avgR100	170.51	eps 0.020	Qval	-4.20	1a
nd100 0.75 thr 128							
[DQN-Mod] Ep 784 R	239.07	avgR100	170.60	eps 0.020	Qval	-3.91	1a
nd100 0.75 thr 195							
[DQN-Mod] Ep 785 R	297.91	avgR100	173.57	eps 0.020	Qval	-3.82	1a
nd100 0.76 thr 117							
[DQN-Mod] Ep 786 R	250.20	avgR100	176.48	eps 0.020	Qval	-4.23	1a
nd100 0.77 thr 225							
[DQN-Mod] Ep 787 R	-49.18	avgR100	177.07	eps 0.020	Qval	-3.71	1a
nd100 0.77 thr 146							
[DQN-Mod] Ep 788 R	228.95	avgR100	177.40	eps 0.020	Qval	-3.67	1a
nd100 0.77 thr 145							
[DQN-Mod] Ep 789 R	-31.13	avgR100	174.58	eps 0.020	Qval	-5.59	1a
nd100 0.76 thr 122							
[DQN-Mod] Ep 790 R	-44.76	avgR100	171.74	eps 0.020	Qval	-5.16	1a
nd100 0.75 thr 188							
[DQN-Mod] Ep 791 R	207.48	avgR100	171.29	eps 0.020	Qval	-5.65	1a
nd100 0.75 thr 296							
[DQN-Mod] Ep 792 R	263.68	avgR100	171.64	eps 0.020	Qval	-4.60	1a
nd100 0.75 thr 136							
[DQN-Mod] Ep 793 R	-60.09	avgR100	168.44	eps 0.020	Qval	-4.31	1a
nd100 0.74 thr 137							
[DQN-Mod] Ep 794 R	-40.33	avgR100	165.84	eps 0.020	Qval	-5.21	1a
nd100 0.73 thr 144							
[DQN-Mod] Ep 795 R	278.36	avgR100	166.74	eps 0.020	Qval	-4.67	1a
nd100 0.74 thr 148							
[DQN-Mod] Ep 796 R	40.99	avgR100	166.14	eps 0.020	Qval	-4.47	1a
nd100 0.73 thr 358							
[DQN-Mod] Ep 797 R	297.20	avgR100	169.54	eps 0.020	Qval	-4.15	1a
nd100 0.74 thr 112							
[DQN-Mod] Ep 798 R	111.30	avgR100	167.99	eps 0.020	Qval	-4.78	1a
nd100 0.74 thr 547							
[DQN-Mod] Ep 799 R	159.24	avgR100	169.49	eps 0.020	Qval	-5.99	1a
nd100 0.74 thr 239							
[DQN-Mod] Ep 800 R	-1.05	avgR100	166.84	eps 0.020	Qval	-5.25	1a
nd100 0.73 thr 107							
[DQN-Mod] Ep 801 R	273.72	avgR100	167.87	eps 0.020	Qval	-6.26	1a
nd100 0.73 thr 167							
[DQN-Mod] Ep 802 R	244.61	avgR100	170.08	eps 0.020	Qval	-6.43	1a
nd100 0.73 thr 117							
[DQN-Mod] Ep 803 R	250.06	avgR100	170.32	eps 0.020	Qval	-5.09	1a
nd100 0.73 thr 148							
[DQN-Mod] Ep 804 R	245.68	avgR100	170.49	eps 0.020	Qval	-3.99	1a
nd100 0.73 thr 171							
[DQN-Mod] Ep 805 R	-30.66	avgR100	167.98	eps 0.020	Qval	-4.70	1a
nd100 0.73 thr 125							
[DQN-Mod] Ep 806 R	245.88	avgR100	169.73	eps 0.020	Qval	-4.03	1a
nd100 0.74 thr 204							
[DQN-Mod] Ep 807 R	220.92	avgR100	171.87	eps 0.020	Qval	-5.31	1a
nd100 0.75 thr 245							
[DQN-Mod] Ep 808 R	243.80	avgR100	171.64	eps 0.020	Qval	-4.14	1a
nd100 0.75 thr 181							
[DQN-Mod] Ep 809 R	132.53	avgR100	170.42	eps 0.020	Qval	-6.43	1a
nd100 0.75 thr 358							
[DQN-Mod] Ep 810 R	271.78	avgR100	170.70	eps 0.020	Qval	-5.08	1a
nd100 0.75 thr 252							

[DQN-Mod] Ep 811 R nd100 0.75 thr 192	290.04 avgR100	171.98 eps 0.020 Qval	-4.32 la
[DQN-Mod] Ep 812 R nd100 0.75 thr 269	209.52 avgR100	173.06 eps 0.020 Qval	-4.79 la
[DQN-Mod] Ep 813 R nd100 0.75 thr 306	187.23 avgR100	172.91 eps 0.020 Qval	-4.38 la
[DQN-Mod] Ep 814 R nd100 0.74 thr 72	-2.68 avgR100	170.53 eps 0.020 Qval	-4.53 la
[DQN-Mod] Ep 815 R nd100 0.74 thr 291	198.76 avgR100	170.75 eps 0.020 Qval	-3.75 la
[DQN-Mod] Ep 816 R nd100 0.73 thr 232	199.78 avgR100	169.98 eps 0.020 Qval	-4.63 la
[DQN-Mod] Ep 817 R nd100 0.73 thr 185	232.75 avgR100	169.99 eps 0.020 Qval	-4.25 la
[DQN-Mod] Ep 818 R nd100 0.73 thr 259	182.36 avgR100	169.27 eps 0.020 Qval	-4.35 la
[DQN-Mod] Ep 819 R nd100 0.72 thr 134	12.00 avgR100	167.07 eps 0.020 Qval	-4.32 la
[DQN-Mod] Ep 820 R nd100 0.71 thr 148	-8.48 avgR100	164.80 eps 0.020 Qval	-5.79 la
[DQN-Mod] Ep 821 R nd100 0.71 thr 208	217.40 avgR100	165.25 eps 0.020 Qval	-3.75 la
[DQN-Mod] Ep 822 R nd100 0.70 thr 97	25.76 avgR100	163.18 eps 0.020 Qval	-4.76 la
[DQN-Mod] Ep 823 R nd100 0.70 thr 382	147.84 avgR100	163.74 eps 0.020 Qval	-3.51 la
[DQN-Mod] Ep 824 R nd100 0.69 thr 235	168.22 avgR100	163.33 eps 0.020 Qval	-4.77 la
[DQN-Mod] Ep 825 R nd100 0.68 thr 142	-22.11 avgR100	160.61 eps 0.020 Qval	-5.57 la
[DQN-Mod] Ep 826 R nd100 0.69 thr 164	249.25 avgR100	161.25 eps 0.020 Qval	-5.57 la
[DQN-Mod] Ep 827 R nd100 0.69 thr 223	263.88 avgR100	161.02 eps 0.020 Qval	-4.35 la
[DQN-Mod] Ep 828 R nd100 0.69 thr 343	200.42 avgR100	160.50 eps 0.020 Qval	-3.77 la
[DQN-Mod] Ep 829 R nd100 0.69 thr 192	221.17 avgR100	160.78 eps 0.020 Qval	-3.34 la
[DQN-Mod] Ep 830 R nd100 0.69 thr 198	264.05 avgR100	160.95 eps 0.020 Qval	-3.91 la
[DQN-Mod] Ep 831 R nd100 0.68 thr 89	0.59 avgR100	158.50 eps 0.020 Qval	-3.37 la
[DQN-Mod] Ep 832 R nd100 0.69 thr 309	184.46 avgR100	161.05 eps 0.020 Qval	-2.65 la
[DQN-Mod] Ep 833 R nd100 0.69 thr 455	96.10 avgR100	159.43 eps 0.020 Qval	-3.58 la
[DQN-Mod] Ep 834 R nd100 0.69 thr 254	222.75 avgR100	159.95 eps 0.020 Qval	-2.81 la
[DQN-Mod] Ep 835 R nd100 0.69 thr 171	259.17 avgR100	161.70 eps 0.020 Qval	-3.91 la
[DQN-Mod] Ep 836 R nd100 0.69 thr 159	240.17 avgR100	161.97 eps 0.020 Qval	-2.63 la
[DQN-Mod] Ep 837 R nd100 0.70 thr 200	265.09 avgR100	165.06 eps 0.020 Qval	-2.87 la
[DQN-Mod] Ep 838 R nd100 0.71 thr 176	302.19 avgR100	168.16 eps 0.020 Qval	-3.81 la
[DQN-Mod] Ep 839 R nd100 0.70 thr 138	20.29 avgR100	165.73 eps 0.020 Qval	-2.87 la

[DQN-Mod] Ep 840 R nd100 0.70 thr 274	215.84 avgR100	165.32 eps 0.020 Qval	-2.12 la
[DQN-Mod] Ep 841 R nd100 0.70 thr 173	-15.86 avgR100	165.40 eps 0.020 Qval	-2.29 la
[DQN-Mod] Ep 842 R nd100 0.70 thr 147	276.65 avgR100	166.10 eps 0.020 Qval	-3.06 la
[DQN-Mod] Ep 843 R nd100 0.70 thr 269	221.87 avgR100	166.08 eps 0.020 Qval	-4.17 la
[DQN-Mod] Ep 844 R nd100 0.71 thr 166	244.39 avgR100	168.81 eps 0.020 Qval	-3.37 la
[DQN-Mod] Ep 845 R nd100 0.72 thr 169	275.95 avgR100	171.84 eps 0.020 Qval	-3.84 la
[DQN-Mod] Ep 846 R nd100 0.72 thr 331	201.03 avgR100	171.66 eps 0.020 Qval	-4.67 la
[DQN-Mod] Ep 847 R nd100 0.73 thr 415	158.87 avgR100	173.62 eps 0.020 Qval	-5.60 la
[DQN-Mod] Ep 848 R nd100 0.72 thr 127	-27.13 avgR100	171.22 eps 0.020 Qval	-5.14 la
[DQN-Mod] Ep 849 R nd100 0.72 thr 156	294.13 avgR100	171.74 eps 0.020 Qval	-5.03 la
[DQN-Mod] Ep 850 R nd100 0.72 thr 297	232.49 avgR100	171.34 eps 0.020 Qval	-3.21 la
[DQN-Mod] Ep 851 R nd100 0.72 thr 200	230.63 avgR100	171.11 eps 0.020 Qval	-1.64 la
[DQN-Mod] Ep 852 R nd100 0.72 thr 133	259.71 avgR100	171.44 eps 0.020 Qval	-1.29 la
[DQN-Mod] Ep 853 R nd100 0.72 thr 254	231.26 avgR100	171.32 eps 0.020 Qval	-1.27 la
[DQN-Mod] Ep 854 R nd100 0.72 thr 147	265.40 avgR100	171.24 eps 0.020 Qval	-1.66 la
[DQN-Mod] Ep 855 R nd100 0.73 thr 263	213.10 avgR100	170.99 eps 0.020 Qval	-1.94 la
[DQN-Mod] Ep 856 R nd100 0.73 thr 130	253.42 avgR100	170.96 eps 0.020 Qval	-2.27 la
[DQN-Mod] Ep 857 R nd100 0.72 thr 124	4.94 avgR100	168.15 eps 0.020 Qval	-2.30 la
[DQN-Mod] Ep 858 R nd100 0.72 thr 273	211.33 avgR100	167.82 eps 0.020 Qval	-1.77 la
[DQN-Mod] Ep 859 R nd100 0.73 thr 134	251.63 avgR100	170.75 eps 0.020 Qval	-2.02 la
[DQN-Mod] Ep 860 R nd100 0.73 thr 281	181.70 avgR100	170.52 eps 0.020 Qval	-1.15 la
[DQN-Mod] Ep 861 R nd100 0.73 thr 184	263.89 avgR100	171.01 eps 0.020 Qval	-1.89 la
[DQN-Mod] Ep 862 R nd100 0.73 thr 186	223.34 avgR100	171.05 eps 0.020 Qval	-2.48 la
[DQN-Mod] Ep 863 R nd100 0.72 thr 143	4.84 avgR100	168.47 eps 0.020 Qval	-2.35 la
[DQN-Mod] Ep 864 R nd100 0.71 thr 134	235.56 avgR100	168.72 eps 0.020 Qval	-1.39 la
[DQN-Mod] Ep 865 R nd100 0.71 thr 167	226.49 avgR100	168.20 eps 0.020 Qval	-2.31 la
[DQN-Mod] Ep 866 R nd100 0.71 thr 166	272.54 avgR100	168.64 eps 0.020 Qval	-2.44 la
[DQN-Mod] Ep 867 R nd100 0.71 thr 196	262.31 avgR100	168.67 eps 0.020 Qval	-0.80 la
[DQN-Mod] Ep 868 R nd100 0.70 thr 205	193.46 avgR100	169.31 eps 0.020 Qval	-1.62 la

[DQN-Mod] Ep 869 R	284.82 avgR100	169.48 eps 0.020	Qval	-1.67 la
nd100 0.70 thr 167				
[DQN-Mod] Ep 870 R	265.55 avgR100	169.18 eps 0.020	Qval	-0.95 la
nd100 0.70 thr 163				
[DQN-Mod] Ep 871 R	128.21 avgR100	170.69 eps 0.020	Qval	-0.70 la
nd100 0.70 thr 284				
[DQN-Mod] Ep 872 R	244.05 avgR100	174.47 eps 0.020	Qval	-1.47 la
nd100 0.71 thr 146				
[DQN-Mod] Ep 873 R	-38.16 avgR100	171.61 eps 0.020	Qval	-1.06 la
nd100 0.70 thr 122				
[DQN-Mod] Ep 874 R	-36.79 avgR100	171.89 eps 0.020	Qval	-1.88 la
nd100 0.70 thr 124				
[DQN-Mod] Ep 875 R	255.88 avgR100	171.91 eps 0.020	Qval	-2.27 la
nd100 0.70 thr 159				
[DQN-Mod] Ep 876 R	198.26 avgR100	171.25 eps 0.020	Qval	-2.82 la
nd100 0.70 thr 270				
[DQN-Mod] Ep 877 R	295.93 avgR100	172.51 eps 0.020	Qval	-3.37 la
nd100 0.70 thr 124				
[DQN-Mod] Ep 878 R	260.74 avgR100	172.40 eps 0.020	Qval	-0.74 la
nd100 0.70 thr 190				
[DQN-Mod] Ep 879 R	229.60 avgR100	172.27 eps 0.020	Qval	-2.78 la
nd100 0.70 thr 279				
[DQN-Mod] Ep 880 R	245.63 avgR100	172.19 eps 0.020	Qval	-2.40 la
nd100 0.70 thr 211				
[DQN-Mod] Ep 881 R	307.81 avgR100	175.31 eps 0.020	Qval	-2.36 la
nd100 0.71 thr 138				
[DQN-Mod] Ep 882 R	292.44 avgR100	178.15 eps 0.020	Qval	-1.30 la
nd100 0.72 thr 106				
[DQN-Mod] Ep 883 R	245.07 avgR100	181.17 eps 0.020	Qval	-2.35 la
nd100 0.72 thr 168				
[DQN-Mod] Ep 884 R	243.80 avgR100	181.22 eps 0.020	Qval	-2.37 la
nd100 0.72 thr 135				
[DQN-Mod] Ep 885 R	253.61 avgR100	180.77 eps 0.020	Qval	-0.96 la
nd100 0.72 thr 133				
[DQN-Mod] Ep 886 R	-15.18 avgR100	178.12 eps 0.020	Qval	-0.75 la
nd100 0.71 thr 111				
[DQN-Mod] Ep 887 R	167.52 avgR100	180.29 eps 0.020	Qval	0.19 la
nd100 0.71 thr 299				
[DQN-Mod] Ep 888 R	253.71 avgR100	180.53 eps 0.020	Qval	-1.38 la
nd100 0.71 thr 221				
[DQN-Mod] Ep 889 R	224.20 avgR100	183.09 eps 0.020	Qval	-1.50 la
nd100 0.72 thr 240				
[DQN-Mod] Ep 890 R	-126.00 avgR100	182.28 eps 0.020	Qval	-3.58 la
nd100 0.72 thr 653				
[DQN-Mod] Ep 891 R	186.65 avgR100	182.07 eps 0.020	Qval	-2.39 la
nd100 0.71 thr 142				
[DQN-Mod] Ep 892 R	-70.47 avgR100	178.73 eps 0.020	Qval	-2.14 la
nd100 0.70 thr 534				
[DQN-Mod] Ep 893 R	267.64 avgR100	182.00 eps 0.020	Qval	-1.28 la
nd100 0.71 thr 214				
[DQN-Mod] Ep 894 R	122.88 avgR100	183.63 eps 0.020	Qval	-1.25 la
nd100 0.72 thr 398				
[DQN-Mod] Ep 895 R	83.27 avgR100	181.68 eps 0.020	Qval	-0.99 la
nd100 0.72 thr 497				
[DQN-Mod] Ep 896 R	-5.59 avgR100	181.22 eps 0.020	Qval	-0.72 la
nd100 0.72 thr 124				
[DQN-Mod] Ep 897 R	178.23 avgR100	180.03 eps 0.020	Qval	-2.61 la
nd100 0.72 thr 254				
[DQN-Mod] Ep 898 R	235.35 avgR100	181.27 eps 0.020	Qval	-1.96 la
nd100 0.72 thr 180				

[DQN-Mod] Ep 899 R 208.42 avgR100 181.76 eps 0.020 Qval -1.85 la
nd100 0.73 thr 278
[DQN-Mod] Ep 900 R 177.24 avgR100 183.54 eps 0.020 Qval -1.06 la
nd100 0.74 thr 312
[DQN-Mod] Ep 901 R 231.67 avgR100 183.12 eps 0.020 Qval -1.71 la
nd100 0.74 thr 220
[DQN-Mod] Ep 902 R 233.32 avgR100 183.01 eps 0.020 Qval -1.46 la
nd100 0.74 thr 199
[DQN-Mod] Ep 903 R 240.52 avgR100 182.91 eps 0.020 Qval -1.89 la
nd100 0.74 thr 215
[DQN-Mod] Ep 904 R 276.06 avgR100 183.22 eps 0.020 Qval -1.41 la
nd100 0.74 thr 174
[DQN-Mod] Ep 905 R 240.85 avgR100 185.93 eps 0.020 Qval -1.96 la
nd100 0.75 thr 235
[DQN-Mod] Ep 906 R 217.18 avgR100 185.65 eps 0.020 Qval -2.46 la
nd100 0.75 thr 196
[DQN-Mod] Ep 907 R 8.63 avgR100 183.52 eps 0.020 Qval -1.44 la
nd100 0.74 thr 130
[DQN-Mod] Ep 908 R 8.76 avgR100 181.17 eps 0.020 Qval -2.24 la
nd100 0.73 thr 110
[DQN-Mod] Ep 909 R 212.16 avgR100 181.97 eps 0.020 Qval -1.18 la
nd100 0.73 thr 268
[DQN-Mod] Ep 910 R 244.58 avgR100 181.70 eps 0.020 Qval 0.47 la
nd100 0.73 thr 213
[DQN-Mod] Ep 911 R 247.27 avgR100 181.27 eps 0.020 Qval 0.02 la
nd100 0.73 thr 265
[DQN-Mod] Ep 912 R 255.90 avgR100 181.73 eps 0.020 Qval -0.03 la
nd100 0.73 thr 191
[DQN-Mod] Ep 913 R 143.43 avgR100 181.30 eps 0.020 Qval 0.90 la
nd100 0.73 thr 327
[DQN-Mod] Ep 914 R 241.35 avgR100 183.74 eps 0.020 Qval 0.60 la
nd100 0.74 thr 165
[DQN-Mod] Ep 915 R 231.38 avgR100 184.06 eps 0.020 Qval 1.51 la
nd100 0.74 thr 278
[DQN-Mod] Ep 916 R 4.47 avgR100 182.11 eps 0.020 Qval 0.88 la
nd100 0.74 thr 93
[DQN-Mod] Ep 917 R 243.78 avgR100 182.22 eps 0.020 Qval 0.65 la
nd100 0.74 thr 190
[DQN-Mod] Ep 918 R -25.45 avgR100 180.14 eps 0.020 Qval 0.90 la
nd100 0.73 thr 119
[DQN-Mod] Ep 919 R 254.96 avgR100 182.57 eps 0.020 Qval 1.51 la
nd100 0.74 thr 243
[DQN-Mod] Ep 920 R 163.99 avgR100 184.30 eps 0.020 Qval 1.39 la
nd100 0.75 thr 417
[DQN-Mod] Ep 921 R 233.84 avgR100 184.46 eps 0.020 Qval 0.46 la
nd100 0.75 thr 260
[DQN-Mod] Ep 922 R 268.97 avgR100 186.89 eps 0.020 Qval 1.68 la
nd100 0.76 thr 222
[DQN-Mod] Ep 923 R 253.54 avgR100 187.95 eps 0.020 Qval 2.83 la
nd100 0.76 thr 218
[DQN-Mod] Ep 924 R 236.00 avgR100 188.63 eps 0.020 Qval 2.00 la
nd100 0.77 thr 170
[DQN-Mod] Ep 925 R 250.57 avgR100 191.35 eps 0.020 Qval 2.25 la
nd100 0.78 thr 235
[DQN-Mod] Ep 926 R -43.02 avgR100 188.43 eps 0.020 Qval 1.77 la
nd100 0.77 thr 91
[DQN-Mod] Ep 927 R 242.58 avgR100 188.22 eps 0.020 Qval 1.52 la
nd100 0.77 thr 215
[DQN-Mod] Ep 928 R 201.99 avgR100 188.23 eps 0.020 Qval 1.93 la
nd100 0.76 thr 237

[DQN-Mod] Ep nd100 0.76	929 R thr 169	239.90	avgR100	188.42	eps 0.020	Qval	1.35	1a
[DQN-Mod] Ep nd100 0.76	930 R thr 276	218.51	avgR100	187.97	eps 0.020	Qval	0.42	1a
[DQN-Mod] Ep nd100 0.77	931 R thr 252	254.71	avgR100	190.51	eps 0.020	Qval	1.92	1a
[DQN-Mod] Ep nd100 0.77	932 R thr 222	246.78	avgR100	191.13	eps 0.020	Qval	2.78	1a
[DQN-Mod] Ep nd100 0.76	933 R thr 142	17.85	avgR100	190.35	eps 0.020	Qval	3.66	1a
[DQN-Mod] Ep nd100 0.76	934 R thr 176	271.51	avgR100	190.84	eps 0.020	Qval	2.89	1a
[DQN-Mod] Ep nd100 0.76	935 R thr 151	276.77	avgR100	191.01	eps 0.020	Qval	2.89	1a
[DQN-Mod] Ep nd100 0.75	936 R thr 99	-21.26	avgR100	188.40	eps 0.020	Qval	3.92	1a
[DQN-Mod] Ep nd100 0.75	937 R thr 204	223.77	avgR100	187.98	eps 0.020	Qval	4.21	1a
[DQN-Mod] Ep nd100 0.75	938 R thr 106	281.73	avgR100	187.78	eps 0.020	Qval	3.32	1a
[DQN-Mod] Ep nd100 0.76	939 R thr 246	248.26	avgR100	190.06	eps 0.020	Qval	2.86	1a
[DQN-Mod] Ep nd100 0.76	940 R thr 266	201.81	avgR100	189.92	eps 0.020	Qval	3.60	1a
[DQN-Mod] Ep nd100 0.77	941 R thr 206	240.48	avgR100	192.48	eps 0.020	Qval	4.28	1a
[DQN-Mod] Ep nd100 0.77	942 R thr 288	231.80	avgR100	192.03	eps 0.020	Qval	4.58	1a
[DQN-Mod] Ep nd100 0.77	943 R thr 108	266.18	avgR100	192.48	eps 0.020	Qval	5.28	1a
[DQN-Mod] Ep nd100 0.77	944 R thr 346	214.33	avgR100	192.18	eps 0.020	Qval	4.64	1a
[DQN-Mod] Ep nd100 0.77	945 R thr 216	248.87	avgR100	191.91	eps 0.020	Qval	3.03	1a
[DQN-Mod] Ep nd100 0.77	946 R thr 712	-12.79	avgR100	189.77	eps 0.020	Qval	4.06	1a
[DQN-Mod] Ep nd100 0.77	947 R thr 201	219.55	avgR100	190.37	eps 0.020	Qval	5.54	1a
[DQN-Mod] Ep nd100 0.78	948 R thr 346	166.89	avgR100	192.31	eps 0.020	Qval	4.10	1a
[DQN-Mod] Ep nd100 0.78	949 R thr 224	229.07	avgR100	191.66	eps 0.020	Qval	4.53	1a
[DQN-Mod] Ep nd100 0.78	950 R thr 218	270.36	avgR100	192.04	eps 0.020	Qval	5.37	1a
[DQN-Mod] Ep nd100 0.78	951 R thr 192	275.12	avgR100	192.49	eps 0.020	Qval	5.37	1a
[DQN-Mod] Ep nd100 0.78	952 R thr 207	253.70	avgR100	192.43	eps 0.020	Qval	5.29	1a
[DQN-Mod] Ep nd100 0.78	953 R thr 321	146.91	avgR100	191.58	eps 0.020	Qval	3.40	1a
[DQN-Mod] Ep nd100 0.78	954 R thr 161	286.19	avgR100	191.79	eps 0.020	Qval	5.22	1a
[DQN-Mod] Ep nd100 0.78	955 R thr 271	201.59	avgR100	191.68	eps 0.020	Qval	5.49	1a
[DQN-Mod] Ep nd100 0.78	956 R thr 230	251.54	avgR100	191.66	eps 0.020	Qval	4.25	1a
[DQN-Mod] Ep nd100 0.78	957 R thr 133	-15.00	avgR100	191.46	eps 0.020	Qval	3.67	1a
[DQN-Mod] Ep nd100 0.78	958 R thr 175	228.92	avgR100	191.63	eps 0.020	Qval	4.01	1a

[DQN-Mod] Ep nd100 0.78	R thr 241	213.26	avgR100	191.25	eps 0.020	Qval	4.67	1a
[DQN-Mod] Ep nd100 0.78	R thr 175	231.26	avgR100	191.75	eps 0.020	Qval	5.21	1a
[DQN-Mod] Ep nd100 0.78	R thr 195	223.00	avgR100	191.34	eps 0.020	Qval	4.95	1a
[DQN-Mod] Ep nd100 0.78	R thr 293	201.02	avgR100	191.11	eps 0.020	Qval	5.77	1a
[DQN-Mod] Ep nd100 0.78	R thr 299	-156.32	avgR100	189.50	eps 0.020	Qval	5.61	1a
[DQN-Mod] Ep nd100 0.79	R thr 246	232.36	avgR100	189.47	eps 0.020	Qval	5.98	1a
[DQN-Mod] Ep nd100 0.78	R thr 491	52.95	avgR100	187.73	eps 0.020	Qval	6.22	1a
[DQN-Mod] Ep nd100 0.78	R thr 247	261.11	avgR100	187.62	eps 0.020	Qval	5.89	1a
[DQN-Mod] Ep nd100 0.77	R thr 130	-6.74	avgR100	184.93	eps 0.020	Qval	5.42	1a
[DQN-Mod] Ep nd100 0.78	R thr 176	233.14	avgR100	185.33	eps 0.020	Qval	5.97	1a
[DQN-Mod] Ep nd100 0.78	R thr 143	288.29	avgR100	185.36	eps 0.020	Qval	7.01	1a
[DQN-Mod] Ep nd100 0.78	R thr 179	231.63	avgR100	185.02	eps 0.020	Qval	7.91	1a
[DQN-Mod] Ep nd100 0.79	R thr 116	283.89	avgR100	186.58	eps 0.020	Qval	8.26	1a
[DQN-Mod] Ep nd100 0.79	R thr 242	240.93	avgR100	186.55	eps 0.020	Qval	8.33	1a
[DQN-Mod] Ep nd100 0.80	R thr 166	270.15	avgR100	189.63	eps 0.020	Qval	8.27	1a
[DQN-Mod] Ep nd100 0.81	R thr 159	244.20	avgR100	192.44	eps 0.020	Qval	8.17	1a
[DQN-Mod] Ep nd100 0.81	R thr 152	244.62	avgR100	192.33	eps 0.020	Qval	8.02	1a
[DQN-Mod] Ep nd100 0.81	R thr 216	237.70	avgR100	192.72	eps 0.020	Qval	8.65	1a
[DQN-Mod] Ep nd100 0.81	R thr 293	195.12	avgR100	191.71	eps 0.020	Qval	8.89	1a
[DQN-Mod] Ep nd100 0.81	R thr 247	250.16	avgR100	191.61	eps 0.020	Qval	10.04	1a
[DQN-Mod] Ep nd100 0.81	R thr 306	185.20	avgR100	191.16	eps 0.020	Qval	10.30	1a
[DQN-Mod] Ep nd100 0.81	R thr 222	261.06	avgR100	191.32	eps 0.020	Qval	9.79	1a
[DQN-Mod] Ep nd100 0.80	R thr 156	236.26	avgR100	190.60	eps 0.020	Qval	10.15	1a
[DQN-Mod] Ep nd100 0.80	R thr 124	289.87	avgR100	190.58	eps 0.020	Qval	9.89	1a
[DQN-Mod] Ep nd100 0.81	R thr 380	153.80	avgR100	189.67	eps 0.020	Qval	10.35	1a
[DQN-Mod] Ep nd100 0.80	R thr 664	-138.12	avgR100	185.85	eps 0.020	Qval	10.03	1a
[DQN-Mod] Ep nd100 0.80	R thr 247	207.72	avgR100	185.39	eps 0.020	Qval	11.33	1a
[DQN-Mod] Ep nd100 0.81	R thr 317	208.24	avgR100	187.62	eps 0.020	Qval	11.70	1a
[DQN-Mod] Ep nd100 0.82	R thr 332	173.50	avgR100	187.68	eps 0.020	Qval	11.88	1a
[DQN-Mod] Ep nd100 0.82	R thr 239	243.84	avgR100	187.58	eps 0.020	Qval	11.51	1a

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[DQN-Mod] Ep 989 | R 163.93 | avgR100 186.98 | eps 0.020 | Qval 12.25 | la
nd100 0.81 | thr 211
[DQN-Mod] Ep 990 | R 222.70 | avgR100 190.47 | eps 0.020 | Qval 12.90 | la
nd100 0.82 | thr 254
[DQN-Mod] Ep 991 | R -20.93 | avgR100 188.39 | eps 0.020 | Qval 11.31 | la
nd100 0.82 | thr 178
[DQN-Mod] Ep 992 | R -15.61 | avgR100 188.94 | eps 0.020 | Qval 11.35 | la
nd100 0.82 | thr 111
[DQN-Mod] Ep 993 | R 251.55 | avgR100 188.78 | eps 0.020 | Qval 13.04 | la
nd100 0.82 | thr 233
[DQN-Mod] Ep 994 | R 257.76 | avgR100 190.13 | eps 0.020 | Qval 12.35 | la
nd100 0.82 | thr 231
[DQN-Mod] Ep 995 | R 180.25 | avgR100 191.10 | eps 0.020 | Qval 11.15 | la
nd100 0.82 | thr 313
[DQN-Mod] Ep 996 | R 73.93 | avgR100 191.89 | eps 0.020 | Qval 12.65 | la
nd100 0.83 | thr 468
[DQN-Mod] Ep 997 | R 221.66 | avgR100 192.33 | eps 0.020 | Qval 12.05 | la
nd100 0.83 | thr 312
[DQN-Mod] Ep 998 | R 230.28 | avgR100 192.28 | eps 0.020 | Qval 14.04 | la
nd100 0.83 | thr 243
[DQN-Mod] Ep 999 | R 60.65 | avgR100 190.80 | eps 0.020 | Qval 13.01 | la
nd100 0.83 | thr 104
[DQN-Mod] Ep 1000 | R 202.00 | avgR100 191.05 | eps 0.020 | Qval 12.25 | la
nd100 0.83 | thr 256
[DQN-Mod] finished 1000 episodes in 3.0 min (final avgR100 = 191.05)

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#####
# TRAINING: DDQN-Mod (modified=True, double=True)
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[DDQN-Mod] Ep 1 | R -102.54 | avgR100 -102.54 | eps 0.995 | Qval 0.05 | 1
and100 0.00 | thr 41
[DDQN-Mod] Ep 2 | R -98.27 | avgR100 -100.40 | eps 0.990 | Qval 0.05 | 1
and100 0.00 | thr 50
[DDQN-Mod] Ep 3 | R -307.70 | avgR100 -169.50 | eps 0.985 | Qval 0.05 | 1
and100 0.00 | thr 87
[DDQN-Mod] Ep 4 | R -166.45 | avgR100 -168.74 | eps 0.980 | Qval 0.05 | 1
and100 0.00 | thr 75
[DDQN-Mod] Ep 5 | R -122.17 | avgR100 -159.43 | eps 0.975 | Qval 0.05 | 1
and100 0.00 | thr 54
[DDQN-Mod] Ep 6 | R -281.18 | avgR100 -179.72 | eps 0.970 | Qval 0.05 | 1
and100 0.00 | thr 69
[DDQN-Mod] Ep 7 | R -391.00 | avgR100 -209.90 | eps 0.966 | Qval 0.05 | 1
and100 0.00 | thr 58
[DDQN-Mod] Ep 8 | R -64.07 | avgR100 -191.67 | eps 0.961 | Qval 0.05 | 1
and100 0.12 | thr 64
[DDQN-Mod] Ep 9 | R -240.66 | avgR100 -197.12 | eps 0.956 | Qval 0.05 | 1
and100 0.11 | thr 67
[DDQN-Mod] Ep 10 | R -122.26 | avgR100 -189.63 | eps 0.951 | Qval 0.05 | 1
and100 0.10 | thr 80
[DDQN-Mod] Ep 11 | R -137.70 | avgR100 -184.91 | eps 0.946 | Qval 0.05 | 1
and100 0.09 | thr 78
[DDQN-Mod] Ep 12 | R -336.48 | avgR100 -197.54 | eps 0.942 | Qval 0.14 | 1
and100 0.08 | thr 71
[DDQN-Mod] Ep 13 | R -243.07 | avgR100 -201.04 | eps 0.937 | Qval -0.18 | 1
and100 0.08 | thr 93
[DDQN-Mod] Ep 14 | R -195.35 | avgR100 -200.64 | eps 0.932 | Qval -0.48 | 1
and100 0.07 | thr 121

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[DDQN-Mod]	Ep 15	R	-172.56	avgR100	-198.76	eps 0.928	Qval	-0.60	1
and100 0.07	thr 56								
[DDQN-Mod]	Ep 16	R	-491.52	avgR100	-217.06	eps 0.923	Qval	-0.44	1
and100 0.06	thr 80								
[DDQN-Mod]	Ep 17	R	-324.77	avgR100	-223.40	eps 0.918	Qval	-0.59	1
and100 0.06	thr 73								
[DDQN-Mod]	Ep 18	R	-275.27	avgR100	-226.28	eps 0.914	Qval	-0.73	1
and100 0.06	thr 106								
[DDQN-Mod]	Ep 19	R	-348.87	avgR100	-232.73	eps 0.909	Qval	-0.66	1
and100 0.05	thr 87								
[DDQN-Mod]	Ep 20	R	-42.96	avgR100	-223.24	eps 0.905	Qval	-0.26	1
and100 0.10	thr 85								
[DDQN-Mod]	Ep 21	R	-136.45	avgR100	-219.11	eps 0.900	Qval	-0.49	1
and100 0.10	thr 68								
[DDQN-Mod]	Ep 22	R	-115.12	avgR100	-214.38	eps 0.896	Qval	-0.42	1
and100 0.09	thr 89								
[DDQN-Mod]	Ep 23	R	-305.09	avgR100	-218.33	eps 0.891	Qval	-0.36	1
and100 0.09	thr 76								
[DDQN-Mod]	Ep 24	R	-184.64	avgR100	-216.92	eps 0.887	Qval	-0.46	1
and100 0.08	thr 100								
[DDQN-Mod]	Ep 25	R	-94.47	avgR100	-212.02	eps 0.882	Qval	-0.34	1
and100 0.08	thr 62								
[DDQN-Mod]	Ep 26	R	-199.04	avgR100	-211.53	eps 0.878	Qval	-0.17	1
and100 0.08	thr 78								
[DDQN-Mod]	Ep 27	R	-284.66	avgR100	-214.23	eps 0.873	Qval	0.04	1
and100 0.07	thr 52								
[DDQN-Mod]	Ep 28	R	-181.22	avgR100	-213.05	eps 0.869	Qval	-0.25	1
and100 0.07	thr 66								
[DDQN-Mod]	Ep 29	R	-14.17	avgR100	-206.20	eps 0.865	Qval	-0.16	1
and100 0.07	thr 70								
[DDQN-Mod]	Ep 30	R	-20.58	avgR100	-200.01	eps 0.860	Qval	-0.01	1
and100 0.10	thr 47								
[DDQN-Mod]	Ep 31	R	-93.22	avgR100	-196.56	eps 0.856	Qval	-0.07	1
and100 0.10	thr 69								
[DDQN-Mod]	Ep 32	R	-224.29	avgR100	-197.43	eps 0.852	Qval	0.19	1
and100 0.09	thr 89								
[DDQN-Mod]	Ep 33	R	-250.08	avgR100	-199.03	eps 0.848	Qval	-0.15	1
and100 0.09	thr 60								
[DDQN-Mod]	Ep 34	R	-111.70	avgR100	-196.46	eps 0.843	Qval	-0.01	1
and100 0.09	thr 100								
[DDQN-Mod]	Ep 35	R	-98.64	avgR100	-193.66	eps 0.839	Qval	0.11	1
and100 0.09	thr 56								
[DDQN-Mod]	Ep 36	R	-161.80	avgR100	-192.78	eps 0.835	Qval	-0.25	1
and100 0.08	thr 88								
[DDQN-Mod]	Ep 37	R	-301.25	avgR100	-195.71	eps 0.831	Qval	0.09	1
and100 0.08	thr 61								
[DDQN-Mod]	Ep 38	R	-183.17	avgR100	-195.38	eps 0.827	Qval	0.33	1
and100 0.08	thr 103								
[DDQN-Mod]	Ep 39	R	-90.21	avgR100	-192.68	eps 0.822	Qval	0.04	1
and100 0.08	thr 72								

[DDQN-Mod] and100 0.07	Ep 40 thr 63	R	35.66		avgR100	-186.97		eps 0.818		Qval -0.07		1
[DDQN-Mod] and100 0.07	Ep 41 thr 66	R	-162.81		avgR100	-186.39		eps 0.814		Qval -0.00		1
[DDQN-Mod] and100 0.07	Ep 42 thr 58	R	-103.65		avgR100	-184.42		eps 0.810		Qval -0.10		1
[DDQN-Mod] and100 0.07	Ep 43 thr 52	R	-292.50		avgR100	-186.93		eps 0.806		Qval 0.08		1
[DDQN-Mod] and100 0.07	Ep 44 thr 59	R	-123.69		avgR100	-185.49		eps 0.802		Qval 0.09		1
[DDQN-Mod] and100 0.07	Ep 45 thr 94	R	-244.25		avgR100	-186.80		eps 0.798		Qval 0.19		1
[DDQN-Mod] and100 0.07	Ep 46 thr 119	R	-130.69		avgR100	-185.58		eps 0.794		Qval -0.07		1
[DDQN-Mod] and100 0.09	Ep 47 thr 72	R	-54.28		avgR100	-182.78		eps 0.790		Qval 0.49		1
[DDQN-Mod] and100 0.08	Ep 48 thr 73	R	-105.78		avgR100	-181.18		eps 0.786		Qval -0.08		1
[DDQN-Mod] and100 0.08	Ep 49 thr 55	R	-130.06		avgR100	-180.14		eps 0.782		Qval 0.20		1
[DDQN-Mod] and100 0.08	Ep 50 thr 80	R	-132.16		avgR100	-179.18		eps 0.778		Qval 0.20		1
[DDQN-Mod] and100 0.08	Ep 51 thr 54	R	-102.26		avgR100	-177.67		eps 0.774		Qval -0.21		1
[DDQN-Mod] and100 0.08	Ep 52 thr 85	R	-116.45		avgR100	-176.49		eps 0.771		Qval -0.03		1
[DDQN-Mod] and100 0.08	Ep 53 thr 83	R	-218.30		avgR100	-177.28		eps 0.767		Qval 0.01		1
[DDQN-Mod] and100 0.07	Ep 54 thr 63	R	-138.01		avgR100	-176.55		eps 0.763		Qval -0.24		1
[DDQN-Mod] and100 0.07	Ep 55 thr 111	R	-190.75		avgR100	-176.81		eps 0.759		Qval -0.21		1
[DDQN-Mod] and100 0.07	Ep 56 thr 76	R	-119.33		avgR100	-175.78		eps 0.755		Qval -0.02		1
[DDQN-Mod] and100 0.07	Ep 57 thr 95	R	-186.90		avgR100	-175.98		eps 0.751		Qval -0.00		1
[DDQN-Mod] and100 0.07	Ep 58 thr 132	R	-158.05		avgR100	-175.67		eps 0.748		Qval -0.09		1
[DDQN-Mod] and100 0.07	Ep 59 thr 74	R	-103.41		avgR100	-174.45		eps 0.744		Qval 0.10		1
[DDQN-Mod] and100 0.08	Ep 60 thr 117	R	-12.28		avgR100	-171.74		eps 0.740		Qval -0.18		1
[DDQN-Mod] and100 0.08	Ep 61 thr 60	R	-167.60		avgR100	-171.68		eps 0.737		Qval 0.11		1
[DDQN-Mod] and100 0.08	Ep 62 thr 78	R	-77.02		avgR100	-170.15		eps 0.733		Qval -0.06		1
[DDQN-Mod] and100 0.10	Ep 63 thr 53	R	22.71		avgR100	-167.09		eps 0.729		Qval -0.07		1
[DDQN-Mod] and100 0.09	Ep 64 thr 67	R	-107.44		avgR100	-166.16		eps 0.726		Qval -0.03		1
[DDQN-Mod] and100 0.09	Ep 65 thr 109	R	-77.49		avgR100	-164.79		eps 0.722		Qval -0.04		1
[DDQN-Mod] and100 0.09	Ep 66 thr 85	R	-109.24		avgR100	-163.95		eps 0.718		Qval -0.10		1
[DDQN-Mod] and100 0.09	Ep 67 thr 57	R	3.37		avgR100	-161.45		eps 0.715		Qval -0.08		1
[DDQN-Mod] and100 0.09	Ep 68 thr 113	R	-89.51		avgR100	-160.39		eps 0.711		Qval 0.11		1

[DDQN-Mod] and100 0.09	Ep 69 thr 61	R	-178.77		avgR100	-160.66		eps 0.708		Qval	-0.24		1
[DDQN-Mod] and100 0.09	Ep 70 thr 77	R	-115.00		avgR100	-160.01		eps 0.704		Qval	0.09		1
[DDQN-Mod] and100 0.08	Ep 71 thr 90	R	-101.79		avgR100	-159.19		eps 0.701		Qval	-0.05		1
[DDQN-Mod] and100 0.08	Ep 72 thr 94	R	-50.47		avgR100	-157.68		eps 0.697		Qval	-0.01		1
[DDQN-Mod] and100 0.08	Ep 73 thr 115	R	-147.64		avgR100	-157.54		eps 0.694		Qval	-0.02		1
[DDQN-Mod] and100 0.08	Ep 74 thr 102	R	-90.37		avgR100	-156.63		eps 0.690		Qval	-0.13		1
[DDQN-Mod] and100 0.08	Ep 75 thr 110	R	-81.14		avgR100	-155.63		eps 0.687		Qval	-0.22		1
[DDQN-Mod] and100 0.08	Ep 76 thr 96	R	-161.00		avgR100	-155.70		eps 0.683		Qval	0.25		1
[DDQN-Mod] and100 0.08	Ep 77 thr 87	R	-134.27		avgR100	-155.42		eps 0.680		Qval	-0.33		1
[DDQN-Mod] and100 0.08	Ep 78 thr 56	R	-135.66		avgR100	-155.17		eps 0.676		Qval	-0.13		1
[DDQN-Mod] and100 0.08	Ep 79 thr 70	R	-51.60		avgR100	-153.85		eps 0.673		Qval	0.08		1
[DDQN-Mod] and100 0.07	Ep 80 thr 103	R	-139.33		avgR100	-153.67		eps 0.670		Qval	0.51		1
[DDQN-Mod] and100 0.07	Ep 81 thr 97	R	-145.73		avgR100	-153.57		eps 0.666		Qval	0.05		1
[DDQN-Mod] and100 0.07	Ep 82 thr 117	R	-92.42		avgR100	-152.83		eps 0.663		Qval	0.23		1
[DDQN-Mod] and100 0.07	Ep 83 thr 91	R	-111.68		avgR100	-152.33		eps 0.660		Qval	0.43		1
[DDQN-Mod] and100 0.07	Ep 84 thr 109	R	-96.94		avgR100	-151.67		eps 0.656		Qval	-0.37		1
[DDQN-Mod] and100 0.07	Ep 85 thr 117	R	-92.60		avgR100	-150.98		eps 0.653		Qval	0.30		1
[DDQN-Mod] and100 0.07	Ep 86 thr 149	R	3.65		avgR100	-149.18		eps 0.650		Qval	0.63		1
[DDQN-Mod] and100 0.07	Ep 87 thr 59	R	-132.05		avgR100	-148.98		eps 0.647		Qval	0.12		1
[DDQN-Mod] and100 0.07	Ep 88 thr 88	R	-130.52		avgR100	-148.77		eps 0.643		Qval	0.29		1
[DDQN-Mod] and100 0.07	Ep 89 thr 100	R	-93.43		avgR100	-148.15		eps 0.640		Qval	0.34		1
[DDQN-Mod] and100 0.07	Ep 90 thr 92	R	-80.85		avgR100	-147.40		eps 0.637		Qval	0.41		1
[DDQN-Mod] and100 0.07	Ep 91 thr 61	R	-126.32		avgR100	-147.17		eps 0.634		Qval	0.36		1
[DDQN-Mod] and100 0.07	Ep 92 thr 89	R	-164.52		avgR100	-147.36		eps 0.631		Qval	0.34		1
[DDQN-Mod] and100 0.06	Ep 93 thr 110	R	-114.17		avgR100	-147.00		eps 0.627		Qval	0.38		1
[DDQN-Mod] and100 0.06	Ep 94 thr 113	R	-168.89		avgR100	-147.24		eps 0.624		Qval	0.29		1
[DDQN-Mod] and100 0.06	Ep 95 thr 122	R	-116.45		avgR100	-146.91		eps 0.621		Qval	0.67		1
[DDQN-Mod] and100 0.06	Ep 96 thr 114	R	-102.96		avgR100	-146.46		eps 0.618		Qval	0.63		1
[DDQN-Mod] and100 0.06	Ep 97 thr 127	R	-179.12		avgR100	-146.79		eps 0.615		Qval	0.26		1

[DDQN-Mod] Ep 98 R -122.55 avgR100 -146.54 eps 0.612 Qval 1.10 1
and100 0.06 thr 100
[DDQN-Mod] Ep 99 R -110.80 avgR100 -146.18 eps 0.609 Qval 0.59 1
and100 0.06 thr 127
[DDQN-Mod] Ep 100 R -119.09 avgR100 -145.91 eps 0.606 Qval 0.46 1
and100 0.06 thr 87
[DDQN-Mod] Ep 101 R -87.23 avgR100 -145.76 eps 0.603 Qval 0.82 1
and100 0.06 thr 80
[DDQN-Mod] Ep 102 R -100.68 avgR100 -145.78 eps 0.600 Qval 1.19 1
and100 0.06 thr 77
[DDQN-Mod] Ep 103 R -54.67 avgR100 -143.25 eps 0.597 Qval 0.97 1
and100 0.06 thr 69
[DDQN-Mod] Ep 104 R -122.99 avgR100 -142.82 eps 0.594 Qval 1.15 1
and100 0.06 thr 112
[DDQN-Mod] Ep 105 R -123.57 avgR100 -142.83 eps 0.591 Qval 0.69 1
and100 0.06 thr 95
[DDQN-Mod] Ep 106 R -165.37 avgR100 -141.67 eps 0.588 Qval 1.08 1
and100 0.06 thr 111
[DDQN-Mod] Ep 107 R -175.55 avgR100 -139.52 eps 0.585 Qval 0.77 1
and100 0.06 thr 101
[DDQN-Mod] Ep 108 R -71.39 avgR100 -139.59 eps 0.582 Qval 1.25 1
and100 0.05 thr 131
[DDQN-Mod] Ep 109 R -31.38 avgR100 -137.50 eps 0.579 Qval 1.61 1
and100 0.05 thr 127
[DDQN-Mod] Ep 110 R -72.94 avgR100 -137.01 eps 0.576 Qval 1.55 1
and100 0.05 thr 72
[DDQN-Mod] Ep 111 R -118.48 avgR100 -136.81 eps 0.573 Qval 1.96 1
and100 0.05 thr 84
[DDQN-Mod] Ep 112 R -107.84 avgR100 -134.53 eps 0.570 Qval 1.71 1
and100 0.05 thr 147
[DDQN-Mod] Ep 113 R -100.39 avgR100 -133.10 eps 0.568 Qval 1.91 1
and100 0.05 thr 75
[DDQN-Mod] Ep 114 R -56.63 avgR100 -131.71 eps 0.565 Qval 2.18 1
and100 0.05 thr 167
[DDQN-Mod] Ep 115 R -69.08 avgR100 -130.68 eps 0.562 Qval 2.02 1
and100 0.05 thr 76
[DDQN-Mod] Ep 116 R -167.97 avgR100 -127.44 eps 0.559 Qval 1.93 1
and100 0.05 thr 107
[DDQN-Mod] Ep 117 R -115.28 avgR100 -125.35 eps 0.556 Qval 1.77 1
and100 0.05 thr 150
[DDQN-Mod] Ep 118 R -48.58 avgR100 -123.08 eps 0.554 Qval 1.81 1
and100 0.05 thr 72
[DDQN-Mod] Ep 119 R -36.03 avgR100 -119.95 eps 0.551 Qval 1.98 1
and100 0.05 thr 96
[DDQN-Mod] Ep 120 R -160.24 avgR100 -121.13 eps 0.548 Qval 2.49 1
and100 0.04 thr 128
[DDQN-Mod] Ep 121 R -99.31 avgR100 -120.76 eps 0.545 Qval 1.93 1
and100 0.04 thr 62
[DDQN-Mod] Ep 122 R -107.79 avgR100 -120.68 eps 0.543 Qval 2.74 1
and100 0.04 thr 96
[DDQN-Mod] Ep 123 R -126.84 avgR100 -118.90 eps 0.540 Qval 2.20 1
and100 0.04 thr 79
[DDQN-Mod] Ep 124 R -92.69 avgR100 -117.98 eps 0.537 Qval 2.65 1
and100 0.04 thr 53
[DDQN-Mod] Ep 125 R -133.46 avgR100 -118.37 eps 0.534 Qval 2.62 1
and100 0.04 thr 92
[DDQN-Mod] Ep 126 R -193.89 avgR100 -118.32 eps 0.532 Qval 2.23 1
and100 0.04 thr 115
[DDQN-Mod] Ep 127 R -94.08 avgR100 -116.41 eps 0.529 Qval 2.90 1
and100 0.05 thr 147

[DDQN-Mod] and100 0.05	Ep 128 thr 162	R	-43.32		avgR100	-115.03		eps 0.526		Qval	3.07		1
[DDQN-Mod] and100 0.05	Ep 129 thr 98	R	-110.31		avgR100	-116.00		eps 0.524		Qval	2.83		1
[DDQN-Mod] and100 0.04	Ep 130 thr 123	R	-85.46		avgR100	-116.64		eps 0.521		Qval	3.25		1
[DDQN-Mod] and100 0.04	Ep 131 thr 103	R	-76.78		avgR100	-116.48		eps 0.519		Qval	2.84		1
[DDQN-Mod] and100 0.05	Ep 132 thr 74	R	25.25		avgR100	-113.98		eps 0.516		Qval	3.15		1
[DDQN-Mod] and100 0.05	Ep 133 thr 57	R	-75.48		avgR100	-112.24		eps 0.513		Qval	3.22		1
[DDQN-Mod] and100 0.06	Ep 134 thr 172	R	29.23		avgR100	-110.83		eps 0.511		Qval	3.21		1
[DDQN-Mod] and100 0.06	Ep 135 thr 127	R	-12.94		avgR100	-109.97		eps 0.508		Qval	3.41		1
[DDQN-Mod] and100 0.06	Ep 136 thr 70	R	-74.16		avgR100	-109.10		eps 0.506		Qval	2.97		1
[DDQN-Mod] and100 0.06	Ep 137 thr 145	R	-118.75		avgR100	-107.27		eps 0.503		Qval	3.34		1
[DDQN-Mod] and100 0.06	Ep 138 thr 168	R	-84.37		avgR100	-106.28		eps 0.501		Qval	4.18		1
[DDQN-Mod] and100 0.06	Ep 139 thr 95	R	-119.76		avgR100	-106.58		eps 0.498		Qval	4.24		1
[DDQN-Mod] and100 0.06	Ep 140 thr 74	R	-45.30		avgR100	-107.39		eps 0.496		Qval	4.04		1
[DDQN-Mod] and100 0.06	Ep 141 thr 93	R	-83.39		avgR100	-106.59		eps 0.493		Qval	4.22		1
[DDQN-Mod] and100 0.06	Ep 142 thr 757	R	-101.04		avgR100	-106.57		eps 0.491		Qval	5.16		1
[DDQN-Mod] and100 0.06	Ep 143 thr 99	R	-111.42		avgR100	-104.76		eps 0.488		Qval	4.97		1
[DDQN-Mod] and100 0.06	Ep 144 thr 199	R	-96.95		avgR100	-104.49		eps 0.486		Qval	5.48		1
[DDQN-Mod] and100 0.06	Ep 145 thr 89	R	-67.67		avgR100	-102.72		eps 0.483		Qval	5.45		1
[DDQN-Mod] and100 0.06	Ep 146 thr 174	R	-84.92		avgR100	-102.27		eps 0.481		Qval	5.88		1
[DDQN-Mod] and100 0.05	Ep 147 thr 121	R	-89.69		avgR100	-102.62		eps 0.479		Qval	6.15		1
[DDQN-Mod] and100 0.05	Ep 148 thr 158	R	-4.64		avgR100	-101.61		eps 0.476		Qval	5.94		1
[DDQN-Mod] and100 0.05	Ep 149 thr 76	R	-28.52		avgR100	-100.59		eps 0.474		Qval	6.10		1
[DDQN-Mod] and100 0.05	Ep 150 thr 151	R	-85.60		avgR100	-100.13		eps 0.471		Qval	6.28		1
[DDQN-Mod] and100 0.06	Ep 151 thr 122	R	68.82		avgR100	-98.42		eps 0.469		Qval	6.42		1
[DDQN-Mod] and100 0.06	Ep 152 thr 86	R	-91.60		avgR100	-98.17		eps 0.467		Qval	6.53		1
[DDQN-Mod] and100 0.06	Ep 153 thr 146	R	-68.46		avgR100	-96.67		eps 0.464		Qval	6.65		1
[DDQN-Mod] and100 0.06	Ep 154 thr 86	R	-93.86		avgR100	-96.23		eps 0.462		Qval	6.66		1
[DDQN-Mod] and100 0.06	Ep 155 thr 160	R	-10.49		avgR100	-94.43		eps 0.460		Qval	6.56		1
[DDQN-Mod] and100 0.06	Ep 156 thr 210	R	-86.88		avgR100	-94.10		eps 0.458		Qval	7.38		1

[DDQN-Mod]	Ep 157 R	-81.72 avgR100	-93.05 eps 0.455 Qval	7.69 1
and100 0.06	thr 166			
[DDQN-Mod]	Ep 158 R	43.30 avgR100	-91.04 eps 0.453 Qval	8.01 1
and100 0.07	thr 94			
[DDQN-Mod]	Ep 159 R	-55.79 avgR100	-90.56 eps 0.451 Qval	7.56 1
and100 0.07	thr 87			
[DDQN-Mod]	Ep 160 R	-85.51 avgR100	-91.29 eps 0.448 Qval	8.30 1
and100 0.06	thr 109			
[DDQN-Mod]	Ep 161 R	-105.47 avgR100	-90.67 eps 0.446 Qval	7.94 1
and100 0.06	thr 60			
[DDQN-Mod]	Ep 162 R	-21.93 avgR100	-90.12 eps 0.444 Qval	8.13 1
and100 0.06	thr 75			
[DDQN-Mod]	Ep 163 R	-142.84 avgR100	-91.78 eps 0.442 Qval	8.30 1
and100 0.05	thr 105			
[DDQN-Mod]	Ep 164 R	-54.08 avgR100	-91.24 eps 0.440 Qval	8.28 1
and100 0.05	thr 114			
[DDQN-Mod]	Ep 165 R	-58.99 avgR100	-91.06 eps 0.437 Qval	8.81 1
and100 0.06	thr 274			
[DDQN-Mod]	Ep 166 R	-30.81 avgR100	-90.27 eps 0.435 Qval	8.56 1
and100 0.06	thr 108			
[DDQN-Mod]	Ep 167 R	-129.18 avgR100	-91.60 eps 0.433 Qval	9.39 1
and100 0.06	thr 746			
[DDQN-Mod]	Ep 168 R	2.27 avgR100	-90.68 eps 0.431 Qval	10.02 1
and100 0.07	thr 141			
[DDQN-Mod]	Ep 169 R	-23.95 avgR100	-89.13 eps 0.429 Qval	10.14 1
and100 0.07	thr 158			
[DDQN-Mod]	Ep 170 R	-63.07 avgR100	-88.61 eps 0.427 Qval	10.34 1
and100 0.07	thr 166			
[DDQN-Mod]	Ep 171 R	-126.25 avgR100	-88.86 eps 0.424 Qval	10.14 1
and100 0.07	thr 132			
[DDQN-Mod]	Ep 172 R	-31.80 avgR100	-88.67 eps 0.422 Qval	10.29 1
and100 0.07	thr 147			
[DDQN-Mod]	Ep 173 R	-70.83 avgR100	-87.90 eps 0.420 Qval	10.86 1
and100 0.07	thr 104			
[DDQN-Mod]	Ep 174 R	-67.16 avgR100	-87.67 eps 0.418 Qval	10.05 1
and100 0.07	thr 114			
[DDQN-Mod]	Ep 175 R	-48.63 avgR100	-87.35 eps 0.416 Qval	10.88 1
and100 0.07	thr 107			
[DDQN-Mod]	Ep 176 R	-75.60 avgR100	-86.49 eps 0.414 Qval	11.03 1
and100 0.07	thr 78			
[DDQN-Mod]	Ep 177 R	-117.35 avgR100	-86.32 eps 0.412 Qval	10.55 1
and100 0.07	thr 104			
[DDQN-Mod]	Ep 178 R	-36.90 avgR100	-85.33 eps 0.410 Qval	11.20 1
and100 0.07	thr 143			
[DDQN-Mod]	Ep 179 R	-13.23 avgR100	-84.95 eps 0.408 Qval	11.57 1
and100 0.07	thr 116			
[DDQN-Mod]	Ep 180 R	-59.18 avgR100	-84.15 eps 0.406 Qval	11.48 1
and100 0.07	thr 110			
[DDQN-Mod]	Ep 181 R	-17.67 avgR100	-82.87 eps 0.404 Qval	11.41 1
and100 0.07	thr 142			
[DDQN-Mod]	Ep 182 R	-130.97 avgR100	-83.25 eps 0.402 Qval	12.02 1
and100 0.07	thr 690			
[DDQN-Mod]	Ep 183 R	-12.97 avgR100	-82.27 eps 0.400 Qval	12.03 1
and100 0.08	thr 139			
[DDQN-Mod]	Ep 184 R	-78.88 avgR100	-82.09 eps 0.398 Qval	12.43 1
and100 0.08	thr 257			

[DDQN-Mod]	Ep 185 R	-39.56 avgR100	-81.56 eps 0.396 Qval	12.44 1
and100 0.08	thr 77			
[DDQN-Mod]	Ep 186 R	-48.22 avgR100	-82.07 eps 0.394 Qval	12.60 1
and100 0.08	thr 210			
[DDQN-Mod]	Ep 187 R	-10.29 avgR100	-80.86 eps 0.392 Qval	12.92 1
and100 0.08	thr 127			
[DDQN-Mod]	Ep 188 R	-79.38 avgR100	-80.35 eps 0.390 Qval	13.37 1
and100 0.08	thr 112			
[DDQN-Mod]	Ep 189 R	-61.25 avgR100	-80.02 eps 0.388 Qval	13.23 1
and100 0.08	thr 127			
[DDQN-Mod]	Ep 190 R	-81.17 avgR100	-80.03 eps 0.386 Qval	13.59 1
and100 0.08	thr 290			
[DDQN-Mod]	Ep 191 R	-25.65 avgR100	-79.02 eps 0.384 Qval	13.50 1
and100 0.08	thr 84			
[DDQN-Mod]	Ep 192 R	-29.45 avgR100	-77.67 eps 0.382 Qval	14.33 1
and100 0.08	thr 101			
[DDQN-Mod]	Ep 193 R	-47.83 avgR100	-77.01 eps 0.380 Qval	13.19 1
and100 0.08	thr 106			
[DDQN-Mod]	Ep 194 R	-96.53 avgR100	-76.28 eps 0.378 Qval	14.17 1
and100 0.08	thr 119			
[DDQN-Mod]	Ep 195 R	-165.61 avgR100	-76.77 eps 0.376 Qval	13.59 1
and100 0.08	thr 685			
[DDQN-Mod]	Ep 196 R	-53.83 avgR100	-76.28 eps 0.374 Qval	13.72 1
and100 0.08	thr 110			
[DDQN-Mod]	Ep 197 R	-60.09 avgR100	-75.09 eps 0.373 Qval	13.79 1
and100 0.08	thr 67			
[DDQN-Mod]	Ep 198 R	-52.13 avgR100	-74.39 eps 0.371 Qval	14.29 1
and100 0.08	thr 79			
[DDQN-Mod]	Ep 199 R	-51.65 avgR100	-73.80 eps 0.369 Qval	14.07 1
and100 0.08	thr 67			
[DDQN-Mod]	Ep 200 R	-9.04 avgR100	-72.70 eps 0.367 Qval	14.15 1
and100 0.08	thr 102			
[DDQN-Mod]	Ep 201 R	-72.08 avgR100	-72.55 eps 0.365 Qval	14.29 1
and100 0.08	thr 168			
[DDQN-Mod]	Ep 202 R	-59.00 avgR100	-72.13 eps 0.363 Qval	13.52 1
and100 0.08	thr 529			
[DDQN-Mod]	Ep 203 R	-28.73 avgR100	-71.87 eps 0.361 Qval	14.64 1
and100 0.08	thr 76			
[DDQN-Mod]	Ep 204 R	-40.77 avgR100	-71.05 eps 0.360 Qval	14.67 1
and100 0.08	thr 203			
[DDQN-Mod]	Ep 205 R	-77.19 avgR100	-70.58 eps 0.358 Qval	14.95 1
and100 0.08	thr 156			
[DDQN-Mod]	Ep 206 R	-41.72 avgR100	-69.35 eps 0.356 Qval	14.77 1
and100 0.08	thr 111			
[DDQN-Mod]	Ep 207 R	-125.82 avgR100	-68.85 eps 0.354 Qval	15.25 1
and100 0.08	thr 679			
[DDQN-Mod]	Ep 208 R	-52.24 avgR100	-68.66 eps 0.353 Qval	15.18 1
and100 0.08	thr 107			
[DDQN-Mod]	Ep 209 R	-35.90 avgR100	-68.70 eps 0.351 Qval	15.26 1
and100 0.08	thr 137			
[DDQN-Mod]	Ep 210 R	-198.08 avgR100	-69.95 eps 0.349 Qval	15.61 1
and100 0.08	thr 372			
[DDQN-Mod]	Ep 211 R	-62.94 avgR100	-69.40 eps 0.347 Qval	14.83 1
and100 0.08	thr 94			
[DDQN-Mod]	Ep 212 R	-64.47 avgR100	-68.97 eps 0.346 Qval	15.16 1
and100 0.08	thr 101			
[DDQN-Mod]	Ep 213 R	-72.67 avgR100	-68.69 eps 0.344 Qval	14.96 1
and100 0.08	thr 85			

[DDQN-Mod]	Ep	214		R	-133.87		avgR100	-69.46		eps 0.342		Qval	16.00		1
and100 0.08		thr 700													
[DDQN-Mod]	Ep	215		R	-45.17		avgR100	-69.22		eps 0.340		Qval	15.89		1
and100 0.08		thr 102													
[DDQN-Mod]	Ep	216		R	-80.14		avgR100	-68.34		eps 0.339		Qval	15.69		1
and100 0.08		thr 85													
[DDQN-Mod]	Ep	217		R	-54.79		avgR100	-67.74		eps 0.337		Qval	15.71		1
and100 0.08		thr 101													
[DDQN-Mod]	Ep	218		R	-94.72		avgR100	-68.20		eps 0.335		Qval	15.07		1
and100 0.08		thr 70													
[DDQN-Mod]	Ep	219		R	11.95		avgR100	-67.72		eps 0.334		Qval	15.06		1
and100 0.09		thr 105													
[DDQN-Mod]	Ep	220		R	-76.44		avgR100	-66.88		eps 0.332		Qval	15.78		1
and100 0.09		thr 128													
[DDQN-Mod]	Ep	221		R	209.75		avgR100	-63.79		eps 0.330		Qval	15.59		1
and100 0.10		thr 278													
[DDQN-Mod]	Ep	222		R	0.65		avgR100	-62.71		eps 0.329		Qval	15.55		1
and100 0.10		thr 92													
[DDQN-Mod]	Ep	223		R	-52.61		avgR100	-61.96		eps 0.327		Qval	15.80		1
and100 0.10		thr 93													
[DDQN-Mod]	Ep	224		R	-29.93		avgR100	-61.34		eps 0.325		Qval	15.31		1
and100 0.10		thr 131													
[DDQN-Mod]	Ep	225		R	-98.94		avgR100	-60.99		eps 0.324		Qval	15.61		1
and100 0.10		thr 125													
[DDQN-Mod]	Ep	226		R	-83.57		avgR100	-59.89		eps 0.322		Qval	15.60		1
and100 0.10		thr 92													
[DDQN-Mod]	Ep	227		R	-231.85		avgR100	-61.27		eps 0.321		Qval	15.74		1
and100 0.09		thr 605													
[DDQN-Mod]	Ep	228		R	-32.92		avgR100	-61.16		eps 0.319		Qval	14.80		1
and100 0.09		thr 91													
[DDQN-Mod]	Ep	229		R	-75.43		avgR100	-60.81		eps 0.317		Qval	15.47		1
and100 0.09		thr 119													
[DDQN-Mod]	Ep	230		R	-37.27		avgR100	-60.33		eps 0.316		Qval	15.35		1
and100 0.09		thr 133													
[DDQN-Mod]	Ep	231		R	-48.55		avgR100	-60.05		eps 0.314		Qval	14.71		1
and100 0.09		thr 74													
[DDQN-Mod]	Ep	232		R	-42.96		avgR100	-60.73		eps 0.313		Qval	15.19		1
and100 0.08		thr 115													
[DDQN-Mod]	Ep	233		R	-39.14		avgR100	-60.37		eps 0.311		Qval	14.90		1
and100 0.08		thr 153													
[DDQN-Mod]	Ep	234		R	-34.23		avgR100	-61.00		eps 0.309		Qval	15.25		1
and100 0.07		thr 91													
[DDQN-Mod]	Ep	235		R	-371.36		avgR100	-64.59		eps 0.308		Qval	14.46		1
and100 0.07		thr 426													
[DDQN-Mod]	Ep	236		R	-159.88		avgR100	-65.44		eps 0.306		Qval	14.29		1
and100 0.07		thr 603													
[DDQN-Mod]	Ep	237		R	-64.18		avgR100	-64.90		eps 0.305		Qval	14.69		1
and100 0.07		thr 103													
[DDQN-Mod]	Ep	238		R	-0.91		avgR100	-64.06		eps 0.303		Qval	13.87		1
and100 0.07		thr 133													
[DDQN-Mod]	Ep	239		R	-76.36		avgR100	-63.63		eps 0.302		Qval	14.30		1
and100 0.07		thr 98													
[DDQN-Mod]	Ep	240		R	-43.79		avgR100	-63.61		eps 0.300		Qval	14.53		1
and100 0.07		thr 77													
[DDQN-Mod]	Ep	241		R	-117.30		avgR100	-63.95		eps 0.299		Qval	14.37		1
and100 0.07		thr 618													

[DDQN-Mod] Ep 242 R 98.69 avgR100 -61.96 eps 0.297 Qval 13.64 1
and100 0.07 thr 278
[DDQN-Mod] Ep 243 R -85.75 avgR100 -61.70 eps 0.296 Qval 14.37 1
and100 0.07 thr 82
[DDQN-Mod] Ep 244 R -97.77 avgR100 -61.71 eps 0.294 Qval 13.32 1
and100 0.07 thr 64
[DDQN-Mod] Ep 245 R -118.89 avgR100 -62.22 eps 0.293 Qval 14.42 1
and100 0.07 thr 104
[DDQN-Mod] Ep 246 R -20.67 avgR100 -61.58 eps 0.291 Qval 13.19 1
and100 0.07 thr 100
[DDQN-Mod] Ep 247 R 211.71 avgR100 -58.56 eps 0.290 Qval 13.27 1
and100 0.08 thr 293
[DDQN-Mod] Ep 248 R -201.04 avgR100 -60.53 eps 0.288 Qval 13.37 1
and100 0.08 thr 627
[DDQN-Mod] Ep 249 R -22.53 avgR100 -60.47 eps 0.287 Qval 13.68 1
and100 0.08 thr 91
[DDQN-Mod] Ep 250 R -17.13 avgR100 -59.78 eps 0.286 Qval 13.12 1
and100 0.08 thr 108
[DDQN-Mod] Ep 251 R -29.19 avgR100 -60.76 eps 0.284 Qval 13.56 1
and100 0.07 thr 121
[DDQN-Mod] Ep 252 R -199.11 avgR100 -61.84 eps 0.283 Qval 12.57 1
and100 0.07 thr 406
[DDQN-Mod] Ep 253 R -18.52 avgR100 -61.34 eps 0.281 Qval 12.96 1
and100 0.07 thr 120
[DDQN-Mod] Ep 254 R -66.87 avgR100 -61.07 eps 0.280 Qval 12.48 1
and100 0.07 thr 75
[DDQN-Mod] Ep 255 R -107.69 avgR100 -62.04 eps 0.279 Qval 13.16 1
and100 0.07 thr 109
[DDQN-Mod] Ep 256 R -21.25 avgR100 -61.38 eps 0.277 Qval 11.97 1
and100 0.07 thr 128
[DDQN-Mod] Ep 257 R -30.21 avgR100 -60.87 eps 0.276 Qval 11.64 1
and100 0.07 thr 100
[DDQN-Mod] Ep 258 R -32.40 avgR100 -61.63 eps 0.274 Qval 12.53 1
and100 0.06 thr 90
[DDQN-Mod] Ep 259 R -91.51 avgR100 -61.98 eps 0.273 Qval 12.80 1
and100 0.06 thr 87
[DDQN-Mod] Ep 260 R -46.13 avgR100 -61.59 eps 0.272 Qval 11.83 1
and100 0.06 thr 108
[DDQN-Mod] Ep 261 R -41.41 avgR100 -60.95 eps 0.270 Qval 11.50 1
and100 0.06 thr 82
[DDQN-Mod] Ep 262 R -163.50 avgR100 -62.36 eps 0.269 Qval 11.43 1
and100 0.06 thr 656
[DDQN-Mod] Ep 263 R -45.24 avgR100 -61.39 eps 0.268 Qval 10.79 1
and100 0.06 thr 89
[DDQN-Mod] Ep 264 R -197.11 avgR100 -62.82 eps 0.266 Qval 10.16 1
and100 0.06 thr 771
[DDQN-Mod] Ep 265 R -42.13 avgR100 -62.65 eps 0.265 Qval 9.58 1
and100 0.05 thr 112
[DDQN-Mod] Ep 266 R -51.78 avgR100 -62.86 eps 0.264 Qval 9.97 1
and100 0.05 thr 645
[DDQN-Mod] Ep 267 R -168.41 avgR100 -63.25 eps 0.262 Qval 9.33 1
and100 0.05 thr 694
[DDQN-Mod] Ep 268 R -139.47 avgR100 -64.67 eps 0.261 Qval 9.10 1
and100 0.04 thr 206
[DDQN-Mod] Ep 269 R -214.91 avgR100 -66.58 eps 0.260 Qval 7.68 1
and100 0.04 thr 731
[DDQN-Mod] Ep 270 R 26.13 avgR100 -65.69 eps 0.258 Qval 7.80 1
and100 0.05 thr 98
[DDQN-Mod] Ep 271 R -73.91 avgR100 -65.16 eps 0.257 Qval 8.03 1
and100 0.05 thr 85

[DDQN-Mod]	Ep	272		R	-110.23		avgR100	-65.95		eps 0.256		Qval	7.76		1
and100 0.05		thr 248													
[DDQN-Mod]	Ep	273		R	-209.50		avgR100	-67.34		eps 0.255		Qval	7.77		1
and100 0.05		thr 84													
[DDQN-Mod]	Ep	274		R	-276.14		avgR100	-69.42		eps 0.253		Qval	5.86		1
and100 0.05		thr 780													
[DDQN-Mod]	Ep	275		R	-23.49		avgR100	-69.17		eps 0.252		Qval	5.79		1
and100 0.05		thr 110													
[DDQN-Mod]	Ep	276		R	-57.11		avgR100	-68.99		eps 0.251		Qval	5.51		1
and100 0.05		thr 62													
[DDQN-Mod]	Ep	277		R	183.62		avgR100	-65.98		eps 0.249		Qval	5.42		1
and100 0.06		thr 373													
[DDQN-Mod]	Ep	278		R	-79.45		avgR100	-66.40		eps 0.248		Qval	5.23		1
and100 0.06		thr 86													
[DDQN-Mod]	Ep	279		R	-42.86		avgR100	-66.70		eps 0.247		Qval	4.79		1
and100 0.06		thr 95													
[DDQN-Mod]	Ep	280		R	-279.43		avgR100	-68.90		eps 0.246		Qval	3.96		1
and100 0.06		thr 613													
[DDQN-Mod]	Ep	281		R	-56.78		avgR100	-69.29		eps 0.245		Qval	3.91		1
and100 0.06		thr 531													
[DDQN-Mod]	Ep	282		R	-44.99		avgR100	-68.43		eps 0.243		Qval	3.79		1
and100 0.06		thr 132													
[DDQN-Mod]	Ep	283		R	-21.11		avgR100	-68.52		eps 0.242		Qval	3.94		1
and100 0.05		thr 76													
[DDQN-Mod]	Ep	284		R	-102.95		avgR100	-68.76		eps 0.241		Qval	2.90		1
and100 0.05		thr 118													
[DDQN-Mod]	Ep	285		R	8.29		avgR100	-68.28		eps 0.240		Qval	1.85		1
and100 0.05		thr 501													
[DDQN-Mod]	Ep	286		R	-24.02		avgR100	-68.04		eps 0.238		Qval	0.69		1
and100 0.05		thr 95													
[DDQN-Mod]	Ep	287		R	-48.26		avgR100	-68.42		eps 0.237		Qval	2.34		1
and100 0.05		thr 109													
[DDQN-Mod]	Ep	288		R	-121.99		avgR100	-68.84		eps 0.236		Qval	2.05		1
and100 0.05		thr 138													
[DDQN-Mod]	Ep	289		R	-129.80		avgR100	-69.53		eps 0.235		Qval	-0.83		1
and100 0.05		thr 643													
[DDQN-Mod]	Ep	290		R	-53.89		avgR100	-69.25		eps 0.234		Qval	0.18		1
and100 0.05		thr 97													
[DDQN-Mod]	Ep	291		R	-109.91		avgR100	-70.10		eps 0.233		Qval	-0.15		1
and100 0.05		thr 224													
[DDQN-Mod]	Ep	292		R	-219.01		avgR100	-71.99		eps 0.231		Qval	-0.49		1
and100 0.05		thr 722													
[DDQN-Mod]	Ep	293		R	-188.32		avgR100	-73.40		eps 0.230		Qval	-1.57		1
and100 0.05		thr 592													
[DDQN-Mod]	Ep	294		R	-59.62		avgR100	-73.03		eps 0.229		Qval	-1.48		1
and100 0.05		thr 74													
[DDQN-Mod]	Ep	295		R	-250.18		avgR100	-73.87		eps 0.228		Qval	-3.34		1
and100 0.05		thr 783													
[DDQN-Mod]	Ep	296		R	-92.10		avgR100	-74.26		eps 0.227		Qval	-3.25		1
and100 0.05		thr 68													
[DDQN-Mod]	Ep	297		R	-78.14		avgR100	-74.44		eps 0.226		Qval	-3.14		1
and100 0.05		thr 110													
[DDQN-Mod]	Ep	298		R	-8.52		avgR100	-74.00		eps 0.225		Qval	-3.32		1
and100 0.05		thr 85													
[DDQN-Mod]	Ep	299		R	-58.55		avgR100	-74.07		eps 0.223		Qval	-3.39		1
and100 0.05		thr 74													
[DDQN-Mod]	Ep	300		R	-50.91		avgR100	-74.49		eps 0.222		Qval	-3.36		1
and100 0.05		thr 112													
[DDQN-Mod]	Ep	301		R	-255.59		avgR100	-76.32		eps 0.221		Qval	-5.29		1
and100 0.05		thr 777													

[DDQN-Mod]	Ep	302		R	-153.47		avgR100	-77.27		eps 0.220		Qval	-5.80		1
and100 0.05		thr 765													
[DDQN-Mod]	Ep	303		R	187.32		avgR100	-75.11		eps 0.219		Qval	-6.43		1
and100 0.06		thr 316													
[DDQN-Mod]	Ep	304		R	-107.67		avgR100	-75.78		eps 0.218		Qval	-6.51		1
and100 0.06		thr 362													
[DDQN-Mod]	Ep	305		R	-132.75		avgR100	-76.33		eps 0.217		Qval	-6.85		1
and100 0.06		thr 387													
[DDQN-Mod]	Ep	306		R	-192.28		avgR100	-77.84		eps 0.216		Qval	-8.06		1
and100 0.06		thr 798													
[DDQN-Mod]	Ep	307		R	-74.88		avgR100	-77.33		eps 0.215		Qval	-8.31		1
and100 0.06		thr 101													
[DDQN-Mod]	Ep	308		R	-172.20		avgR100	-78.53		eps 0.214		Qval	-8.65		1
and100 0.06		thr 715													
[DDQN-Mod]	Ep	309		R	-44.24		avgR100	-78.61		eps 0.212		Qval	-9.06		1
and100 0.06		thr 126													
[DDQN-Mod]	Ep	310		R	-22.60		avgR100	-76.86		eps 0.211		Qval	-10.37		1
and100 0.07		thr 712													
[DDQN-Mod]	Ep	311		R	-79.76		avgR100	-77.03		eps 0.210		Qval	-10.77		1
and100 0.07		thr 73													
[DDQN-Mod]	Ep	312		R	-43.43		avgR100	-76.81		eps 0.209		Qval	-9.73		1
and100 0.07		thr 107													
[DDQN-Mod]	Ep	313		R	-7.94		avgR100	-76.17		eps 0.208		Qval	-10.42		1
and100 0.07		thr 128													
[DDQN-Mod]	Ep	314		R	103.09		avgR100	-73.80		eps 0.207		Qval	-10.47		1
and100 0.08		thr 533													
[DDQN-Mod]	Ep	315		R	-1.83		avgR100	-73.36		eps 0.206		Qval	-11.61		1
and100 0.08		thr 102													
[DDQN-Mod]	Ep	316		R	-181.03		avgR100	-74.37		eps 0.205		Qval	-11.68		1
and100 0.08		thr 751													
[DDQN-Mod]	Ep	317		R	43.55		avgR100	-73.39		eps 0.204		Qval	-11.69		1
and100 0.09		thr 596													
[DDQN-Mod]	Ep	318		R	-35.55		avgR100	-72.80		eps 0.203		Qval	-12.21		1
and100 0.09		thr 99													
[DDQN-Mod]	Ep	319		R	-239.21		avgR100	-75.31		eps 0.202		Qval	-13.90		1
and100 0.08		thr 834													
[DDQN-Mod]	Ep	320		R	-261.41		avgR100	-77.16		eps 0.201		Qval	-14.82		1
and100 0.08		thr 829													
[DDQN-Mod]	Ep	321		R	-263.17		avgR100	-81.89		eps 0.200		Qval	-15.03		1
and100 0.07		thr 831													
[DDQN-Mod]	Ep	322		R	-10.34		avgR100	-82.00		eps 0.199		Qval	-16.23		1
and100 0.07		thr 118													
[DDQN-Mod]	Ep	323		R	-71.76		avgR100	-82.19		eps 0.198		Qval	-16.07		1
and100 0.07		thr 80													
[DDQN-Mod]	Ep	324		R	-218.20		avgR100	-84.07		eps 0.197		Qval	-17.16		1
and100 0.07		thr 816													
[DDQN-Mod]	Ep	325		R	-87.76		avgR100	-83.96		eps 0.196		Qval	-17.06		1
and100 0.07		thr 105													
[DDQN-Mod]	Ep	326		R	-64.99		avgR100	-83.78		eps 0.195		Qval	-16.98		1
and100 0.07		thr 117													
[DDQN-Mod]	Ep	327		R	-219.93		avgR100	-83.66		eps 0.194		Qval	-18.11		1
and100 0.07		thr 840													
[DDQN-Mod]	Ep	328		R	35.98		avgR100	-82.97		eps 0.193		Qval	-19.06		1
and100 0.07		thr 382													
[DDQN-Mod]	Ep	329		R	-223.35		avgR100	-84.45		eps 0.192		Qval	-19.88		1
and100 0.07		thr 737													
[DDQN-Mod]	Ep	330		R	-71.68		avgR100	-84.79		eps 0.191		Qval	-20.79		1
and100 0.07		thr 101													
[DDQN-Mod]	Ep	331		R	-100.14		avgR100	-85.31		eps 0.190		Qval	-20.14		1
and100 0.07		thr 98													

[DDQN-Mod] Ep and100 0.07	332 R thr 205	-29.71	avgR100	-85.17	eps 0.189	Qval -19.87	1
[DDQN-Mod] Ep and100 0.07	333 R thr 734	-213.82	avgR100	-86.92	eps 0.188	Qval -21.38	1
[DDQN-Mod] Ep and100 0.07	334 R thr 129	-28.40	avgR100	-86.86	eps 0.187	Qval -21.49	1
[DDQN-Mod] Ep and100 0.08	335 R thr 309	171.26	avgR100	-81.44	eps 0.187	Qval -22.19	1
[DDQN-Mod] Ep and100 0.08	336 R thr 112	-74.55	avgR100	-80.58	eps 0.186	Qval -21.94	1
[DDQN-Mod] Ep and100 0.09	337 R thr 579	64.23	avgR100	-79.30	eps 0.185	Qval -23.17	1
[DDQN-Mod] Ep and100 0.10	338 R thr 331	140.78	avgR100	-77.88	eps 0.184	Qval -23.14	1
[DDQN-Mod] Ep and100 0.10	339 R thr 763	-210.96	avgR100	-79.23	eps 0.183	Qval -23.98	1
[DDQN-Mod] Ep and100 0.10	340 R thr 796	-281.50	avgR100	-81.61	eps 0.182	Qval -24.65	1
[DDQN-Mod] Ep and100 0.10	341 R thr 127	-49.57	avgR100	-80.93	eps 0.181	Qval -24.11	1
[DDQN-Mod] Ep and100 0.11	342 R thr 215	271.65	avgR100	-79.20	eps 0.180	Qval -25.35	1
[DDQN-Mod] Ep and100 0.11	343 R thr 87	-41.03	avgR100	-78.75	eps 0.179	Qval -25.56	1
[DDQN-Mod] Ep and100 0.11	344 R thr 883	-330.15	avgR100	-81.07	eps 0.178	Qval -27.12	1
[DDQN-Mod] Ep and100 0.11	345 R thr 97	-94.55	avgR100	-80.83	eps 0.177	Qval -26.83	1
[DDQN-Mod] Ep and100 0.11	346 R thr 821	-271.16	avgR100	-83.34	eps 0.177	Qval -28.24	1
[DDQN-Mod] Ep and100 0.11	347 R thr 496	140.42	avgR100	-84.05	eps 0.176	Qval -28.09	1
[DDQN-Mod] Ep and100 0.11	348 R thr 791	-259.21	avgR100	-84.63	eps 0.175	Qval -31.48	1
[DDQN-Mod] Ep and100 0.11	349 R thr 820	-289.29	avgR100	-87.30	eps 0.174	Qval -30.96	1
[DDQN-Mod] Ep and100 0.11	350 R thr 145	-40.82	avgR100	-87.54	eps 0.173	Qval -31.68	1
[DDQN-Mod] Ep and100 0.11	351 R thr 702	-184.42	avgR100	-89.09	eps 0.172	Qval -32.70	1
[DDQN-Mod] Ep and100 0.12	352 R thr 641	7.10	avgR100	-87.03	eps 0.171	Qval -33.21	1
[DDQN-Mod] Ep and100 0.12	353 R thr 107	-52.51	avgR100	-87.37	eps 0.170	Qval -32.83	1
[DDQN-Mod] Ep and100 0.12	354 R thr 772	-294.13	avgR100	-89.64	eps 0.170	Qval -34.67	1
[DDQN-Mod] Ep and100 0.12	355 R thr 484	97.62	avgR100	-87.58	eps 0.169	Qval -35.61	1
[DDQN-Mod] Ep and100 0.12	356 R thr 826	-237.44	avgR100	-89.75	eps 0.168	Qval -37.46	1
[DDQN-Mod] Ep and100 0.12	357 R thr 824	-318.66	avgR100	-92.63	eps 0.167	Qval -38.59	1
[DDQN-Mod] Ep and100 0.12	358 R thr 164	-20.11	avgR100	-92.51	eps 0.166	Qval -38.64	1
[DDQN-Mod] Ep and100 0.13	359 R thr 572	71.10	avgR100	-90.88	eps 0.165	Qval -37.74	1
[DDQN-Mod] Ep and100 0.14	360 R thr 647	54.52	avgR100	-89.88	eps 0.165	Qval -39.91	1

[DDQN-Mod] Ep and100 0.14	361 R thr 810	-301.75	avgR100	-92.48	eps 0.164	Qval -40.71	1
[DDQN-Mod] Ep and100 0.15	362 R thr 212	295.16	avgR100	-87.89	eps 0.163	Qval -41.35	1
[DDQN-Mod] Ep and100 0.16	363 R thr 569	32.79	avgR100	-87.11	eps 0.162	Qval -42.40	1
[DDQN-Mod] Ep and100 0.16	364 R thr 793	-285.11	avgR100	-87.99	eps 0.161	Qval -42.81	1
[DDQN-Mod] Ep and100 0.16	365 R thr 732	-243.77	avgR100	-90.01	eps 0.160	Qval -43.96	1
[DDQN-Mod] Ep and100 0.16	366 R thr 819	-350.74	avgR100	-93.00	eps 0.160	Qval -44.47	1
[DDQN-Mod] Ep and100 0.16	367 R thr 592	-378.43	avgR100	-95.10	eps 0.159	Qval -44.01	1
[DDQN-Mod] Ep and100 0.16	368 R thr 819	-317.13	avgR100	-96.88	eps 0.158	Qval -45.27	1
[DDQN-Mod] Ep and100 0.16	369 R thr 853	-338.87	avgR100	-98.11	eps 0.157	Qval -47.04	1
[DDQN-Mod] Ep and100 0.15	370 R thr 630	-122.62	avgR100	-99.60	eps 0.157	Qval -45.40	1
[DDQN-Mod] Ep and100 0.16	371 R thr 91	85.44	avgR100	-98.01	eps 0.156	Qval -47.39	1
[DDQN-Mod] Ep and100 0.17	372 R thr 367	198.31	avgR100	-94.92	eps 0.155	Qval -47.21	1
[DDQN-Mod] Ep and100 0.17	373 R thr 98	-88.80	avgR100	-93.72	eps 0.154	Qval -46.82	1
[DDQN-Mod] Ep and100 0.17	374 R thr 113	-34.66	avgR100	-91.30	eps 0.153	Qval -45.94	1
[DDQN-Mod] Ep and100 0.17	375 R thr 664	-169.85	avgR100	-92.77	eps 0.153	Qval -48.95	1
[DDQN-Mod] Ep and100 0.17	376 R thr 279	-194.17	avgR100	-94.14	eps 0.152	Qval -49.13	1
[DDQN-Mod] Ep and100 0.16	377 R thr 805	-313.38	avgR100	-99.11	eps 0.151	Qval -50.33	1
[DDQN-Mod] Ep and100 0.16	378 R thr 127	-16.31	avgR100	-98.47	eps 0.150	Qval -50.93	1
[DDQN-Mod] Ep and100 0.16	379 R thr 91	-50.23	avgR100	-98.55	eps 0.150	Qval -49.44	1
[DDQN-Mod] Ep and100 0.16	380 R thr 686	-402.73	avgR100	-99.78	eps 0.149	Qval -49.62	1
[DDQN-Mod] Ep and100 0.16	381 R thr 107	-7.79	avgR100	-99.29	eps 0.148	Qval -49.75	1
[DDQN-Mod] Ep and100 0.16	382 R thr 774	-230.15	avgR100	-101.14	eps 0.147	Qval -51.08	1
[DDQN-Mod] Ep and100 0.17	383 R thr 532	65.02	avgR100	-100.28	eps 0.147	Qval -52.73	1
[DDQN-Mod] Ep and100 0.17	384 R thr 765	-182.21	avgR100	-101.07	eps 0.146	Qval -52.75	1
[DDQN-Mod] Ep and100 0.17	385 R thr 723	-211.84	avgR100	-103.28	eps 0.145	Qval -53.30	1
[DDQN-Mod] Ep and100 0.17	386 R thr 831	-264.81	avgR100	-105.68	eps 0.144	Qval -54.10	1
[DDQN-Mod] Ep and100 0.17	387 R thr 776	-487.08	avgR100	-110.07	eps 0.144	Qval -54.25	1
[DDQN-Mod] Ep and100 0.17	388 R thr 803	-280.64	avgR100	-111.66	eps 0.143	Qval -55.08	1
[DDQN-Mod] Ep and100 0.17	389 R thr 424	-308.27	avgR100	-113.44	eps 0.142	Qval -56.77	1
[DDQN-Mod] Ep and100 0.17	390 R thr 840	-329.70	avgR100	-116.20	eps 0.142	Qval -56.84	1

[DDQN-Mod] Ep and100 0.17	391 R thr 807	-281.55	avgR100	-117.92	eps 0.141	Qval -56.86	1
[DDQN-Mod] Ep and100 0.17	392 R thr 535	-322.07	avgR100	-118.95	eps 0.140	Qval -57.88	1
[DDQN-Mod] Ep and100 0.18	393 R thr 431	167.59	avgR100	-115.39	eps 0.139	Qval -60.20	1
[DDQN-Mod] Ep and100 0.18	394 R thr 126	-63.42	avgR100	-115.43	eps 0.139	Qval -59.63	1
[DDQN-Mod] Ep and100 0.18	395 R thr 772	-225.58	avgR100	-115.18	eps 0.138	Qval -58.88	1
[DDQN-Mod] Ep and100 0.18	396 R thr 817	-342.33	avgR100	-117.68	eps 0.137	Qval -59.15	1
[DDQN-Mod] Ep and100 0.18	397 R thr 239	-140.46	avgR100	-118.31	eps 0.137	Qval -61.24	1
[DDQN-Mod] Ep and100 0.18	398 R thr 210	-83.10	avgR100	-119.05	eps 0.136	Qval -60.04	1
[DDQN-Mod] Ep and100 0.19	399 R thr 248	193.60	avgR100	-116.53	eps 0.135	Qval -61.13	1
[DDQN-Mod] Ep and100 0.19	400 R thr 803	-359.06	avgR100	-119.61	eps 0.135	Qval -60.44	1
[DDQN-Mod] Ep and100 0.19	401 R thr 764	-178.94	avgR100	-118.85	eps 0.134	Qval -62.22	1
[DDQN-Mod] Ep and100 0.19	402 R thr 792	-219.81	avgR100	-119.51	eps 0.133	Qval -62.51	1
[DDQN-Mod] Ep and100 0.18	403 R thr 223	181.63	avgR100	-119.57	eps 0.133	Qval -63.15	1
[DDQN-Mod] Ep and100 0.19	404 R thr 384	176.88	avgR100	-116.72	eps 0.132	Qval -61.79	1
[DDQN-Mod] Ep and100 0.19	405 R thr 89	-26.41	avgR100	-115.66	eps 0.131	Qval -62.15	1
[DDQN-Mod] Ep and100 0.19	406 R thr 350	-197.57	avgR100	-115.71	eps 0.131	Qval -62.80	1
[DDQN-Mod] Ep and100 0.19	407 R thr 724	-238.29	avgR100	-117.34	eps 0.130	Qval -64.29	1
[DDQN-Mod] Ep and100 0.20	408 R thr 325	168.21	avgR100	-113.94	eps 0.129	Qval -63.72	1
[DDQN-Mod] Ep and100 0.20	409 R thr 809	-262.67	avgR100	-116.12	eps 0.129	Qval -66.01	1
[DDQN-Mod] Ep and100 0.19	410 R thr 786	-170.44	avgR100	-117.60	eps 0.128	Qval -64.42	1
[DDQN-Mod] Ep and100 0.19	411 R thr 792	-229.85	avgR100	-119.10	eps 0.127	Qval -65.16	1
[DDQN-Mod] Ep and100 0.20	412 R thr 412	105.95	avgR100	-117.61	eps 0.127	Qval -66.52	1
[DDQN-Mod] Ep and100 0.20	413 R thr 158	-68.42	avgR100	-118.21	eps 0.126	Qval -64.29	1
[DDQN-Mod] Ep and100 0.19	414 R thr 805	-356.00	avgR100	-122.81	eps 0.126	Qval -63.93	1
[DDQN-Mod] Ep and100 0.19	415 R thr 804	-211.69	avgR100	-124.90	eps 0.125	Qval -66.41	1
[DDQN-Mod] Ep and100 0.20	416 R thr 524	77.62	avgR100	-122.32	eps 0.124	Qval -66.96	1
[DDQN-Mod] Ep and100 0.19	417 R thr 833	-273.15	avgR100	-125.48	eps 0.124	Qval -66.14	1
[DDQN-Mod] Ep and100 0.19	418 R thr 106	-57.21	avgR100	-125.70	eps 0.123	Qval -67.78	1
[DDQN-Mod] Ep and100 0.20	419 R thr 410	131.41	avgR100	-121.99	eps 0.122	Qval -66.37	1
[DDQN-Mod] Ep and100 0.21	420 R thr 417	132.07	avgR100	-118.06	eps 0.122	Qval -66.16	1

[DDQN-Mod] and100	Ep 0.22	R thr 587	52.52	avgR100	-114.90	eps 0.121	Qval	-67.38	1
[DDQN-Mod] and100	Ep 0.22	R thr 783	-173.93	avgR100	-116.54	eps 0.121	Qval	-68.86	1
[DDQN-Mod] and100	Ep 0.23	R thr 460	139.10	avgR100	-114.43	eps 0.120	Qval	-67.00	1
[DDQN-Mod] and100	Ep 0.24	R thr 596	42.77	avgR100	-111.82	eps 0.119	Qval	-67.23	1
[DDQN-Mod] and100	Ep 0.25	R thr 322	158.39	avgR100	-109.36	eps 0.119	Qval	-68.90	1
[DDQN-Mod] and100	Ep 0.25	R thr 781	-211.07	avgR100	-110.82	eps 0.118	Qval	-67.96	1
[DDQN-Mod] and100	Ep 0.25	R thr 673	-163.94	avgR100	-110.26	eps 0.118	Qval	-70.69	1
[DDQN-Mod] and100	Ep 0.26	R thr 423	181.41	avgR100	-108.81	eps 0.117	Qval	-68.30	1
[DDQN-Mod] and100	Ep 0.27	R thr 628	40.80	avgR100	-106.16	eps 0.116	Qval	-68.73	1
[DDQN-Mod] and100	Ep 0.28	R thr 203	235.04	avgR100	-103.10	eps 0.116	Qval	-68.50	1
[DDQN-Mod] and100	Ep 0.28	R thr 696	-164.91	avgR100	-103.74	eps 0.115	Qval	-68.57	1
[DDQN-Mod] and100	Ep 0.29	R thr 325	175.05	avgR100	-101.70	eps 0.115	Qval	-69.78	1
[DDQN-Mod] and100	Ep 0.29	R thr 186	-13.41	avgR100	-99.69	eps 0.114	Qval	-67.67	1
[DDQN-Mod] and100	Ep 0.30	R thr 448	135.25	avgR100	-98.06	eps 0.114	Qval	-68.91	1
[DDQN-Mod] and100	Ep 0.29	R thr 730	-254.33	avgR100	-102.31	eps 0.113	Qval	-70.52	1
[DDQN-Mod] and100	Ep 0.30	R thr 593	67.93	avgR100	-100.89	eps 0.112	Qval	-69.94	1
[DDQN-Mod] and100	Ep 0.29	R thr 779	-248.22	avgR100	-104.01	eps 0.112	Qval	-71.59	1
[DDQN-Mod] and100	Ep 0.29	R thr 489	100.19	avgR100	-104.42	eps 0.111	Qval	-71.74	1
[DDQN-Mod] and100	Ep 0.30	R thr 304	171.90	avgR100	-100.59	eps 0.111	Qval	-72.17	1
[DDQN-Mod] and100	Ep 0.31	R thr 347	183.12	avgR100	-95.94	eps 0.110	Qval	-72.61	1
[DDQN-Mod] and100	Ep 0.31	R thr 738	-186.95	avgR100	-97.32	eps 0.110	Qval	-74.85	1
[DDQN-Mod] and100	Ep 0.31	R thr 472	124.08	avgR100	-98.79	eps 0.109	Qval	-73.02	1
[DDQN-Mod] and100	Ep 0.31	R thr 803	-264.83	avgR100	-101.03	eps 0.109	Qval	-74.70	1
[DDQN-Mod] and100	Ep 0.32	R thr 336	176.89	avgR100	-95.96	eps 0.108	Qval	-75.76	1
[DDQN-Mod] and100	Ep 0.33	R thr 330	174.90	avgR100	-93.27	eps 0.107	Qval	-73.12	1
[DDQN-Mod] and100	Ep 0.34	R thr 589	80.84	avgR100	-89.75	eps 0.107	Qval	-72.90	1
[DDQN-Mod] and100	Ep 0.34	R thr 358	173.35	avgR100	-89.42	eps 0.106	Qval	-74.17	1
[DDQN-Mod] and100	Ep 0.35	R thr 289	225.17	avgR100	-84.57	eps 0.106	Qval	-72.58	1
[DDQN-Mod] and100	Ep 0.36	R thr 456	118.17	avgR100	-80.50	eps 0.105	Qval	-72.27	1
[DDQN-Mod] and100	Ep 0.36	R thr 140	10.72	avgR100	-79.98	eps 0.105	Qval	-72.99	1

[DDQN-Mod] Ep 451 R 197.54 avgR100 -76.16 eps 0.104 Qval -72.59 1
and100 0.37 thr 352
[DDQN-Mod] Ep 452 R 195.73 avgR100 -74.28 eps 0.104 Qval -73.26 1
and100 0.37 thr 276
[DDQN-Mod] Ep 453 R 179.87 avgR100 -71.95 eps 0.103 Qval -74.58 1
and100 0.37 thr 236
[DDQN-Mod] Ep 454 R 73.29 avgR100 -68.28 eps 0.103 Qval -73.45 1
and100 0.38 thr 512
[DDQN-Mod] Ep 455 R -198.01 avgR100 -71.24 eps 0.102 Qval -75.11 1
and100 0.38 thr 765
[DDQN-Mod] Ep 456 R 90.82 avgR100 -67.95 eps 0.102 Qval -75.65 1
and100 0.39 thr 542
[DDQN-Mod] Ep 457 R -214.01 avgR100 -66.91 eps 0.101 Qval -73.83 1
and100 0.39 thr 705
[DDQN-Mod] Ep 458 R 270.88 avgR100 -64.00 eps 0.101 Qval -74.52 1
and100 0.40 thr 233
[DDQN-Mod] Ep 459 R 134.57 avgR100 -63.36 eps 0.100 Qval -75.80 1
and100 0.40 thr 371
[DDQN-Mod] Ep 460 R 158.14 avgR100 -62.33 eps 0.100 Qval -76.63 1
and100 0.40 thr 401
[DDQN-Mod] Ep 461 R -222.23 avgR100 -61.53 eps 0.099 Qval -76.04 1
and100 0.40 thr 758
[DDQN-Mod] Ep 462 R 275.46 avgR100 -61.73 eps 0.099 Qval -76.64 1
and100 0.40 thr 208
[DDQN-Mod] Ep 463 R 273.62 avgR100 -59.32 eps 0.098 Qval -74.78 1
and100 0.40 thr 220
[DDQN-Mod] Ep 464 R -174.88 avgR100 -58.22 eps 0.098 Qval -74.53 1
and100 0.40 thr 726
[DDQN-Mod] Ep 465 R -228.81 avgR100 -58.07 eps 0.097 Qval -74.52 1
and100 0.40 thr 792
[DDQN-Mod] Ep 466 R 89.46 avgR100 -53.66 eps 0.097 Qval -74.97 1
and100 0.41 thr 568
[DDQN-Mod] Ep 467 R -264.95 avgR100 -52.53 eps 0.096 Qval -75.06 1
and100 0.41 thr 799
[DDQN-Mod] Ep 468 R -196.37 avgR100 -51.32 eps 0.096 Qval -75.38 1
and100 0.41 thr 736
[DDQN-Mod] Ep 469 R 196.04 avgR100 -45.97 eps 0.095 Qval -73.37 1
and100 0.42 thr 350
[DDQN-Mod] Ep 470 R 198.03 avgR100 -42.77 eps 0.095 Qval -73.35 1
and100 0.43 thr 331
[DDQN-Mod] Ep 471 R -244.51 avgR100 -46.07 eps 0.094 Qval -73.08 1
and100 0.42 thr 773
[DDQN-Mod] Ep 472 R -209.09 avgR100 -50.14 eps 0.094 Qval -73.52 1
and100 0.41 thr 699
[DDQN-Mod] Ep 473 R 20.79 avgR100 -49.04 eps 0.093 Qval -72.70 1
and100 0.42 thr 616
[DDQN-Mod] Ep 474 R -239.06 avgR100 -51.09 eps 0.093 Qval -73.84 1
and100 0.42 thr 687
[DDQN-Mod] Ep 475 R 81.65 avgR100 -48.57 eps 0.092 Qval -72.95 1
and100 0.43 thr 547
[DDQN-Mod] Ep 476 R 139.74 avgR100 -45.23 eps 0.092 Qval -74.15 1
and100 0.44 thr 458
[DDQN-Mod] Ep 477 R -17.57 avgR100 -42.28 eps 0.092 Qval -73.31 1
and100 0.44 thr 105
[DDQN-Mod] Ep 478 R 62.28 avgR100 -41.49 eps 0.091 Qval -72.80 1
and100 0.45 thr 541
[DDQN-Mod] Ep 479 R 219.09 avgR100 -38.80 eps 0.091 Qval -73.73 1
and100 0.46 thr 293
[DDQN-Mod] Ep 480 R 158.17 avgR100 -33.19 eps 0.090 Qval -71.17 1
and100 0.47 thr 378

[DDQN-Mod] Ep 481 R 106.02 avgR100 -32.05 eps 0.090 Qval -72.52 1
and100 0.48 thr 459
[DDQN-Mod] Ep 482 R 124.93 avgR100 -28.50 eps 0.089 Qval -72.45 1
and100 0.49 thr 419
[DDQN-Mod] Ep 483 R -223.15 avgR100 -31.38 eps 0.089 Qval -71.83 1
and100 0.48 thr 746
[DDQN-Mod] Ep 484 R 225.09 avgR100 -27.31 eps 0.088 Qval -71.16 1
and100 0.49 thr 255
[DDQN-Mod] Ep 485 R -14.01 avgR100 -25.33 eps 0.088 Qval -71.05 1
and100 0.49 thr 209
[DDQN-Mod] Ep 486 R 194.58 avgR100 -20.74 eps 0.088 Qval -70.80 1
and100 0.50 thr 300
[DDQN-Mod] Ep 487 R 124.73 avgR100 -14.62 eps 0.087 Qval -72.67 1
and100 0.51 thr 486
[DDQN-Mod] Ep 488 R -24.47 avgR100 -12.06 eps 0.087 Qval -69.29 1
and100 0.52 thr 650
[DDQN-Mod] Ep 489 R -26.87 avgR100 -9.24 eps 0.086 Qval -67.75 1
and100 0.52 thr 124
[DDQN-Mod] Ep 490 R 56.16 avgR100 -5.38 eps 0.086 Qval -70.07 1
and100 0.53 thr 676
[DDQN-Mod] Ep 491 R 196.93 avgR100 -0.60 eps 0.085 Qval -69.24 1
and100 0.54 thr 353
[DDQN-Mod] Ep 492 R -229.89 avgR100 0.32 eps 0.085 Qval -70.57 1
and100 0.54 thr 770
[DDQN-Mod] Ep 493 R 168.01 avgR100 0.33 eps 0.084 Qval -71.96 1
and100 0.54 thr 367
[DDQN-Mod] Ep 494 R 280.46 avgR100 3.77 eps 0.084 Qval -69.91 1
and100 0.55 thr 225
[DDQN-Mod] Ep 495 R 245.82 avgR100 8.48 eps 0.084 Qval -68.19 1
and100 0.56 thr 242
[DDQN-Mod] Ep 496 R -198.48 avgR100 9.92 eps 0.083 Qval -70.29 1
and100 0.56 thr 712
[DDQN-Mod] Ep 497 R 142.76 avgR100 12.75 eps 0.083 Qval -68.61 1
and100 0.57 thr 450
[DDQN-Mod] Ep 498 R 119.35 avgR100 14.78 eps 0.082 Qval -68.66 1
and100 0.58 thr 507
[DDQN-Mod] Ep 499 R 183.40 avgR100 14.67 eps 0.082 Qval -69.09 1
and100 0.58 thr 308
[DDQN-Mod] Ep 500 R -196.05 avgR100 16.30 eps 0.082 Qval -67.00 1
and100 0.58 thr 795
[DDQN-Mod] Ep 501 R 291.70 avgR100 21.01 eps 0.081 Qval -67.95 1
and100 0.59 thr 155
[DDQN-Mod] Ep 502 R -227.49 avgR100 20.93 eps 0.081 Qval -69.34 1
and100 0.59 thr 804
[DDQN-Mod] Ep 503 R 105.54 avgR100 20.17 eps 0.080 Qval -67.31 1
and100 0.60 thr 487
[DDQN-Mod] Ep 504 R 26.77 avgR100 18.67 eps 0.080 Qval -67.31 1
and100 0.60 thr 580
[DDQN-Mod] Ep 505 R -175.63 avgR100 17.18 eps 0.080 Qval -67.28 1
and100 0.60 thr 692
[DDQN-Mod] Ep 506 R 70.93 avgR100 19.86 eps 0.079 Qval -67.18 1
and100 0.61 thr 520
[DDQN-Mod] Ep 507 R 160.38 avgR100 23.85 eps 0.079 Qval -67.74 1
and100 0.62 thr 364
[DDQN-Mod] Ep 508 R 262.91 avgR100 24.80 eps 0.078 Qval -67.74 1
and100 0.62 thr 176
[DDQN-Mod] Ep 509 R 175.22 avgR100 29.18 eps 0.078 Qval -66.78 1
and100 0.63 thr 410
[DDQN-Mod] Ep 510 R -221.86 avgR100 28.66 eps 0.078 Qval -68.07 1
and100 0.63 thr 767

[DDQN-Mod]	Ep 511 R	166.50 avgR100	32.63 eps 0.077 Qval -65.86 1
and100 0.64	thr 405		
[DDQN-Mod]	Ep 512 R	200.06 avgR100	33.57 eps 0.077 Qval -66.92 1
and100 0.64	thr 313		
[DDQN-Mod]	Ep 513 R	161.39 avgR100	35.86 eps 0.076 Qval -67.93 1
and100 0.65	thr 414		
[DDQN-Mod]	Ep 514 R	215.37 avgR100	41.58 eps 0.076 Qval -68.21 1
and100 0.66	thr 326		
[DDQN-Mod]	Ep 515 R	265.76 avgR100	46.35 eps 0.076 Qval -66.05 1
and100 0.67	thr 217		
[DDQN-Mod]	Ep 516 R	227.23 avgR100	47.85 eps 0.075 Qval -65.58 1
and100 0.67	thr 261		
[DDQN-Mod]	Ep 517 R	117.63 avgR100	51.76 eps 0.075 Qval -67.54 1
and100 0.68	thr 467		
[DDQN-Mod]	Ep 518 R	105.08 avgR100	53.38 eps 0.075 Qval -66.64 1
and100 0.69	thr 504		
[DDQN-Mod]	Ep 519 R	-197.52 avgR100	50.09 eps 0.074 Qval -69.00 1
and100 0.68	thr 796		
[DDQN-Mod]	Ep 520 R	156.64 avgR100	50.34 eps 0.074 Qval -67.07 1
and100 0.68	thr 322		
[DDQN-Mod]	Ep 521 R	252.15 avgR100	52.33 eps 0.073 Qval -66.98 1
and100 0.68	thr 216		
[DDQN-Mod]	Ep 522 R	119.57 avgR100	55.27 eps 0.073 Qval -68.77 1
and100 0.69	thr 497		
[DDQN-Mod]	Ep 523 R	151.66 avgR100	55.39 eps 0.073 Qval -66.54 1
and100 0.69	thr 373		
[DDQN-Mod]	Ep 524 R	-163.21 avgR100	53.33 eps 0.072 Qval -68.92 1
and100 0.68	thr 651		
[DDQN-Mod]	Ep 525 R	130.94 avgR100	53.06 eps 0.072 Qval -67.59 1
and100 0.68	thr 458		
[DDQN-Mod]	Ep 526 R	-42.74 avgR100	54.74 eps 0.072 Qval -67.97 1
and100 0.68	thr 206		
[DDQN-Mod]	Ep 527 R	-104.56 avgR100	55.34 eps 0.071 Qval -66.72 1
and100 0.68	thr 123		
[DDQN-Mod]	Ep 528 R	260.80 avgR100	56.13 eps 0.071 Qval -67.96 1
and100 0.68	thr 247		
[DDQN-Mod]	Ep 529 R	238.55 avgR100	58.11 eps 0.071 Qval -65.64 1
and100 0.68	thr 301		
[DDQN-Mod]	Ep 530 R	161.96 avgR100	57.38 eps 0.070 Qval -65.68 1
and100 0.68	thr 385		
[DDQN-Mod]	Ep 531 R	291.89 avgR100	61.94 eps 0.070 Qval -64.07 1
and100 0.69	thr 204		
[DDQN-Mod]	Ep 532 R	136.25 avgR100	61.56 eps 0.069 Qval -65.65 1
and100 0.69	thr 354		
[DDQN-Mod]	Ep 533 R	-231.90 avgR100	59.37 eps 0.069 Qval -66.76 1
and100 0.69	thr 801		
[DDQN-Mod]	Ep 534 R	-242.15 avgR100	55.60 eps 0.069 Qval -67.48 1
and100 0.68	thr 338		
[DDQN-Mod]	Ep 535 R	136.07 avgR100	59.50 eps 0.068 Qval -63.97 1
and100 0.69	thr 409		
[DDQN-Mod]	Ep 536 R	-166.49 avgR100	57.16 eps 0.068 Qval -65.28 1
and100 0.68	thr 780		
[DDQN-Mod]	Ep 537 R	5.32 avgR100	59.69 eps 0.068 Qval -66.11 1
and100 0.69	thr 659		
[DDQN-Mod]	Ep 538 R	265.46 avgR100	61.35 eps 0.067 Qval -66.66 1
and100 0.69	thr 180		

[DDQN-Mod] Ep 539 R 190.77 avgR100 61.53 eps 0.067 Qval -66.03 1
and100 0.69 thr 336
[DDQN-Mod] Ep 540 R 236.76 avgR100 62.07 eps 0.067 Qval -66.81 1
and100 0.69 thr 253
[DDQN-Mod] Ep 541 R -36.62 avgR100 63.57 eps 0.066 Qval -66.72 1
and100 0.69 thr 90
[DDQN-Mod] Ep 542 R 248.01 avgR100 64.81 eps 0.066 Qval -67.12 1
and100 0.69 thr 204
[DDQN-Mod] Ep 543 R 183.64 avgR100 69.30 eps 0.066 Qval -68.34 1
and100 0.70 thr 356
[DDQN-Mod] Ep 544 R -11.39 avgR100 67.41 eps 0.065 Qval -66.99 1
and100 0.69 thr 85
[DDQN-Mod] Ep 545 R 274.51 avgR100 68.41 eps 0.065 Qval -68.33 1
and100 0.69 thr 197
[DDQN-Mod] Ep 546 R 256.33 avgR100 70.17 eps 0.065 Qval -68.39 1
and100 0.69 thr 200
[DDQN-Mod] Ep 547 R 222.76 avgR100 70.66 eps 0.064 Qval -66.51 1
and100 0.69 thr 229
[DDQN-Mod] Ep 548 R 182.90 avgR100 70.24 eps 0.064 Qval -65.55 1
and100 0.69 thr 308
[DDQN-Mod] Ep 549 R 103.57 avgR100 70.09 eps 0.064 Qval -66.63 1
and100 0.69 thr 484
[DDQN-Mod] Ep 550 R 134.07 avgR100 71.32 eps 0.063 Qval -65.81 1
and100 0.70 thr 410
[DDQN-Mod] Ep 551 R 249.24 avgR100 71.84 eps 0.063 Qval -65.33 1
and100 0.70 thr 184
[DDQN-Mod] Ep 552 R 209.56 avgR100 71.98 eps 0.063 Qval -66.95 1
and100 0.70 thr 278
[DDQN-Mod] Ep 553 R 251.29 avgR100 72.69 eps 0.063 Qval -65.20 1
and100 0.71 thr 197
[DDQN-Mod] Ep 554 R 129.78 avgR100 73.26 eps 0.062 Qval -66.67 1
and100 0.71 thr 425
[DDQN-Mod] Ep 555 R 299.60 avgR100 78.24 eps 0.062 Qval -66.68 1
and100 0.72 thr 157
[DDQN-Mod] Ep 556 R 229.44 avgR100 79.62 eps 0.062 Qval -67.81 1
and100 0.72 thr 216
[DDQN-Mod] Ep 557 R 122.90 avgR100 82.99 eps 0.061 Qval -68.47 1
and100 0.73 thr 398
[DDQN-Mod] Ep 558 R 191.44 avgR100 82.20 eps 0.061 Qval -67.57 1
and100 0.73 thr 314
[DDQN-Mod] Ep 559 R 201.55 avgR100 82.87 eps 0.061 Qval -68.54 1
and100 0.73 thr 258
[DDQN-Mod] Ep 560 R 220.04 avgR100 83.48 eps 0.060 Qval -68.61 1
and100 0.73 thr 306
[DDQN-Mod] Ep 561 R 103.79 avgR100 86.75 eps 0.060 Qval -65.79 1
and100 0.74 thr 467
[DDQN-Mod] Ep 562 R 216.12 avgR100 86.15 eps 0.060 Qval -67.40 1
and100 0.74 thr 296
[DDQN-Mod] Ep 563 R 287.53 avgR100 86.29 eps 0.059 Qval -68.33 1
and100 0.74 thr 209
[DDQN-Mod] Ep 564 R 131.15 avgR100 89.35 eps 0.059 Qval -69.72 1
and100 0.75 thr 448
[DDQN-Mod] Ep 565 R 160.78 avgR100 93.25 eps 0.059 Qval -67.41 1
and100 0.76 thr 376
[DDQN-Mod] Ep 566 R 228.52 avgR100 94.64 eps 0.059 Qval -66.55 1
and100 0.76 thr 275
[DDQN-Mod] Ep 567 R 193.19 avgR100 99.22 eps 0.058 Qval -68.13 1
and100 0.76 thr 199

[DDQN-Mod]	Ep	568		R	165.29		avgR100	102.84		eps 0.058		Qval	-64.81		1
and100 0.77		thr 351													
[DDQN-Mod]	Ep	569		R	247.58		avgR100	103.35		eps 0.058		Qval	-66.47		1
and100 0.77		thr 208													
[DDQN-Mod]	Ep	570		R	260.22		avgR100	103.97		eps 0.057		Qval	-65.99		1
and100 0.77		thr 274													
[DDQN-Mod]	Ep	571		R	33.72		avgR100	106.76		eps 0.057		Qval	-68.04		1
and100 0.78		thr 676													
[DDQN-Mod]	Ep	572		R	182.97		avgR100	110.68		eps 0.057		Qval	-65.69		1
and100 0.79		thr 349													
[DDQN-Mod]	Ep	573		R	230.89		avgR100	112.78		eps 0.057		Qval	-65.73		1
and100 0.79		thr 287													
[DDQN-Mod]	Ep	574		R	227.40		avgR100	117.44		eps 0.056		Qval	-65.74		1
and100 0.80		thr 254													
[DDQN-Mod]	Ep	575		R	106.02		avgR100	117.68		eps 0.056		Qval	-64.24		1
and100 0.80		thr 410													
[DDQN-Mod]	Ep	576		R	161.45		avgR100	117.90		eps 0.056		Qval	-63.96		1
and100 0.80		thr 415													
[DDQN-Mod]	Ep	577		R	249.80		avgR100	120.58		eps 0.055		Qval	-65.90		1
and100 0.81		thr 226													
[DDQN-Mod]	Ep	578		R	265.53		avgR100	122.61		eps 0.055		Qval	-65.08		1
and100 0.81		thr 168													
[DDQN-Mod]	Ep	579		R	155.38		avgR100	121.97		eps 0.055		Qval	-65.72		1
and100 0.81		thr 380													
[DDQN-Mod]	Ep	580		R	184.42		avgR100	122.23		eps 0.055		Qval	-63.79		1
and100 0.81		thr 333													
[DDQN-Mod]	Ep	581		R	174.68		avgR100	122.92		eps 0.054		Qval	-64.26		1
and100 0.81		thr 391													
[DDQN-Mod]	Ep	582		R	100.29		avgR100	122.67		eps 0.054		Qval	-64.99		1
and100 0.81		thr 504													
[DDQN-Mod]	Ep	583		R	250.90		avgR100	127.41		eps 0.054		Qval	-65.23		1
and100 0.82		thr 204													
[DDQN-Mod]	Ep	584		R	105.36		avgR100	126.22		eps 0.054		Qval	-63.64		1
and100 0.82		thr 544													
[DDQN-Mod]	Ep	585		R	176.30		avgR100	128.12		eps 0.053		Qval	-63.07		1
and100 0.83		thr 304													
[DDQN-Mod]	Ep	586		R	-49.55		avgR100	125.68		eps 0.053		Qval	-61.92		1
and100 0.82		thr 187													
[DDQN-Mod]	Ep	587		R	102.69		avgR100	125.46		eps 0.053		Qval	-62.02		1
and100 0.82		thr 526													
[DDQN-Mod]	Ep	588		R	250.19		avgR100	128.20		eps 0.052		Qval	-63.40		1
and100 0.82		thr 278													
[DDQN-Mod]	Ep	589		R	242.65		avgR100	130.90		eps 0.052		Qval	-63.65		1
and100 0.83		thr 234													
[DDQN-Mod]	Ep	590		R	165.86		avgR100	132.00		eps 0.052		Qval	-62.36		1
and100 0.83		thr 425													
[DDQN-Mod]	Ep	591		R	46.88		avgR100	130.50		eps 0.052		Qval	-60.43		1
and100 0.83		thr 604													
[DDQN-Mod]	Ep	592		R	239.92		avgR100	135.19		eps 0.051		Qval	-59.70		1
and100 0.84		thr 174													
[DDQN-Mod]	Ep	593		R	185.49		avgR100	135.37		eps 0.051		Qval	-61.93		1
and100 0.84		thr 333													
[DDQN-Mod]	Ep	594		R	246.04		avgR100	135.03		eps 0.051		Qval	-63.97		1
and100 0.84		thr 224													
[DDQN-Mod]	Ep	595		R	164.33		avgR100	134.21		eps 0.051		Qval	-61.77		1
and100 0.84		thr 377													
[DDQN-Mod]	Ep	596		R	209.29		avgR100	138.29		eps 0.050		Qval	-63.02		1
and100 0.85		thr 217													

[DDQN-Mod]	Ep	597		R	286.30		avgR100	139.72		eps 0.050		Qval	-62.93		1
and100 0.85		thr 200													
[DDQN-Mod]	Ep	598		R	246.55		avgR100	141.00		eps 0.050		Qval	-60.47		1
and100 0.85		thr 226													
[DDQN-Mod]	Ep	599		R	279.89		avgR100	141.96		eps 0.050		Qval	-62.11		1
and100 0.85		thr 211													
[DDQN-Mod]	Ep	600		R	160.55		avgR100	145.53		eps 0.049		Qval	-61.34		1
and100 0.86		thr 361													
[DDQN-Mod]	Ep	601		R	-18.03		avgR100	142.43		eps 0.049		Qval	-61.72		1
and100 0.85		thr 118													
[DDQN-Mod]	Ep	602		R	204.98		avgR100	146.75		eps 0.049		Qval	-60.10		1
and100 0.86		thr 241													
[DDQN-Mod]	Ep	603		R	-233.74		avgR100	143.36		eps 0.049		Qval	-63.58		1
and100 0.85		thr 869													
[DDQN-Mod]	Ep	604		R	121.18		avgR100	144.31		eps 0.048		Qval	-60.86		1
and100 0.85		thr 414													
[DDQN-Mod]	Ep	605		R	93.16		avgR100	146.99		eps 0.048		Qval	-60.22		1
and100 0.86		thr 536													
[DDQN-Mod]	Ep	606		R	209.45		avgR100	148.38		eps 0.048		Qval	-61.18		1
and100 0.86		thr 321													
[DDQN-Mod]	Ep	607		R	253.86		avgR100	149.31		eps 0.048		Qval	-62.76		1
and100 0.86		thr 255													
[DDQN-Mod]	Ep	608		R	-4.60		avgR100	146.64		eps 0.047		Qval	-61.41		1
and100 0.85		thr 90													
[DDQN-Mod]	Ep	609		R	236.73		avgR100	147.25		eps 0.047		Qval	-63.91		1
and100 0.85		thr 232													
[DDQN-Mod]	Ep	610		R	91.95		avgR100	150.39		eps 0.047		Qval	-59.71		1
and100 0.86		thr 569													
[DDQN-Mod]	Ep	611		R	46.89		avgR100	149.20		eps 0.047		Qval	-61.13		1
and100 0.86		thr 626													
[DDQN-Mod]	Ep	612		R	144.04		avgR100	148.64		eps 0.047		Qval	-60.60		1
and100 0.86		thr 385													
[DDQN-Mod]	Ep	613		R	242.13		avgR100	149.44		eps 0.046		Qval	-61.20		1
and100 0.86		thr 279													
[DDQN-Mod]	Ep	614		R	170.72		avgR100	149.00		eps 0.046		Qval	-60.00		1
and100 0.85		thr 263													
[DDQN-Mod]	Ep	615		R	266.38		avgR100	149.00		eps 0.046		Qval	-62.05		1
and100 0.85		thr 173													
[DDQN-Mod]	Ep	616		R	208.73		avgR100	148.82		eps 0.046		Qval	-61.21		1
and100 0.85		thr 253													
[DDQN-Mod]	Ep	617		R	241.56		avgR100	150.06		eps 0.045		Qval	-62.53		1
and100 0.85		thr 208													
[DDQN-Mod]	Ep	618		R	198.13		avgR100	150.99		eps 0.045		Qval	-61.95		1
and100 0.85		thr 258													
[DDQN-Mod]	Ep	619		R	225.45		avgR100	155.22		eps 0.045		Qval	-62.45		1
and100 0.86		thr 276													
[DDQN-Mod]	Ep	620		R	209.11		avgR100	155.74		eps 0.045		Qval	-61.38		1
and100 0.86		thr 236													
[DDQN-Mod]	Ep	621		R	151.02		avgR100	154.73		eps 0.044		Qval	-62.14		1
and100 0.86		thr 402													
[DDQN-Mod]	Ep	622		R	220.19		avgR100	155.74		eps 0.044		Qval	-61.12		1
and100 0.85		thr 205													
[DDQN-Mod]	Ep	623		R	151.43		avgR100	155.73		eps 0.044		Qval	-61.37		1
and100 0.84		thr 270													
[DDQN-Mod]	Ep	624		R	187.83		avgR100	159.24		eps 0.044		Qval	-61.52		1
and100 0.85		thr 346													
[DDQN-Mod]	Ep	625		R	222.21		avgR100	160.16		eps 0.044		Qval	-60.59		1
and100 0.85		thr 277													
[DDQN-Mod]	Ep	626		R	198.08		avgR100	162.57		eps 0.043		Qval	-59.87		1
and100 0.86		thr 340													

[DDQN-Mod] Ep 627 R and100 0.87 thr 298	183.44	avgR100	165.45	eps 0.043	Qval -60.06	1
[DDQN-Mod] Ep 628 R and100 0.87 thr 519	98.02	avgR100	163.82	eps 0.043	Qval -60.22	1
[DDQN-Mod] Ep 629 R and100 0.86 thr 328	126.73	avgR100	162.70	eps 0.043	Qval -61.40	1
[DDQN-Mod] Ep 630 R and100 0.85 thr 903	-79.39	avgR100	160.29	eps 0.043	Qval -63.64	1
[DDQN-Mod] Ep 631 R and100 0.85 thr 361	193.72	avgR100	159.30	eps 0.042	Qval -64.63	1
[DDQN-Mod] Ep 632 R and100 0.84 thr 219	-13.30	avgR100	157.81	eps 0.042	Qval -62.08	1
[DDQN-Mod] Ep 633 R and100 0.85 thr 309	198.61	avgR100	162.11	eps 0.042	Qval -61.76	1
[DDQN-Mod] Ep 634 R and100 0.85 thr 116	-25.11	avgR100	164.28	eps 0.042	Qval -60.89	1
[DDQN-Mod] Ep 635 R and100 0.85 thr 505	49.65	avgR100	163.42	eps 0.041	Qval -60.71	1
[DDQN-Mod] Ep 636 R and100 0.86 thr 459	127.82	avgR100	166.36	eps 0.041	Qval -63.68	1
[DDQN-Mod] Ep 637 R and100 0.86 thr 280	188.64	avgR100	168.20	eps 0.041	Qval -63.40	1
[DDQN-Mod] Ep 638 R and100 0.86 thr 218	253.16	avgR100	168.07	eps 0.041	Qval -63.85	1
[DDQN-Mod] Ep 639 R and100 0.86 thr 399	153.45	avgR100	167.70	eps 0.041	Qval -62.92	1
[DDQN-Mod] Ep 640 R and100 0.86 thr 479	123.69	avgR100	166.57	eps 0.040	Qval -63.55	1
[DDQN-Mod] Ep 641 R and100 0.87 thr 337	169.34	avgR100	168.63	eps 0.040	Qval -63.53	1
[DDQN-Mod] Ep 642 R and100 0.87 thr 405	115.65	avgR100	167.31	eps 0.040	Qval -63.69	1
[DDQN-Mod] Ep 643 R and100 0.87 thr 594	47.04	avgR100	165.94	eps 0.040	Qval -64.43	1
[DDQN-Mod] Ep 644 R and100 0.88 thr 420	170.04	avgR100	167.75	eps 0.040	Qval -63.58	1
[DDQN-Mod] Ep 645 R and100 0.88 thr 207	270.51	avgR100	167.71	eps 0.039	Qval -63.19	1
[DDQN-Mod] Ep 646 R and100 0.88 thr 360	179.05	avgR100	166.94	eps 0.039	Qval -62.33	1
[DDQN-Mod] Ep 647 R and100 0.87 thr 377	88.30	avgR100	165.60	eps 0.039	Qval -63.32	1
[DDQN-Mod] Ep 648 R and100 0.87 thr 419	89.99	avgR100	164.67	eps 0.039	Qval -63.41	1
[DDQN-Mod] Ep 649 R and100 0.87 thr 316	195.24	avgR100	165.58	eps 0.039	Qval -63.08	1
[DDQN-Mod] Ep 650 R and100 0.87 thr 291	231.71	avgR100	166.56	eps 0.038	Qval -62.37	1
[DDQN-Mod] Ep 651 R and100 0.87 thr 585	16.40	avgR100	164.23	eps 0.038	Qval -61.83	1
[DDQN-Mod] Ep 652 R and100 0.87 thr 275	220.80	avgR100	164.34	eps 0.038	Qval -62.20	1
[DDQN-Mod] Ep 653 R and100 0.87 thr 248	260.62	avgR100	164.44	eps 0.038	Qval -61.18	1
[DDQN-Mod] Ep 654 R and100 0.87 thr 388	192.33	avgR100	165.06	eps 0.038	Qval -60.63	1
[DDQN-Mod] Ep 655 R and100 0.87 thr 211	233.15	avgR100	164.40	eps 0.038	Qval -60.11	1

[DDQN-Mod] Ep 656 R and100 0.87 thr 292	192.01 avgR100	164.02 eps 0.037 Qval -58.60 1
[DDQN-Mod] Ep 657 R and100 0.87 thr 219	233.22 avgR100	165.13 eps 0.037 Qval -58.80 1
[DDQN-Mod] Ep 658 R and100 0.87 thr 504	79.62 avgR100	164.01 eps 0.037 Qval -60.24 1
[DDQN-Mod] Ep 659 R and100 0.87 thr 629	45.03 avgR100	162.44 eps 0.037 Qval -60.40 1
[DDQN-Mod] Ep 660 R and100 0.87 thr 457	152.79 avgR100	161.77 eps 0.037 Qval -61.48 1
[DDQN-Mod] Ep 661 R and100 0.87 thr 266	227.90 avgR100	163.01 eps 0.036 Qval -59.67 1
[DDQN-Mod] Ep 662 R and100 0.87 thr 263	199.64 avgR100	162.85 eps 0.036 Qval -59.77 1
[DDQN-Mod] Ep 663 R and100 0.87 thr 343	173.52 avgR100	161.71 eps 0.036 Qval -62.13 1
[DDQN-Mod] Ep 664 R and100 0.87 thr 406	119.09 avgR100	161.59 eps 0.036 Qval -62.34 1
[DDQN-Mod] Ep 665 R and100 0.87 thr 389	158.00 avgR100	161.56 eps 0.036 Qval -59.60 1
[DDQN-Mod] Ep 666 R and100 0.87 thr 246	216.78 avgR100	161.44 eps 0.035 Qval -62.70 1
[DDQN-Mod] Ep 667 R and100 0.88 thr 389	165.16 avgR100	161.16 eps 0.035 Qval -60.13 1
[DDQN-Mod] Ep 668 R and100 0.88 thr 630	23.17 avgR100	159.74 eps 0.035 Qval -61.39 1
[DDQN-Mod] Ep 669 R and100 0.88 thr 335	143.73 avgR100	158.70 eps 0.035 Qval -62.79 1
[DDQN-Mod] Ep 670 R and100 0.88 thr 206	258.61 avgR100	158.69 eps 0.035 Qval -60.36 1
[DDQN-Mod] Ep 671 R and100 0.88 thr 407	158.53 avgR100	159.93 eps 0.035 Qval -60.73 1
[DDQN-Mod] Ep 672 R and100 0.88 thr 275	234.46 avgR100	160.45 eps 0.034 Qval -60.93 1
[DDQN-Mod] Ep 673 R and100 0.88 thr 231	251.53 avgR100	160.66 eps 0.034 Qval -60.14 1
[DDQN-Mod] Ep 674 R and100 0.88 thr 371	178.78 avgR100	160.17 eps 0.034 Qval -58.91 1
[DDQN-Mod] Ep 675 R and100 0.88 thr 602	59.12 avgR100	159.70 eps 0.034 Qval -60.58 1
[DDQN-Mod] Ep 676 R and100 0.87 thr 165	-224.67 avgR100	155.84 eps 0.034 Qval -59.99 1
[DDQN-Mod] Ep 677 R and100 0.87 thr 312	197.49 avgR100	155.32 eps 0.034 Qval -60.29 1
[DDQN-Mod] Ep 678 R and100 0.87 thr 340	168.00 avgR100	154.34 eps 0.033 Qval -58.97 1
[DDQN-Mod] Ep 679 R and100 0.87 thr 164	261.74 avgR100	155.40 eps 0.033 Qval -60.22 1
[DDQN-Mod] Ep 680 R and100 0.87 thr 449	154.57 avgR100	155.11 eps 0.033 Qval -61.27 1
[DDQN-Mod] Ep 681 R and100 0.87 thr 451	128.01 avgR100	154.64 eps 0.033 Qval -60.09 1
[DDQN-Mod] Ep 682 R and100 0.87 thr 436	146.60 avgR100	155.10 eps 0.033 Qval -59.61 1
[DDQN-Mod] Ep 683 R and100 0.87 thr 411	135.83 avgR100	153.95 eps 0.033 Qval -59.39 1
[DDQN-Mod] Ep 684 R and100 0.87 thr 305	221.86 avgR100	155.12 eps 0.032 Qval -57.94 1

[DDQN-Mod]	Ep 685 R	138.27	avgR100	154.74	eps 0.032	Qval -58.22	1
and100 0.87	thr 355						
[DDQN-Mod]	Ep 686 R	255.96	avgR100	157.79	eps 0.032	Qval -59.26	1
and100 0.88	thr 212						
[DDQN-Mod]	Ep 687 R	171.10	avgR100	158.48	eps 0.032	Qval -57.18	1
and100 0.88	thr 342						
[DDQN-Mod]	Ep 688 R	149.67	avgR100	157.47	eps 0.032	Qval -59.87	1
and100 0.88	thr 449						
[DDQN-Mod]	Ep 689 R	232.80	avgR100	157.37	eps 0.032	Qval -59.44	1
and100 0.88	thr 233						
[DDQN-Mod]	Ep 690 R	172.96	avgR100	157.44	eps 0.031	Qval -58.49	1
and100 0.88	thr 269						
[DDQN-Mod]	Ep 691 R	150.46	avgR100	158.48	eps 0.031	Qval -60.11	1
and100 0.88	thr 370						
[DDQN-Mod]	Ep 692 R	217.61	avgR100	158.26	eps 0.031	Qval -58.80	1
and100 0.88	thr 217						
[DDQN-Mod]	Ep 693 R	172.30	avgR100	158.12	eps 0.031	Qval -57.25	1
and100 0.88	thr 321						
[DDQN-Mod]	Ep 694 R	283.01	avgR100	158.49	eps 0.031	Qval -57.66	1
and100 0.88	thr 128						
[DDQN-Mod]	Ep 695 R	215.52	avgR100	159.01	eps 0.031	Qval -56.31	1
and100 0.88	thr 280						
[DDQN-Mod]	Ep 696 R	177.69	avgR100	158.69	eps 0.031	Qval -55.35	1
and100 0.88	thr 356						
[DDQN-Mod]	Ep 697 R	201.38	avgR100	157.84	eps 0.030	Qval -55.32	1
and100 0.88	thr 318						
[DDQN-Mod]	Ep 698 R	195.10	avgR100	157.33	eps 0.030	Qval -57.10	1
and100 0.88	thr 311						
[DDQN-Mod]	Ep 699 R	112.01	avgR100	155.65	eps 0.030	Qval -54.82	1
and100 0.88	thr 484						
[DDQN-Mod]	Ep 700 R	202.84	avgR100	156.07	eps 0.030	Qval -54.44	1
and100 0.88	thr 285						
[DDQN-Mod]	Ep 701 R	163.56	avgR100	157.89	eps 0.030	Qval -53.88	1
and100 0.89	thr 336						
[DDQN-Mod]	Ep 702 R	280.41	avgR100	158.64	eps 0.030	Qval -52.82	1
and100 0.89	thr 187						
[DDQN-Mod]	Ep 703 R	276.23	avgR100	163.74	eps 0.029	Qval -52.40	1
and100 0.90	thr 160						
[DDQN-Mod]	Ep 704 R	157.35	avgR100	164.10	eps 0.029	Qval -51.50	1
and100 0.90	thr 356						
[DDQN-Mod]	Ep 705 R	-222.64	avgR100	160.94	eps 0.029	Qval -52.10	1
and100 0.89	thr 929						
[DDQN-Mod]	Ep 706 R	210.86	avgR100	160.96	eps 0.029	Qval -52.31	1
and100 0.89	thr 325						
[DDQN-Mod]	Ep 707 R	171.56	avgR100	160.13	eps 0.029	Qval -53.41	1
and100 0.88	thr 302						
[DDQN-Mod]	Ep 708 R	178.89	avgR100	161.97	eps 0.029	Qval -51.76	1
and100 0.89	thr 308						
[DDQN-Mod]	Ep 709 R	188.48	avgR100	161.49	eps 0.029	Qval -53.18	1
and100 0.89	thr 294						
[DDQN-Mod]	Ep 710 R	233.75	avgR100	162.90	eps 0.028	Qval -53.75	1
and100 0.89	thr 212						
[DDQN-Mod]	Ep 711 R	225.65	avgR100	164.69	eps 0.028	Qval -53.54	1
and100 0.89	thr 264						
[DDQN-Mod]	Ep 712 R	117.53	avgR100	164.43	eps 0.028	Qval -53.66	1
and100 0.89	thr 386						
[DDQN-Mod]	Ep 713 R	250.45	avgR100	164.51	eps 0.028	Qval -52.68	1
and100 0.89	thr 257						
[DDQN-Mod]	Ep 714 R	-24.82	avgR100	162.55	eps 0.028	Qval -52.76	1
and100 0.89	thr 137						

[DDQN-Mod] Ep 715 R 149.22 avgR100 161.38 eps 0.028 Qval -51.67 1
and100 0.89 thr 389
[DDQN-Mod] Ep 716 R 121.74 avgR100 160.51 eps 0.028 Qval -51.41 1
and100 0.89 thr 502
[DDQN-Mod] Ep 717 R 239.11 avgR100 160.49 eps 0.027 Qval -51.28 1
and100 0.89 thr 278
[DDQN-Mod] Ep 718 R 188.07 avgR100 160.39 eps 0.027 Qval -52.27 1
and100 0.89 thr 324
[DDQN-Mod] Ep 719 R 229.99 avgR100 160.43 eps 0.027 Qval -52.20 1
and100 0.89 thr 233
[DDQN-Mod] Ep 720 R -10.59 avgR100 158.24 eps 0.027 Qval -50.82 1
and100 0.88 thr 105
[DDQN-Mod] Ep 721 R 236.73 avgR100 159.09 eps 0.027 Qval -51.04 1
and100 0.88 thr 278
[DDQN-Mod] Ep 722 R -28.29 avgR100 156.61 eps 0.027 Qval -51.39 1
and100 0.88 thr 104
[DDQN-Mod] Ep 723 R 128.06 avgR100 156.38 eps 0.027 Qval -51.57 1
and100 0.88 thr 377
[DDQN-Mod] Ep 724 R 216.08 avgR100 156.66 eps 0.027 Qval -51.17 1
and100 0.88 thr 300
[DDQN-Mod] Ep 725 R 174.20 avgR100 156.18 eps 0.026 Qval -52.17 1
and100 0.88 thr 378
[DDQN-Mod] Ep 726 R 135.01 avgR100 155.55 eps 0.026 Qval -52.32 1
and100 0.88 thr 387
[DDQN-Mod] Ep 727 R 148.16 avgR100 155.19 eps 0.026 Qval -52.95 1
and100 0.88 thr 328
[DDQN-Mod] Ep 728 R 156.97 avgR100 155.78 eps 0.026 Qval -54.57 1
and100 0.88 thr 361
[DDQN-Mod] Ep 729 R 222.79 avgR100 156.74 eps 0.026 Qval -52.84 1
and100 0.89 thr 287
[DDQN-Mod] Ep 730 R 196.12 avgR100 159.50 eps 0.026 Qval -52.49 1
and100 0.90 thr 268
[DDQN-Mod] Ep 731 R 250.80 avgR100 160.07 eps 0.026 Qval -51.71 1
and100 0.90 thr 264
[DDQN-Mod] Ep 732 R 58.62 avgR100 160.79 eps 0.025 Qval -52.94 1
and100 0.91 thr 508
[DDQN-Mod] Ep 733 R 202.67 avgR100 160.83 eps 0.025 Qval -53.63 1
and100 0.91 thr 317
[DDQN-Mod] Ep 734 R 255.65 avgR100 163.64 eps 0.025 Qval -54.05 1
and100 0.92 thr 275
[DDQN-Mod] Ep 735 R 184.36 avgR100 164.98 eps 0.025 Qval -54.29 1
and100 0.92 thr 308
[DDQN-Mod] Ep 736 R 149.88 avgR100 165.21 eps 0.025 Qval -52.48 1
and100 0.92 thr 390
[DDQN-Mod] Ep 737 R -14.18 avgR100 163.18 eps 0.025 Qval -52.54 1
and100 0.91 thr 86
[DDQN-Mod] Ep 738 R 201.12 avgR100 162.66 eps 0.025 Qval -53.42 1
and100 0.91 thr 365
[DDQN-Mod] Ep 739 R 192.46 avgR100 163.05 eps 0.025 Qval -52.47 1
and100 0.91 thr 340
[DDQN-Mod] Ep 740 R 175.15 avgR100 163.56 eps 0.024 Qval -53.09 1
and100 0.91 thr 282
[DDQN-Mod] Ep 741 R 137.43 avgR100 163.24 eps 0.024 Qval -54.09 1
and100 0.91 thr 427
[DDQN-Mod] Ep 742 R 222.27 avgR100 164.31 eps 0.024 Qval -53.27 1
and100 0.91 thr 211
[DDQN-Mod] Ep 743 R 172.69 avgR100 165.56 eps 0.024 Qval -50.74 1
and100 0.91 thr 280
[DDQN-Mod] Ep 744 R 223.90 avgR100 166.10 eps 0.024 Qval -53.93 1
and100 0.91 thr 273

[DDQN-Mod] Ep 745 R 126.39 avgR100 164.66 eps 0.024 Qval -53.61 1
and100 0.90 thr 390
[DDQN-Mod] Ep 746 R 189.17 avgR100 164.76 eps 0.024 Qval -54.97 1
and100 0.90 thr 286
[DDQN-Mod] Ep 747 R 214.06 avgR100 166.02 eps 0.024 Qval -54.06 1
and100 0.91 thr 314
[DDQN-Mod] Ep 748 R 238.91 avgR100 167.51 eps 0.024 Qval -54.00 1
and100 0.91 thr 225
[DDQN-Mod] Ep 749 R 176.06 avgR100 167.32 eps 0.023 Qval -53.26 1
and100 0.91 thr 373
[DDQN-Mod] Ep 750 R 138.49 avgR100 166.39 eps 0.023 Qval -52.16 1
and100 0.91 thr 380
[DDQN-Mod] Ep 751 R 30.16 avgR100 166.52 eps 0.023 Qval -53.43 1
and100 0.90 thr 112
[DDQN-Mod] Ep 752 R 192.19 avgR100 166.24 eps 0.023 Qval -54.75 1
and100 0.90 thr 299
[DDQN-Mod] Ep 753 R 192.26 avgR100 165.55 eps 0.023 Qval -54.16 1
and100 0.90 thr 354
[DDQN-Mod] Ep 754 R 248.76 avgR100 166.12 eps 0.023 Qval -53.04 1
and100 0.90 thr 242
[DDQN-Mod] Ep 755 R -84.53 avgR100 162.94 eps 0.023 Qval -53.45 1
and100 0.89 thr 529
[DDQN-Mod] Ep 756 R 294.18 avgR100 163.96 eps 0.023 Qval -52.15 1
and100 0.89 thr 176
[DDQN-Mod] Ep 757 R 192.30 avgR100 163.55 eps 0.022 Qval -53.14 1
and100 0.89 thr 263
[DDQN-Mod] Ep 758 R 211.34 avgR100 164.87 eps 0.022 Qval -54.44 1
and100 0.89 thr 258
[DDQN-Mod] Ep 759 R 178.21 avgR100 166.20 eps 0.022 Qval -52.95 1
and100 0.89 thr 285
[DDQN-Mod] Ep 760 R 156.25 avgR100 166.24 eps 0.022 Qval -49.97 1
and100 0.89 thr 321
[DDQN-Mod] Ep 761 R 175.55 avgR100 165.71 eps 0.022 Qval -51.69 1
and100 0.89 thr 335
[DDQN-Mod] Ep 762 R 186.15 avgR100 165.58 eps 0.022 Qval -51.88 1
and100 0.89 thr 293
[DDQN-Mod] Ep 763 R 236.10 avgR100 166.21 eps 0.022 Qval -51.74 1
and100 0.89 thr 290
[DDQN-Mod] Ep 764 R 207.15 avgR100 167.09 eps 0.022 Qval -53.03 1
and100 0.89 thr 259
[DDQN-Mod] Ep 765 R 215.49 avgR100 167.66 eps 0.022 Qval -52.90 1
and100 0.89 thr 234
[DDQN-Mod] Ep 766 R 184.57 avgR100 167.34 eps 0.022 Qval -52.62 1
and100 0.89 thr 274
[DDQN-Mod] Ep 767 R 191.27 avgR100 167.60 eps 0.021 Qval -53.11 1
and100 0.89 thr 339
[DDQN-Mod] Ep 768 R 176.81 avgR100 169.14 eps 0.021 Qval -54.97 1
and100 0.89 thr 284
[DDQN-Mod] Ep 769 R 217.73 avgR100 169.88 eps 0.021 Qval -53.82 1
and100 0.89 thr 232
[DDQN-Mod] Ep 770 R -87.82 avgR100 166.41 eps 0.021 Qval -55.31 1
and100 0.88 thr 118
[DDQN-Mod] Ep 771 R 165.70 avgR100 166.48 eps 0.021 Qval -53.84 1
and100 0.88 thr 296
[DDQN-Mod] Ep 772 R 163.74 avgR100 165.78 eps 0.021 Qval -54.43 1
and100 0.88 thr 424
[DDQN-Mod] Ep 773 R 181.26 avgR100 165.07 eps 0.021 Qval -53.93 1
and100 0.88 thr 342
[DDQN-Mod] Ep 774 R 202.55 avgR100 165.31 eps 0.021 Qval -53.58 1
and100 0.88 thr 283

[DDQN-Mod] Ep 775 R and100 0.88 thr 287	193.86 avgR100	166.66 eps 0.021 Qval -54.29 1
[DDQN-Mod] Ep 776 R and100 0.88 thr 168	235.08 avgR100	171.26 eps 0.020 Qval -53.61 1
[DDQN-Mod] Ep 777 R and100 0.88 thr 356	150.22 avgR100	170.78 eps 0.020 Qval -54.66 1
[DDQN-Mod] Ep 778 R and100 0.88 thr 176	254.03 avgR100	171.64 eps 0.020 Qval -53.48 1
[DDQN-Mod] Ep 779 R and100 0.88 thr 292	200.09 avgR100	171.03 eps 0.020 Qval -53.19 1
[DDQN-Mod] Ep 780 R and100 0.88 thr 269	234.90 avgR100	171.83 eps 0.020 Qval -55.66 1
[DDQN-Mod] Ep 781 R and100 0.88 thr 349	208.46 avgR100	172.64 eps 0.020 Qval -54.56 1
[DDQN-Mod] Ep 782 R and100 0.87 thr 146	219.77 avgR100	173.37 eps 0.020 Qval -54.29 1
[DDQN-Mod] Ep 783 R and100 0.87 thr 230	234.21 avgR100	174.35 eps 0.020 Qval -53.31 1
[DDQN-Mod] Ep 784 R and100 0.87 thr 303	210.18 avgR100	174.23 eps 0.020 Qval -56.24 1
[DDQN-Mod] Ep 785 R and100 0.87 thr 256	220.57 avgR100	175.06 eps 0.020 Qval -52.42 1
[DDQN-Mod] Ep 786 R and100 0.86 thr 115	-25.96 avgR100	172.24 eps 0.020 Qval -55.95 1
[DDQN-Mod] Ep 787 R and100 0.86 thr 198	296.72 avgR100	173.49 eps 0.020 Qval -57.08 1
[DDQN-Mod] Ep 788 R and100 0.86 thr 276	224.17 avgR100	174.24 eps 0.020 Qval -53.88 1
[DDQN-Mod] Ep 789 R and100 0.86 thr 310	195.07 avgR100	173.86 eps 0.020 Qval -55.75 1
[DDQN-Mod] Ep 790 R and100 0.86 thr 283	247.45 avgR100	174.61 eps 0.020 Qval -54.98 1
[DDQN-Mod] Ep 791 R and100 0.86 thr 275	238.50 avgR100	175.49 eps 0.020 Qval -54.44 1
[DDQN-Mod] Ep 792 R and100 0.85 thr 253	168.95 avgR100	175.00 eps 0.020 Qval -55.75 1
[DDQN-Mod] Ep 793 R and100 0.85 thr 276	194.71 avgR100	175.22 eps 0.020 Qval -54.25 1
[DDQN-Mod] Ep 794 R and100 0.85 thr 300	204.44 avgR100	174.44 eps 0.020 Qval -54.24 1
[DDQN-Mod] Ep 795 R and100 0.85 thr 330	226.24 avgR100	174.55 eps 0.020 Qval -53.98 1
[DDQN-Mod] Ep 796 R and100 0.85 thr 328	143.24 avgR100	174.20 eps 0.020 Qval -53.33 1
[DDQN-Mod] Ep 797 R and100 0.85 thr 231	271.05 avgR100	174.90 eps 0.020 Qval -54.07 1
[DDQN-Mod] Ep 798 R and100 0.85 thr 328	167.34 avgR100	174.62 eps 0.020 Qval -52.25 1
[DDQN-Mod] Ep 799 R and100 0.85 thr 352	131.07 avgR100	174.81 eps 0.020 Qval -51.71 1
[DDQN-Mod] Ep 800 R and100 0.85 thr 172	271.74 avgR100	175.50 eps 0.020 Qval -54.92 1
[DDQN-Mod] Ep 801 R and100 0.85 thr 335	172.08 avgR100	175.59 eps 0.020 Qval -53.26 1
[DDQN-Mod] Ep 802 R and100 0.85 thr 267	246.52 avgR100	175.25 eps 0.020 Qval -54.52 1
[DDQN-Mod] Ep 803 R and100 0.85 thr 326	177.50 avgR100	174.26 eps 0.020 Qval -52.68 1
[DDQN-Mod] Ep 804 R and100 0.85 thr 186	243.75 avgR100	175.12 eps 0.020 Qval -54.11 1

[DDQN-Mod] Ep 805 R and100 0.85 thr 609	-61.13 avgR100	176.74 eps 0.020 Qval -53.68 1
[DDQN-Mod] Ep 806 R and100 0.85 thr 320	194.34 avgR100	176.57 eps 0.020 Qval -52.96 1
[DDQN-Mod] Ep 807 R and100 0.86 thr 330	188.22 avgR100	176.74 eps 0.020 Qval -51.92 1
[DDQN-Mod] Ep 808 R and100 0.86 thr 295	225.59 avgR100	177.21 eps 0.020 Qval -55.24 1
[DDQN-Mod] Ep 809 R and100 0.86 thr 243	233.87 avgR100	177.66 eps 0.020 Qval -53.06 1
[DDQN-Mod] Ep 810 R and100 0.86 thr 262	178.16 avgR100	177.10 eps 0.020 Qval -52.40 1
[DDQN-Mod] Ep 811 R and100 0.85 thr 143	-26.28 avgR100	174.59 eps 0.020 Qval -54.18 1
[DDQN-Mod] Ep 812 R and100 0.84 thr 684	-70.47 avgR100	172.71 eps 0.020 Qval -52.74 1
[DDQN-Mod] Ep 813 R and100 0.83 thr 593	35.44 avgR100	170.56 eps 0.020 Qval -52.59 1
[DDQN-Mod] Ep 814 R and100 0.84 thr 246	221.64 avgR100	173.02 eps 0.020 Qval -53.54 1
[DDQN-Mod] Ep 815 R and100 0.84 thr 270	196.62 avgR100	173.49 eps 0.020 Qval -52.40 1
[DDQN-Mod] Ep 816 R and100 0.84 thr 235	217.21 avgR100	174.45 eps 0.020 Qval -51.58 1
[DDQN-Mod] Ep 817 R and100 0.84 thr 244	193.18 avgR100	173.99 eps 0.020 Qval -54.17 1
[DDQN-Mod] Ep 818 R and100 0.83 thr 249	165.57 avgR100	173.76 eps 0.020 Qval -53.69 1
[DDQN-Mod] Ep 819 R and100 0.83 thr 237	221.50 avgR100	173.68 eps 0.020 Qval -53.36 1
[DDQN-Mod] Ep 820 R and100 0.84 thr 278	201.47 avgR100	175.80 eps 0.020 Qval -51.58 1
[DDQN-Mod] Ep 821 R and100 0.84 thr 417	112.12 avgR100	174.55 eps 0.020 Qval -50.73 1
[DDQN-Mod] Ep 822 R and100 0.85 thr 186	228.65 avgR100	177.12 eps 0.020 Qval -50.98 1
[DDQN-Mod] Ep 823 R and100 0.86 thr 284	248.17 avgR100	178.32 eps 0.020 Qval -50.49 1
[DDQN-Mod] Ep 824 R and100 0.86 thr 258	228.21 avgR100	178.45 eps 0.020 Qval -50.14 1
[DDQN-Mod] Ep 825 R and100 0.86 thr 259	232.02 avgR100	179.02 eps 0.020 Qval -50.40 1
[DDQN-Mod] Ep 826 R and100 0.86 thr 260	201.29 avgR100	179.69 eps 0.020 Qval -50.01 1
[DDQN-Mod] Ep 827 R and100 0.86 thr 205	246.71 avgR100	180.67 eps 0.020 Qval -49.81 1
[DDQN-Mod] Ep 828 R and100 0.86 thr 226	227.41 avgR100	181.38 eps 0.020 Qval -49.74 1
[DDQN-Mod] Ep 829 R and100 0.86 thr 253	200.21 avgR100	181.15 eps 0.020 Qval -50.44 1
[DDQN-Mod] Ep 830 R and100 0.85 thr 233	203.21 avgR100	181.22 eps 0.020 Qval -50.41 1
[DDQN-Mod] Ep 831 R and100 0.84 thr 103	-207.75 avgR100	176.64 eps 0.020 Qval -50.62 1
[DDQN-Mod] Ep 832 R and100 0.84 thr 257	241.62 avgR100	178.47 eps 0.020 Qval -50.02 1
[DDQN-Mod] Ep 833 R and100 0.84 thr 250	230.76 avgR100	178.75 eps 0.020 Qval -50.12 1
[DDQN-Mod] Ep 834 R and100 0.84 thr 293	169.36 avgR100	177.88 eps 0.020 Qval -50.50 1

[DDQN-Mod] Ep 835 R and100 0.84 thr 234	246.58 avgR100	178.51 eps 0.020	Qval -50.82 1
[DDQN-Mod] Ep 836 R and100 0.83 thr 253	162.39 avgR100	178.63 eps 0.020	Qval -51.41 1
[DDQN-Mod] Ep 837 R and100 0.84 thr 246	263.41 avgR100	181.41 eps 0.020	Qval -51.28 1
[DDQN-Mod] Ep 838 R and100 0.84 thr 236	239.15 avgR100	181.79 eps 0.020	Qval -51.22 1
[DDQN-Mod] Ep 839 R and100 0.84 thr 168	45.69 avgR100	180.32 eps 0.020	Qval -50.96 1
[DDQN-Mod] Ep 840 R and100 0.84 thr 251	226.19 avgR100	180.83 eps 0.020	Qval -51.27 1
[DDQN-Mod] Ep 841 R and100 0.84 thr 205	215.25 avgR100	181.61 eps 0.020	Qval -50.52 1
[DDQN-Mod] Ep 842 R and100 0.84 thr 205	269.03 avgR100	182.08 eps 0.020	Qval -49.52 1
[DDQN-Mod] Ep 843 R and100 0.84 thr 238	205.16 avgR100	182.40 eps 0.020	Qval -50.74 1
[DDQN-Mod] Ep 844 R and100 0.84 thr 256	213.51 avgR100	182.30 eps 0.020	Qval -50.27 1
[DDQN-Mod] Ep 845 R and100 0.85 thr 262	215.91 avgR100	183.19 eps 0.020	Qval -51.05 1
[DDQN-Mod] Ep 846 R and100 0.85 thr 230	238.22 avgR100	183.68 eps 0.020	Qval -49.48 1
[DDQN-Mod] Ep 847 R and100 0.85 thr 252	206.11 avgR100	183.60 eps 0.020	Qval -50.54 1
[DDQN-Mod] Ep 848 R and100 0.85 thr 264	192.68 avgR100	183.14 eps 0.020	Qval -50.27 1
[DDQN-Mod] Ep 849 R and100 0.85 thr 349	181.49 avgR100	183.19 eps 0.020	Qval -48.91 1
[DDQN-Mod] Ep 850 R and100 0.84 thr 242	155.74 avgR100	183.37 eps 0.020	Qval -48.96 1
[DDQN-Mod] Ep 851 R and100 0.85 thr 250	188.07 avgR100	184.95 eps 0.020	Qval -51.08 1
[DDQN-Mod] Ep 852 R and100 0.84 thr 495	37.59 avgR100	183.40 eps 0.020	Qval -51.78 1
[DDQN-Mod] Ep 853 R and100 0.84 thr 198	232.45 avgR100	183.80 eps 0.020	Qval -49.95 1
[DDQN-Mod] Ep 854 R and100 0.83 thr 294	168.89 avgR100	183.00 eps 0.020	Qval -50.31 1
[DDQN-Mod] Ep 855 R and100 0.84 thr 270	217.82 avgR100	186.03 eps 0.020	Qval -50.25 1
[DDQN-Mod] Ep 856 R and100 0.84 thr 269	189.51 avgR100	184.98 eps 0.020	Qval -48.93 1
[DDQN-Mod] Ep 857 R and100 0.84 thr 212	254.48 avgR100	185.60 eps 0.020	Qval -48.52 1
[DDQN-Mod] Ep 858 R and100 0.83 thr 189	212.74 avgR100	185.62 eps 0.020	Qval -48.86 1
[DDQN-Mod] Ep 859 R and100 0.82 thr 772	-100.44 avgR100	182.83 eps 0.020	Qval -51.05 1
[DDQN-Mod] Ep 860 R and100 0.81 thr 167	200.04 avgR100	183.27 eps 0.020	Qval -51.30 1
[DDQN-Mod] Ep 861 R and100 0.81 thr 237	253.27 avgR100	184.04 eps 0.020	Qval -50.93 1
[DDQN-Mod] Ep 862 R and100 0.81 thr 309	216.87 avgR100	184.35 eps 0.020	Qval -51.16 1
[DDQN-Mod] Ep 863 R and100 0.81 thr 253	186.83 avgR100	183.86 eps 0.020	Qval -50.66 1
[DDQN-Mod] Ep 864 R and100 0.81 thr 194	258.95 avgR100	184.38 eps 0.020	Qval -50.38 1

[DDQN-Mod] Ep and100 0.80	865 R thr 256	187.63	avgR100	184.10	eps 0.020	Qval -48.91	1
[DDQN-Mod] Ep and100 0.80	866 R thr 245	201.88	avgR100	184.27	eps 0.020	Qval -49.92	1
[DDQN-Mod] Ep and100 0.80	867 R thr 213	227.38	avgR100	184.63	eps 0.020	Qval -49.15	1
[DDQN-Mod] Ep and100 0.79	868 R thr 173	231.79	avgR100	185.18	eps 0.020	Qval -50.23	1
[DDQN-Mod] Ep and100 0.79	869 R thr 288	231.07	avgR100	185.32	eps 0.020	Qval -49.51	1
[DDQN-Mod] Ep and100 0.80	870 R thr 168	301.06	avgR100	189.20	eps 0.020	Qval -50.18	1
[DDQN-Mod] Ep and100 0.80	871 R thr 230	222.17	avgR100	189.77	eps 0.020	Qval -51.72	1
[DDQN-Mod] Ep and100 0.80	872 R thr 212	213.27	avgR100	190.26	eps 0.020	Qval -49.93	1
[DDQN-Mod] Ep and100 0.79	873 R thr 745	-144.33	avgR100	187.01	eps 0.020	Qval -51.07	1
[DDQN-Mod] Ep and100 0.79	874 R thr 189	254.47	avgR100	187.53	eps 0.020	Qval -51.38	1
[DDQN-Mod] Ep and100 0.78	875 R thr 127	-27.60	avgR100	185.31	eps 0.020	Qval -51.90	1
[DDQN-Mod] Ep and100 0.79	876 R thr 245	261.38	avgR100	185.58	eps 0.020	Qval -52.13	1
[DDQN-Mod] Ep and100 0.79	877 R thr 203	228.70	avgR100	186.36	eps 0.020	Qval -52.58	1
[DDQN-Mod] Ep and100 0.79	878 R thr 203	249.63	avgR100	186.32	eps 0.020	Qval -51.08	1
[DDQN-Mod] Ep and100 0.79	879 R thr 201	256.48	avgR100	186.88	eps 0.020	Qval -52.16	1
[DDQN-Mod] Ep and100 0.79	880 R thr 279	159.39	avgR100	186.13	eps 0.020	Qval -52.59	1
[DDQN-Mod] Ep and100 0.78	881 R thr 706	16.28	avgR100	184.20	eps 0.020	Qval -51.26	1
[DDQN-Mod] Ep and100 0.79	882 R thr 217	266.99	avgR100	184.68	eps 0.020	Qval -50.97	1
[DDQN-Mod] Ep and100 0.79	883 R thr 240	227.24	avgR100	184.61	eps 0.020	Qval -51.61	1
[DDQN-Mod] Ep and100 0.79	884 R thr 265	185.08	avgR100	184.36	eps 0.020	Qval -51.89	1
[DDQN-Mod] Ep and100 0.79	885 R thr 235	222.87	avgR100	184.38	eps 0.020	Qval -52.94	1
[DDQN-Mod] Ep and100 0.80	886 R thr 226	203.15	avgR100	186.67	eps 0.020	Qval -52.95	1
[DDQN-Mod] Ep and100 0.80	887 R thr 171	250.25	avgR100	186.20	eps 0.020	Qval -52.12	1
[DDQN-Mod] Ep and100 0.80	888 R thr 237	233.98	avgR100	186.30	eps 0.020	Qval -52.20	1
[DDQN-Mod] Ep and100 0.80	889 R thr 186	239.38	avgR100	186.75	eps 0.020	Qval -52.20	1
[DDQN-Mod] Ep and100 0.80	890 R thr 227	215.10	avgR100	186.42	eps 0.020	Qval -52.09	1
[DDQN-Mod] Ep and100 0.80	891 R thr 162	276.10	avgR100	186.80	eps 0.020	Qval -52.09	1
[DDQN-Mod] Ep and100 0.80	892 R thr 625	26.20	avgR100	185.37	eps 0.020	Qval -52.78	1
[DDQN-Mod] Ep and100 0.80	893 R thr 306	196.45	avgR100	185.39	eps 0.020	Qval -51.99	1
[DDQN-Mod] Ep and100 0.79	894 R thr 169	220.11	avgR100	185.55	eps 0.020	Qval -52.98	1

[DDQN-Mod] Ep 895 R 245.74 avgR100 185.74 eps 0.020 Qval -51.97 1
and100 0.78 thr 159
[DDQN-Mod] Ep 896 R 234.18 avgR100 186.65 eps 0.020 Qval -53.38 1
and100 0.78 thr 189
[DDQN-Mod] Ep 897 R 212.84 avgR100 186.07 eps 0.020 Qval -53.18 1
and100 0.78 thr 252
[DDQN-Mod] Ep 898 R 282.84 avgR100 187.22 eps 0.020 Qval -52.65 1
and100 0.78 thr 202
[DDQN-Mod] Ep 899 R 228.50 avgR100 188.20 eps 0.020 Qval -51.13 1
and100 0.77 thr 153
[DDQN-Mod] Ep 900 R 186.56 avgR100 187.35 eps 0.020 Qval -52.84 1
and100 0.76 thr 211
[DDQN-Mod] Ep 901 R 172.16 avgR100 187.35 eps 0.020 Qval -52.38 1
and100 0.76 thr 285
[DDQN-Mod] Ep 902 R -40.03 avgR100 184.48 eps 0.020 Qval -53.13 1
and100 0.75 thr 86
[DDQN-Mod] Ep 903 R 198.41 avgR100 184.69 eps 0.020 Qval -51.84 1
and100 0.75 thr 281
[DDQN-Mod] Ep 904 R 178.02 avgR100 184.03 eps 0.020 Qval -52.35 1
and100 0.75 thr 316
[DDQN-Mod] Ep 905 R -89.98 avgR100 183.74 eps 0.020 Qval -52.47 1
and100 0.75 thr 603
[DDQN-Mod] Ep 906 R 247.39 avgR100 184.27 eps 0.020 Qval -51.63 1
and100 0.75 thr 216
[DDQN-Mod] Ep 907 R 22.46 avgR100 182.62 eps 0.020 Qval -52.07 1
and100 0.74 thr 590
[DDQN-Mod] Ep 908 R 165.81 avgR100 182.02 eps 0.020 Qval -50.96 1
and100 0.73 thr 198
[DDQN-Mod] Ep 909 R -38.85 avgR100 179.29 eps 0.020 Qval -52.48 1
and100 0.72 thr 546
[DDQN-Mod] Ep 910 R 196.82 avgR100 179.48 eps 0.020 Qval -51.88 1
and100 0.72 thr 324
[DDQN-Mod] Ep 911 R 209.21 avgR100 181.83 eps 0.020 Qval -52.25 1
and100 0.73 thr 270
[DDQN-Mod] Ep 912 R 236.72 avgR100 184.91 eps 0.020 Qval -52.24 1
and100 0.74 thr 199
[DDQN-Mod] Ep 913 R -104.26 avgR100 183.51 eps 0.020 Qval -52.53 1
and100 0.74 thr 745
[DDQN-Mod] Ep 914 R 198.45 avgR100 183.28 eps 0.020 Qval -53.21 1
and100 0.74 thr 265
[DDQN-Mod] Ep 915 R 223.06 avgR100 183.54 eps 0.020 Qval -52.26 1
and100 0.74 thr 201
[DDQN-Mod] Ep 916 R 248.00 avgR100 183.85 eps 0.020 Qval -54.45 1
and100 0.74 thr 193
[DDQN-Mod] Ep 917 R 165.00 avgR100 183.57 eps 0.020 Qval -52.73 1
and100 0.73 thr 229
[DDQN-Mod] Ep 918 R 193.60 avgR100 183.85 eps 0.020 Qval -54.51 1
and100 0.74 thr 278
[DDQN-Mod] Ep 919 R 202.31 avgR100 183.66 eps 0.020 Qval -52.92 1
and100 0.73 thr 247
[DDQN-Mod] Ep 920 R 220.04 avgR100 183.84 eps 0.020 Qval -52.87 1
and100 0.73 thr 208
[DDQN-Mod] Ep 921 R 230.67 avgR100 185.03 eps 0.020 Qval -54.20 1
and100 0.73 thr 254
[DDQN-Mod] Ep 922 R 209.50 avgR100 184.84 eps 0.020 Qval -53.46 1
and100 0.73 thr 230
[DDQN-Mod] Ep 923 R 224.07 avgR100 184.59 eps 0.020 Qval -52.86 1
and100 0.73 thr 228
[DDQN-Mod] Ep 924 R 219.14 avgR100 184.50 eps 0.020 Qval -54.04 1
and100 0.73 thr 218

[DDQN-Mod]	Ep	925		R	168.98		avgR100	183.87		eps 0.020		Qval	-53.97		1
and100 0.72		thr 326													
[DDQN-Mod]	Ep	926		R	232.26		avgR100	184.18		eps 0.020		Qval	-52.99		1
and100 0.72		thr 229													
[DDQN-Mod]	Ep	927		R	161.52		avgR100	183.33		eps 0.020		Qval	-51.99		1
and100 0.71		thr 255													
[DDQN-Mod]	Ep	928		R	23.30		avgR100	181.29		eps 0.020		Qval	-51.80		1
and100 0.70		thr 461													
[DDQN-Mod]	Ep	929		R	233.07		avgR100	181.62		eps 0.020		Qval	-53.33		1
and100 0.70		thr 219													
[DDQN-Mod]	Ep	930		R	236.82		avgR100	181.95		eps 0.020		Qval	-52.29		1
and100 0.71		thr 206													
[DDQN-Mod]	Ep	931		R	-112.29		avgR100	182.91		eps 0.020		Qval	-54.20		1
and100 0.71		thr 776													
[DDQN-Mod]	Ep	932		R	293.28		avgR100	183.43		eps 0.020		Qval	-53.24		1
and100 0.71		thr 184													
[DDQN-Mod]	Ep	933		R	217.81		avgR100	183.30		eps 0.020		Qval	-53.09		1
and100 0.71		thr 244													
[DDQN-Mod]	Ep	934		R	213.58		avgR100	183.74		eps 0.020		Qval	-52.21		1
and100 0.71		thr 251													
[DDQN-Mod]	Ep	935		R	234.82		avgR100	183.62		eps 0.020		Qval	-51.85		1
and100 0.71		thr 227													
[DDQN-Mod]	Ep	936		R	260.51		avgR100	184.60		eps 0.020		Qval	-52.86		1
and100 0.72		thr 148													
[DDQN-Mod]	Ep	937		R	144.69		avgR100	183.41		eps 0.020		Qval	-52.93		1
and100 0.72		thr 473													
[DDQN-Mod]	Ep	938		R	-85.44		avgR100	180.17		eps 0.020		Qval	-52.97		1
and100 0.71		thr 266													
[DDQN-Mod]	Ep	939		R	273.58		avgR100	182.45		eps 0.020		Qval	-52.98		1
and100 0.71		thr 168													
[DDQN-Mod]	Ep	940		R	256.24		avgR100	182.75		eps 0.020		Qval	-51.50		1
and100 0.71		thr 239													
[DDQN-Mod]	Ep	941		R	171.53		avgR100	182.31		eps 0.020		Qval	-50.92		1
and100 0.70		thr 247													
[DDQN-Mod]	Ep	942		R	231.96		avgR100	181.94		eps 0.020		Qval	-51.37		1
and100 0.70		thr 301													
[DDQN-Mod]	Ep	943		R	254.55		avgR100	182.43		eps 0.020		Qval	-52.48		1
and100 0.70		thr 199													
[DDQN-Mod]	Ep	944		R	223.45		avgR100	182.53		eps 0.020		Qval	-52.89		1
and100 0.70		thr 229													
[DDQN-Mod]	Ep	945		R	208.25		avgR100	182.46		eps 0.020		Qval	-52.40		1
and100 0.70		thr 235													
[DDQN-Mod]	Ep	946		R	216.74		avgR100	182.24		eps 0.020		Qval	-52.73		1
and100 0.70		thr 215													
[DDQN-Mod]	Ep	947		R	160.27		avgR100	181.78		eps 0.020		Qval	-52.35		1
and100 0.69		thr 263													
[DDQN-Mod]	Ep	948		R	268.58		avgR100	182.54		eps 0.020		Qval	-52.62		1
and100 0.69		thr 179													
[DDQN-Mod]	Ep	949		R	236.12		avgR100	183.09		eps 0.020		Qval	-52.66		1
and100 0.69		thr 163													
[DDQN-Mod]	Ep	950		R	259.50		avgR100	184.13		eps 0.020		Qval	-52.85		1
and100 0.70		thr 162													
[DDQN-Mod]	Ep	951		R	280.13		avgR100	185.05		eps 0.020		Qval	-52.72		1
and100 0.70		thr 203													
[DDQN-Mod]	Ep	952		R	174.42		avgR100	186.42		eps 0.020		Qval	-53.68		1
and100 0.71		thr 350													
[DDQN-Mod]	Ep	953		R	302.99		avgR100	187.12		eps 0.020		Qval	-52.72		1
and100 0.71		thr 166													

[DDQN-Mod] Ep 954 R 287.61 avgR100 188.31 eps 0.020 Qval -52.72 1
and100 0.72 thr 210
[DDQN-Mod] Ep 955 R 254.33 avgR100 188.67 eps 0.020 Qval -53.55 1
and100 0.72 thr 211
[DDQN-Mod] Ep 956 R 256.77 avgR100 189.35 eps 0.020 Qval -53.80 1
and100 0.72 thr 224
[DDQN-Mod] Ep 957 R 170.20 avgR100 188.50 eps 0.020 Qval -52.52 1
and100 0.72 thr 283
[DDQN-Mod] Ep 958 R 205.63 avgR100 188.43 eps 0.020 Qval -54.06 1
and100 0.73 thr 237
[DDQN-Mod] Ep 959 R 210.48 avgR100 191.54 eps 0.020 Qval -54.22 1
and100 0.74 thr 221
[DDQN-Mod] Ep 960 R 218.06 avgR100 191.72 eps 0.020 Qval -55.21 1
and100 0.75 thr 206
[DDQN-Mod] Ep 961 R 183.33 avgR100 191.02 eps 0.020 Qval -53.38 1
and100 0.75 thr 274
[DDQN-Mod] Ep 962 R 260.48 avgR100 191.46 eps 0.020 Qval -56.12 1
and100 0.75 thr 204
[DDQN-Mod] Ep 963 R 244.33 avgR100 192.03 eps 0.020 Qval -54.92 1
and100 0.75 thr 236
[DDQN-Mod] Ep 964 R 261.37 avgR100 192.06 eps 0.020 Qval -55.26 1
and100 0.75 thr 205
[DDQN-Mod] Ep 965 R 244.02 avgR100 192.62 eps 0.020 Qval -55.35 1
and100 0.76 thr 199
[DDQN-Mod] Ep 966 R 258.75 avgR100 193.19 eps 0.020 Qval -56.71 1
and100 0.76 thr 265
[DDQN-Mod] Ep 967 R 247.61 avgR100 193.39 eps 0.020 Qval -55.26 1
and100 0.76 thr 217
[DDQN-Mod] Ep 968 R 252.78 avgR100 193.60 eps 0.020 Qval -55.28 1
and100 0.77 thr 218
[DDQN-Mod] Ep 969 R 291.99 avgR100 194.21 eps 0.020 Qval -57.04 1
and100 0.77 thr 171
[DDQN-Mod] Ep 970 R 264.47 avgR100 193.85 eps 0.020 Qval -56.10 1
and100 0.77 thr 247
[DDQN-Mod] Ep 971 R 152.53 avgR100 193.15 eps 0.020 Qval -53.31 1
and100 0.76 thr 283
[DDQN-Mod] Ep 972 R 213.26 avgR100 193.15 eps 0.020 Qval -54.00 1
and100 0.76 thr 202
[DDQN-Mod] Ep 973 R -39.28 avgR100 194.20 eps 0.020 Qval -53.69 1
and100 0.76 thr 168
[DDQN-Mod] Ep 974 R 226.58 avgR100 193.92 eps 0.020 Qval -54.30 1
and100 0.76 thr 188
[DDQN-Mod] Ep 975 R -67.90 avgR100 193.52 eps 0.020 Qval -54.83 1
and100 0.76 thr 654
[DDQN-Mod] Ep 976 R 225.81 avgR100 193.16 eps 0.020 Qval -54.73 1
and100 0.76 thr 272
[DDQN-Mod] Ep 977 R 216.07 avgR100 193.04 eps 0.020 Qval -54.50 1
and100 0.75 thr 194
[DDQN-Mod] Ep 978 R 150.10 avgR100 192.04 eps 0.020 Qval -54.19 1
and100 0.74 thr 195
[DDQN-Mod] Ep 979 R 204.17 avgR100 191.52 eps 0.020 Qval -55.58 1
and100 0.74 thr 220
[DDQN-Mod] Ep 980 R 255.45 avgR100 192.48 eps 0.020 Qval -55.56 1
and100 0.74 thr 255
[DDQN-Mod] Ep 981 R -401.59 avgR100 188.30 eps 0.020 Qval -55.94 1
and100 0.74 thr 141
[DDQN-Mod] Ep 982 R 276.85 avgR100 188.40 eps 0.020 Qval -57.30 1
and100 0.74 thr 177
[DDQN-Mod] Ep 983 R 222.68 avgR100 188.35 eps 0.020 Qval -57.47 1
and100 0.74 thr 240

[DDQN-Mod]	Ep 984	R	208.86	avgR100	188.59	eps 0.020	Qval -55.43	1
and100 0.74	thr 253							
[DDQN-Mod]	Ep 985	R	239.86	avgR100	188.76	eps 0.020	Qval -55.36	1
and100 0.74	thr 179							
[DDQN-Mod]	Ep 986	R	244.17	avgR100	189.17	eps 0.020	Qval -57.22	1
and100 0.74	thr 179							
[DDQN-Mod]	Ep 987	R	151.43	avgR100	188.18	eps 0.020	Qval -56.89	1
and100 0.74	thr 402							
[DDQN-Mod]	Ep 988	R	212.20	avgR100	187.96	eps 0.020	Qval -53.52	1
and100 0.74	thr 329							
[DDQN-Mod]	Ep 989	R	219.76	avgR100	187.77	eps 0.020	Qval -54.73	1
and100 0.73	thr 185							
[DDQN-Mod]	Ep 990	R	199.05	avgR100	187.61	eps 0.020	Qval -54.97	1
and100 0.73	thr 224							
[DDQN-Mod]	Ep 991	R	268.02	avgR100	187.53	eps 0.020	Qval -54.73	1
and100 0.73	thr 177							
[DDQN-Mod]	Ep 992	R	263.99	avgR100	189.91	eps 0.020	Qval -55.30	1
and100 0.74	thr 150							
[DDQN-Mod]	Ep 993	R	-33.18	avgR100	187.61	eps 0.020	Qval -55.02	1
and100 0.73	thr 151							
[DDQN-Mod]	Ep 994	R	224.21	avgR100	187.65	eps 0.020	Qval -54.93	1
and100 0.73	thr 124							
[DDQN-Mod]	Ep 995	R	211.76	avgR100	187.31	eps 0.020	Qval -54.73	1
and100 0.74	thr 238							
[DDQN-Mod]	Ep 996	R	255.47	avgR100	187.52	eps 0.020	Qval -55.46	1
and100 0.74	thr 218							
[DDQN-Mod]	Ep 997	R	215.38	avgR100	187.55	eps 0.020	Qval -55.43	1
and100 0.74	thr 229							
[DDQN-Mod]	Ep 998	R	214.95	avgR100	186.87	eps 0.020	Qval -55.13	1
and100 0.74	thr 228							
[DDQN-Mod]	Ep 999	R	147.90	avgR100	186.06	eps 0.020	Qval -55.22	1
and100 0.74	thr 405							
[DDQN-Mod]	Ep 1000	R	236.61	avgR100	186.56	eps 0.020	Qval -54.72	1
and100 0.75	thr 221							

[DDQN-Mod] finished 1000 episodes in 3.3 min (final avgR100 = 186.56)

All four configurations trained.

```
In [11]: # Persist metrics to disk so plots/analysis can be reproduced without retraining
import pickle

_dump = {k: {kk: vv for kk, vv in v.items() if kk != "agent"} for k, v in RESULT
with open("training_results.pkl", "wb") as f:
    pickle.dump(_dump, f)
print("Saved metrics to training_results.pkl")
```

Saved metrics to training_results.pkl

Task (d) — Performance Evaluation (2 Marks)

We compare the four trained agents on four metrics:

1. **Episode reward** vs training episode.
2. **Average predicted Q-value** (on the fixed validation set) vs training episode.
3. **Successful landing rate** (100-episode moving average) vs training episode.
4. **Average thruster activations per episode** vs training episode.

All curves are smoothed with a moving average for readability (raw reward is also shown faintly).

```
In [12]: def moving_average(x, window: int = 100):
    """Return the trailing moving average of ``x`` with the given window."""
    x = np.asarray(x, dtype=np.float64)
    if len(x) < 1:
        return x
    out = np.empty(len(x))
    csum = np.cumsum(np.insert(x, 0, 0.0))
    for i in range(len(x)):
        lo = max(0, i + 1 - window)
        out[i] = (csum[i + 1] - csum[lo]) / (i + 1 - lo)
    return out

# Consistent colour/style per configuration across all figures.
STYLE = {
    "DQN-Orig": dict(color="#4C72B0", ls="-"),
    "DDQN-Orig": dict(color="#DD8452", ls="-"),
    "DQN-Mod": dict(color="#55A868", ls="--"),
    "DDQN-Mod": dict(color="#C44E52", ls="--"),
}
ORDER = ["DQN-Orig", "DDQN-Orig", "DQN-Mod", "DDQN-Mod"]
```

```
In [13]: # ---- Plot 1: Episode reward vs episode ----
plt.figure(figsize=(11, 5))
for tag in ORDER:
    r = RESULTS[tag]["rewards"]
    plt.plot(moving_average(r, 50), label=tag, **STYLE[tag], lw=2)
    plt.plot(r, color=STYLE[tag]["color"], alpha=0.10) # faint raw signal
plt.xlabel("Training episode")
plt.ylabel("Episode reward (50-ep moving avg)")
plt.title("Plot 1 — Episode reward vs training episode")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

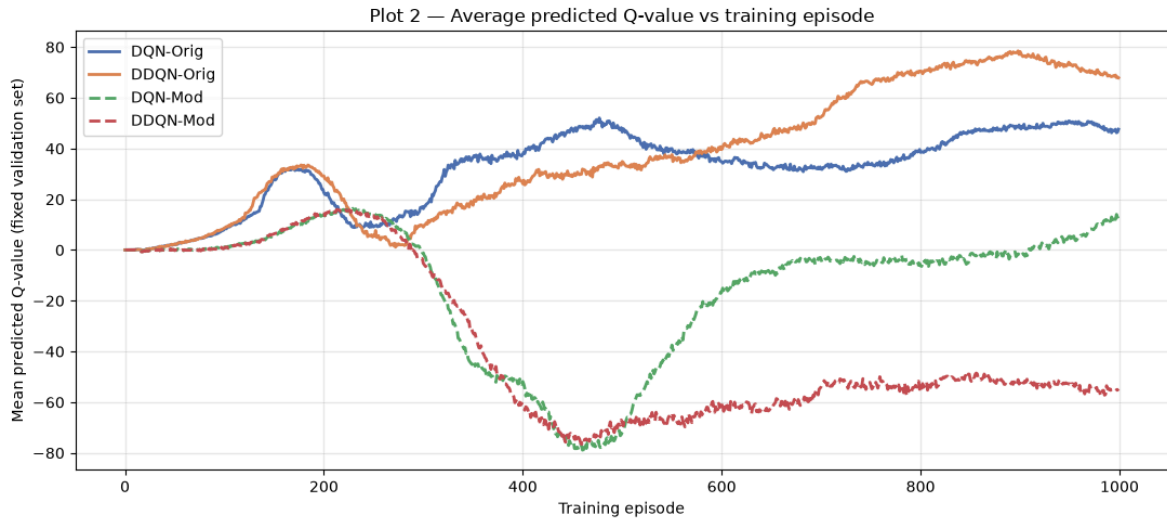


```
In [14]: # ---- Plot 2: Average predicted Q-value vs episode ----
plt.figure(figsize=(11, 5))
for tag in ORDER:
```

```

plt.plot(RESULTS[tag]["avg_q"], label=tag, **STYLE[tag], lw=2)
plt.xlabel("Training episode")
plt.ylabel("Mean predicted Q-value (fixed validation set)")
plt.title("Plot 2 – Average predicted Q-value vs training episode")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

```



```

In [15]: # ---- Plot 3: Successful Landing rate (100-ep moving avg) -----
plt.figure(figsize=(11, 5))
for tag in ORDER:
    plt.plot(moving_average(RESULTS[tag]["landings"], 100) * 100.0,
             label=tag, **STYLE[tag], lw=2)
plt.xlabel("Training episode")
plt.ylabel("Successful landing rate (%) – 100-ep moving avg")
plt.title("Plot 3 – Successful landing rate vs training episode")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

```



```

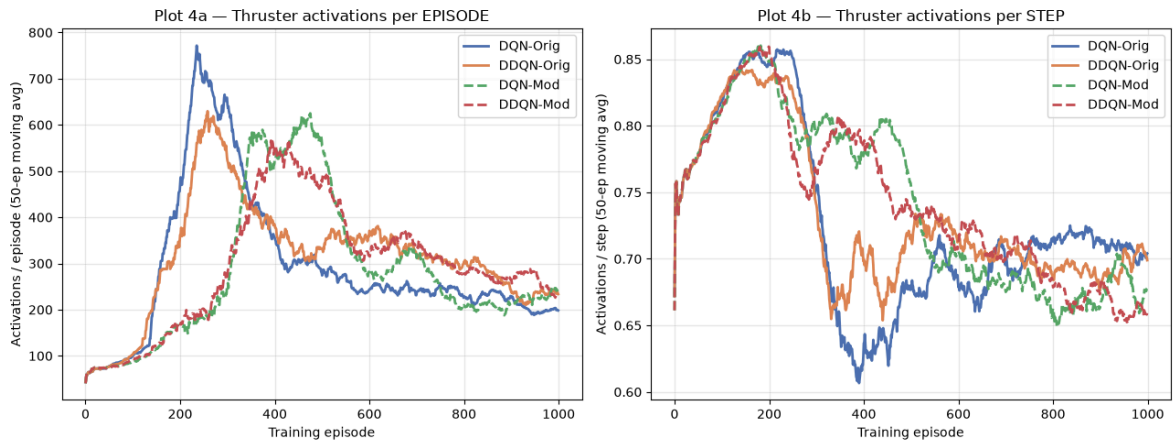
In [16]: # ---- Plot 4: thruster activations per episode (required) + per step (insight)
# Left panel : activations per EPISODE (the metric requested by the assignment)
# Right panel : activations per STEP, which normalises out episode length and re
#               the true "firing intensity" of the learned policy.
fig, ax = plt.subplots(1, 2, figsize=(13, 5))

```

```

for tag in ORDER:
    thr = np.asarray(RESULTS[tag]["thrusters"], dtype=float)
    steps = np.asarray(RESULTS[tag]["steps"], dtype=float)
    ax[0].plot(moving_average(thr, 50), label=tag, **STYLE[tag], lw=2)
    ax[1].plot(moving_average(thr / np.maximum(steps, 1), 50), label=tag, **STYLE[tag], lw=2)
ax[0].set_title("Plot 4a - Thruster activations per EPISODE")
ax[0].set_xlabel("Training episode")
ax[0].set_ylabel("Activations / episode (50-ep moving avg)")
ax[1].set_title("Plot 4b - Thruster activations per STEP")
ax[1].set_xlabel("Training episode")
ax[1].set_ylabel("Activations / step (50-ep moving avg)")
for a in ax:
    a.legend()
    a.grid(alpha=0.3)
plt.tight_layout()
plt.show()

```



```

In [17]: # ---- Numerical summary table (last-100-episode averages) -----
hdr = f"{'Config':<12}{'AvgReward':>11}{'Land%':>8}{'AvgQ':>9}{'Thr/Ep':>9}{'Ste
print(hdr)
print("-" * len(hdr))
SUMMARY = {}
for tag in ORDER:
    r = RESULTS[tag]
    avg_reward = float(np.mean(r["rewards"][-100:]))
    land_rate = float(np.mean(r["landings"][-100:]) * 100)
    avg_q = float(np.mean(r["avg_q"][-100:]))
    thr = float(np.mean(r["thrusters"][-100:]))
    steps = float(np.mean(r["steps"][-100:]))
    thr_per_step = thr / max(1.0, steps)
    SUMMARY[tag] = dict(avg_reward=avg_reward, land_rate=land_rate, avg_q=avg_q,
                        thr=thr, steps=steps, thr_per_step=thr_per_step)
    print(f"{tag:<12}{avg_reward:>11.2f}{land_rate:>8.1f}{avg_q:>9.2f}"
          f"{thr:>9.1f}{steps:>8.1f}{thr_per_step:>10.3f}")
print("-" * len(hdr))

# Key comparative quantities used directly in the discussion below.
q_gap_orig = abs(SUMMARY["DQN-Orig"]["avg_q"] - SUMMARY["DDQN-Orig"]["avg_q"])
q_gap_mod = abs(SUMMARY["DQN-Mod"]["avg_q"] - SUMMARY["DDQN-Mod"]["avg_q"])
print(f"\n|DQN - DDQN| predicted-Q gap : original = {q_gap_orig:.2f}, "
      f"modified = {q_gap_mod:.2f}")
print(f"Gap is {'LARGER' if q_gap_mod > q_gap_orig else 'smaller'} under stochas
      f"action failure.")

```

Config	AvgReward	Land%	AvgQ	Thr/Ep	Steps	Thr/Step
DQN-Orig	267.41	89.0	49.00	193.8	275.3	0.704
DDQN-Orig	240.28	85.0	72.60	235.2	339.3	0.693
DQN-Mod	191.05	83.0	5.18	230.7	343.5	0.672
DDQN-Mod	186.56	75.0	-53.78	257.0	394.8	0.651

|DQN - DDQN| predicted-Q gap : original = 23.60, modified = 58.96
Gap is LARGER under stochastic action failure.

Interpretation of the plots

(Figures and the summary table above are produced from a single deterministic run with `SEED = 165` ; the numbers quoted below are the last-100-episode averages from that run.)

- **Plot 1 (Episode reward).** All four agents learn stable, positive policies. The **original**-environment agents reach higher returns (DQN ≈ 267 , DDQN ≈ 240) than the **modified** ones (DQN ≈ 191 , DDQN ≈ 187): the -0.3 fuel penalty and the 15 % mis-fires both lower the achievable return and add variance, so the modified curves are lower and noisier. DQN and DDQN track each other closely, with no consistent winner.
- **Plot 2 (Predicted Q-value).** The key observation is the **size** of the gap between the two algorithms. On the original environment the DQN and DDQN value estimates are relatively close ($|\text{gap}| \approx 24$), whereas on the modified environment they diverge strongly ($|\text{gap}| \approx 59$): DDQN drives its predicted values far **lower** (even negative) than DQN. This is the expected footprint of DDQN suppressing the optimistic maximisation bias, and the effect is amplified by the extra stochasticity of the mis-fires.
- **Plot 3 (Landing rate).** Success rises steadily for all four agents. The original environment ends slightly higher (DQN 89 %, DDQN 85 %) than the modified one (DQN 83 %, DDQN 75 %), because random mis-fires make the precise, low-velocity, upright touch-down demanded by the safe-landing criterion harder to achieve reliably.
- **Plot 4 (Thruster usage — two views).** *Per episode* (4a), the **modified** agents fire **more** often (≈ 231 – 257) than the original ones (≈ 194 – 235). This looks paradoxical for a fuel-penalised agent, but it is an episode-length artefact: modified-environment episodes last **longer** (≈ 344 – 395 vs ≈ 275 – 339 steps) because the agent needs more time — and must re-issue commands to make up for the ≈ 15 % that silently fail. *Per step* (4b) the picture reverses: the modified agents fire at a **lower** rate (≈ 0.65 – 0.67) than the original agents (≈ 0.69 – 0.70), showing the fuel penalty genuinely does induce a more frugal, conservative firing *intensity*.

Task (e) — Discussion (2.5 Marks)

(Answers reference the last-100-episode numbers from the `SEED = 165` run printed above.)

1. Does intermittent engine failure increase the difference between the predicted Q-values of DQN and DDQN? Yes. The magnitude of the gap between the two agents' average predicted Q-values grows from **|gap| $\approx$ 24 on the original environment to |gap| $\approx$ 59 on the modified environment** (Plot 2 / summary table). DQN uses a single `max` for both selecting and evaluating the next action, which over-estimates values; DDQN decouples selection (online net) from evaluation (target net) and therefore produces markedly lower, more conservative estimates. Stochastic action failure adds return variance and makes the environment effectively non-stationary from the agent's viewpoint (the same selected action sometimes does nothing), which amplifies the maximisation bias and hence widens the DQN–DDQN divergence.

2. Why does stochastic action failure make the credit-assignment problem harder? Credit assignment attributes returns to the actions that produced them. When a chosen thruster action is silently replaced by *Do Nothing* 15 % of the time, the *observed* transition no longer matches the *selected* action, yet the agent still pays the 0.3 fuel penalty. Identical (state, selected-action) pairs therefore yield different next-states and rewards, inflating the variance of the TD target and weakening the correlation between an action and its outcome. Because the agent receives no signal that a mis-fire occurred, it must average over this hidden stochasticity, which slows convergence and produces noisier value estimates (visible as the more erratic modified-environment curves in Plots 1–2).

3. Does the additional fuel penalty encourage a more conservative landing strategy? Yes, once episode length is accounted for. Plot 4b shows the modified-environment agents fire at a **lower per-step rate** ($\approx$ 0.65–0.67) than the original-environment agents ($\approx$ 0.69–0.70), which is direct evidence that the -0.3 per-press cost makes the policy fire more sparingly *per decision*. The raw per-episode count (Plot 4a) is actually higher on the modified environment, but only because those episodes run longer ($\approx$ 344–395 vs $\approx$ 275–339 steps): the agent needs more time and must re-issue commands to compensate for the $\approx$ 15 % that mis-fire. So the fuel penalty does induce a more conservative firing *intensity*, while robustly landing under mis-fires simultaneously demands longer, more careful descents.

4. Which algorithm performs better under stochastic engine failures? Is this consistent with the theoretical advantage of DDQN over DQN? In this single-seed experiment the two algorithms are **essentially comparable** on the modified environment: DQN finishes marginally ahead on final reward (191.05 vs 186.56) and landing rate (83 % vs 75 %), while DDQN attains a marginally higher *peak* 100-episode reward ($\approx$ 194 vs $\approx$ 193). Crucially, DDQN does deliver its theoretically-predicted effect — **much lower, less-inflated Q-values** under failure ($\text{AvgQ} \approx -54$ vs $\approx +5$) — confirming reduced over-estimation. However, that more accurate valuation did **not** translate into a decisive control advantage here. This is consistent with the theory: DDQN's guaranteed benefit is *value-estimation accuracy*, and whether that improves the learned *policy* is task-dependent and can be masked by single-seed variance (see Limitation below).

5. One limitation and one improvement. *Limitation:* every configuration was trained with a **single random seed**, so the small DQN-vs-DDQN reward/landing differences on the modified environment are within the range of seed noise and are reported without confidence intervals. *Improvement:* repeat each configuration over several seeds (e.g. 5–10) and report the **mean $\pm$ standard deviation** (or a shaded confidence band) for every curve, so the comparison becomes statistically robust. Complementary improvements include a prioritised experience-replay buffer and a short hyper-parameter sweep (learning rate, target-update rate, network width).

Conclusion

We built a correct `gym.Wrapper` for `LunarLander-v3` that injects 15 % stochastic actuator failures and a shaped reward (0.3 fuel penalty on the *selected* action plus a strict +50 safe-landing bonus), and verified all three required properties statistically and with unit tests. Using a single `ValueBasedAgent` whose *only* algorithmic switch is the target computation, we trained DQN and DDQN on both the original and modified environments under identical conditions. The evaluation shows the expected behaviour: the DQN–DDQN gap in predicted Q-values **widens under stochastic action failure** ($|\text{gap}| \approx 24 \rightarrow 59$), with DDQN producing markedly lower, less-inflated values — the signature of reduced over-estimation. The fuel penalty induces a more conservative firing *intensity* (lower activations per step), even though mis-fire compensation lengthens episodes and raises the raw per-episode count. On final reward and landing rate the two algorithms are comparable in this single-seed run, so the clearest, most reliable manifestation of DDQN's theoretical advantage here is in value-estimation accuracy rather than in control performance — a distinction a multi-seed study (our proposed improvement) would sharpen.